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New

SURE SHOT QUESTIONS 2026

Chapter – 01 (Questions)

Relations and Functions

Questions

1. Is the function $F: \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) = x^2 + x$ injective, surjective or bijective?
2. If $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = 4x^3 + 7$, show that $f(x)$ is bijection.
3. Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^3 + 1$ for all $x \in \mathbb{R}$ is bijective.
4. Check whether the relation R defined on the set $A = \{1, 2, 3, 4, 5, 6\}$ as $R = \{(a, b) : b = a + 1\}$ is reflexive, symmetric or transitive.
5. Let $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$. Show that that $R = \{(a, b) : a, b \in A, |a - b| \text{ is divisible by } 4\}$ is an equivalence relation. Find the set of all elements related to 1. Also write the equivalence class.
6. Show that $f: \mathbb{N} \rightarrow \mathbb{N}$, given by $f(x) = \begin{cases} x + 1, & \text{if } x \text{ is odd} \\ x - 1, & \text{if } x \text{ is even} \end{cases}$ is one-one and onto.
7. A relation R is defined on $\mathbb{N} \times \mathbb{N}$ (where $\mathbb{N}$ is the set of natural numbers) as:
 $(a, b)R(c, d) \Leftrightarrow a - c = b - d$
Show that R is an equivalence relation.
8. A relation R is defined on $\mathbb{N} \times \mathbb{N}$ (where $\mathbb{N}$ is the set of natural numbers) as $(a, b)R(c, d) \Leftrightarrow \frac{a}{c} = \frac{b}{d}$.
Show that R is an equivalence relation.
9. Check whether the relation R defined on the set $A = \{1, 2, 3, 4, 5, 6\}$ as
 $R = \{(a, b) : b = a + 1\}$ is reflexive, symmetric or transitive.
10. Show that the relation S in the set $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$ given by
 $S = \{(a, b) : a, b \in \mathbb{Z} \mid a - b \text{ is divisible by } 3\}$ is an equivalence relation.
11. If $y = f(x) = \frac{x^2}{1+x^2}$, is the function one-one and onto provided $f: \mathbb{R} \rightarrow \mathbb{R}$?
12. Give an example of relation R on $A = \{1, 2, 3\}$ which is reflexive but neither symmetric nor transitive.
13. State the reason why the relation $R = \{(a, b) : a \leq b^2\}$ on the set $\mathbb{R}$ of real numbers is not reflexive.
14. Show that the relation R in the set $\mathbb{N} \times \mathbb{N}$ defined by
 $(a, b)R(c, d)$ if $a^2 + d^2 = b^2 + c^2 \forall a, b, c, d \in \mathbb{N}$, is an equivalence relation.
15. Show that the relation S defined on set $\mathbb{N} \times \mathbb{N}$ by $(a, b)S(c, d)$ if $a + d = b + c$ is an equivalence relation.
16. Let $\mathbb{N}$ denote the set of all natural numbers and R be the relation on $\mathbb{N} \times \mathbb{N}$ defined by
 $(a, b)R(c, d)$ if $ad(b + c) = bc(a + d)$. Show that R is an equivalence relation.
17. Prove that the relation R in the set $A = \{1, 2, 3, 4, 5\}$ given by
 $R = \{(a, b) : |a - b| \text{ is even}\}$ is an equivalence relation.
18. Let $A = \mathbb{R} - \{3\}$, $B = \mathbb{R} - \{1\}$. Let $f: A \rightarrow B$ defined by
 $f(x) = \frac{x-2}{x-3} \forall x \in A$. Then show that f is bijective.
19. Show that the function f in $A = \mathbb{R} - \{\frac{2}{3}\}$ defined as
 $f(x) = \frac{4x+3}{6x-4}$ is one-one and onto.
20. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by (i) $f(x) = x + |x|$ (ii) $f(x) = x + 1$. Determine whether or not f is onto.

21. Write the domain of the relation R defined on the set Z of integers as follows: $(a, b) \in R \Leftrightarrow a^2 + b^2 = 25$

22. Let $f: R \rightarrow R$ be defined by $f(x) = x^2 + 1$. Find the pre-image of 17 and (-3).

23. Show that $f: R^+ \rightarrow R^+$ defined by $f(x) = \frac{1}{2x}$ is bijective, where R^+ is the set of all non zero positive real number.

24. Let $A = \{x: x \in R, -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}\}$ and $B = \{y: y \in R, -1 \leq y \leq 1\}$. Show that the function $f: A \rightarrow B$ such that $f(x) = \sin x$ is bijective.

25. Show that the function $f: R \rightarrow R$ defined by $x^3 + x$ is a bijection.

26. Let N be the set of all natural numbers and R be a relation in N defined by $R = \{(a, b): a \text{ is a factor of } b\}$, then show that R is reflexive and transitive but not symmetric.

27. Classify the following functions as injective, surjective or bijective. (i) $f: R \rightarrow R$ defined by $f(x) = \sin x$ (ii) $f: R \rightarrow R$ defined by $f(x) = \sin^2 x + \cos^2 x$

28. If $A = \{1, 2, 3, 4\}$, define relations on A which have properties of being: (i) reflexive, transitive but not symmetric. (ii) symmetric but neither reflexive nor transitive. (iii) reflexive, symmetric and transitive.

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New

SURE SHOT QUESTIONS 2026

Chapter – 01 (Solutions)

Relations and Functions

Questions

1. .

Soln.

f is not injective, since

$$f(x) = f(y)$$

$$\Rightarrow x^2 + x = y^2 + y$$

$$\Rightarrow x^2 + x + \frac{1}{4} = y^2 + y + \frac{1}{4}$$

$$\Rightarrow \left(x + \frac{1}{2}\right)^2 = \left(y + \frac{1}{2}\right)^2$$

$$\Rightarrow x + \frac{1}{2} = \pm \left(y + \frac{1}{2}\right)$$

$$\Rightarrow x + \frac{1}{2} = y + \frac{1}{2}$$

$$\text{or } x + \frac{1}{2} = -y - \frac{1}{2}$$

Thus, $f(x) = f(y)$ does not give the unique solution $x = y$

but also gives $x = -y - 1$.

Thus, f is not injective.

f is not surjective, since clearly $f(x) = x^2 + x \geq 0$ for all $x \in \mathbb{Z}$.

Thus, negative integers do not have pre-images in $\mathbb{Z}$.

Therefore, f is not an onto function.

2. .Soln.

The given function is $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) = 4x^3 + 7$$

One – one: Given that

$$f(x) = 4x^3 + 7$$

$$\text{Let } f(x_1) = f(x_2), \forall x_1, x_2 \in \mathbb{R}$$

$$\Rightarrow 4x_1^3 + 7 = 4x_2^3 + 7$$

$$\Rightarrow 4x_1^3 = 4x_2^3$$

$$\Rightarrow x_1^3 - x_2^3 = 0$$

$$\Rightarrow (x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2) = 0$$

$$\Rightarrow \text{Either } x_1 - x_2 = 0 \quad \dots\dots(i)$$

$$\text{or } x_1^2 + x_1x_2 + x_2^2 = 0 \quad \dots\dots(ii)$$

But equation (ii) is not possible as $x_1, x_2 \in \mathbb{R}$

$$\therefore x_1 - x_2 = 0$$

$$\Rightarrow x_1 = x_2$$

$\therefore$ We have shown that

$$f(x_1) = f(x_2)$$

$$\Rightarrow x_1 = x_2, \forall x_1, x_2 \in \mathbb{R}$$

$\therefore f(x)$ is a one – one function.

3. .

Soln.

$f: \mathbb{R} \rightarrow \mathbb{R}$ is given by

$$f(x) = x^3 + 1$$

For injective:

$$\text{Let } f(x) = f(y)$$

$$\Rightarrow x^3 + 1 = y^3 + 1$$

$$\Rightarrow x^3 = y^3$$

$$\Rightarrow x = y$$

$$\text{Thus, } f(x) = f(y)$$

$$\Rightarrow x = y \forall x, y \in \mathbb{R}$$

For surjective:

$$\text{Let } y \in \mathbb{R}$$

$$\text{Then, } f(x) = y$$

$$\Rightarrow x^3 + 1 = y$$

$$\Rightarrow x^3 = y - 1$$

Thus, for any $y \in \mathbb{R}$,

$$x = (y - 1)^{1/3} \text{ is real number and } f(x) = f[(y - 1)^{1/3}]$$

$$= [(y - 1)^{1/3}]^3 + 1$$

$$= (y - 1) + 1 = y$$

$$\Rightarrow f \text{ is surjective.}$$

Therefore, f is bijective.

4. .Soln.

$$\text{Here, } R = \{(a, b) : b = a + 1\}$$

$$\therefore R = \{(a, a + 1) : a, a + 1 \in \{1, 2, 3, 4, 5, 6\}\}$$

$$\Rightarrow R = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$$

$$(i) \quad R \text{ is not reflexive as } (a, a) \notin R \forall a$$

$$(ii) \quad R \text{ is not symmetric as } (1, 2) \in R \text{ but } (2, 1) \notin R$$

R is not transitive as $(1, 2) \in R, (2, 3) \in R$ but

$$(1, 3) \notin R$$

5. Soln.

Given, $R = \{(a, b) : a, b \in A, |a - b| \text{ is divisible by } 4\}$

Reflexivity: For any $a \in A$

$|a - a| = 0$, which is divisible by 4

$$(a, a) \in R$$

So, R is reflexive.

Symmetry: Let $(a, b) \in R$

$$\Rightarrow |a - b| \text{ is divisible by } 4$$

$$\Rightarrow |b - a| \text{ is divisible by } 4$$

$$[\because |a - b| = |b - a|]$$

$$\Rightarrow (b, a) \in R$$

So, R is symmetric.

Transitive: Let $(a, b) \in R$ and $(b, c) \in R$

$$\Rightarrow |a - b| \text{ is divisible by } 4$$

$$\Rightarrow |a - b| = 4k$$

$$\therefore a - b = \pm 4k, k \in \mathbb{Z} \quad \text{.....(i)}$$

Also, $|b - c|$ is divisible by 4

$$\Rightarrow |b - c| = 4m$$

$$\therefore b - c = \pm 4m, m \in \mathbb{Z} \quad \text{.....(ii)}$$

Adding equations (i) and (ii)

$$a - b + b - c = \pm 4(k + m)$$

$$\Rightarrow a - c = \pm 4(k + m)$$

$$|a - c| \text{ is divisible by } 4,$$

$$\Rightarrow (a, c) \in R$$

So, R is transitive.

$\Rightarrow$ R is reflexive, symmetric and transitive.

$\therefore$ R is an equivalence relation.

Let x be an element of R such that $(x, 1) \in R$

Then $|x - 1|$ is divisible by 4

$$x - 1 = 0, 4, 8, 12, \dots$$

$$\Rightarrow x = 1, 5, 9 \quad (\because x \leq 12)$$

$\therefore$ Set of all elements of A which are related to 1 are $\{1, 5, 9\}$.

Equivalence class of 2 i.e.,

$$[2] = \{(a, 2) : a \in A, |a - 2| \text{ is divisible by } 4\}$$

$$\Rightarrow |a - 2| = 4k \text{ (k is whole number, } k \leq 3)$$

$$\Rightarrow a = 2, 6, 10$$

6. .

Soln.

Let $x, y \in R$ such that $f(x) = f(y)$

$$\therefore f(x) = f(y)$$

If x and y are odd, then

$$f(x) = f(y)$$

$$\Rightarrow x + 1 = y + 1$$

$$\Rightarrow x = y$$

If x and y are even, then

$$f(x) = f(y)$$

$$\Rightarrow x - 1 = y - 1$$

$$\Rightarrow x = y$$

If x is odd and y is even, then

$f(x) = x + 1$ is even and $f(y) = y + 1$ is odd.

$$\therefore x \neq y \Rightarrow f(x) \neq f(y)$$

Hence, $f : N \rightarrow N$ is one - one

$$\text{Also, } f(1) = 1 + 1 = 2$$

$$f(1) = 2 \quad (\because 1 \text{ is odd})$$

If x is odd number, then $\exists$ an even natural number,

$$x + 1 \in N \text{ such that,}$$

$$f(x + 1) = x + 1 - 1$$

$$= x$$

If x is even number, then there exist a odd natural

$$\text{number } x - 1 \in N \text{ such that,}$$

$$f(x - 1) = x - 1 + 1$$

$$= x$$

Hence for every $y \in N \exists x \in N$ such that $f(x) = y$, so f is onto.

Hence f is both one - one and onto.

7. Sol. For reflexive : Let $(a, b) R (a, b)$

$$\Leftrightarrow a - a = b - b \Leftrightarrow 0 = 0$$

$\therefore$ R is reflexive

For symmetric : Let $(a, b) R (c, d)$

$$\Rightarrow a - c = b - d \Rightarrow c - a = d - b$$

$$\Rightarrow (c, d) R (a, b) \therefore R \text{ is symmetric.}$$

For transitive : Let $(a, b) R (c, d)$ and $(c, d) R (e, f)$

$$\Rightarrow a - c = b - d$$

$$\text{and } c - e = d - f$$

Adding (i) and (ii), we get

$$a - c + c - e = b - d + d - f$$

$$\Rightarrow a - e = b - f \Rightarrow (a, b) R (e, f)$$

$\therefore$ R is transitive

So, R is an equivalence relation.

8. Sol. Reflexive: Let $(a, b) \in N \times N$. Then $ab =$

ba (By commutativity of multiplication of natural number)

$$\Rightarrow (a, b) R (a, b)$$

Thus, $(a, b) R (a, b)$ for all $(a, b) \in N \times N$

So, R is reflexive.

Symmetric : Let $(a, b), (c, d) \in N \times N$ such that $(a, b) R (c, d)$

$$\Rightarrow ad = bc \Rightarrow cb = da$$

(By commutativity of multiplication of natural numbers)

$$\Rightarrow (c, d) R (a, b)$$

Thus, $(a, b) R (c, d) \Rightarrow (c, d) R (a, b)$ for $(a, b), (c, d) \in N \times N$

So, R is symmetric.

Transitive : Let $(a, b), (c, d), (e, f) \in N \times N$ such that $(a, b) R (c, d)$ and $(c, d) R (e, f)$

$$\text{Now, } (a, b) R (c, d) \Rightarrow ad = bc \quad \dots(i)$$

$$\text{and } (c, d) R (e, f) \Rightarrow cf = de \quad \dots(ii)$$

Multiplying (i) and (ii), we get $ad \cdot cf = bc \cdot de$

$$\Rightarrow af = be \Rightarrow (a, b) R (e, f)$$

Thus, $(a, b) R (c, d)$ and $(c, d) R (e, f) \Rightarrow (a, b) R (e, f)$

So, R is transitive.

$\therefore R$ is an equivalence relation.

9. .

Soln. We have, $A = \{1, 2, 3, 4, 5, 6\}$ and a relation R on

A defined as $R = \{(a, b) : b = a + 1\}$

Reflexive: Let $(a, a) \in R$

$$\Rightarrow a = a + 1 \Rightarrow a - a = 1 \Rightarrow 0 = 1, \text{ which is not possible.}$$

$\therefore (a, a) \notin R \Rightarrow R$ is not reflexive.

Symmetric: Let $(a, b) \in R \Rightarrow b = a + 1 \dots(i)$

Now, if $(b, a) \in R$

$$\Rightarrow a = b + 1 \Rightarrow b = b + 1 + 1 \text{ (using (i))}$$

$$\Rightarrow b = b + 2 \Rightarrow b - b = 2 \Rightarrow 0 = 2,$$

which is not possible

$$\Rightarrow (b, a) \notin R \Rightarrow R \text{ is not symmetric.}$$

Transitive: Let $(a, b) \in R$ and $(b, c) \in R$

$$\Rightarrow b = a + 1 \text{ and } c = b + 1 \Rightarrow c = a + 1 + 1$$

$$\Rightarrow c = a + 2 \neq a + 1 \Rightarrow (a, c) \notin R \Rightarrow R \text{ is not transitive.}$$

10. .

Soln.

We have $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$

$$\therefore A = \{0, 1, 2, 3, \dots, 12\}$$

Also, $S = \{(a, b) : a, b \in \mathbb{Z}, |a - b| \text{ is divisible by } 3\}$

(i) Reflexive : For any $a \in A$,

$$|a - a| = 0, \text{ which is divisible by } 3$$

Thus, $(a, a) \in S \therefore S$ is reflexive.

(ii) Symmetric: Let $(a, b) \in S$

$$\Rightarrow |a - b| \text{ is divisible by } 3.$$

$$\Rightarrow |b - a| \text{ is divisible by } 3 \Rightarrow (b, a) \in S \text{ i.e.,}$$

$$(a, b) \in S \Rightarrow (b, a) \in S$$

$\therefore S$ is symmetric.

(iii) Transitive:

Let $(a, b) \in S$ and $(b, c) \in S$

$$\Rightarrow |a - b| \text{ is divisible by } 3 \text{ and } |b - c| \text{ is divisible by } 3.$$

$$\Rightarrow (a - b) = \pm 3k_1 \text{ and } (b - c) = \pm 3k_2, \forall k_1, k_2 \in \mathbb{N}$$

$$\Rightarrow (a - b) + (b - c) = \pm 3(k_1 + k_2)$$

$$\Rightarrow (a - c) = \pm 3(k_1 + k_2); \forall k_1, k_2 \in \mathbb{N}$$

$$\Rightarrow |a - c| \text{ is divisible by } 3 \Rightarrow (a, c) \in S \therefore S \text{ is}$$

Transitive.

Hence, S is an equivalence relation.

11. .

Soln.

$f(x)$ is not one-one (i.e., injective)

$$\text{As } f(-x) = \frac{(-x)^2}{1 + (-x)^2} = \frac{x^2}{1 + x^2} = f(x)$$

But $x \neq -x$

For the function to be an onto function, let

$$y = f(x)$$

$$\Rightarrow \frac{x^2}{1 + x^2} = y$$

$$\Rightarrow x^2 = \frac{y}{1 - y}$$

$$\Rightarrow \frac{y}{1 - y} \geq 0, y \neq 1$$

$$\Rightarrow 0 \leq y < 1$$

$\Rightarrow$ Range of $f(x)$ is not equal to its codomain.

Hence, it is not onto.

That is, the given function is neither one - one nor an onto function.

12. .Soln.

$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (3, 1)\}$

Clearly R is reflexive.

R is not symmetric since

$$(1, 2) \in R \text{ but } (2, 1) \notin R$$

$$(2, 3) \in R \text{ but } (3, 2) \notin R$$

$$(3, 1) \in R \text{ but } (1, 3) \notin R$$

Also, R is not transitive.

$$\text{Since } (1, 2) \in R, (2, 3) \in R \text{ but } (1, 3) \notin R.$$

13. .

Soln.

$$\frac{1}{2} \neq \left(\frac{1}{2}\right)^2 = \left(\frac{1}{2}, \frac{1}{2}\right) \neq R.$$

Hence, R is not reflexive.

14. .Soln.

Let $(a, b) \in N \times N$

Then,

$$\therefore a^2 + b^2 = a^2 + b^2$$

$$\therefore (a, b) R (a, b)$$

Hence R is reflexive.

Let $(a, b), (c, d) \in N \times N$ be such that

$$(a, b) R (c, d)$$

$$\Rightarrow a^2 + d^2 = b^2 + c^2$$

$$\Rightarrow c^2 + b^2 = d^2 + a^2$$

$$\Rightarrow (c, d) R (a, b)$$

Hence, R is symmetric.

Let $(a, b), (c, d), (e, f) \in N \times N$ be such that

$$(a, b) R (c, d), (c, d) R (e, f).$$

$$\Rightarrow a^2 + d^2 = b^2 + c^2 \quad \dots(i)$$

$$\text{and } c^2 + f^2 = d^2 + e^2 \quad \dots(ii)$$

Adding eqn. (i) and (ii),

$$\Rightarrow a^2 + d^2 + c^2 + f^2 = b^2 + c^2 + d^2 + e^2$$

$$\Rightarrow a^2 + f^2 = b^2 + e^2$$

$$\Rightarrow (a, b) R (e, f)$$

Hence, R is transitive

Since, R is reflexive, symmetric and transitive.

15. .Soln.

Given a relation S in $N \times N$, defined as

$$(a, b) S (c, d), \text{ if } a + d = b + c.$$

Reflexive Let (a, b) be any arbitrary element of $N \times N$

$$\text{i.e., } (a, b) \in N \times N, \text{ where } a, b \in N$$

$$\text{Now, as } a + b = b + a$$

[$\because$ addition is commutative]

$$\therefore (a, b) S (a, b)$$

So, S is reflexive.

Symmetric Let $(a, b), (c, d) \in N \times N$, such that (a, b)

$$S (c, d). \text{ Then, } a + d = b + c$$

$$\Rightarrow b + c = a + d \Rightarrow c + b = d + a$$

$$\Rightarrow (c, d) S (a, b)$$

So, S is symmetric.

Transitive Let $(a, b), (c, d), (e, f) \in N \times N$ such that

$$(a, b) S (c, d) \text{ and } (c, d) S (e, f).$$

$$\text{Then, } a + d = b + c \text{ and } c + f = d + e$$

On adding the above equations, we get

$$a + d + c + f = b + c + d + e$$

$$\Rightarrow a + f = b + e \Rightarrow (a, b) S (e, f)$$

So, S is transitive.

Thus, S is reflexive, symmetric and transitive. Hence, S is an equivalence relation.

16. .Soln.

Given $(a, b) R (c, d)$ as $ad(b+c) = bc(a+d)$

$$\therefore \forall a, b \in N,$$

$$\text{Or } ab(b+a) = ba(a+b)$$

$$\text{Or } (a, b) R (a, b)$$

$$\therefore R \text{ is reflexive } \dots(i)$$

Let $(a, b) R (c, d)$ for $(a, b), (c, d) \in N \times N$

$$\therefore ad(b+c) = bc(a+d) \quad \dots(ii)$$

$$\text{Also, } (c, d) R (a, b)$$

$$\therefore cb(d+a) = da(c+b) \text{ [By commutation of addition and multiplication on } N]$$

$$\therefore R \text{ is symmetric. } \dots(iii)$$

Let $(a, b) R (c, d)$ and $(c, d) R (e, f)$ for $a, b, c, d, e, f \in N$

$$\therefore ad(b+c) = bc(a+d) \quad \dots(iv)$$

$$\text{and } cf(d+e) = de(c+f) \quad \dots(v)$$

Dividing eqn. (iv) by abcd and eqn. (v) by cdef

$$\text{i.e., } \frac{1}{c} + \frac{1}{b} = \frac{1}{d} + \frac{1}{a}$$

$$\text{and } \frac{1}{e} + \frac{1}{d} = \frac{1}{f} + \frac{1}{c}$$

On adding, we get

$$\frac{1}{c} + \frac{1}{b} + \frac{1}{e} + \frac{1}{d} = \frac{1}{d} + \frac{1}{a} + \frac{1}{f} + \frac{1}{c}$$

$$\text{Or } af(b+e) = be(a+f)$$

Hence

$$(a, b) R (e, f)$$

$$\therefore R \text{ is transitive.}$$

From equations (i), (iii) and (iv), R is an equivalence relation.

17. .Soln.

The given relation is $R = \{(a, b) : |a - b| \text{ is even}\}$

defined on set $A = \{1, 2, 3, 4, 5\}$.

Reflexive As $|x - x| = 0$ is even, $\forall x \in A$

Therefore, R is reflexive.

Symmetric Let $\{x, y\} \in R \Rightarrow |x - y|$ is even

[by the definition of given relation]

$$\Rightarrow |y - x| \text{ is also even } [\because |a| = |-a| \forall a \in R]$$

$$\Rightarrow (y, x) \in R$$

$$\text{Thus, } (x, y) \in R$$

$$\Rightarrow (y, x) \in R, \forall x, y \in A$$

Therefore, R is symmetric.

Transitive Let $(x, y) \in R$ and $(y, z) \in R$

$$\Rightarrow |x - y| \text{ is even and } |y - z| \text{ is even.}$$

[by using definition of given relation]

Now, $|x - z|$ is even.

$\Rightarrow x$ and y both are even or odd.

And $|y - x|$ is even.

$\Rightarrow y$ and z both are even or odd.

Clearly, two cases arises.

Case I When y is even. Then, both x and z are even.

$\Rightarrow |x - z|$ is even.

[$\because$ difference of two even numbers is even]

$\Rightarrow (x, z) \in R$

Case II When y is odd.

Then, both x and z are odd.

$\Rightarrow |x - z|$ is even.

[$\because$ difference of two odd numbers is even]

$\Rightarrow (x, z) \in R$

Thus, $(x, y) \in R$ and $(y, z) \in R$

$\Rightarrow (x, z) \in R, \forall x, y, z \in A$

Therefore, R is transitive.

Since, R is reflexive, symmetric and transitive, so it is an equivalence relation.

18.

Soln.

$$f(x_1) = f(x_2)$$

$$\text{Or } \left(\frac{x_1 - 2}{x_1 - 3} \right) = \left(\frac{x_2 - 2}{x_2 - 3} \right)$$

$$\text{Or } x_1 = x_2$$

$\therefore f$ is a one - one function.

$$\text{Let, } y = f(x) = \frac{x-2}{x-3}$$

$$\text{Or } x = \frac{3y-2}{y-1}, \text{ where } y \neq 1 \text{ and } x \neq 3$$

$\therefore$ For each $y \in B$ there exists $x \in A$ such that $f(x) = y$ or f is onto

f is a one - one and onto function.

Or f is bijective function.

$$f^{-1}: B \rightarrow A \text{ with } f^{-1}(x) = \frac{3x-2}{x-1}$$

$$19. \text{Soln. Given } f(x) = \frac{4x+3}{6x-4}$$

To show f is one - one:

$$\text{Let, } f(x_1) = f(x_2),$$

$$\text{Then } \frac{4x_1+3}{6x_1-4} = \frac{4x_2+3}{6x_2-4}$$

$$\text{Or } (4x_1+3)(6x_2-4) = (6x_1-4)(4x_2+3)$$

Or

$$24x_1x_2 - 16x_1 + 18x_2 - 12 = 24x_1x_2 + 18x_1 - 16x_2 - 12$$

$$\text{Or } -16x_1 + 18x_2 = 18x_1 - 16x_2$$

$$\text{Or } -16x_1 - 18x_2 = -18x_2 - 16x_2$$

$$\text{Or } -34x_1 = -34x_2$$

$$\text{Or } x_1 = x_2$$

Or f is one - one.

To show f is onto:

$$\text{Let, } y \in B$$

$$\therefore y = f(x)$$

$$\text{Or } y = \frac{4x+3}{6x-4}$$

$$\text{Or } y(6x-4) = 4x+3$$

$$\text{Or } 6xy - 4y = 4x+3$$

$$\text{Or } 6xy - 4x = 4y+3$$

$$\text{Or } x(6y-4) = 4y+3$$

$$\text{Or } x = \frac{4y+3}{6y-4} \in B = R - \left\{ \frac{2}{3} \right\}$$

Or For every value of y except $y = \left\{ \frac{2}{3} \right\}$, there is a pre-

$$\text{image } x = \frac{4y+3}{6y-4} = g(y).$$

$$\text{Or } x \in A$$

Or f is onto.

To find f^{-1} :

Since, f is one - one and onto, therefore f is invertible.

$$\text{Thus, } f^{-1}(x) = \frac{4x+3}{6x-4} = g(x)$$

20. Soln.

$$(i) f(x) = x + |x|$$

$$= \begin{cases} x+x, & \text{if } x \geq 0 \\ x-x, & \text{if } x < 0 \end{cases} = \begin{cases} 2x, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

$$\text{Thus, } f(x) = 2x \geq 0 \forall x \geq 0 \text{ and } f(x) = 0 \forall x < 0$$

$\therefore f(x)$ can't be negative for any $x \in R$

Thus, f is not onto.

$$(ii) \text{ We have } f(x) = x+1 \forall x \in R$$

$$\text{For any } y \in R, y = f(x) \Rightarrow y = x+1 \Rightarrow x = y-1$$

$$\therefore f(y-1) = y-1+1 = y$$

Hence, f is onto.

21. Soln.

$$R = \{(a, b) : a^2 + b^2 = 25\} \text{ be a relation on } Z.$$

To domain of R is the value of $a \in Z$ that satisfy the relation $a^2 + b^2 = 25$

$$\Rightarrow a^2 = 25 - b^2 \Rightarrow a = \pm \sqrt{25 - b^2}$$

$$\text{Domain of } R = \{0, \pm 3, \pm 4, \pm 5\}$$

22. Soln.

If $f: A \rightarrow B$ is such that $y \in B$, then

$$f^{-1}(y) = \{x \in A : f(x) = y\}$$

In other words, $f^{-1}(y)$ is the set of pre-images of y .

$$\text{Let } f^{-1}(17) = x$$

$$\Rightarrow f(x) = 17 \Rightarrow x^2 + 1 = 17$$

$$\Rightarrow x^2 = 17 - 1 = 16 \Rightarrow x = \pm 4$$

$$\therefore f^{-1}(17) = \{-4, 4\}$$

Again, let $f^{-1}(-3) = x$, then

$$f(x) = -3 \Rightarrow x^2 + 1 = -3$$

$$\Rightarrow x^2 = -4 \Rightarrow x = \sqrt{-4}$$

Clearly no solution is available in R .

$$\text{So, } f^{-1}(-3) = \phi.$$

23. Soln.

$$\text{Here, } f : R^+ \rightarrow R^+ \text{ defined by } f(x) = \frac{1}{2x}$$

One-One: Let $x_1, x_2 \in R^+$ (domain)

$$\text{Now, } f(x_1) = f(x_2) \Rightarrow \frac{1}{2x_1} = \frac{1}{2x_2}$$

$$\Rightarrow 2x_1 = 2x_2 \Rightarrow x_1 = x_2$$

$\therefore f$ is one-one.

Onto: Let $y \in R^+$ (co-domain) be any arbitrary element then $y \neq 0$

$$\text{Let } y = f(x)$$

$$\Rightarrow y = \frac{1}{2x} \Rightarrow x = \frac{1}{2y} \in R^+$$

$\therefore f$ is onto.

Hence, f is bijective where $\frac{1}{2y}$ is non zero real number.

Hence, each element of co-domain (R^+) is the image of some element of domain (R^+).

24. Soln.

$$\text{Here, } A = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \text{ and } B = [-1, 1]$$

Also $f : A \rightarrow B$ such that $f(x) = \sin x$

$\therefore f$ is one-one.

$$\therefore f(x_1) = f(x_2) \Rightarrow \sin x_1 = \sin x_2$$

$$\Rightarrow x_1 = x_2 \quad \left[\because x_1, x_2 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \right]$$

Also, range (f) = $[-1, 1] = B$. So, f is onto.

Thus, f is one-one and onto and hence bijective.

25. Soln.

$$\text{We have } f : R \rightarrow R \text{ defined by } f(x) = x^3 + x$$

$$\text{Let } x_1, x_2 \in R \text{ such that } f(x_1) = f(x_2)$$

$$\Rightarrow x_1^3 + x_1 = x_2^3 + x_2 \Rightarrow x_1^3 - x_2^3 + x_1 - x_2 = 0$$

$$\Rightarrow (x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2 + 1) = 0$$

$$\Rightarrow x_1 - x_2 = 0$$

$$[\because x_1^2 + x_1x_2 + x_2^2 \geq 0 \text{ for all } x_1, x_2 \in R]$$

$$\therefore x_1^2 + x_1x_2 + x_2^2 + 1 \geq 1 \text{ for all } x_1, x_2 \in R]$$

$$\Rightarrow x_1 = x_2$$

$\Rightarrow f$ is one-one

Let y be any arbitrary element of R , then there exists $x \in R$ such that $f(x) = y$.

$$\Rightarrow x^3 + x = y \Rightarrow x^3 + x - y = 0$$

Since, odd degree equation has atleast one real root.

Thus, for every value of y , the equation

$$x^3 + x - y = 0 \text{ has a real root } \alpha, \text{ such that}$$

$$\alpha^3 + \alpha - y = 0$$

$$\Rightarrow f(\alpha) = y$$

Thus, for every $y \in R, \exists \alpha \in R$ such that $f(\alpha) = y$

So, f is onto function.

Hence, $f : R \rightarrow R$ is a bijection.

26. Soln.

Here, $R = \{(a, b) : a \text{ is a factor of } b \text{ for } a, b \in N\}$

Reflexive: Let a be an arbitrary element of N then, clearly, a is a factor of a .

$\therefore (a, a) \in R \forall a \in N \therefore R$ is reflexive.

Symmetric: Clearly 2 and 6 are natural numbers and 2 is a factor of 6.

$\therefore (2, 6) \in R$ but 6 is not a factor of 2 $\Rightarrow (6, 2) \notin R$

Thus, $(2, 6) \in R$ but $(6, 2) \notin R \therefore R$ is not symmetric.

Transitive: Let $a, b, c \in N$

Now $(a, b) \in R$ and $(b, c) \in R$

$\Rightarrow (a \text{ is a factor of } b) \text{ and } (b \text{ is a factor of } c)$

$\Rightarrow b = ad \text{ and } c = be \text{ for some } d, e \in N$

$\Rightarrow c = (ad)e = a(de)$ [By associative law]

$\Rightarrow a \text{ is a factor of } c \Rightarrow (a, c) \in R$

Thus, $(a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R$

$\therefore R$ is transitive.

27. Soln.

(i) Here, $f : R \rightarrow R$ given by $f(x) = \sin x$

Let $x_1, x_2 \in R$ such that $f(x_1) = f(x_2)$

$$\Rightarrow \sin x_1 = \sin x_2 \Rightarrow x_1 = n\pi + (-1)^n x_2 \Rightarrow x_1 = x_2$$

$\therefore f$ is not one-one.

Let $y \in R$ be any arbitrary element, then there exists

$$x \in R \text{ such that } f(x) = y$$

$$\Rightarrow \sin x = y \Rightarrow x = \sin^{-1} y$$

$\therefore$ For $y > 1, x \notin R$ (domain)

$\therefore$ f is not onto.

Hence, f is not a bijective function.

(ii) $f: R \rightarrow R$ defined by $f(x) = \sin^2 x + \cos^2 x$

$$\text{Since, } f(x) = \sin^2 x + \cos^2 x = 1$$

Now, $f(x) = 1$ is a constant function and we know that constant function is neither injective nor surjective.

$\therefore f$ is neither one - one nor onto.

28. Soln.

We have given, $A = \{1, 2, 3, 4\}$

(i) Consider,

$$R_1 = \{(1, 1), (1, 2), (2, 3), (2, 2), (1, 3), (3, 3)\}$$

As, $(1, 1), (2, 2), (3, 3)$ lie in R_1

$\therefore R_1$ is reflexive.

Also, $(1, 2) \in R_1, (2, 3) \in R_1 \Rightarrow (1, 3) \in R_1$

So, R_1 is also transitive.

Since, $(2, 3) \in R_1$ but $(3, 2) \notin R_1$.

So, it is not symmetric.

(ii) Consider, $R_2 = \{(1, 2), (2, 1)\}$

As, $(1, 2) \in R_2$ and $(2, 1) \in R_2$

So, it is symmetric but it is neither reflexive nor transitive.

(iii) Consider,

$$R_3 = \{(1, 2), (2, 1), (1, 1), (2, 2), (3, 3), (1, 3), (3, 1), (2, 3), (3, 2)\}$$

Hence, R_3 is reflexive, symmetric and transitive.

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SURE SHOT QUESTIONS 2026

Chapter – 02 (Questions)

Inverse Trigonometric Functions

Questions

1. find the number of triplets (x, y, z) satisfying the equation

$$\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{3\pi}{2}.$$

2. Solve for x : $\cos(\tan^{-1}x) = \sin\left(\cot^{-1}\frac{3}{4}\right).$

3. Find the value of $\sin^{-1}\left(\cos\frac{43\pi}{5}\right).$

4. Write the principle value of $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$

5. Write the simplest form of $\tan^{-1}\left[\frac{\sqrt{1+x^2}-1}{x}\right].$

6. Express in the simplest form:

$$\tan^{-1}\left[\frac{\cos x - \sin x}{\cos x + \sin x}\right], -\frac{\pi}{4} < x < \frac{\pi}{4}$$

7. Write the value of $\sin\left(2\sin^{-1}\frac{3}{5}\right).$

8. Evaluate: $\cos\left[\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right) + \frac{\pi}{6}\right]$

9. Find the value of $\sec^2(\tan^{-1}2) + \operatorname{cosec}^2(\cot^{-1}3).$

10. Find the range of $f(x) = \sin^{-1}x + \tan^{-1}x + \sec^{-1}x.$

11. If $\operatorname{cosec}^{-1}x + \operatorname{cosec}^{-1}y + \operatorname{cosec}^{-1}z = -\frac{3\pi}{2},$
find the 2 value of $\frac{x}{y} + \frac{y}{z} + \frac{z}{x}.$

12. Find the value of $\tan(\cos^{-1}x)$ and hence evaluate $\tan\left(\cos^{-1}\left(\frac{8}{17}\right)\right).$

13. Find the value of $\tan^{-1}\left(\tan\frac{5\pi}{6}\right) + \cos^{-1}\left(\cos\frac{13\pi}{6}\right).$

14. Evaluate: $\frac{1}{\pi}\left\{216\sin^{-1}\left(\sin\frac{7\pi}{6}\right) + 27\cos^{-1}\left(\cos\frac{2\pi}{3}\right) + 28\tan^{-1}\left(\tan\frac{5\pi}{4}\right) + 200\cot^{-1}\left(\cot\frac{-\pi}{4}\right)\right\}$

15. Show that $\sin^{-1}\frac{5}{13} + \cos^{-1}\frac{3}{5} = \tan^{-1}\frac{63}{16}.$

16. Find the value of

$$\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right]$$

17. Find the domain of the function

$$f(x) = \sin^{-1}(x^2 - 4). \text{ Also, find its range.}$$

18. Evaluate: $\cos^{-1}\left[\cos\left(-\frac{7\pi}{3}\right)\right]$

19. Prove that: $\frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\left(\frac{1}{3}\right) = \frac{9}{4}\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right).$

20. Prove that: $2\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{7} = \tan^{-1}\frac{31}{17}.$

21. Prove that:

$$\cos^{-1}\left(\frac{12}{13}\right) + \sin^{-1}\left(\frac{3}{5}\right) = \sin^{-1}\left(\frac{56}{65}\right).$$

22. Prove that:

$$\tan^{-1}\left(\frac{\sqrt{1+x} + \sqrt{1-x}}{\sqrt{1+x} - \sqrt{1-x}}\right) = \frac{\pi}{4} - \frac{1}{2}\cos^{-1}x$$

$$; -\frac{1}{\sqrt{2}} \leq x \leq 1.$$

23. Find the principal value branch of $\cos^{-1}x.$

24. Find the value of $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$.

25. Find the domain and range of $\sin^{-1}x$.

26. Find the number of solutions of the equation

$$2\cos^{-1}x + \sin^{-1}x = \frac{11\pi}{6}, \text{ if } \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$$

27. Evaluate: $\cos\left\{\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right) + \frac{\pi}{6}\right\}$.

28. Find the value of $\cos^{-1}\left(\cos\frac{14\pi}{3}\right)$

29. Simplify : $\cos^{-1}\left(\frac{3}{5}\cos x + \frac{4}{5}\sin x\right)$.

30. What is the domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$?

31. Find the domain of the function $\cos^{-1}(2x - 1)$.

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SURE SHOT QUESTIONS 2026

Chapter – 02 (Solutions)

Inverse Trigonometric Functions

Questions

1. Soln. We have, $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$

$$\because -\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}, -\frac{\pi}{2} \leq \sin^{-1} y \leq \frac{\pi}{2}$$

$$\text{and } -\frac{\pi}{2} \leq \sin^{-1} z \leq \frac{\pi}{2}$$

$\therefore$ The above condition will true if

$$\sin^{-1} x = \sin^{-1} y = \sin^{-1} z = \frac{\pi}{2} \Rightarrow x = y = z = 1$$

Thus, there is only one triplet.

2. Soln. Let $\cot^{-1} \frac{3}{4} = \theta \Rightarrow \cot \theta = \frac{3}{4}$

$$\therefore \operatorname{cosec} \theta = \sqrt{1 + \cot^2 \theta} = \sqrt{1 + \frac{9}{16}} = \frac{5}{4}$$

$$\therefore \sin \theta = \frac{4}{5} \Rightarrow \theta = \sin^{-1} \frac{4}{5}$$

$$\text{So, } \sin \left(\cot^{-1} \frac{3}{4} \right) = \sin \left(\sin^{-1} \frac{4}{5} \right) = \frac{4}{5}$$

Let $\tan^{-1} X = \phi$. Then, $\tan \phi = X$.

$$\therefore \sec \phi = \sqrt{1 + \tan^2 \phi} = \sqrt{1 + X^2}$$

$$\therefore \cos \phi = \frac{1}{\sqrt{1 + X^2}}$$

$$\cos(\tan^{-1} X) = \cos \phi = \frac{1}{\sqrt{1 + X^2}}$$

$$\text{So, } \frac{1}{\sqrt{1 + X^2}} = \frac{4}{5} \Rightarrow \frac{1}{1 + X^2} = \frac{16}{25} \Rightarrow 16X^2 = 9$$

$$\Rightarrow X^2 = \frac{9}{16} \Rightarrow X = \pm \frac{3}{4}$$

3. Soln. $\sin^{-1} \left(\cos \left(8\pi + \frac{3\pi}{5} \right) \right) =$

$$\sin^{-1} \left(\cos \frac{3\pi}{5} \right)$$

$$= \sin^{-1} \left(\sin \left(\frac{\pi}{2} - \frac{3\pi}{5} \right) \right)$$

$$= \sin^{-1} \left(\sin \left(-\frac{\pi}{10} \right) \right)$$

$$= -\frac{\pi}{10} \because -\frac{\pi}{10} \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

4. Soln. We have, $\cos^{-1} \left(\frac{1}{2} \right) = \cos^{-1} \left(\cos \frac{\pi}{3} \right)$
 $= \frac{\pi}{3} \because \frac{\pi}{3} \in [0, \pi]$

$$\text{Also } \sin^{-1} \left(\frac{1}{2} \right) = \sin^{-1} \left(\sin \frac{\pi}{6} \right)$$

$$= \frac{\pi}{6} \because \frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

$$\therefore \cos^{-1} \left(\frac{1}{2} \right) + 2 \sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{3} + 2 \left(\frac{\pi}{6} \right)$$

$$= \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$$

5. Soln. Let $X = \tan \theta \Rightarrow \theta = \tan^{-1} X$

$$\Rightarrow \tan^{-1} \left[\frac{\sqrt{1 + X^2} - 1}{X} \right] = \tan^{-1} \left[\frac{\sqrt{1 + \tan^2 \theta} - 1}{\tan \theta} \right]$$

$$= \tan^{-1} \left[\frac{\sqrt{\sec^2 \theta} - 1}{\tan \theta} \right] = \tan^{-1} \left[\frac{\sec \theta - 1}{\tan \theta} \right]$$

$$= \tan^{-1} \left[\frac{1}{\frac{\cos \theta}{\sin \theta}} - 1 \right] = \tan^{-1} \left[\frac{1 - \cos \theta}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} \right]$$

$$= \tan^{-1} \left[\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2}} \right] = \tan^{-1} \left(\tan \frac{\theta}{2} \right)$$

$$= \frac{\theta}{2} = \frac{1}{2} (\tan^{-1} X)$$

6. Soln. We have, $\tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right)$

$$= \tan^{-1} \left(\frac{1 - \tan x}{1 + \tan x} \right)$$

$$= \tan^{-1} \left\{ \tan \left(\frac{\pi}{4} - x \right) \right\} = \frac{\pi}{4} - x$$

7. Soln. $\sin \left(2 \sin^{-1} \frac{3}{5} \right) = \sin \left(2 \tan^{-1} \frac{3}{4} \right)$

$$= \sin \left[\tan^{-1} \left(\frac{2 \times \frac{3}{4}}{1 - \frac{9}{16}} \right) \right]$$

$$= \sin \left[\tan^{-1} \left(\frac{3}{2} \times \frac{16}{7} \right) \right]$$

$$= \sin \left[\tan^{-1} \frac{24}{7} \right]$$

$$= \sin \left[\sin^{-1} \frac{24}{25} \right]$$

$$= \frac{24}{25}$$

$$\left[\because \sin^{-1}(\sin \theta) = \theta \forall \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) \right]$$

8. .Soln. Since, $\cos \left[\cos^{-1} \left(\frac{-\sqrt{3}}{2} \right) + \frac{\pi}{6} \right]$

$$= \cos \left[\cos^{-1} \left(\cos \frac{5\pi}{6} \right) + \frac{\pi}{6} \right]$$

$$\left[\because \cos \frac{5\pi}{6} = \cos \left(\pi - \frac{\pi}{6} \right) = \frac{-\sqrt{3}}{2} \right]$$

$$= \cos \left(\frac{5\pi}{6} + \frac{\pi}{6} \right) \{ \because \cos^{-1}(\cos \theta) = \theta; \theta \in [0, \pi] \}$$

$$= \cos(\pi) = -1$$

9. .Soln. $\sec^2(\tan^{-1} 2) + \operatorname{cosec}^2(\cot^{-1} 3)$

$$= [\sec(\tan^{-1} 2)]^2 + [\operatorname{cosec}(\cot^{-1} 3)]^2$$

$$= [\sec(\sec^{-1} \sqrt{5})]^2 + [\operatorname{cosec}(\operatorname{cosec}^{-1} \sqrt{10})]^2$$

$$= (\sqrt{5})^2 + (\sqrt{10})^2 = 5 + 10 = 15$$

10. .Soln.

Given, $f(x) = \sin^{-1} x + \tan^{-1} x + \sec^{-1} x$

Domain of $\sin^{-1} x = [-1, 1]$

Domain of $\tan^{-1} x = (-\infty, \infty)$

Domain of $\sec^{-1} x = (-\infty, \infty) - (-1, 1)$

Domain of $f(x) = [-1, 1] \cap (-\infty, \infty) \cap [(-\infty, \infty) - (-1, 1)]$

$$= \{-1, 1\}$$

Now, $f(-1) = \sin^{-1}(-1) + \tan^{-1}(-1) + \sec^{-1}(-1)$

$$= -\frac{\pi}{2} - \frac{\pi}{4} + \pi = \frac{\pi}{4}$$

And $f(1) = \sin^{-1}(1) + \tan^{-1}(1) + \sec^{-1}(1)$

$$= \frac{\pi}{2} + \frac{\pi}{4} + 0 = \frac{3\pi}{4}$$

Range of $f(x) = \left\{ \frac{\pi}{4}, \frac{3\pi}{4} \right\}$

11. .Soln. We know that the minimum value of

$$\operatorname{cosec}^{-1} x \text{ is } -\frac{\pi}{2} \text{ which is attained at } x = -1.$$

$$\therefore \operatorname{cosec}^{-1} x + \operatorname{cosec}^{-1} y + \operatorname{cosec}^{-1} z = -\frac{3\pi}{2}$$

$$\Rightarrow \operatorname{cosec}^{-1} x + \operatorname{cosec}^{-1} y + \operatorname{cosec}^{-1} z =$$

$$\left(-\frac{\pi}{2} \right) + \left(-\frac{\pi}{2} \right) + \left(-\frac{\pi}{2} \right)$$

$$\Rightarrow \operatorname{cosec}^{-1} x = -\frac{\pi}{2}, \operatorname{cosec}^{-1} y = -\frac{\pi}{2}, \operatorname{cosec}^{-1} z = -\frac{\pi}{2}$$

$$\Rightarrow x = -1, y = -1, z = -1$$

$$\therefore \frac{x}{y} + \frac{y}{z} + \frac{z}{x} = \frac{(-1)}{(-1)} + \frac{(-1)}{(-1)} + \frac{(-1)}{(-1)} = 3$$

12. .Soln. Let $\cos^{-1} x = \theta$, then $\cos \theta = x$, where $\theta \in [0, \pi]$

$$\therefore \tan(\cos^{-1} x) = \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sqrt{1 - \cos^2 \theta}}{\cos \theta}$$

$$= \frac{\sqrt{1 - x^2}}{x}$$

$$\text{Hence, } \tan \left(\cos^{-1} \left(\frac{8}{17} \right) \right) = \frac{\sqrt{1 - (8/17)^2}}{8/17} = \frac{15}{8}$$

13. .Soln. As we know, $\tan^{-1}(\tan x) = x; x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$

And $\cos^{-1}(\cos x) = x; x \in [0, \pi]$

$$\therefore \tan^{-1} \left(\tan \frac{5\pi}{6} \right) + \cos^{-1} \left(\cos \frac{13\pi}{6} \right)$$

$$= \tan^{-1} \left[\tan \left(\pi - \frac{\pi}{6} \right) \right] + \cos^{-1} \left[\cos \left(\pi + \frac{7\pi}{6} \right) \right]$$

$$\left\{ \because \frac{5\pi}{6} \notin \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) \text{ and } \frac{13\pi}{6} \notin [0, \pi] \right\}$$

$$= \tan^{-1} \left(-\tan \frac{\pi}{6} \right) + \cos^{-1} \left(-\cos \frac{7\pi}{6} \right)$$

$$\left[\because \tan(\pi - \theta) = -\tan \theta \text{ and } \cos(\pi + \theta) = -\cos \theta \right]$$

$$= -\tan^{-1} \left(\tan \frac{\pi}{6} \right) + \pi - \left[\cos^{-1} \cos \left(\frac{7\pi}{6} \right) \right]$$

$$\left\{ \because \tan^{-1}(-x) = -\tan^{-1} x; x \in \mathbb{R} \text{ and } \cos^{-1}(-x) = \pi - \cos^{-1} x; x \in [-1, 1] \right\}$$

$$= -\tan^{-1} \left(\tan \frac{\pi}{6} \right) + \pi - \cos^{-1} \left[\cos \left(\pi + \frac{\pi}{6} \right) \right]$$

$$= -\tan^{-1} \left(\tan \frac{\pi}{6} \right) + \pi - \left[\cos^{-1} \left(-\cos \frac{\pi}{6} \right) \right]$$

$$= -\tan^{-1} \left(\tan \frac{\pi}{6} \right) + \pi - \pi + \cos^{-1} \left(\cos \frac{\pi}{6} \right)$$

$$= -\frac{\pi}{6} + 0 + \frac{\pi}{6} = 0$$

14. .

Soln. Since, $\sin^{-1} \left(\sin \frac{7\pi}{6} \right)$

$$= \sin^{-1} \left(\sin \left(\pi + \frac{\pi}{6} \right) \right) = \sin^{-1} \left(-\sin \frac{\pi}{6} \right)$$

$$= \sin^{-1} \left(\sin \left(-\frac{\pi}{6} \right) \right) = -\frac{\pi}{6}$$

$$\cos^{-1} \left(\cos \frac{2\pi}{3} \right) = \frac{2\pi}{3}, \tan^{-1} \left(\tan \frac{5\pi}{4} \right)$$

$$= \tan^{-1} \tan \left(\pi + \frac{\pi}{4} \right)$$

$$= \tan^{-1} \left[\tan \left(\frac{\pi}{4} \right) \right] = \frac{\pi}{4} \text{ and } \cot^{-1} \left\{ \cot \left(-\frac{\pi}{4} \right) \right\}$$

$$= \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

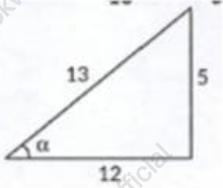
Hence, required value is

$$\frac{1}{\pi} \left\{ 216 \times -\frac{\pi}{6} + 27 \times \frac{2\pi}{3} + 28 \times \frac{\pi}{4} + 200 \times \frac{3\pi}{4} \right\}$$

$$= -36 + 18 + 7 + 150 = 139$$

15. Soln. We have to prove,

$$\sin^{-1} \frac{5}{13} + \cos^{-1} \frac{3}{5} = \tan^{-1} \frac{63}{16}$$



$$\sin^{-1} \frac{5}{13} = \alpha$$

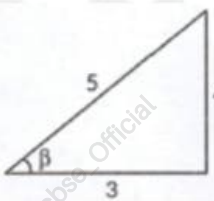
Consider,

$$\Rightarrow \sin \alpha = \frac{5}{13} \text{ and } \cos \alpha = \frac{12}{13}$$

$$\therefore \tan \alpha = \frac{5}{12} \dots\dots\dots(i)$$

$$\text{Again, consider } \cos^{-1} \frac{3}{5} = \beta \Rightarrow \cos \beta = \frac{3}{5}$$

$$\sin \beta = \frac{4}{5}, \tan \beta = \frac{4}{3}$$



As we know,

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{\frac{5}{12} + \frac{4}{3}}{1 - \frac{5}{12} \times \frac{4}{3}}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{\frac{15+48}{36}}{\frac{36-20}{36}}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{63}{16} \Rightarrow \alpha + \beta = \tan^{-1} \frac{63}{16}$$

$$\Rightarrow \sin^{-1} \frac{5}{13} + \cos^{-1} \frac{3}{5} = \tan^{-1} \frac{63}{16}$$

16. Ans.

We have,

$$\tan^{-1} \left(\frac{-1}{\sqrt{3}} \right) + \cot^{-1} \left(\frac{1}{\sqrt{3}} \right) + \tan^{-1} \left[\sin \left(\frac{-\pi}{2} \right) \right]$$

$$= -\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) + \cot^{-1} \left(\frac{1}{\sqrt{3}} \right) + \tan^{-1} (-1)$$

$$= -\frac{\pi}{6} + \frac{\pi}{3} - \frac{\pi}{4} = -\frac{\pi}{12}$$

17. Ans.

For domain, $-1 \leq x^2 - 4 \leq 1$

$$\Rightarrow 3 \leq x^2 \leq 5 \Rightarrow \sqrt{3} \leq |x| \leq \sqrt{5} \Rightarrow x \in [-\sqrt{5}, -\sqrt{3}] \cup [\sqrt{3}, \sqrt{5}]$$

$$\text{Let } y = \sin^{-1}(x^2 - 4) \Rightarrow \sin y + 4 = x^2$$

$$\text{Now, } 3 \leq x^2 \leq 5 \Rightarrow 3 \leq \sin y + 4 \leq 5 \Rightarrow -1 \leq \sin y \leq 1$$

$$\Rightarrow -\frac{\pi}{2} \leq y \leq \frac{\pi}{2} \therefore R_f = \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

$$18. \text{Ans. } \cos^{-1} \left[\cos \left(-\frac{7\pi}{3} \right) \right] = \cos^{-1} \left[\cos \left(\frac{7\pi}{3} \right) \right]$$

$$(\because \cos(-\theta) = \cos \theta)$$

$$= \cos^{-1} \left(\cos \left(2\pi + \frac{\pi}{3} \right) \right) = \cos^{-1} \left(\cos \left(\frac{\pi}{3} \right) \right)$$

$$= \frac{\pi}{3} (\because \cos^{-1}(\cos x) = x \forall 0 \leq x \leq \pi)$$

19. Ans. Consider L.H.S.

$$= \frac{9\pi}{8} - \frac{9}{4} \sin^{-1} \left(\frac{1}{3} \right) = \frac{9}{4} \left[\frac{\pi}{2} - \sin^{-1} \left(\frac{1}{3} \right) \right]$$

$$= \frac{9}{4} \cos^{-1} \left(\frac{1}{3} \right) \dots\dots\dots(i)$$

$$\left(\because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \right)$$

$$\text{Let } a = \cos^{-1} \left(\frac{1}{3} \right)$$

$$= \cos a = \frac{1}{3} \Rightarrow \sin a = \sqrt{1 - \cos^2 a}$$

$$[\because \sin^2 \theta + \cos^2 \theta = 1]$$

$$\Rightarrow \sin a = \sqrt{1 - \left(\frac{1}{3} \right)^2} = \sqrt{1 - \frac{1}{9}} = \sqrt{\frac{8}{9}} = \frac{2\sqrt{2}}{3}$$

$$\Rightarrow a = \sin^{-1} \left(\frac{2\sqrt{2}}{3} \right)$$

$$\text{So, L.H.S.} = \frac{9}{4} \sin^{-1} \left(\frac{2\sqrt{2}}{3} \right) = R.H.S.$$

$$20. \text{Ans. Consider, L.H.S.} = 2 \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7}$$

$$= \tan^{-1} \frac{2 \times \frac{1}{2}}{1 - \left(\frac{1}{2} \right)^2} + \tan^{-1} \frac{1}{7} = \tan^{-1} \frac{1}{\frac{3}{4}} + \tan^{-1} \frac{1}{7}$$

$$= \tan^{-1} \frac{4}{3} + \tan^{-1} \frac{1}{7} = \tan^{-1} \frac{\left(\frac{4}{3} + \frac{1}{7} \right)}{1 - \frac{4}{21}} = \tan^{-1} \frac{\frac{31}{21}}{\frac{17}{21}}$$

$$\begin{aligned} & \left[\because \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right) \right] \\ & = \tan^{-1} \frac{31}{17} = R.H.S. \end{aligned}$$

Hence proved.

21. .Ans. Let $x = \cos^{-1} \left(\frac{12}{13} \right)$ and $y = \sin^{-1} \left(\frac{3}{5} \right)$

Or $\cos x = \frac{12}{13}$ and $\sin y = \frac{3}{5}$

Now, $\sin x = \sqrt{1 - \cos^2 x}$ and $\cos y = \sqrt{1 - \sin^2 y}$

$$\Rightarrow \sin x = \sqrt{1 - \frac{144}{169}} \text{ and } \cos y = \sqrt{1 - \frac{9}{25}}$$

$$\Rightarrow \sin x = \frac{5}{13} \text{ and } \cos y = \frac{4}{5}$$

We know that,

$$\sin(x+y) = \frac{5}{13} \times \frac{4}{5} + \frac{12}{13} \times \frac{3}{5} = \frac{20}{65} + \frac{36}{65} = \frac{56}{65}$$

$$\Rightarrow x+y = \sin^{-1} \left(\frac{56}{65} \right)$$

$$\text{Or, } \cos^{-1} \left(\frac{12}{13} \right) + \sin^{-1} \left(\frac{3}{5} \right) = \sin^{-1} \left(\frac{56}{65} \right)$$

22. .Ans. Consider L.H.S. = $\tan^{-1} \left(\frac{\sqrt{1+x} + \sqrt{1-x}}{\sqrt{1+x} - \sqrt{1-x}} \right)$

Put $X = \cos \theta$, we get

$$\tan^{-1} \left[\frac{\sqrt{1+\cos \theta} - \sqrt{1-\cos \theta}}{\sqrt{1+\cos \theta} + \sqrt{1-\cos \theta}} \right]$$

L.H.S. =

$$= \tan^{-1} \left[\frac{\left| \sqrt{2} \cos \frac{\theta}{2} \right| + \left| \sqrt{2} \sin \frac{\theta}{2} \right|}{\left| \sqrt{2} \cos \frac{\theta}{2} \right| - \left| \sqrt{2} \sin \frac{\theta}{2} \right|} \right]$$

$$[\because 1 + \cos^2 \theta = 2 \cos^2 \theta, 1 - \cos^2 \theta = 2 \sin^2 \theta]$$

$$= \tan^{-1} \left[\frac{-\sqrt{2} \cos \frac{\theta}{2} + \sqrt{2} \sin \frac{\theta}{2}}{-\sqrt{2} \cos \frac{\theta}{2} - \sqrt{2} \sin \frac{\theta}{2}} \right] = \tan^{-1} \left[\frac{-\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{-\cos \frac{\theta}{2} - \sin \frac{\theta}{2}} \right]$$

23. .Soln.

We define,

$$\cos^{-1} x: [-1, 1] \rightarrow [0, \pi]$$

For every $x \in [-1, 1]$ there exists a unique $\theta \in [0, \pi]$ such that $\cos \theta = x$. Thus θ is called a principal value of $\cos^{-1} x$. However we could also define.

$$\cos^{-1} x: [-1, 1] \rightarrow [-\pi, 0]$$

$$\text{or } \cos^{-1} x: [-1, 1] \rightarrow [\pi, 2\pi]$$

Such that $\cos \theta = x$, Infact we can find infinitely many such intervals. But only the value of θ lying in $[0, \pi]$ is called the principal value.

That is, the principal value of $\cos^{-1} x$ is the numerically least among all the values of $\cos^{-1} x$. The principal value branch of $\cos^{-1} x$ is $[0, \pi]$.

24. .Soln.

We know that the range of principal value branch of

$\cos^{-1}$ and $\sin^{-1}$ are $[0, \pi]$ and $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$ respectively.

$$\cos^{-1} \left(\frac{1}{2} \right) = x \Rightarrow \frac{1}{2} = \cos x$$

Let

$$\frac{1}{2} = \cos \left(\frac{\pi}{3} \right), \text{ where } \frac{\pi}{3} \in [0, \pi]$$

Then,

$$\sin^{-1} \left(\frac{1}{2} \right) = y \Rightarrow \frac{1}{2} = \sin y$$

Let

$$\frac{1}{2} = \sin \left(\frac{\pi}{6} \right), \text{ where } \frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

Then,

$$\therefore \cos^{-1} \left(\frac{1}{2} \right) + 2 \sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{3} + 2 \cdot \frac{\pi}{6} = \frac{2\pi}{3}$$

25. .Soln.

$\sin: \mathbb{R} \rightarrow \mathbb{R}$ such that

$\sin \theta = x$ for all $\theta \in \mathbb{R}$ is a many - one into function.

This function cannot have an inverse.

Therefore, we restrict the domain to the interval

$\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$ and codomain to the interval $[-1, 1]$.

Then,

$\sin: \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] \rightarrow [-1, 1]$ is a one - one, onto

function and is therefore, invertible.

The inverse of the sine function is defined as

$$\sin^{-1}: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

Such that $\sin^{-1} x = \theta$

$$\Leftrightarrow \sin \theta = x$$

Thus, the domain of the function is $[-1, 1]$ and the range

is $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$.

26. .Soln. Given equation is $2 \cos^{-1} x + \sin^{-1} x = \frac{11\pi}{6}$

$$\Rightarrow \cos^{-1} x + (\cos^{-1} x + \sin^{-1} x) = \frac{11\pi}{6}$$

$$\Rightarrow \cos^{-1} x + \frac{\pi}{2} = \frac{11\pi}{6} \quad \left(\text{Given } \cos^{-1} x + \sin^{-1} x = \frac{\pi}{2} \right)$$

$$\Rightarrow \cos^{-1} x = \frac{4\pi}{3}$$

Which is not possible as $\cos^{-1} x \in [0, \pi]$.

Thus, given equation has no solution.

27. .Soln. We have,

$$\cos \left\{ \cos^{-1} \left(\frac{-\sqrt{3}}{2} \right) + \frac{\pi}{6} \right\}$$

Now let

$$\cos^{-1} \left(\frac{-\sqrt{3}}{2} \right) = \theta$$

$$\Rightarrow \cos \theta = \frac{-\sqrt{3}}{2}$$

$$= -\cos \frac{\pi}{6}$$

$$= \cos \left(\pi - \frac{\pi}{6} \right)$$

$$= \cos \frac{5\pi}{6}$$

$$\Rightarrow \theta = \frac{5\pi}{6}$$

$$\therefore \cos^{-1} \left(\frac{-\sqrt{3}}{2} \right) = \frac{5\pi}{6}$$

$$\therefore \cos \left\{ \cos^{-1} \left(\frac{-\sqrt{3}}{2} \right) + \frac{\pi}{6} \right\} = \cos \left(\frac{5\pi}{6} + \frac{\pi}{6} \right)$$

$$= \cos \pi$$

$$= -1$$

28. .Soln.

$$\text{Since, } \cos^{-1} \left(\cos \frac{14\pi}{3} \right) = \cos^{-1} \cos \left(4\pi + \frac{2\pi}{3} \right)$$

$$= \cos^{-1} \cos \frac{2\pi}{3} = \frac{2\pi}{3}$$

$$\{ \because \cos^{-1}(\cos x) = x, x \in [0, \pi] \}$$

29. .

$$\text{Soln. Here, } \left(\frac{3}{5} \right)^2 + \left(\frac{4}{5} \right)^2 = 1$$

$$\text{Put } \frac{3}{5} = \cos \theta \text{ and } \frac{4}{5} = \sin \theta \Rightarrow \tan \theta = \frac{4}{3}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{4}{3} \right)$$

$$\therefore \cos^{-1} \left(\frac{3}{5} \cos x + \frac{4}{5} \sin x \right)$$

$$= \cos^{-1} (\cos \theta \cos x + \sin \theta \sin x)$$

$$= \cos^{-1} \{ \cos(x - \theta) \} = x - \theta = x - \tan^{-1} \left(\frac{4}{3} \right)$$

30. .

$$\text{Soln. Given, } f(x) = \sin^{-1} \sqrt{x-1}$$

$$\text{Since, } x-1 \geq 0 \text{ and } -1 \leq \sqrt{x-1} \leq 1$$

$$\therefore 0 \leq x-1 \leq 1 \Rightarrow 1 \leq x \leq 2$$

31. .Soln. We have given, $f(x) = \cos^{-1}(2x-1)$

$$\text{Since, } -1 \leq 2x-1 \leq 1$$

$$\Rightarrow 0 \leq 2x \leq 2$$

$$\Rightarrow 0 \leq x \leq 1$$

New

SURE SHOT QUESTIONS 2026

Chapter – 03 (Solutions)

Matrices

Questions

1. . Soln:
Given,
 $(A-I)^3 + (A+I)^3 - 7A$
 $= A^3 - I^3 - 3A^2I + 3AI^2 + A^3 + I^3 + 3A^2I + 3AI^2 - 7A$
 $= 2A^3 + 6AI^2 - 7A$
 $= 2AA^2 + 6AI^2 - 7A$
 $= 2AI + 6AI - 7A$
 $= 8A - 7A$
 $= A$

2. . Soln. The matrix $A = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ x & -3 & 0 \end{bmatrix}$ is skew-symmetric.

$\therefore A' = -A \Rightarrow \begin{bmatrix} 0 & -1 & x \\ 1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & -3 \\ -x & 3 & 0 \end{bmatrix} \Rightarrow x = 2$

3. . Soln. Since, $A^2 = A$,
 $7A - (I+A)^3 = 7A - I^3 - 3A^2I - 3AI^2 - A^3$
 $= 7A - I - 3A - 3A - A^2A$
 $= 7A - I - 3A - 3A - A.A$
 $= 7A - I - 3A - 3A - A$
 $= 7A - I - 7A$
 $= -I$

4. . Soln. $\begin{bmatrix} a+4 & 3b \\ 8 & -6 \end{bmatrix} = \begin{bmatrix} 2a+2 & b+2 \\ 8 & a-8b \end{bmatrix}$

By equating, $a+4 = 2a+2$ or $a = 2$

$3b = b+2$ or $b = 1$

$\therefore a-2b = 2-2(1)$

$= 2-2$

$= 0$

5. . Soln. Given $A^2 = \lambda A$, where $A = \begin{bmatrix} 3 & -3 \\ -3 & 3 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 3 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 3 & -3 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 9+9 & -9-9 \\ -9-9 & 9+9 \end{bmatrix} = \begin{bmatrix} 18 & -18 \\ -18 & 18 \end{bmatrix}$$

Since $A^2 = \lambda A$ or $\begin{bmatrix} 18 & -18 \\ -18 & 18 \end{bmatrix} = \begin{bmatrix} 3\lambda & -3\lambda \\ -3\lambda & 3\lambda \end{bmatrix}$

Or $18 = 3\lambda$ or $\lambda = 6$

6. . Soln. We have, $X + Y = \begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix}$, $X - Y = \begin{bmatrix} 6 & 5 \\ 7 & 3 \end{bmatrix}$

$(X+Y) + (X-Y) = \begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix} + \begin{bmatrix} 6 & 5 \\ 7 & 3 \end{bmatrix}$

$2X = \begin{bmatrix} 8 & 8 \\ 12 & 4 \end{bmatrix}$

Or $X = \begin{bmatrix} 4 & 4 \\ 6 & 2 \end{bmatrix}$

$\therefore X = \begin{bmatrix} 4 & 4 \\ 6 & 2 \end{bmatrix}$

And $Y = \begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 4 \\ 6 & 2 \end{bmatrix}$

$= \begin{bmatrix} -2 & -1 \\ -1 & -1 \end{bmatrix}$

7. . Soln. Getting $A^2 = A.A$

$= \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix}$

$\therefore A^2 - 5A + 4I = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - 5 \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$

$+ 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} + \begin{bmatrix} -10 & 0 & -5 \\ -10 & -5 & -15 \\ -5 & 5 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

$= \begin{bmatrix} -1 & -1 & -3 \\ -1 & -3 & -10 \\ -5 & 4 & 2 \end{bmatrix}$

Getting $X = -(A^2 - 5A + 4I)$

i.e., $X = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{bmatrix}$

8. . Soln. We have, $A = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$

$A' = \begin{bmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{bmatrix}$

$$A + A' = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Or } \frac{1}{2}(A + A') = \frac{1}{2} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

$$\text{And } A - A' = \begin{bmatrix} 0 & 2a & 2b \\ -2a & 0 & 2c \\ -2b & -2c & 0 \end{bmatrix}$$

$$\text{Or } \frac{1}{2}(A - A') = \frac{1}{2} \begin{bmatrix} 0 & 2a & 2b \\ -2a & 0 & 2c \\ -2b & -2c & 0 \end{bmatrix}$$

$$\text{Or } \frac{1}{2}(A - A') = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$$

9.

Soln. We know that a square matrix A can be written as

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

Out of which $\frac{1}{2}(A + A^T)$ is symmetric and $\frac{1}{2}(A - A^T)$

is skew symmetric matrix.

∴ For the given matrix

$$A = \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix} \text{ and } A^T = \begin{bmatrix} 2 & 7 & 1 \\ 4 & 3 & -2 \\ -6 & 5 & 4 \end{bmatrix}$$

$$\therefore A + A^T = \begin{bmatrix} 4 & 11 & -5 \\ 11 & 6 & 3 \\ -5 & 3 & 8 \end{bmatrix} \text{ and } A - A^T = \begin{bmatrix} 0 & -3 & -7 \\ 3 & 0 & 7 \\ 7 & -7 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -3 & -7 \\ 3 & 0 & 7 \\ 7 & -7 & 0 \end{bmatrix}$$

$$\text{Hence, } A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

$$= \begin{bmatrix} 2 & 11/2 & -5/2 \\ 11/2 & 3 & 3/2 \\ -5/2 & 3/2 & 4 \end{bmatrix} + \begin{bmatrix} 0 & -3/2 & -7/2 \\ 3/2 & 0 & 7/2 \\ 7/2 & -7/2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix} = A$$

10. Soln. We know that

$$A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$$

Here, $\frac{1}{2}(A + A')$ is symmetric matrix and

$\frac{1}{2}(A - A')$ is skew symmetric matrix.

$$\text{Now, } A = \begin{bmatrix} 3 & -2 & -4 \\ 3 & -2 & -5 \\ -1 & 1 & 2 \end{bmatrix} \Rightarrow A' =$$

$$\begin{bmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{bmatrix}$$

$$\therefore \frac{1}{2}(A + A') = \frac{1}{2} \begin{bmatrix} 3+3 & -2+3 & -4-1 \\ 3-2 & -2-2 & -5+1 \\ -1-4 & 1-5 & 2+2 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 6 & 1 & -5 \\ 1 & -4 & -4 \\ -5 & -4 & 4 \end{bmatrix} \text{ which is symmetric}$$

$$\frac{1}{2}(A - A') = \frac{1}{2} \begin{bmatrix} 3-3 & -2-3 & -4+1 \\ 3+2 & -2+2 & -5-1 \\ -1+4 & 1+5 & 2-2 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 0 & -5 & -3 \\ 5 & 0 & -6 \\ 3 & 6 & 0 \end{bmatrix} \text{ which is symmetric.}$$

$$\therefore A = \frac{1}{2} \begin{bmatrix} 6 & 1 & -5 \\ 1 & -4 & -4 \\ -5 & -4 & 4 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 0 & -5 & -3 \\ 5 & 0 & -6 \\ 3 & 6 & 0 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 3 & 1/2 & -5/2 \\ 1/2 & -2 & -2 \\ -5/2 & -2 & 2 \end{bmatrix} + \begin{bmatrix} 0 & -5/2 & -3/2 \\ 5/2 & 0 & -3 \\ 3/2 & 3 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -2 & -4 \\ 3 & -2 & -5 \\ -1 & 1 & 2 \end{bmatrix}$$

11. Soln. Case I: Let A be a symmetric matrix. Then $A^T = A$.

$$\text{Now, } (B^T A B)^T = B^T A^T (B^T)^T \quad [\text{By reversal law}]$$

$$= B^T A^T B \quad [\because (B^T)^T = B]$$

$$\text{Or } (B^T A B)^T = B^T A B \quad [\because A^T = A]$$

$$\therefore B^T A B \text{ is a symmetric matrix.}$$

Case II: Let A be a skew-symmetric matrix. Then, $A^T = -A$.

$$\text{Now, } (B^T A B)^T = B^T A^T (B^T)^T \quad [\text{By reversal law}]$$

$$\text{Or } (B^T A B)^T = B^T A^T B \quad [\because (B^T)^T = B]$$

$$\text{Or } (B^T A B)^T = B^T (-A) B \quad [\because A^T = -A]$$

$$\text{Or } (B^T A B)^T = -B^T A B$$

∴ $B^T A B$ is a skew-symmetric matrix.

12. Soln. We shall prove the result by using principle of mathematical induction.

$$P(n): A^n = \begin{bmatrix} 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \end{bmatrix}$$

Let

$$P(1): A^1 = \begin{bmatrix} 3^0 & 3^0 & 3^0 \\ 3^0 & 3^0 & 3^0 \\ 3^0 & 3^0 & 3^0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Now,

The result is true for $n = 1$.

Let the result be true for $n = k$.

$$A^k = \begin{bmatrix} 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \end{bmatrix}$$

So,

Now, we prove that $P(k+1)$ is true.

Now, $A^{k+1} = AA^k$

$$\begin{aligned} &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \end{bmatrix} \\ &= \begin{bmatrix} 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} \\ 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} \\ 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} \end{bmatrix} \\ &= \begin{bmatrix} 3^k & 3^k & 3^k \\ 3^k & 3^k & 3^k \\ 3^k & 3^k & 3^k \end{bmatrix} \\ &= A^{k+1} \end{aligned}$$

Hence, it is true $n = k + 1$.

Hence, by principle of mathematical induction $P(n)$ is true for all $n \in \mathbb{N}$.

13. . Soln. We have, $\begin{bmatrix} x^2 - 4x & x^2 \\ x^2 & x^2 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -x+2 & 1 \end{bmatrix}$

Equating the corresponding elements of two matrices, we get

$$x^2 - 4x = -3 \Rightarrow x^2 - 4x + 3 = 0 \Rightarrow (x-1)(x-3) = 0$$

$$\Rightarrow x = 1, 3$$

$$x^2 = 1 \Rightarrow x = \pm 1$$

$$x^2 = -x + 2 \Rightarrow x^2 + x - 2 = 0$$

$$\Rightarrow (x-1)(x+2) = 0 \Rightarrow x = 1, -2$$

$$x^3 = 1 \Rightarrow x^3 - 1 = 0 \Rightarrow (x-1)(x^2 + x + 1) = 0$$

$$\Rightarrow x = 1, \omega, \omega^2,$$

$$\omega = \frac{-1 + \sqrt{3}i}{2}, \omega^2 = \frac{-1 - \sqrt{3}i}{2}$$

Where,

Since, common value of x is 1.

$$\therefore x = 1$$

14. .

Soln. We have, $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \Rightarrow A' = \begin{bmatrix} p & r \\ q & s \end{bmatrix}$

$$\therefore A + A' = \begin{bmatrix} p & q \\ r & s \end{bmatrix} + \begin{bmatrix} p & r \\ q & s \end{bmatrix} = \begin{bmatrix} 2p & q+r \\ q+r & 2s \end{bmatrix}$$

$$\Rightarrow \frac{1}{2}(A + A') = \frac{1}{2} \begin{bmatrix} 2p & q+r \\ q+r & 2s \end{bmatrix} = \begin{bmatrix} p & \frac{q+r}{2} \\ \frac{q+r}{2} & s \end{bmatrix}$$

$$\text{Also, } A - A' = \begin{bmatrix} p & q \\ r & s \end{bmatrix} - \begin{bmatrix} p & r \\ q & s \end{bmatrix} = \begin{bmatrix} 0 & q-r \\ r-q & 0 \end{bmatrix}$$

$$\Rightarrow \frac{1}{2}(A - A') = \frac{1}{2} \begin{bmatrix} 0 & q-r \\ r-q & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{q-r}{2} \\ \frac{r-q}{2} & 0 \end{bmatrix}$$

$$\therefore A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$$

$$= \begin{bmatrix} p & \frac{q+r}{2} \\ \frac{q+r}{2} & s \end{bmatrix} + \begin{bmatrix} 0 & \frac{q-r}{2} \\ \frac{r-q}{2} & 0 \end{bmatrix}$$

15. .Ans. We have, $A^2 + I = kA$

$$\Rightarrow \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = k \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 11 & -8 \\ -4 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = k \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 12 & -8 \\ -4 & 4 \end{bmatrix} = k \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$$

$$\Rightarrow -4 \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix} = k \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$$

On comparing, we get $k = -4$

16. .Ans. Given, $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$

Now, $A^2 - 5A + 4I$

$$= \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} - 5 \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$$

$$+ 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & -1 & 2 \\ 9 & 2 & 5 \\ 0 & -1 & 2 \end{bmatrix} - \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 & -3 \\ -1 & -3 & -10 \\ -5 & 4 & 2 \end{bmatrix}$$

Since,

$$A^2 - 5A + 4I + X = O \Rightarrow X = -(A^2 - 5A + 4I)$$

$$\therefore X = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{bmatrix}$$

17. .

$$\text{Ans. } A^2 = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} = \begin{bmatrix} 28 & 38 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix}$$

$$\text{Now, } A^3 - 4A^2 - 3A + 11I = \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} -$$

$$4 \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix}$$

$$- 3 \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} + 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} - \begin{bmatrix} 36 & 28 & 20 \\ 4 & 16 & 4 \\ 32 & 36 & 36 \end{bmatrix} - \begin{bmatrix} 3 & 9 & 6 \\ 6 & 0 & -3 \\ 3 & 6 & 9 \end{bmatrix}$$

$$+ \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Hence, } A^3 - 4A^2 - 3A + 11I = O$$

$$\text{Now, } A^{-1}[A^3 - 4A^2 - 3A + 11I] = A^{-1}O$$

$$\Rightarrow A^2 - 4A - 3A^{-1}A + 11A^{-1} = O$$

$$\Rightarrow A^2 - 4A - 3I + 11A^{-1} = O$$

$$\Rightarrow A^{-1} = \frac{-A^2 + 4A + 3I}{11}$$

$$\Rightarrow A^{-1} = \frac{1}{11} \left(\begin{bmatrix} -9 & -7 & -5 \\ -1 & -4 & -1 \\ -8 & -9 & -9 \end{bmatrix} + \begin{bmatrix} 4 & 12 & 8 \\ 8 & 0 & -4 \\ 4 & 8 & 12 \end{bmatrix} \right.$$

$$\left. + \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \right)$$

$$\Rightarrow A^{-1} = \frac{1}{11} \left(\begin{bmatrix} -2 & 5 & 3 \\ 7 & -1 & -5 \\ -4 & -1 & 6 \end{bmatrix} \right)$$

$$= \begin{bmatrix} -2/11 & 5/11 & 3/11 \\ 7/11 & -1/11 & -5/11 \\ -4/11 & -1/11 & 6/11 \end{bmatrix}$$

18. .

$$\text{Ans. Let, } A = \begin{bmatrix} 0 & 6-5x \\ x^2 & x+3 \end{bmatrix}$$

A is symmetric, then $A' = A$

$$\therefore \begin{bmatrix} 0 & x^2 \\ 6-5x & x+3 \end{bmatrix} = \begin{bmatrix} 0 & 6-5x \\ x^2 & x+3 \end{bmatrix}$$

On comparing both sides, we get

$$\Rightarrow x^2 = 6 - 5x \Rightarrow x^2 + 5x - 6 = 0$$

$$\Rightarrow (x+6)(x-1) = 0 \Rightarrow x = -6, 1$$

$$19. \text{ .Ans. Given, } A = \begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix}$$

$$(i) A' = \begin{bmatrix} 1 & 6 \\ 5 & 7 \end{bmatrix}$$

$$A + A' = \begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 6 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 11 \\ 11 & 14 \end{bmatrix}$$

$$\therefore (A + A')' = \begin{bmatrix} 2 & 11 \\ 11 & 14 \end{bmatrix}$$

$$\Rightarrow (A + A')' = A + A'$$

$\therefore (A + A')$ is a symmetric matrix.

$$(ii) A - A' = \begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix} - \begin{bmatrix} 1 & 6 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$(A - A')' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = -(A - A')$$

$\Rightarrow (A - A')$ is a skew symmetric matrix.

20. .Soln.

$$\text{Given } X = \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}$$

$$X^2 = X.X = \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 16-1 & 4+2 \\ -4-2 & -1+4 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & 6 \\ -6 & 3 \end{bmatrix}$$

$$\therefore 6X - X^2 = 6 \begin{bmatrix} 4 & 1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} 15 & 6 \\ -6 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 24 & 6 \\ -6 & 12 \end{bmatrix} - \begin{bmatrix} 15 & 6 \\ -6 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 24-15 & 6-6 \\ -6+6 & 12-3 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix}$$

$$= 9 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 9I$$

21. .

Soln.

Given, $2X + Y = \begin{bmatrix} 2 & -3 & 1 \\ 1 & 2 & 3 \end{bmatrix}$

And $X - Y = \begin{bmatrix} 0 & 3 & 5 \\ -2 & -4 & 1 \end{bmatrix}$

For finding the value of X, add $2X + Y$ and $X - Y$.

$$2X + Y + X - Y = \begin{bmatrix} 2 & -3 & 1 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 3 & 5 \\ -2 & -4 & 1 \end{bmatrix}$$

$$3X = \begin{bmatrix} 2 & 0 & 6 \\ -1 & -2 & 4 \end{bmatrix}$$

$$\therefore X = \frac{1}{3} \begin{bmatrix} 2 & 0 & 6 \\ -1 & -2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2/3 & 0 & 2 \\ -1/3 & -2/3 & 4/3 \end{bmatrix}$$

On putting the value of X in $(X - Y)$,

$$\begin{bmatrix} 2/3 & 0 & 2 \\ -1/3 & -2/3 & 4/3 \end{bmatrix} - Y = \begin{bmatrix} 0 & 3 & 5 \\ -2 & -4 & 1 \end{bmatrix}$$

$$Y = \begin{bmatrix} 2/3 & 0 & 2 \\ -1/3 & -2/3 & 4/3 \end{bmatrix} - \begin{bmatrix} 0 & 3 & 5 \\ -2 & -4 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2/3 & -3 & -3 \\ 5/3 & 10/3 & 1/3 \end{bmatrix}$$

Hence, $X = \begin{bmatrix} 2/3 & 0 & 2 \\ -1/3 & -2/3 & 4/3 \end{bmatrix}$

$$Y = \begin{bmatrix} 2/3 & -3 & -3 \\ 5/3 & 10/3 & 1/3 \end{bmatrix}$$

22. .Soln. Given,

$$\lambda \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} + 2 \begin{bmatrix} 2 & 5 & 6 \\ 1 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 7 & 19 & 27 \\ 8 & 18 & 28 \end{bmatrix}$$

Multiply by non-zero scalar λ to the corresponding matrix and then add to another matrix,

$$\begin{bmatrix} \lambda & 3\lambda & 0 \\ 2\lambda & 4\lambda & 6\lambda \end{bmatrix} + \begin{bmatrix} 1 & 0 & 12 \\ 5 & 6 & 10 \end{bmatrix} = \begin{bmatrix} 7 & 19 & 27 \\ 8 & 18 & 28 \end{bmatrix}$$

$$\begin{bmatrix} \lambda & 3\lambda & 5\lambda \\ 2\lambda & 4\lambda & 6\lambda \end{bmatrix} = \begin{bmatrix} 7 & 19 & 27 \\ 8 & 18 & 28 \end{bmatrix} - \begin{bmatrix} 4 & 10 & 12 \\ 2 & 6 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 19 & 27 \\ 8 & 18 & 28 \end{bmatrix} + \begin{bmatrix} -4 & -10 & -12 \\ -2 & -6 & -10 \end{bmatrix}$$

$$= \begin{bmatrix} 7-4 & 19-10 & 27-12 \\ 8-2 & 18-6 & 28-10 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 9 & 15 \\ 6 & 12 & 18 \end{bmatrix}$$

On equating the corresponding elements, we

23. .Soln.

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$$

Given,

$$A^2 = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$$

Now,

$$A^2 = \begin{bmatrix} 4+0+1 & 0+0-1 & 2+0+0 \\ 4+2+3 & 0+1-3 & 2+3+0 \\ 2-2+0 & 0-1-0 & 1-3+0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix}$$

$$5A = 5 \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix}$$

And

$$\therefore A^2 - 5A = \begin{bmatrix} 2 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix}$$

$$A^2 - 5A = \begin{bmatrix} -8 & -1 & -3 \\ -1 & -7 & -10 \\ -5 & 4 & -2 \end{bmatrix}$$

24. .Soln.

We have, $\begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{bmatrix}_{3 \times 2} A = \begin{bmatrix} -1 & -8 \\ 1 & -2 \\ 9 & 22 \end{bmatrix}_{3 \times 2}$

$\therefore A$ is of order 2×2

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\text{Then, } \begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} -1 & -8 \\ 1 & -2 \\ 9 & 22 \end{bmatrix}$$

$$\begin{bmatrix} 2a-c & 2b-d \\ a & b \\ -3a+4c & -3b+4d \end{bmatrix} = \begin{bmatrix} -1 & -8 \\ 1 & -2 \\ 9 & 22 \end{bmatrix}$$

By equality of matrices, on comparing, we get

$$2a - c = -1$$

$$2b - d = -8$$

$$a = 1$$

$$b = -2$$

$$-3a + 4c = 9$$

$$-3b + 4d = 22$$

On solving the equations, we get

$$a = 1, b = -2, c = 3, d = 4$$

Hence, $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ Ans.

25...

Soln.

Given, $B = \begin{bmatrix} -2 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 0 & -2 \\ 7 & 1 & 6 \end{bmatrix}$

And $2A - 3B + 5C = 0$

$$\Rightarrow 2A = -5C + 3B$$

$$\Rightarrow A = \frac{1}{2}[3B - 5C]$$

$$\Rightarrow A = \frac{1}{2} \left(3 \begin{bmatrix} -2 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix} - 5 \begin{bmatrix} 2 & 0 & -2 \\ 7 & 1 & 6 \end{bmatrix} \right)$$

$$\Rightarrow A = \frac{1}{2} \left(\begin{bmatrix} -6 & 6 & 0 \\ 9 & 3 & 12 \end{bmatrix} - \begin{bmatrix} 10 & 0 & -10 \\ 35 & 5 & 30 \end{bmatrix} \right)$$

$$\Rightarrow A = \frac{1}{2} \begin{bmatrix} -16 & 6 & 10 \\ -26 & -2 & -18 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} -8 & 3 & 5 \\ -13 & -1 & -9 \end{bmatrix} \text{ Ans.}$$

26. Soln. We have, $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Now, $4A + 2B = 4 \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix} + 2 \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$$= \begin{bmatrix} 8 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 12 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 10 \end{bmatrix}$$

$$\therefore 4A + 2B = \text{diag}[14 - 4 10]$$

27. Soln. Given, $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+4 & 0+0+0 & 2+0+6 \\ 0+0+2 & 0+4+0 & 0+2+3 \\ 2+2+6 & 0+0+0 & 4+0+9 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$$A^3 = A \cdot A^2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$$= \begin{bmatrix} 5+0+16 & 0+0+0 & 8+0+26 \\ 0+4+8 & 0+8+0 & 0+10+13 \\ 10+0+24 & 0+0+0 & 16+0+39 \end{bmatrix}$$

$$= \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix}$$

Since, $A^3 - 6A^2 + 7A + kI_3 = O$

$$\Rightarrow \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} - 6 \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

$$+ k \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} - \begin{bmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 14 \\ 0 & 14 & 7 \\ 14 & 0 & 21 \end{bmatrix}$$

$$+ \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2+k & 0 & 0 \\ 0 & -2+k & 0 \\ 0 & 0 & -2+k \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

On equating the corresponding elements, we get $-2 + k = 0 \Rightarrow k = 2$

28. Soln. We have, $A = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$

$$\begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$$

$$\Rightarrow A' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix} \text{ and } B' = \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix}$$

(i) $(A')' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix} = A$

(ii) $A + B = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix} + \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$

$$= \begin{bmatrix} 3+2 & \sqrt{3}-1 & 2+2 \\ 4+1 & 2+2 & 0+4 \end{bmatrix} = \begin{bmatrix} 5 & \sqrt{3}-1 & 4 \\ 5 & 4 & 4 \end{bmatrix}$$

$$\Rightarrow (A+B)' = \begin{bmatrix} 5 & \sqrt{3}-1 & 4 \\ 5 & 4 & 4 \end{bmatrix}' =$$

$$\begin{bmatrix} 5 & 5 \\ \sqrt{3}-1 & 4 \\ 4 & 4 \end{bmatrix} \text{ Also, } A' + B' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix} =$$

Thus, $(A+B)' = A' + B'$

$$(iii) \quad kB = k \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2k & -k & 2k \\ k & 2k & 4k \end{bmatrix} \text{ And}$$

$$(kB)' = \begin{bmatrix} 2k & -k & 2k \\ k & 2k & 4k \end{bmatrix}' = \begin{bmatrix} 2k & k \\ -k & 2k \\ 2k & 4k \end{bmatrix} =$$

$$k \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix} = kB' \text{ Thus } (kB)' = kB'$$

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New

SURE SHOT QUESTIONS 2026

Chapter – 04 (Questions)

Determinants

➤ Questions

1. If $A = \begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix}$, find $\text{adj } A$.

2. If A is a square matrix satisfying $A' A = I$, write the value of $|A|$.

3. If A is a 3×3 matrix, $|A| \neq 0$, $|3A| = k|A|$, then write the value of k .

4. If $A = \begin{pmatrix} 4 & 6 \\ 7 & 5 \end{pmatrix}$, then what is the value of $A \cdot (\text{adj } A)$?

5. For what value of x , the given matrix $A = \begin{pmatrix} 3-2x & x+1 \\ 2 & 4 \end{pmatrix}$ is singular matrix?

6. For what value of x , the matrix $\begin{pmatrix} 1+x & 7 \\ 3-x & 8 \end{pmatrix}$ is a singular matrix?

7. If A, B are square matrices of the same order, then prove that

$$\text{adj}(AB) = (\text{adj } B)(\text{adj } A).$$

8. If $A = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$ find $\text{adj } A$ and verify that $A(\text{adj } A) = (\text{adj } A)A = |A|I_3$.

9. If $A = \begin{bmatrix} 1 & \cot x \\ -\cot x & 1 \end{bmatrix}$, show that $A' A^{-1} = \begin{bmatrix} -\cos 2x & -\sin 2x \\ \sin 2x & -\cos 2x \end{bmatrix}$.

10. If $A = \begin{bmatrix} 2 & 1 & -3 \\ 3 & 2 & 1 \\ 1 & 2 & -1 \end{bmatrix}$, find A^{-1} and hence solve the

following system of equations :

$$2x + y - 3z = 13$$

$$3x + 2y + z = 4$$

$$x + 2y - z = 8$$

11. For the matrix $A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$, verify the following:

$$A(\text{adj } A) = (\text{adj } A)A = |A|I.$$

12. Given $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$, compute I^{-1} and show that

$$2A^{-1} = 9I - A.$$

13. If A is a skew-symmetric matrix of order 3, then prove that $\det A = 0$.

14. Using matrix method, solve $x - 2y = 4$ and $-3x + 5y = -7$.

15. Let $A = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} - & 13 \\ & 1 \end{bmatrix}$, find a matrix X such that $AX = B$.

16. If $A = \begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix}$, find matrix B such that $AB = I$.

17. If $A = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -1 & 2 \\ 7 & 3 & -3 \end{bmatrix}$, then find A^{-1} and hence solve the following system of equation: $3x + 4y + 7z = 14$, $2x - y + 3z = 4$, $x + 2y - 3z = 0$

18. If $A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 1 \\ 1 & -3 & 1 \end{bmatrix}$, find A^{-1} . Hence solve the system of equations:
 $x + 2y + z = 4, -x + y + z = 0, x - 3y + z = 4.$

19. If $A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$ are square matrices, find $A \cdot B$ and hence solve the system of equations:
 $x - y = 3, 2x + 3y + 4z = 17$ and $y + 2z = 7.$

20. If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$ find $\text{adj } A$ and verify that $A(\text{adj } A) = (\text{adj } A)A = |A|I_3.$

21. If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and I is the identity matrix of order 2, then show that $A^2 = 4A - 3I$. Hence find $A^{-1}.$

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@sureshotquestions

New

SURE SHOT QUESTIONS 2026

Chapter – 04 (Solutions)

Determinants

Questions

1.

Soln. Let C_{ij} be the cofactor of a_{ij} in $A = [a_{ij}]$. Then,

$$C_{11} = \text{cofactor of } a_{11} = (-1)^{1+1} 7 = 7$$

$$C_{12} = \text{cofactor of } a_{12} = (-1)^{1+2} 6 = -6$$

$$C_{21} = \text{cofactor of } a_{21} = (-1)^{2+1} 5 = -5$$

$$C_{22} = \text{cofactor of } a_{22} = (-1)^{2+2} 1 = 1$$

So adj A is

$$\text{adj} A = \begin{bmatrix} 6 & -6 \\ -5 & 1 \end{bmatrix}^T$$

$$\text{adj} A = \begin{bmatrix} 7 & -5 \\ -6 & 1 \end{bmatrix}$$

2. ..

Soln. Given, $A'A = I$

Then $|A'A| = |I|$

$$\Rightarrow |A'| |A| = |I|$$

$$\Rightarrow |A| |A| = |I| \quad [\because |A'| = |A|]$$

$$\Rightarrow |A|^2 = 1 \quad [\because |I| = 1]$$

$$\Rightarrow |A| = \pm 1 \quad \text{Ans.}$$

3. ..

Soln. Since $|kA| = k^n |A|$, where n is the order of matrix. $= 27 |A|$

$$\therefore |3A| = 3^3 |A| \quad \text{or} \quad k = 27$$

4. .

$$\text{Soln. } A = \begin{pmatrix} 4 & 6 \\ 7 & 5 \end{pmatrix}$$

$$\text{adj} A = \begin{pmatrix} 5 & -6 \\ -7 & 4 \end{pmatrix}$$

$$A. (\text{adj} A) = \begin{pmatrix} 4 & 6 \\ 7 & 5 \end{pmatrix} \begin{pmatrix} 5 & -6 \\ -7 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 20 - 42 & -24 + 24 \\ 35 - 35 & -42 + 20 \end{pmatrix}$$

$$= \begin{pmatrix} -22 & 0 \\ 0 & -22 \end{pmatrix}$$

5. .

$$\text{Soln. } A = \begin{vmatrix} 3 - 2x & x + 1 \\ 2 & 4 \end{vmatrix}$$

Since A is a singular matrix. i.e., $|A| = 0$

$$\begin{vmatrix} 3 - 2x & x + 1 \\ 2 & 4 \end{vmatrix} = 0$$

$$\text{Or, } 4(3 - 2x) - 2(x + 1) = 0$$

$$\text{Or, } 12 - 8x - 2x - 2 = 0$$

$$\text{Or, } -10x + 10 = 0$$

$$\text{Or, } x = 1$$

6. .

Soln. Since, A is a singular matrix

$$\begin{vmatrix} 1 + x & 7 \\ 3 - x & 8 \end{vmatrix} = 0$$

$$\text{Or, } 8(1 + x) - 7(3 - x) = 0$$

$$\text{Or, } 8 + 8x - 21 + 7x = 0$$

$$\text{Or, } 15x - 13 = 0$$

$$\text{Or } x = \frac{13}{15}$$

7. .

Soln. We know that

$$(AB) \text{adj}(AB) = |AB| \text{adj}(AB) \quad \dots\dots\dots (i)$$

$$(AB)(\text{adj} B \text{adj} A)$$

$$= A B \text{adj} B \text{adj} A = A(B \text{adj} B) \text{adj} A$$

$$= A(|B| I) \text{adj} A \quad [\because B \text{adj} B = |B| I]$$

$$= |B| (A \cdot \text{adj} A)$$

$$= |B| |A| I \quad [\because A \text{adj} A = |A| I]$$

$$= |A| |B| I$$

$$= |AB| I \quad \dots\dots\dots (ii)$$

From (i) and (ii), we get

$$AB(\text{adj} AB) = AB(\text{adj} B \cdot \text{adj} A)$$

Pre-multiplying both sides by $(AB)^{-1}$, we get

$$(AB)^{-1}[(AB) \text{adj} AB]$$

$$= (AB)^{-1}[(AB) \text{adj} B \cdot \text{adj} A]$$

$$\text{Or } \text{adj} AB = \text{adj} B \cdot \text{adj} A$$

8. ..

Soln. We have, $A = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Clearly, the cofactors of elements of $|A|$ are given by

$$A_{11} = \cos \alpha; A_{12} = -\sin \alpha; A_{13} = 0;$$

$$A_{21} = \sin \alpha; A_{22} = \cos \alpha; A_{23} = 0$$

$$A_{31} = 0; A_{32} = 0 \text{ and } A_{33} = 1$$

$$\therefore \text{adj}(A) = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$$

$$= \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, $A(\text{adj } A)$

$$= \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos^2 \alpha + \sin^2 \alpha & 0 & 0 \\ 0 & \sin^2 \alpha + \cos^2 \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \dots (i)$$

$$(\text{adj } A) \cdot (A) = \begin{pmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos^2 \alpha + \sin^2 \alpha & 0 & 0 \\ 0 & \sin^2 \alpha + \cos^2 \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \dots (ii)$$

$$\text{And } |A| = \begin{vmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= 1 \cdot (\cos^2 \alpha + \sin^2 \alpha) = 1 \dots (iii)$$

From eqs. (i), (ii) and (iii), we get

$$A(\text{adj } A) = (\text{adj } A) \cdot A = |A| I_3$$

9. .

Ans. We have,

$$A = \begin{bmatrix} 1 & \cot x \\ -\cot x & 1 \end{bmatrix} \Rightarrow A' = \begin{bmatrix} 1 & -\cot x \\ \cot x & 1 \end{bmatrix}$$

Also, $|A| = 1 + \cot^2 x = \text{cosec}^2 x \neq 0 \therefore A^{-1}$ exists

$$\Rightarrow \text{adj } A = \begin{bmatrix} 1 & \cot x \\ -\cot x & 1 \end{bmatrix}' = \begin{bmatrix} 1 & -\cot x \\ \cot x & 1 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{\text{cosec}^2 x} \begin{bmatrix} 1 & -\cot x \\ \cot x & 1 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} \frac{1}{1+\cot^2 x} & \frac{-\cot x}{1+\cot^2 x} \\ \frac{\cot x}{1+\cot^2 x} & \frac{1}{1+\cot^2 x} \end{bmatrix}$$

$$\text{Now, } A' A^{-1} = \begin{bmatrix} 1 & -\cot x \\ \cot x & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{1+\cot^2 x} & \frac{-\cot x}{1+\cot^2 x} \\ \frac{\cot x}{1+\cot^2 x} & \frac{1}{1+\cot^2 x} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1-\cot^2 x}{1+\cot^2 x} & \frac{-\cot x - \cot x}{1+\cot^2 x} \\ \frac{\cot x + \cot x}{1+\cot^2 x} & \frac{-\cot^2 x + 1}{1+\cot^2 x} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sin^2 x - \cos^2 x}{\sin^2 x + \cos^2 x} & \frac{-2\cos x \sin x}{\sin^2 x + \cos^2 x} \\ \frac{2\sin x \cos x}{\sin^2 x + \cos^2 x} & \frac{-\cos^2 x + \sin^2 x}{\cos^2 x + \sin^2 x} \end{bmatrix}$$

$$= \begin{bmatrix} -\cos 2x & -\sin 2x \\ \sin 2x & -\cos 2x \end{bmatrix}$$

Hence proved.

10. .Ans. We have, $A = \begin{bmatrix} 2 & 1 & -3 \\ 3 & 2 & 1 \\ 1 & 2 & -1 \end{bmatrix}$

$$\Rightarrow |A| = 2(-2-2) - 1(-3-1) - 3(6-2)$$

$$= -8 + 4 - 12 = -16 \neq 0$$

$\therefore A^{-1}$ exists.

$$\text{Adj } A = \begin{bmatrix} -4 & -5 & 7 \\ 4 & 1 & -11 \\ 4 & -3 & 1 \end{bmatrix} \Rightarrow A^{-1} = \frac{-1}{16} \begin{bmatrix} -4 & -5 & 7 \\ 4 & 1 & -11 \\ 4 & -3 & 1 \end{bmatrix}$$

The system of equations :

$$2x + y - 3z = 13$$

$$3x + 2y + z = 4$$

$$x + 2y - z = 8$$

can be written as $AX = B$

$$\text{i.e., } \begin{bmatrix} 2 & 1 & -3 \\ 3 & 2 & 1 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 13 \\ 4 \\ 8 \end{bmatrix}$$

$$\text{Now, } X = A^{-1}B = \frac{-1}{16} \begin{bmatrix} -4 & -5 & 7 \\ 4 & 1 & -11 \\ 4 & -3 & 1 \end{bmatrix} \begin{bmatrix} 13 \\ 4 \\ 8 \end{bmatrix}$$

$$= \frac{-1}{16} \begin{bmatrix} -52 - 20 + 56 \\ 52 + 4 - 88 \\ 52 - 12 + 8 \end{bmatrix} = \frac{-1}{16} \begin{bmatrix} -16 \\ -32 \\ 48 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

$$\therefore x = 1, y = 2, z = -3$$

11. .

$$A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \Rightarrow |A| = -12 + 12 = 0$$

Ans. Given,

$\therefore A$ is a singular matrix

$$\text{adj } A = \begin{bmatrix} -6 & 4 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$$

$$\text{Now, } A(\text{adj } A) = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

.....(i)

$$\text{And } (\text{adj } A)A = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{.....(ii)}$$

$$\text{Similarly, } |A|I = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \text{.....(iii) } (\because |A| = 0)$$

From equation (i), (ii) and (iii);

$$\text{We have } A(\text{adj } A) = (\text{adj } A)A = |A|I$$

12. .

$$A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$$

Ans. We have,

$$\therefore |A| = \begin{vmatrix} 2 & -3 \\ -4 & 7 \end{vmatrix} = 14 - 12 = 2 \neq 0$$

So, A is a non-singular matrix and therefore it is invertible.

$$\therefore \text{adj } A = \begin{bmatrix} 7 & 4 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$$

$$\text{Hence, } A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{2} \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$$

$$\Rightarrow 2A^{-1} = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix} \text{.....(i)}$$

$$\text{Now, } 9I - A = 9 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix} \text{ (From (ii))}$$

$$\text{Hence, } 2A^{-1} = 9I - A.$$

13. .

Soln.

Given, A is a skew-symmetric matrix of order 3.

$$\text{So, } A = -A^T$$

$$\text{Now, } |A| = |-A^T|$$

$$|A| = (-1)^3 |A^T|$$

$$\therefore |kA| = k^n |A|$$

where n is order of A

$$|A| = -|A| \quad [\because |A^T| = |A|]$$

$$|A| = -|A|$$

$$|A| + |A| = 0$$

$$\therefore |A| = 0$$

$$\text{i.e., } \det A = 0$$

14. .

Soln. Given,

$$x - 2y = 4$$

$$-3x + 5y = -7$$

The above system of equations can be represented in the matrix form as

$$\begin{bmatrix} 1 & -2 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$$

$$\text{Or } AX = B,$$

$$\text{Where } A = \begin{bmatrix} 1 & -2 \\ -3 & 5 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\text{And } B = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$$

$$\text{Now, } |A| = \begin{vmatrix} 1 & -2 \\ -3 & 5 \end{vmatrix} = 5 - 6 = -1 \neq 0$$

$\therefore A^{-1}$ exists.

Hence, the system has unique solution given by

$$X = A^{-1}B$$

$$\text{Now } \text{adj } A = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$$

So $A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{-11} \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$

$$= \begin{bmatrix} -5 & -2 \\ -3 & -1 \end{bmatrix}$$

Now, $X = A^{-1}B$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -5 & -2 \\ -3 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ -7 \end{bmatrix}$$

$$= \begin{bmatrix} -20+14 \\ -12+7 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -6 \\ -5 \end{bmatrix}$$

$\therefore x = -6 \text{ and } y = -5.$

15..

Soln. Given, $A = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} -13 \\ 1 \end{bmatrix}$

$AX = B$

Multiply by A^{-1} on both sides

$$A^{-1}AX = A^{-1}B$$

$$IX = A^{-1}B \quad [\because A^{-1}A = I]$$

$$\Rightarrow X = A^{-1}B \quad \dots\dots\dots(i)$$

Now, $A = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix}$

$$|A| = \begin{vmatrix} 5 & 2 \\ 3 & -1 \end{vmatrix} = -5 - 6 = -11 \neq 0$$

So A is a non-singular matrix.

As a result A^{-1} exists.

Now cofactors of A are

$$C_{11} = -1, C_{12} = -3$$

$$C_{21} = -2, C_{22} = 5$$

Then, $\text{adj}(A) = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}^T$

$$= \begin{bmatrix} -1 & -3 \\ -2 & 5 \end{bmatrix}^T = \begin{bmatrix} -1 & -2 \\ -3 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$= \frac{1}{-11} \begin{bmatrix} -1 & -2 \\ -3 & 5 \end{bmatrix}$$

Now by equation (i)

$$X = \frac{1}{-11} \begin{bmatrix} -1 & -2 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} -13 \\ 1 \end{bmatrix}$$

$$= \frac{1}{-11} \begin{bmatrix} 13 & -2 \\ -39 & 5 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 11 \\ -34 \end{bmatrix} = \begin{bmatrix} -1 \\ 34/11 \end{bmatrix}$$

$\therefore x = -1 \text{ and } y = 34/11$

[comparing corresponding elements]

$$\Rightarrow X = \begin{bmatrix} -1 \\ 34/11 \end{bmatrix}$$

16..

Soln.

Given, $A = \begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix}$

And $AB = I$

$$\Rightarrow A^{-1}AB = A^{-1}I$$

$$IB = A^{-1}I$$

$$B = A^{-1}I$$

Now, $A = \begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix}$

$$|A| = \begin{vmatrix} 3 & -4 \\ -1 & 2 \end{vmatrix} = 3 \times 2 - (-1) \times (-4) \\ = 6 - 4 \\ = 2 \neq 0$$

So A^{-1} can be calculated.

Cofactor of matrix A,

$$C_{11} = 2, C_{12} = -(-1) = 1$$

$$C_{21} = -(-4) = 4, C_{22} = 3$$

$$\text{adj } A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}^T = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} (\text{adj } A)$$

$$= \frac{1}{2} \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 1/2 & 3/2 \end{bmatrix}$$

17..

Soln. $|A| = 3(3 - 6) + (-2)(-12 - 14) + 1(12 + 7) = 62 \neq 0$

Hence, A^{-1} exists. Let C_{ij} represent the cofactor of (i, j)

i th element of A . Then,

$$C_{11} = 3, C_{12} = 26, C_{13} = 19, C_{21} = 9, C_{22} = -16, C_{23} = 5,$$

$$C_{31} = 5, C_{32} = -2, C_{33} = -11.$$

$$\text{adj } A = \begin{bmatrix} -3 & 9 & 5 \\ 26 & -16 & -2 \\ 19 & 5 & -11 \end{bmatrix}$$

$$A^{-1} = \frac{1}{62} \begin{bmatrix} -3 & 9 & 5 \\ 26 & -16 & -2 \\ 19 & 5 & -11 \end{bmatrix}$$

Then given system of equations is equivalent to the matrix equation

$$AX = B, \text{ where } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 14 \\ 4 \\ 0 \end{bmatrix}$$

$$\text{Or } X = A^{-1}B$$

$$= \frac{1}{62} \begin{bmatrix} -3 & 26 & 19 \\ 9 & -16 & 5 \\ 5 & -2 & -11 \end{bmatrix} \begin{bmatrix} 14 \\ 4 \\ 0 \end{bmatrix}$$

$$= \frac{1}{62} \begin{bmatrix} 62 \\ 62 \\ 62 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{Hence, } x = 1, y = 1, z = 1$$

18. .

$$\text{Soln. Given } A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 1 \\ 1 & -3 & 1 \end{bmatrix}$$

$$|A| = 1(1+3) - 2(-1-1) + 1(3-1)$$

$$= 4 + 4 + 2 = 10$$

Co-factor Matrix is:

$$= \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

$\therefore \text{adj } A = \text{transpose of above matrix}$

$$= \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$$= \frac{1}{10} \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 2/5 & -1/2 & 1/10 \\ 1/5 & 0 & -1/5 \\ 1/5 & 1/2 & 3/10 \end{bmatrix}$$

Given set of equations are:

$$x + 2y + z = 4$$

$$-x + y + z = 0$$

$$x - 3y + z = 4$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 1 \\ 1 & -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 4 \end{bmatrix}$$

$$\Rightarrow AX = B$$

Multiplying both sides by A^{-1} , we get

$$A^{-1}AX = A^{-1}B$$

$$\Rightarrow X = A^{-1}B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2/5 & -1/2 & 1/10 \\ 1/5 & 0 & -1/5 \\ 1/5 & 1/2 & 3/10 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{8}{5} + \frac{2}{5} \\ \frac{4}{5} - \frac{4}{5} \\ \frac{4}{5} + \frac{6}{5} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$$

$$\therefore x = 2; y = 0, z = 2$$

19. .

$$\text{Soln. Given } A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\text{And } B = \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} 2+4 & 2-2 & -4+4 \\ 4-12+8 & 4+6-4 & -8-12+20 \\ -4+4 & 2-2 & -4+10 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} = 6I$$

$$\therefore AB = 6I$$

Premultiplying by A^{-1}

$$A^{-1}AB = 6A^{-1}I$$

$$\Rightarrow IB = 6A^{-1}I \quad (\because A^{-1}A = I)$$

$$\Rightarrow B = 6A^{-1} \quad (\because IX = X)$$

$$\Rightarrow A^{-1} = \frac{1}{6}B$$

Given equations are:

$$x - y = 3$$

$$2x + 3y + 4z = 17$$

$$y + 2z = 7$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix}$$

$$AX = C, \text{ where } C = \begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow A^{-1}AX = A^{-1}C$$

$$\Rightarrow X = A^{-1}C$$

$$\Rightarrow X = \frac{1}{6} \begin{bmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 17 \\ 7 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 12 \\ -6 \\ 24 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

$$\therefore x = 2, y = -1 \text{ and } z = 4$$

20. .

Soln.

$$\text{Given, } A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Now, } A_{11} = \cos \alpha, A_{12} = -\sin \alpha, A_{13} = 0,$$

$$A_{21} = \sin \alpha, A_{22} = \cos \alpha, A_{23} = 0,$$

$$A_{31} = 0, A_{32} = 0, A_{33} = 1$$

$$\therefore \text{adj}(A) = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A \cdot \text{adj}(A)$$

$$= \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \dots\dots\dots(i)$$

$$\text{adj}(A) \cdot A = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \dots\dots\dots(ii)$$

$$|A| = \begin{vmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \cos \alpha (\cos \alpha - 0) + \sin \alpha (\sin \alpha - 0) + 0 = 1 \dots\dots(iii)$$

From (i), (ii) and (iii), we get

$$A(\text{adj} A) = (\text{adj} A)A = |A| I_3$$

21..

$$\text{Soln. } A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix} \dots\dots\dots(i)$$

$$\text{Now, } 4A - 3I = 4 \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow 4A - 3I = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix} \dots\dots\dots(ii)$$

From (i) and (ii), we get

$$A^2 = 4A - 3I$$

Pre-multiplying by A^{-1} on both sides, we get

$$A^{-1}(A^2) = 4A^{-1}A - 3A^{-1}I$$

$$\Rightarrow A = 4I - 3A^{-1} [\because AA^{-1} = I]$$

$$\Rightarrow 3A^{-1} = 4I - A$$

$$\Rightarrow A^{-1} = \frac{4}{3}I - \frac{1}{3}A$$

$$\frac{4}{3} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{bmatrix}$$

New

SURE SHOT QUESTIONS 2026

Chapter – 05 (Questions)

Continuity and Differentiability

Questions

1. Find the value of a , if the function $f(x)$ is defined

$$\text{by: } f(x) = \begin{cases} 2x - 1, & x < 2 \\ a, & x = 2 \\ x + 1, & x > 2 \end{cases} \text{ is continuous at } x = 2.$$

2. Show that the function $f(x) = |x - 3|$, $x \in R$, is continuous but not differentiable at $x = 3$.

3. Determine the value of the constant ' k ' so that the function

$$f(x) = \begin{cases} \frac{kx}{|x|}, & \text{if } x < 0 \\ 3, & \text{if } x \geq 0 \end{cases} \text{ is continuous at } x = 0.$$

4. Examine the continuity of the following function:

$$f(x) = \begin{cases} \frac{x}{2|x|}, & x \neq 0 \\ \frac{1}{2}, & x = 0 \end{cases} \text{ at } x=0.$$

5. Find the values of k so that the function f is continuous at the indicated point for: $f(x) =$

$$\begin{cases} \frac{k \cos x}{\pi - 2x}, & \text{if } x \neq \frac{\pi}{2} \\ 3, & \text{if } x = \frac{\pi}{2} \end{cases} \text{ at } x = \frac{\pi}{2}.$$

6. Show that the function $f(x) = 2x - |x|$ is continuous but not differentiable at $x=0$.

7. Show that the function ' f ' defined by $f(x) =$

$$\begin{cases} 3x - 2, & 0 < x \leq 1 \\ 2x^2 - x, & 1 < x \leq 2 \\ 5x - 4, & x > 2 \end{cases} \text{ is continuous at } x=2, \text{ but not differentiable.}$$

8. If $f(x) = |\tan 2x|$, then find the value of $f'(x)$ at $x = \frac{\pi}{3}$.

9. Find $\frac{dy}{dx}$, if $(\cos x)^y = (\cos y)^x$.

10. Find the value(s) of ' λ ', if the function

$$f(x) = \begin{cases} \frac{\sin^2 \lambda x}{x^2}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases} \text{ is continuous at } x = 0.$$

Ans. We have, $f(x) = \begin{cases} \frac{\sin^2 \lambda x}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ is continuous at $x = 0$

$$\therefore \text{L.H.L.} = \text{R.H.L.}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin^2 \lambda x}{x^2} = 1 \Rightarrow \lim_{x \rightarrow 0} \frac{\sin^2 \lambda x}{\lambda^2 x^2} \cdot \lambda^2 = 1$$

$$\Rightarrow \lambda^2 \lim_{\lambda x \rightarrow 0} \left(\frac{\sin \lambda x}{\lambda x} \right)^2 = 1 \Rightarrow \lambda^2 \cdot 1 = 1$$

$$\Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$$

11. If $y = \sqrt{a + \sqrt{a + x}}$, then find $\frac{dy}{dx}$.

12. Find the values of p and q , for which

$$f(x) = \begin{cases} \frac{1 - \sin^3 x}{3 \cos^2 x}, & \text{if } x < \pi/2 \\ p, & \text{if } x = \pi/2 \\ \frac{q(1 - \sin x)}{(\pi - 2x)^2}, & \text{if } x > \pi/2 \end{cases} \text{ is continuous}$$

$$\text{at } x = \pi/2.$$

[Delhi 2016]

13. Find the values of a and b , if the function f defined

$$\text{by } f(x) = \begin{cases} x^2 + 3x + a, & x \leq 1 \\ bx + 2, & x > 1 \end{cases}$$

is differentiable at $x = 1$. [Foreign 2016]

14. If $y = e^{x^2 \cos x} + (\cos x)^x$, then find $\frac{dy}{dx}$.

15. If $\log(x^2 + y^2) = 2 \tan^{-1} \left(\frac{y}{x} \right)$, show that

$$\frac{dy}{dx} = \frac{x + y}{x - y}.$$

16. If $y = (x)^{\cos x} + (\cos x)^{\sin x}$, then find $\frac{dy}{dx}$.

17. Differentiate $\tan^{-1} \left[\frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}} \right]$ with respect to $\cot^{-1} x^2$.

18. Differentiate $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ w.r.t. $\sin^{-1} \frac{2x}{1+x^2}$, if $X \in (-1, 1)$.

19. Find the relationship between a and b so that the function 'f' defined by:

$$f(x) = \begin{cases} ax + 1, & \text{if } x \leq 3 \\ bx + 3, & \text{if } x > 3 \end{cases} \text{ is continuous at } x = 3.$$

20. Show that the function $f(x) = |x - 1| + |x + 1|$, for all $x \in R$, is not differentiable at the points $x = -1$ and $x = 1$.

21. If $x^y = e^{x-y}$, then show that $\frac{dy}{dx} = \frac{\log x}{\{\log(xe)\}^2}$.

22. If $(\cos x)^y = (\cos y)^x$, then find $\frac{dy}{dx}$.

23. If $y = \cot^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right]$, then find the value of $\frac{dy}{dx}$.

24. If $y = x^{\sin x} + (\sin x)^{\cos x}$, then find $\frac{dy}{dx}$.

25. Differentiate the following function with respect to

$$x: y = (\sin x)^x + \sin^{-1} \sqrt{x}$$

26. If $x = a \cos \theta + b \sin \theta$ and $y = a \sin \theta - b \cos \theta$, then show that $y^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = 0$.

27. If $y = e^{x+e^{x+e^{x+1000}}}$, prove that $\frac{dy}{dx} = \frac{y}{1-y}$.

28. Differentiate $\tan^{-1} \left(\frac{\sqrt{x}-x}{1+x^{3/2}} \right)$ w.r.t. x .

29. If $y = Ae^{-kt} \cos(pt + c)$, prove that $\frac{d^2 y}{dt^2} + 2k \frac{dy}{dt} + n^2 y = 0$, where $n^2 = p^2 + k^2$.

30. If $x = a \left(\cos t + \log \tan \frac{t}{2} \right)$, $y = a \sin t$, then find $\frac{d^2 y}{dt^2}$ and $\frac{d^2 y}{dx^2}$.

New

SURE SHOT QUESTIONS 2026

Chapter – 05 (Solutions)

Continuity and Differentiability

Questions

1.

Soln. Since $f(x)$ is continuous at $x = 2$, we have

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2) \quad \dots (i)$$

$$\begin{aligned} \text{Now } \lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^-} (2x - 1) \\ &= \lim_{h \rightarrow 0} [2(2 - h) - 1] = 3 \end{aligned}$$

Also, we are given

$$f(2) = a$$

Therefore, from (i), we can write:

$$f(2) = 3 \text{ or } a = 3.$$

2.

Soln. Given, $f(x) = |x - 3|, x \in \mathbb{R}$

$$= \begin{cases} x - 3, & \text{if } x \geq 3 \\ 3 - x, & \text{if } x < 3 \end{cases}$$

When $x > 3$, $f(x) = x - 3$ and it is a polynomial, so it is continuous.

When $x < 3$, $f(x) = 3 - x$. Again it is a polynomial, so it is continuous.

$$\begin{aligned} \text{Also } f(3 - 0) &= 0 = f(3 + 0) = f(3) \\ \Rightarrow f(x) &\text{ is continuous at } x = 3. \end{aligned}$$

$$\begin{aligned} \text{Now, LHD} &= f'(3 - 0) = \lim_{x \rightarrow 3-h} \frac{f(x) - f(3)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{3 - (3 - h) - 0}{-h} \\ &= \lim_{h \rightarrow 0} \frac{h}{-h} = -1. \end{aligned}$$

$$\begin{aligned} \text{And RHD} &= f'(3 + 0) = \lim_{x \rightarrow 3+h} \frac{f(x) - f(3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(3 + h) - 3 - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} = 1. \end{aligned}$$

Since $LHD \neq RHD$

$\Rightarrow f$ is not differentiable at $x = 3$. Hence Proved

3.

$$f(x) = \begin{cases} \frac{kx}{|x|}, & \text{if } x < 0 \\ 3, & \text{if } x \geq 0 \end{cases} \text{ is continuous at } x = 0.$$

Soln. Given, that the function is continuous at $x = 0$

$$\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0) \quad \dots (i)$$

$$\begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} \frac{kx}{|x|} \\ &= k \lim_{x \rightarrow 0^-} \frac{x}{-x} \\ &= k(-1) = -k \end{aligned}$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 3 = 3$$

From equation (i),

$$-k = 3$$

$$\text{Or } k = -3$$

4.

$$f(x) = \begin{cases} \frac{x}{2|x|}, & x \neq 0 \\ \frac{1}{2}, & x = 0 \end{cases}$$

Soln. Given,

For continuity at $x = 0$, we have

$$f(0) = \frac{1}{2}$$

$$LHL = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x}{2|x|} = \lim_{x \rightarrow 0^-} \frac{x}{-2x} = -\frac{1}{2}$$

$$RHL = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x}{2|x|} = \lim_{x \rightarrow 0^+} \frac{x}{2x} = \frac{1}{2}$$

Hence, $LHL \neq RHL$

So, $f(x)$ is discontinuous at $x = 0$.

5.

Soln. We have, $f(x) = \begin{cases} \frac{k \cos x}{\pi - 2x}, & \text{if } x \neq \frac{\pi}{2} \\ 3, & \text{if } x = \frac{\pi}{2} \end{cases}$

For $f(x)$ to be continuous at $x = 0$,

$$LHL = RHL = f\left(\frac{\pi}{2}\right)$$

Now, $f\left(\frac{\pi}{2}\right) = 3$ (i)

$$LHL = \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{k \cos x}{\pi - 2x}$$

$$= \lim_{h \rightarrow 0} \frac{k \cos\left(\frac{\pi}{2} - h\right)}{\pi - 2\left(\frac{\pi}{2} - h\right)} \quad [\text{We assume } x = \frac{\pi}{2} - h \text{ so that}]$$

$$x \rightarrow \frac{\pi}{2}, h \rightarrow 0]$$

$$= \lim_{h \rightarrow 0} \frac{k \sinh}{2h} = \lim_{h \rightarrow 0} \frac{k}{2} \times \left(\frac{\sinh}{h}\right) = \frac{k}{2}$$

$$\left(\because \lim_{h \rightarrow 0} \frac{\sinh}{h} = 1\right) \quad \text{.....(ii)}$$

$$RHL = \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^+} \frac{k \cos x}{\pi - 2x}$$

$$= \lim_{h \rightarrow 0} \frac{k \cos\left(\frac{\pi}{2} + h\right)}{\pi - 2\left(\frac{\pi}{2} + h\right)}$$

$$\left[\text{Let } x = \frac{\pi}{2} + h, h \rightarrow 0 \text{ as } x \rightarrow \frac{\pi}{2} \right]$$

$$= \lim_{h \rightarrow 0} \frac{-k \sinh}{-2h} = \lim_{h \rightarrow 0} \frac{k}{2} \times \frac{\sinh}{h} = \frac{k}{2} \quad \text{.....(iii)}$$

Hence, (i), (ii) and (iii) $\Rightarrow \frac{k}{2} = 3$ or $k = 6$

6. Soln.

Here $f(x) = 2x - |x|$

For continuity at $x = 0$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0 + h) = \lim_{h \rightarrow 0} f(h)$$

$$= \lim_{h \rightarrow 0} \{2h - |h|\} = \lim_{h \rightarrow 0} (2h - h)$$

$$= \lim_{h \rightarrow 0} h$$

$$= 0 \quad \text{.....(i)}$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0 - h) = \lim_{h \rightarrow 0} f(-h)$$

$$= \lim_{h \rightarrow 0} \{2(-h) - |-h|\} = \lim_{h \rightarrow 0} \{-2h - h\}$$

$$= \lim_{h \rightarrow 0} (-3h)$$

$$= 0 \quad \text{.....(ii)}$$

$$= \text{Also } f(0) = 2 \times 0 - |0| = 0 \quad \text{.....(iii)}$$

$$(i), (ii) \text{ and } (iii) \Rightarrow \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = f(0)$$

Hence, $f(x)$ is continuous at $x = 0$

For differentiability at $x = 0$

$$LHD = \lim_{h \rightarrow 0} \frac{f(0 - h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{f(-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{(2(-h) - |-h|) - \{2 \times 0 - |0|\}}{-h} = \lim_{h \rightarrow 0} \frac{-2h - h - 0}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-3h}{-h} = \lim_{h \rightarrow 0} 3$$

$$LHD = 3 \quad \text{.....(iv)}$$

$$\text{Again } RHD = \lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{2h - |h| - 2 \times 0 - |0|}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2h - h}{h} = \lim_{h \rightarrow 0} \frac{h}{h}$$

$$= \lim_{h \rightarrow 0} 1$$

$$RHD = 1 \quad \text{.....(v)}$$

From (iv) and (v)

$$LHD \neq RHD$$

i.e., function $f(x) = 2x - |x|$ is not differentiable at $x = 0$

Hence, $f(x)$ is continuous but not differentiable at $x = 0$.

7.

Soln. For continuity:

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{h \rightarrow 0} f(2 - h)$$

$$[\text{Let } x = 2 - h \Rightarrow x \rightarrow 2^- \Rightarrow h \rightarrow 0]$$

$$= \lim_{h \rightarrow 0} 2(2-h)^2 - (2-h)$$

$$= \lim_{h \rightarrow 0} 2\{4 + h^2 - 4h\} - (2-h)$$

$$= \lim_{h \rightarrow 0} (8 + 2h^2 - 8h - 2 + h) = 6$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{h \rightarrow 0} f(2+h)$$

$$[\text{Let } x = 2+h \Rightarrow x \rightarrow 2^+ \Rightarrow h \rightarrow 0]$$

$$= \lim_{h \rightarrow 0} 5(2+h) - 4 = 6$$

$$f(2) = 2.2^2 - 2 = 6$$

$$\text{i.e., } \lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} f(x) = f(2)$$

$$\Rightarrow f(x) \text{ is continuous at } x = 2.$$

For Differentiability:

$$\text{LHD (at } x = 2) = \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{2(2-h)^2 - (2-h) - \{2.2^2 - 2\}}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{8 + 2h^2 - 8h - 2 + h - 6}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{2h^2 - 7h}{-h} = \lim_{h \rightarrow 0} \frac{2h - 7}{-1} = 7$$

$$\text{RHD (at } x = 2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{5(2+h) - 4 - \{2.2^2 - 2\}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{10 + 5h - 4 - 6}{h} = \lim_{h \rightarrow 0} 5 = 5$$

$$\therefore \text{LHD} \neq \text{RHD (at } x = 2)$$

Hence, $f(x)$ is not differentiable at $x = 2$.

$$8. \text{Ans. We have, } f(x) = |\tan 2x|$$

$$\Rightarrow f'(x) = \frac{\tan 2x}{|\tan 2x|} \cdot \frac{d}{dx}(\tan 2x) = \frac{\tan 2x}{|\tan 2x|} \cdot 2\sec^2 2x$$

$$\therefore f'\left(\frac{\pi}{3}\right) = \frac{\tan 2\pi/3}{|\tan 2\pi/3|} \times 2\sec^2\left(2 \cdot \frac{\pi}{3}\right)$$

$$= \frac{-\sqrt{3}}{\sqrt{3}} \times 2 \cdot (-2)^2 = -8$$

$$9. \text{Ans. Given, } (\cos x)^y = (\cos y)^x$$

On taking log both sides, we get

$$y \log(\cos x) = x \log(\cos y)$$

On differentiating both sides w. r. t. x , we get

$$y \frac{1}{\cos x} \frac{d}{dx}(\cos x) + \log(\cos x) \frac{dy}{dx}$$

$$= x \frac{1}{\cos y} \frac{d}{dx}(\cos y) + \log(\cos y)(1)$$

$$\Rightarrow y \frac{(-\sin x)}{\cos x} + \log(\cos x) \frac{dy}{dx} = x \frac{(-\sin y)}{\cos y} \frac{dy}{dx} + \log(\cos y)$$

$$\Rightarrow -y \tan x + \log(\cos x) \frac{dy}{dx} = -x \tan y \frac{dy}{dx} + \log(\cos y)$$

$$\Rightarrow [x \tan y + \log(\cos x)] \frac{dy}{dx} = \log(\cos y) + y \tan x$$

$$\therefore \frac{dy}{dx} = \frac{\log(\cos y) + y \tan x}{x \tan y + \log(\cos x)}$$

$$10. \text{Ans. We have, } f(x) = \begin{cases} \frac{\sin^2 \lambda x}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases} \text{ is continuous}$$

at $x = 0$

$$\therefore \text{L.H.L.} = \text{R.H.L.}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin^2 \lambda x}{x^2} = 1 \Rightarrow \lim_{x \rightarrow 0} \frac{\sin^2 \lambda x}{\lambda^2 x^2} \cdot \lambda^2 = 1$$

$$\Rightarrow \lambda^2 \lim_{\lambda x \rightarrow 0} \left(\frac{\sin \lambda x}{\lambda x} \right)^2 = 1 \Rightarrow \lambda^2 \cdot 1 = 1$$

$$\Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$$

$$11. \text{Ans. We have, } y = \sqrt{a + \sqrt{a + x}}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{a + \sqrt{a + x}}} \times \frac{1}{2\sqrt{a + x}} = \frac{1}{4} \cdot \frac{1}{\sqrt{a}\sqrt{(a+x)} + a+x}$$

12. .

Ans. $\therefore f(x)$ is continuous at $\pi/2$.

$$\therefore \lim_{x \rightarrow \pi/2} f(x) = \lim_{x \rightarrow \pi/2} f(x) = f(\pi/2) \dots (1)$$

$$\text{Now, } \lim_{x \rightarrow \pi/2} f(x) = \lim_{h \rightarrow 0} f\left(\frac{\pi}{2} - h\right)$$

$$= \lim_{h \rightarrow 0} \frac{1 - \sin^3\left(\frac{\pi}{2} - h\right)}{3\cos^2\left(\frac{\pi}{2} - h\right)} = \lim_{h \rightarrow 0} \frac{1 - \cos^3 h}{3\sin^2 h}$$

$$= \lim_{h \rightarrow 0} \frac{(1 - \cosh)(1 + \cos^2 h + \cosh)}{3(1 - \cosh)(1 + \cosh)}$$

$$= \lim_{h \rightarrow 0} \frac{(1 + \cos^2 h + \cosh)}{3(1 + \cosh)} = \frac{1+1+1}{3(1+1)} = \frac{1}{2}$$

$$\text{And } \lim_{x \rightarrow \pi/2} f(x) = \lim_{h \rightarrow 0} f\left(\frac{\pi}{2} + h\right)$$

$$= \lim_{h \rightarrow 0} \frac{q \left[1 - \sin \left(\frac{\pi}{2} + h \right) \right]}{\left[\pi - 2 \left(\frac{\pi}{2} + h \right) \right]^2} = \lim_{h \rightarrow 0} \frac{q(1 - \cosh)}{4h^2}$$

$$= \frac{q}{4} \times \lim_{h \rightarrow 0} \frac{2 \sin^2 \frac{h}{2}}{h^2 \times 4} = \frac{q}{4} \times \frac{2}{4} = \frac{q}{8} \text{ and } f(\pi/2) = p$$

$$\therefore \frac{1}{2} = \frac{q}{8} = p$$

$$\Rightarrow p = \frac{1}{2} \text{ and } q = 4$$

13. . Ans. Given that $f(x)$ is differentiable at $x = 1$.
Therefore, $f(x)$ is continuous at $x = 1$

$$\Rightarrow \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} f(x) = f(1)$$

$$\Rightarrow \lim_{x \rightarrow 1} (x^2 + 3x + a) = \lim_{x \rightarrow 1} (bx + 2)$$

$$\Rightarrow 1 + 3 + a = b + 2 \Rightarrow a - b + 2 = 0 \dots\dots\dots(1)$$

Again, $f(x)$ is differentiable at $x = 1$. So,
(L.H.D. at $x = 1$) = (R.H.D. at $x = 1$)

$$\Rightarrow \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{(x^2 + 3x + a) - (4 + a)}{x - 1} = \lim_{x \rightarrow 1} \frac{(bx + 2) - (4 + a)}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^2 + 3x - 4}{x - 1} = \lim_{x \rightarrow 1} \frac{bx - 2 - a}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{(x+4)(x-1)}{x-1} = \lim_{x \rightarrow 1} \frac{bx - b}{x - 1} \quad [\text{From (1)}]$$

$$\Rightarrow \lim_{x \rightarrow 1} (x+4) = \lim_{x \rightarrow 1} \frac{b(x-1)}{x-1} \Rightarrow 5 = b$$

Putting $b = 5$ in (1), we get $a = 3$.

Hence, $a = 3$ and $b = 5$

14. . Ans. We have, $y = e^{x^2 \cos x} + (\cos x)^x$

$$= e^{x^2 \cos x} + e^{x(\ln \cos x)}$$

$$\therefore \frac{dy}{dx} = e^{x^2 \cos x} \frac{d}{dx} (x^2 \cos x) + e^{x \ln \cos x} \frac{d}{dx} (x \ln \cos x)$$

$$= e^{x^2 \cos x} (2x \cos x - x^2 \sin x)$$

$$+ e^{x \ln \cos x} \left(\ln \cos x - \frac{x}{\cos x} \sin x \right)$$

$$= e^{x^2 \cos x} (2x \cos x - x^2 \sin x) + (\cos x)^x (\ln \cos x - x \tan x)$$

15. .

Ans. Given, $\log(x^2 + y^2) = 2 \tan^{-1} \left(\frac{y}{x} \right)$

On differentiating w.r.t. x on both sides, we get

$$\frac{1}{x^2 + y^2} \left(2x + 2y \frac{dy}{dx} \right) = 2 \times \frac{1}{1 + \frac{y^2}{x^2}} \frac{d}{dx} \left(\frac{y}{x} \right)$$

$$\Rightarrow \frac{2x}{x^2 + y^2} + \frac{2y}{x^2 + y^2} \frac{dy}{dx} = \frac{2x^2}{x^2 + y^2} \left(\frac{1}{x} \frac{dy}{dx} + y \left(\frac{-1}{x^2} \right) \right)$$

$$\Rightarrow \frac{dy}{dx} \left[\frac{2y}{x^2 + y^2} - \frac{2x}{x^2 + y^2} \right] = \frac{2x^2}{x^2 + y^2} \left[\frac{-y}{x^2} - \frac{1}{x} \right]$$

$$\Rightarrow \frac{2(y-x)}{x^2 + y^2} \frac{dy}{dx} = \frac{-2x^2}{x^2 + y^2} \left(\frac{y+x}{x^2} \right) \Rightarrow \frac{dy}{dx} = \frac{x+y}{x-y}$$

16. . Ans. Given $y = (x)^{\cos x} + (\cos x)^{\sin x}$

Let $u = (x)^{\cos x}$, $v = (\cos x)^{\sin x} \therefore y = u + v$

$$\Rightarrow \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \dots\dots\dots(i)$$

Now, $u = x^{\cos x}$

$$\Rightarrow \log u = \cos x \log x$$

Differentiating with respect to x , we get

$$\frac{1}{u} \frac{du}{dx} = \cos x \cdot \frac{1}{x} + \log x (-\sin x)$$

$$\Rightarrow \frac{du}{dx} = x^{\cos x} \left[\frac{\cos x}{x} - \sin x \log x \right] \dots\dots(ii)$$

Now, $v = (\cos x)^{\sin x}$

$$\Rightarrow \log v = \sin x \log \cos x$$

Differentiating with respect to x , we get

$$\frac{1}{v} \frac{dv}{dx} = \sin x \cdot \frac{1}{\cos x} (-\sin x) + \log \cos x \cdot \cos x$$

$$\Rightarrow \frac{dv}{dx} = (\cos x)^{\sin x} \left[\frac{-\sin^2 x + \cos^2 x \log \cos x}{\cos x} \right] \dots\dots\dots(iii)$$

Putting value of (ii) and (iii) into (i), we get

$$\frac{dy}{dx} = x^{\cos x} \left[\frac{\cos x}{x} - \sin x \log x \right]$$

$$+ (\cos x)^{\sin x} \left[\frac{-\sin^2 x + \cos^2 x \log \cos x}{\cos x} \right]$$

17. .

Ans. Let $u = \tan^{-1} \left[\frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}} \right] \dots\dots\dots(i)$

Putting $x^2 = \cos 2\theta$ in (i), we get

$$u = \tan^{-1} \left[\frac{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}} \right]$$

$$= \tan^{-1} \left[\frac{\sqrt{2\cos^2 \theta} - \sqrt{2\sin^2 \theta}}{\sqrt{2\cos^2 \theta} + \sqrt{2\sin^2 \theta}} \right]$$

$$= \tan^{-1} \left[\frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} \right] = \tan^{-1} \left[\frac{1 - \tan \theta}{1 + \tan \theta} \right]$$

$$= \tan^{-1} \left[\tan \left(\frac{\pi}{4} - \theta \right) \right] = \frac{\pi}{4} - \theta$$

$$= \frac{\pi}{4} - \frac{\cos^{-1}(x^2)}{2}$$

$$\left(\because x^2 = \cos 2\theta \Rightarrow \theta = \frac{\cos^{-1} x^2}{2} \right)$$

$$\Rightarrow \frac{du}{dx} = \frac{1}{2} \frac{1}{\sqrt{1-x^4}} 2x = \frac{x}{\sqrt{1-x^4}} \dots\dots\dots(ii)$$

$$\text{Let } v = \cos^{-1}(x^2) \Rightarrow \frac{dv}{dx} = \frac{-2x}{\sqrt{1-x^4}} \dots\dots\dots(iii)$$

$$\text{Dividing (ii) by (iii), we get } \frac{du}{dv} = -\frac{1}{2}.$$

18. .

$$\text{Ans. Let } u = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$$

$$\text{Put } x = \tan \theta \Rightarrow \theta = \tan^{-1} x$$

$$\therefore u = \tan^{-1} \left(\frac{\sqrt{1+\tan^2 \theta}-1}{\tan \theta} \right)$$

$$\Rightarrow u = \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) \Rightarrow u = \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right)$$

$$\Rightarrow u = \tan^{-1} \left(\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right) \Rightarrow u = \tan^{-1} \left(\tan \frac{\theta}{2} \right)$$

$$\therefore u = \frac{\theta}{2} \Rightarrow u = \frac{1}{2} \tan^{-1} x$$

$$\left(\because 2 \sin^2 \frac{x}{2} = 1 - \cos x, 2 \sin \frac{x}{2} \cos \frac{x}{2} = \sin x \right)$$

$$\text{Differentiating w.r.t. } x, \text{ we get } \frac{du}{dx} = \frac{1}{2(1+x^2)}$$

$$\text{Also, let } v = \sin^{-1} \left(\frac{2x}{1+x^2} \right) \Rightarrow v = 2 \tan^{-1} x$$

$$\frac{dv}{dx} = \frac{2}{1+x^2} \therefore \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{\frac{1}{2(1+x^2)}}{\frac{2}{1+x^2}} \Rightarrow \frac{du}{dv} = \frac{1}{4}$$

19. .

Soln. Since $f(x)$ is continuous at $x = 3$, we have:

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = f(3) \dots\dots(i)$$

$$\text{Now, } \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (ax+1)$$

$$\lim_{h \rightarrow 0} [a(3-h)+1] = 3a+1$$

Similarly,

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{h \rightarrow 0} [b(3+h)+3]$$

$$= 3b+3$$

From (i), we can equate:

$$3a+1 = 3b+3$$

$$\Rightarrow 3(a-b) = 2$$

$$\text{or } a-b = \frac{2}{3}$$

20. .

Soln. Here, given function is

$$f(x) = |x-1| + |x+1|$$

$$\Rightarrow f(x) = \begin{cases} -(x-1) - (x+1), & x < -1 \\ 2, & x = -1 \\ -(x-1) + (x+1), & -1 < x < 1 \\ 2, & x = 1 \\ (x-1) + (x+1) & x > 1 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} -2x, & \text{if } x < -1 \\ 2, & \text{if } x = -1 \\ 2, & \text{if } -1 < x < 1 \\ 2, & \text{if } x = 1 \\ 2x, & \text{if } x > 1 \end{cases} \Rightarrow f(x) = \begin{cases} -2x & \text{if } x < -1 \\ 2 & \text{if } -1 \leq x \leq 1 \\ 2x & \text{if } x > 1 \end{cases}$$

For $x = -1$

$$\text{Now, RHD} = \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2-2}{h} = \lim_{h \rightarrow 0} 0 = 0$$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(-1-h) - f(-1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-2(-1-h)-2}{-h} = \lim_{h \rightarrow 0} \frac{2+2h-2}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{2h}{-h} = \lim_{h \rightarrow 0} -2 = -2.$$

i.e., $RHD \neq LHD$.

Hence, $f(x)$ is not differentiable at $x = -1$.

For $x = 1$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(1+h)-2}{h} = \lim_{h \rightarrow 0} \frac{2+2h-2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2h}{h} = \lim_{h \rightarrow 0} 2 = 2$$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{2-2}{-h} = \lim_{h \rightarrow 0} \frac{0}{-h}$$

$$= \lim_{h \rightarrow 0} 0 = 0$$

$RHD \neq LHD$.

Hence, $f(x)$ not differentiable at $x = 1$.

21. .

Soln. Given, $x^y = e^{x^y}$

Taking log both sides, we get

$$\Rightarrow \log x^y = \log e^{x-y}$$

$$\Rightarrow y \cdot \log x = (x-y) \log e \quad [\because \log e = 1]$$

$$\Rightarrow y = \frac{x}{1 + \log x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(1 + \log x) \cdot 1 - x \left(0 + \frac{1}{x}\right)}{(1 + \log x)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1 + \log x - 1}{(1 + \log x)^2} = \frac{\log x}{(\log e + \log x)^2} \quad [\because 1 = \log e]$$

$$\Rightarrow \frac{dy}{dx} = \frac{\log x}{(\log ex)^2} \Rightarrow \frac{dy}{dx} = \frac{\log x}{\{\log(ex)\}^2}$$

22.

Soln. Given, $(\cos x)^y = (\cos y)^x$

Taking logarithm both sides, we get

$$\log(\cos x)^y = \log(\cos y)^x$$

$$\Rightarrow y \cdot \log(\cos x) = x \cdot \log(\cos y)$$

$$[\because \log m^n = n \log m]$$

Differentiating both sides, we get

$$y \cdot \frac{1}{\cos x} (-\sin x) + \log(\cos x) \cdot \frac{dy}{dx}$$

$$= x \cdot \frac{1}{\cos y} (-\sin y) + \log(\cos y)$$

$$\Rightarrow -\frac{y \sin x}{\cos x} + \log(\cos x) \cdot \frac{dy}{dx} = -\frac{x \sin y}{\cos y} + \log(\cos y)$$

$$\Rightarrow \log(\cos x) \cdot \frac{dy}{dx} + \frac{x \sin y}{\cos y} \cdot \frac{dy}{dx} = \log(\cos y) + \frac{y \sin x}{\cos x}$$

$$\Rightarrow \frac{dy}{dx} \left[\log(\cos x) + \frac{x \sin y}{\cos y} \right] = \log(\cos y) + \frac{y \sin x}{\cos x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\log(\cos y) + \frac{y \sin x}{\cos x}}{\log(\cos x) + \frac{x \sin y}{\cos y}} = \frac{\log(\cos y) + y \tan x}{\log(\cos x) + x \tan y}$$

23.

Soln. Consider $\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}$

$$= \frac{\sqrt{1+2\sin\frac{x}{2}\cos\frac{x}{2}} + \sqrt{1-2\sin\frac{x}{2}\cos\frac{x}{2}}}{\sqrt{1+2\sin\frac{x}{2}\cos\frac{x}{2}} - \sqrt{1-2\sin\frac{x}{2}\cos\frac{x}{2}}}$$

$$\left[\because \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2} \right]$$

$$= \frac{\sqrt{\cos^2\frac{x}{2} + \sin^2\frac{x}{2} + 2\sin\frac{x}{2}\cos\frac{x}{2}} + \sqrt{\sin^2\frac{x}{2} + \cos^2\frac{x}{2} - 2\sin\frac{x}{2}\cos\frac{x}{2}}}{\sqrt{\cos^2\frac{x}{2} + \sin^2\frac{x}{2} + 2\sin\frac{x}{2}\cos\frac{x}{2}} - \sqrt{\sin^2\frac{x}{2} + \cos^2\frac{x}{2} - 2\sin\frac{x}{2}\cos\frac{x}{2}}}$$

$$= \frac{\cos\frac{x}{2} + \sin\frac{x}{2} + \cos\frac{x}{2} - \sin\frac{x}{2}}{\cos\frac{x}{2} + \sin\frac{x}{2} - \cos\frac{x}{2} + \sin\frac{x}{2}} = \frac{2\cos\left(\frac{x}{2}\right)}{2\sin\left(\frac{x}{2}\right)} = \cot\frac{x}{2}$$

$$\therefore y = \cot^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right]$$

$$= \cot^{-1} \left(\cot \frac{x}{2} \right) = \frac{x}{2} \Rightarrow \frac{dy}{dx} = \frac{1}{2}$$

24.

Soln. Let $y = x^{\sin x} + (\sin x)^{\cos x}$, $u = x^{\sin x}$, $v = (\sin x)^{\cos x}$

Now, $y = u + v$

And $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$ (i)

We have, $u = x^{\sin x}$

Taking log on both sides, we get

$$\log u = \sin x \log x$$

Differentiating both sides with respect to x, we get

$$\frac{1}{u} \frac{du}{dx} = \sin x \times \frac{1}{x} + \log x \cdot \cos x$$

$$\frac{du}{dx} = u \left(\frac{\sin x}{x} + \log x \cdot \cos x \right)$$

$$= x^{\sin x} \left[\frac{\sin x}{x} + \log x \cdot \cos x \right] \text{(ii)}$$

Again, $v = (\sin x)^{\cos x}$

Taking log on both sides with respect to x, we get

$$\log v = \cos x \log \sin x$$

Differentiating both sides with respect to x, we get

$$\frac{1}{v} \frac{dv}{dx} = \cos x \times \frac{1}{\sin x} \times \cos x + \log \sin x (-\sin x)$$

$$= \cos x \cot x + \log \sin x \cdot (-\sin x)$$

$$= \cos x \cdot \cot x - \sin x \cdot \log \sin x$$

$$\therefore \frac{dv}{dx} = v [\cos x \cdot \cot x - \sin x \cdot \log \sin x]$$

$$\frac{dv}{dx} = (\sin x)^{\cos x} [\cos x \cdot \cot x - \sin x \log \sin x] \text{(iii)}$$

From (i), (ii) and (iii), we get

$$\frac{dy}{dx} = x^{\sin x} \left[\frac{\sin x}{x} + \log x \cdot \cos x \right] + (\sin x)^{\cos x} [\cos x \cot x - \sin x \log \sin x]$$

25.

Soln. Given, $y = (\sin x)^x + \sin^{-1} \sqrt{x}$

$y = u + v$, where $u = (\sin x)^x$, $v = \sin^{-1} \sqrt{x}$

$$\therefore \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \text{(i)}$$

Now, $u = (\sin x)^x$

Taking log both sides, we get

$$\log u = \log(\sin x)^x \Rightarrow \log u = x \cdot \log(\sin x)$$

Differentiating both sides with respect to x we get

$$\frac{1}{u} \cdot \frac{du}{dx} = x \cdot \frac{1}{\sin x} \cos x + \log \sin x$$

$$\Rightarrow \frac{du}{dx} = u \{ x \cot x + \log \sin x \}$$

$$\Rightarrow \frac{du}{dx} = (\sin x)^x \{ x \cot x + \log \sin x \} \dots\dots\dots(ii)$$

$$\text{Also, } v = \sin^{-1} \sqrt{x}$$

$$\frac{dv}{dx} = \frac{1}{\sqrt{1-(\sqrt{x})^2}} \times \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x(1-x)}} \dots\dots\dots(iii)$$

From (i), (ii) and (iii), we get

$$\therefore \frac{dy}{dx} = (\sin x)^x \{ x \cot x + \log \sin x \} + \frac{1}{2\sqrt{x(1-x)}}$$

26.

$$\text{Soln. Given, } x = a \cos \theta + b \sin \theta$$

$$\Rightarrow \frac{dx}{d\theta} = -a \sin \theta + b \cos \theta \dots\dots\dots(i)$$

$$\text{Also, } y = a \sin \theta - b \cos \theta$$

$$\frac{dy}{d\theta} = a \cos \theta + b \sin \theta \dots\dots\dots(ii)$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a \cos \theta + b \sin \theta}{-a \sin \theta + b \cos \theta} \quad [\text{From (i) and (ii)}]$$

$$\Rightarrow \frac{dy}{dx} = \frac{a \cos \theta + b \sin \theta}{b \cos \theta - a \sin \theta}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{y} \dots\dots\dots(iii)$$

Differentiating again with respect to x, we get

$$\frac{d^2y}{dx^2} = -\frac{y - x \cdot \frac{dy}{dx}}{y^2}$$

$$\Rightarrow y^2 \frac{d^2y}{dx^2} = -y + x \frac{dy}{dx}$$

$$\Rightarrow y^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = 0$$

27.

$$\text{Soln. Given, } y = e^{x+e^{x+e^{x+\dots}}}$$

$$\therefore y = e^{x+y} \dots\dots\dots(i)$$

$$\text{Taking logarithm, we get } \log y = (x+y) \log_e e$$

$$\text{Or } \log y = x+y \quad [\because \log_e e = 1]$$

Differentiating w.r.t. x, we get

$$\frac{1}{y} \cdot \frac{dy}{dx} = 1 + \frac{dy}{dx} \Rightarrow \left(\frac{1}{y} - 1\right) \frac{dy}{dx} = 1$$

$$\Rightarrow \left(\frac{1-y}{y}\right) \frac{dy}{dx} = 1$$

$$\therefore \frac{dy}{dx} = \frac{y}{1-y}$$

28.

$$\text{Soln. Let } y = \tan^{-1} \left(\frac{\sqrt{x-x^3}}{1+x^{3/2}} \right)$$

$$\text{Then, } y = \tan^{-1} \left(\frac{\sqrt{x-x^3}}{1+\sqrt{x} \cdot x} \right) = \tan^{-1}(\sqrt{x}) - \tan^{-1}(x)$$

$$\therefore \frac{dy}{dx} = \frac{d}{dx} (\tan^{-1} \sqrt{x}) - \frac{d}{dx} (\tan^{-1} x)$$

$$= \frac{d(\tan^{-1} \sqrt{x})}{d\sqrt{x}} \cdot \frac{d\sqrt{x}}{dx} - \frac{1}{1+x^2}$$

$$= \frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}} - \frac{1}{1+x^2}$$

$$= \frac{1}{2\sqrt{x}(1+x)} - \frac{1}{1+x^2}$$

29.

$$\text{Soln. Here, } y = Ae^{-kt} \cos(pt+c)$$

Differentiating w.r.t. t, we get

$$\frac{dy}{dt} = -kAe^{-kt} \cos(pt+c) - pAe^{-kt} \sin(pt+c)$$

$$\text{Or } \frac{dy}{dt} = -ky - pAe^{-kt} \sin(pt+c) \dots\dots\dots(i)$$

Differentiating (i) again w.r.t. 't', we get

$$\frac{d^2y}{dt^2} = -k \frac{dy}{dt} + pAe^{-kt} \sin(pt+c) -$$

$$p^2 Ae^{-kt} \cos(pt+c)$$

$$= -k \frac{dy}{dt} + pAe^{-kt} \sin(pt+c) - p^2 y$$

$$\Rightarrow \frac{d^2y}{dt^2} = -k \frac{dy}{dt} + k \left(-ky - \frac{dy}{dt} \right) - p^2 y \quad [\text{From (i)}]$$

$$\Rightarrow \frac{d^2y}{dt^2} = -2k \frac{dy}{dt} - k^2 y - p^2 y$$

$$\Rightarrow \frac{d^2y}{dt^2} + 2k \frac{dy}{dt} + y(p^2 + k^2) = 0$$

$$\Rightarrow \frac{d^2y}{dt^2} + 2k \frac{dy}{dt} + n^2 y = 0,$$

$$\text{where } n^2 = p^2 + k^2$$

30.

$$\text{Soln. Given, } x = a \left(\cos t + \log \tan \frac{t}{2} \right)$$

Differentiating with respect to t, we get

$$\frac{dx}{dt} = a \left(-\sin t + \frac{1}{\tan \frac{t}{2}} \cdot \sec^2 \frac{t}{2} \cdot \frac{1}{2} \right)$$

$$= a \left\{ -\sin t + \frac{1}{2 \sin \frac{t}{2} \cdot \cos \frac{t}{2}} \right\} = a \left\{ -\sin t + \frac{1}{\sin t} \right\}$$

$$\frac{dx}{dt} = a \left(\frac{1 - \sin^2 t}{\sin t} \right) = a \frac{\cos^2 t}{\sin t}$$

$$\therefore y = a \sin t$$

Differentiating with respect to t, we get

$$\frac{dy}{dt} = a \cos t \Rightarrow \frac{d^2y}{dt^2} = -a \sin t$$

$$\therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a \cos t \cdot \sin t}{a \cos^2 t} = \tan t$$

$$\therefore \frac{d^2y}{dx^2} = \sec^2 t \cdot \frac{dt}{dx} = \sec^2 t \cdot \frac{1 \times \sin t}{a \cos^2 t} = \frac{1}{a} \sec^4 t \cdot \sin t$$

Hence, $\frac{d^2y}{dt^2} = -a \sin t$ and $\frac{d^2y}{dx^2} = \frac{\sec^4 t \sin t}{a}$

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New

SURE SHOT QUESTIONS 2026

Chapter – 06 (Questions)

Application of Derivatives

Questions

1. Find the intervals in which the function

$$f(x) = \frac{x^4}{4} - x^3 - 5x^2 + 24x + 12 \text{ is (i) strictly increasing, (ii) strictly decreasing.}$$

2. An expensive square piece of golden colour board of side 24 centimetres, is to be made into a box by cutting a square from each corner and folding the flaps to form a box. What should be the side of the square piece to be cut from each corner of the board to hold maximum volume and minimize the wastage?

3. An open tank with a square base and vertical sides is to be constructed from a metal sheet so as to hold a given quantity of water. Show that the cost of material will be least when depth of the tank is half of its width.

4. AB is the diameter of a circle and C is any point on the circle. Show that the area of triangle ABC is maximum, when it is an isosceles triangle.

5. Find the intervals in which the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is (a) strictly increasing (b) strictly decreasing.

6. Show that the function $f(x) = x^3 - 3x^2 + 3x$, $x \in R$ is increasing on R.

7. A tank with rectangular base and rectangular sides open at the top is to be constructed so that its depth is 3 m and volume is 75 cm^3 . If building of tank costs Rs. 100 per square metre for the base and Rs. 50 per square metre for the sides, find the cost of least expensive tank.

8. The volume of a cube is increasing at the rate of $6 \text{ cm}^3/\text{s}$. How fast is the surface area of cube increasing, when the length of an edge is 8 cm?

9. If M and m denote the local maximum and local minimum values of the function $f(x) = x + \frac{1}{x}$ ($x \neq 0$) respectively, find the value of (M - m).

10. A ladder 13 m long is leaning against a vertical wall. The bottom of the ladder is dragged away from the wall along the ground at the rate of 2 cm/sec. How fast is the height on the wall decreasing when the foot of the ladder is 5 m away from the wall?

11. Find the intervals in which the function

$$f(x) = \frac{x^4}{4} - x^3 - 5x^2 + 24x + 12 \text{ is}$$

- (a) Strictly increasing
(b) Strictly decreasing

12. Find the intervals in which the function

$$f(x) = 3x^4 - 4x^3 - 12x^2 + 5 \text{ is}$$

[Delhi 2014]

- (a) Strictly increasing
(b) Strictly decreasing

13. Show that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius R is $\frac{2R}{\sqrt{3}}$. Also, find the maximum volume.

14. If the sum of lengths of the hypotenuse and a side of a right angled triangle is given, show that the area of the triangle is maximum, when the angle between them is $\frac{\pi}{3}$.
15. Find the maximum and minimum values of $f(x) = -|x - 1| + 5$ for all $x \in R$.
16. A figure consists of a semi-circle with a rectangle on its diameter. Given the perimeter of the figure, find its dimensions in order that the area may be maximum.
17. Show that semi-vertical angle of a cone of maximum volume and given slant height is $\cos^{-1}(1/\sqrt{3})$
18. Prove that the least perimeter of an isosceles triangle in which a circle of radius r can be inscribed is $6\sqrt{3} r$.
19. Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius r is $4r/3$. Also show that the maximum volume of the cone is $8/27$ of the volume of the sphere.
20. Prove that the height of the cylinder of maximum volume, that can be inscribed in a sphere of radius R is $2R/\sqrt{3}$. Also find the maximum volume.
21. Show that the right-circular cone of least curved surface and given volume has an altitude equal to $\sqrt{2}$ times the radius of the base.
22. x and y are the sides of two squares such that $y = x - x^2$. Find the rate of change of the area of second square with respect to the area of first square.
23. If lengths of three sides of a trapezium other than base are equal to 10cm, then find the area of the trapezium when it is maximum.
24. A ladder 13 m long is leaning against a vertical wall. The bottom of the ladder is dragged away from the wall along the ground at the rate of 2 cm/sec. How fast is the height on the wall decreasing when the foot of the ladder is 5 m away from the wall?
25. The side of an equilateral triangle is increasing at the rate of 2 cm/s. At what rate is its area increasing when the side of the triangle is 20 cm?
26. The sides of an equilateral triangle are increasing at the rate of 2 cm/sec. Find the rate at which the area increases, when the side is 10 cm.
27. The volume of a sphere is increasing at the rate of 3 cubic centimetre per second. Find the rate of increase of its surface area, when the radius is 2 cm.
28. Find the intervals in which the function $f(x) = \frac{x^4}{4} - x^3 + 5x^2 + 24x + 12$ is
(a) strictly increasing (b) strictly decreasing
29. Prove that the function f defined by $f(x) = x^2 - x + 1$ is neither increasing nor decreasing in $(-1,1)$. Hence, find the intervals in which $f(x)$ is
(i) strictly increasing (ii) strictly decreasing.

New

SURE SHOT QUESTIONS 2026

Chapter – 06 (Solutions)

Application of Derivatives

Questions

1.

Soln. Given function is

$$f(x) = \frac{x^4}{4} - x^3 - 5x^2 + 24x + 12$$

$$\Rightarrow f'(x) = \frac{4x^3}{4} - 3x^2 - 10x + 24$$

For critical points, put $f'(x) = 0$

$$\therefore x^3 - 3x^2 - 10x + 24 = 0$$

$$(x-2)(x^2 - x - 12) = 0$$

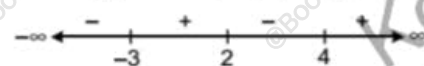
$$(x-2)(x-4)(x+3) = 0$$

$$\Rightarrow x = 2, 4, -3$$

Therefore, we have the intervals

$$(-\infty, -3), (-3, 2), (2, 4) \text{ and } (4, \infty)$$

Since $f'(x) > 0$ in $(-3, 2) \cup (4, \infty)$



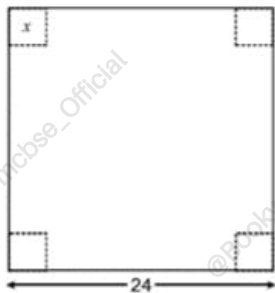
$\therefore f(x)$ is increasing in interval $(-3, 2) \cup (4, \infty)$

And $f'(x) < 0$ in $(-\infty, -3) \cup (2, 4)$

$\therefore f(x)$ is decreasing in $(-\infty, -3) \cup (2, 4)$.

2.

Sol. Let length of a side of the square cut out = x cm



Volume, $V = (24 - 2x)^2 x$

$$\frac{dV}{dx} = 2(24 - 2x)(-2)x + (24 - 2x)^2$$

$$= (24 - 2x)(-4x + 24 - 2x)$$

$$= (24 - 2x)(24 - 6x)$$

$$= 12(12 - x)(4 - x)$$

Put $\frac{dV}{dx} = 0$ i.e., $(12 - x)(4 - x) = 0$

Since, $x \neq 12$,

$$\therefore x = 4$$

Hence, the volume will be maximum when length of a side of the square cut out = 4 cm.

3.

Soln. Let the length, breadth and height of the open tank be x , x and y units respectively.

Then, Volume (V) = $x^2 y$ (i)

Total surface area (S) = $x^2 + 4xy$ (ii)

$$S = x^2 + 4x \frac{V}{x^2} \quad [\text{using (i)}]$$

$$\Rightarrow S = x^2 + \frac{4V}{x}$$

$$\Rightarrow \frac{dS}{dx} = 2x - \frac{4V}{x^2}$$

For critical points, put

$$\frac{dS}{dx} = 0$$

$$\Rightarrow 2x - \frac{4V}{x^2} = 0$$

$$\Rightarrow 2x^3 = 4V$$

$$\Rightarrow 2x^3 = 4x^2 y \quad [\text{using (i)}]$$

$$\Rightarrow x = 2y \quad \text{.....(iii)}$$

Now, $\frac{d^2S}{dx^2} = 2 + \frac{8V}{x^3}$

$$= 2 + \frac{8V}{8y^3} \quad [\text{using (iii)}]$$

$$= 2 + \frac{V}{y^3} > 0$$

Area is minimum, thus cost is minimum when $x = 2y$.

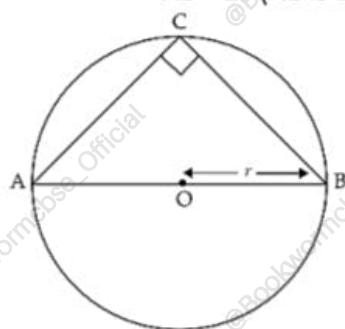
i.e., depth of tank is half of the width.

Value: Any relevant value.

4.

Soln. Let r be the radius of the circle then,

$AB = 2r$ (AB is diameter)



Let, $BC = x$ units

We know that angle subtended by diameter in a circle is right angle

$$\therefore \angle C = 90^\circ$$

$$\text{Then, } AC = \sqrt{(AB)^2 - (BC)^2}$$

$$AC = \sqrt{(2r)^2 - (x)^2} = \sqrt{4r^2 - x^2} \quad \dots\dots(i)$$

Now, area of $\triangle ABC$

$$A = \frac{1}{2} (AC)(BC)$$

$$A = \frac{1}{2} \sqrt{4r^2 - x^2} (x)$$

Differentiating A w.r.t. x , we get

$$\Rightarrow \frac{dA}{dx} = \frac{1}{2} \left\{ \sqrt{4r^2 - x^2} + x \cdot \frac{1}{2\sqrt{4r^2 - x^2}} \cdot \frac{d}{dx} (4r^2 - x^2) \right\}$$

$$\Rightarrow \frac{dA}{dx} = \frac{1}{2} \left\{ \sqrt{4r^2 - x^2} - \frac{x^2}{\sqrt{4r^2 - x^2}} \right\}$$

$$= \frac{1}{2} \left\{ \frac{4r^2 - x^2 - x^2}{\sqrt{4r^2 - x^2}} \right\}$$

$$\Rightarrow \frac{dA}{dx} = \frac{1}{2} \left\{ \frac{4r^2 - 2x^2}{\sqrt{4r^2 - x^2}} \right\}$$

The critical numbers of x are given by $\frac{dA}{dx} = 0$

$$\Rightarrow \frac{1}{2} \left\{ \frac{4r^2 - 2x^2}{\sqrt{4r^2 - x^2}} \right\} = 0$$

$$\Rightarrow 4r^2 - 2x^2 = 0$$

$$\Rightarrow 4r^2 = 2x^2$$

$$\therefore x = \sqrt{2}r$$

$$\text{Now, } \frac{dA}{dx} = \frac{1}{2} \left\{ \frac{4r^2 - 2x^2}{\sqrt{4r^2 - x^2}} \right\}$$

Again, differentiating w.r.t. x , we get

$$\begin{aligned} \frac{d^2A}{dx^2} &= \frac{1}{2} \left\{ (-4x) \frac{1}{\sqrt{4r^2 - x^2}} + (4r^2 - 2x^2) \frac{d}{dx} \left(\frac{-1}{\sqrt{4r^2 - x^2}} \right) \right\} \\ &= \frac{d^2A}{dx^2} = \frac{1}{2} \left\{ \frac{-4x}{\sqrt{4r^2 - x^2}} + \frac{x(4r^2 - 2x^2)}{(4r^2 - x^2)^{3/2}} \right\} \\ &\Rightarrow \left(\frac{d^2A}{dx^2} \right)_{x=\sqrt{2}r} = \frac{1}{2} \left\{ \frac{-4(\sqrt{2}r)}{\sqrt{4r^2 - 2r^2}} + \frac{\sqrt{2}r(4r^2 - 4r^2)}{(4r^2 - 2r^2)^{3/2}} \right\} \\ &= \frac{-2\sqrt{2}r}{\sqrt{2}r} = -2 < 0 \end{aligned}$$

Thus, A is maximum when $x = \sqrt{2}r$.

Putting $x = \sqrt{2}r$ in (i),

$$AC = \sqrt{4r^2 - (\sqrt{2}r)^2}$$

$$\therefore AC = \sqrt{2}r$$

$$\therefore BC = AC = \sqrt{2}r$$

Hence, A is maximum when the triangle is isosceles.

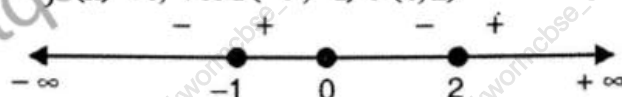
5.

Soln. $f'(x) = 12x^3 - 12x^2 - 24x$

$$= 12x(x+1)(x-2)$$

$$f'(x) > 0, \forall x \in (-1, 0) \cup (2, \infty)$$

$$f'(x) < 0, \forall x \in (-\infty, -1) \cup (0, 2)$$



$\therefore f(x)$ is strictly increasing in $(-1, 0) \cup (2, \infty)$

And strictly decreasing in $(-\infty, -1) \cup (0, 2)$

6.

Soln. We know that, a function $y = f(x)$ is said to be

increasing on R , if $\frac{dy}{dx} > 0, \forall x \in R$

$$\text{Given, } y = x^3 - 3x^2 + 3x$$

On differentiating both sides w.r.t. x , we get

$$\frac{dy}{dx} = 3x^2 - 6x + 3$$

$$\frac{dy}{dx} = 3(x^2 - 2x + 1)$$

Or

$$\frac{dy}{dx} = 3(x-1)^2$$

Or

Now, $3(x-1)^2 > 0$ for all real values of x , i.e., $\forall x \in R$

$$\frac{dy}{dx} > 0, \forall x \in R$$

Hence, the given function is increasing on R .

7. .

Soln. Let l, b, h be the length, breadth and depth of the tank, respectively.

$$\therefore l \times b \times 3 = 75$$

$$\text{or } l \times b = 25$$

Let C be the cost, then

$$\begin{aligned} C &= 100(l \times b) + 50 \times 2[h(b+l)] \\ &= 100\left(l \times \frac{25}{l}\right) + 300\left(\frac{25}{l} + l\right) \\ &= 2500 + 300\left(\frac{25}{l} + l\right) \end{aligned}$$

Differentiating w.r.t. l ,

$$\therefore \frac{dC}{dl} = 0 + 300\left(\frac{-25}{l^2} + 1\right)$$

$$\text{Putting } \frac{dC}{dl} = 0,$$

$$\text{or } 300\left(-\frac{25}{l^2} + 1\right) = 0$$

$$\text{or } l^2 = 25 \text{ or } l = 5$$

$$\text{Getting } \frac{d^2C}{dl^2} = 300\left(\frac{50}{l^3}\right)$$

$$\text{Or } \left(\frac{d^2C}{dl^2}\right)_{at l=5} = \frac{15000}{125} > 0$$

i.e., C is minimum when $l = 5$

$$\Rightarrow b = 5$$

$$\begin{aligned} \therefore C &= 100(25) + 300(10) \\ &= 2,500 + 3,000 \\ &= 5,500 \end{aligned}$$

Hence the minimum cost is Rs.5,500.

8. .

Ans. Volume of cube (V) = a^3

$$\Rightarrow \frac{dV}{dt} = 3a^2 \frac{da}{dt} \Rightarrow 6 = 3a^2 \frac{da}{dt} \quad \left[\because \frac{dV}{dt} = 6 \text{ cm}^3/\text{s} \right]$$

$$\Rightarrow \frac{da}{dt} = \frac{6}{3a^2} \quad \dots(i)$$

Now, surface area of cube (s) = $6a^2$

$$\Rightarrow \frac{ds}{dt} = 12a \frac{da}{dt}$$

$$= 12a \times \frac{6}{3a^2} = \frac{24}{a} \quad [\text{From (i)}]$$

$$\therefore \frac{ds}{dt} \text{ at } a = 8 \text{ cm is } \frac{24}{8} = 3 \text{ cm}^2/\text{sec.}$$

9. .

Ans. We have, $f(x) = x + \frac{1}{x}$

$$\Rightarrow f'(x) = 1 - \frac{1}{x^2} \text{ and } f''(x) = \frac{2}{x^3}$$

For maxima/minima, put $f'(x) = 0$

$$\Rightarrow 1 - \frac{1}{x^2} = 0 \Rightarrow \frac{1}{x^2} = 1$$

$$\Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$\text{At } x = 1, f''(x) = \frac{2}{1} = 2 > 0 \quad (\text{Minimum})$$

$$\text{At } x = -1, f''(x) = \frac{2}{-1} < 0 \quad (\text{Maximum})$$

$$\therefore M = -1 + \frac{1}{(-1)} = -2 \text{ and } m = 1 + \frac{1}{1} = 2$$

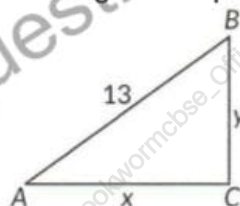
$$\text{So, } M - m = -2 - 2 = -4$$

10. .

Ans. Let foot of the ladder is at a distance x m from the wall and height on the wall is y m.

Here, $x^2 + y^2 = (13)^2$ [Using Pythagoras theorem]

Differentiating with respect to t , we get



$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\Rightarrow \frac{dy}{dt} = \frac{-x}{y} \frac{dx}{dt}$$

$$\text{When } x = 5 \text{ m, } y^2 = (13)^2 - (5)^2 = 169 - 25 = 144$$

$$\therefore y = 12 \text{ m}$$

$$\frac{dx}{dt} = 2 \text{ cm/sec}$$

Also, $\frac{dx}{dt} = 2 \text{ cm/sec}$ [Given]

$$\therefore \frac{dy}{dt} = \frac{-5}{12} \times 2 = \frac{-5}{6} \text{ cm/sec}$$

11. .

(a)

Ans. We have, $f(x) = \frac{x^4}{4} - x^3 - 5x^2 + 24x + 12$

.....(i)

$f(x)$ being polynomial function is continuous and derivable on $\mathbb{R}$.

Differentiating (i) w.r.t. x , we get

$$f'(x) = \frac{4x^3}{4} - 3x^2 - 10x + 24$$

$$= x^3 - 3x^2 - 10x + 24$$

$$= (x-2)(x^2 - x - 12) = (x-2)(x-4)(x+3)$$

(a) For strictly increasing, $f'(x) > 0$

$$\Rightarrow (x-2)(x-4)(x+3) > 0$$

$$\Rightarrow x \in (-3, 2) \cup (4, \infty)$$

(b) For strictly decreasing, $f'(x) < 0$

$$\Rightarrow (x-2)(x-4)(x+3) < 0$$

$$\Rightarrow x \in (-\infty, -3) \cup (2, 4)$$

12.

(a)

Ans. We have, $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$

$$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x^2 - x - 2)$$

$$\Rightarrow f'(x) = 12x(x+1)(x-2)$$

Now, $f'(x) = 0$

$$\Rightarrow 12x(x+1)(x-2) = 0$$

$$\Rightarrow x = -1, x = 0 \text{ or } x = 2$$

Hence, these points divide the whole real line into four disjoint open intervals namely $(-\infty, -1)$, $(-1, 0)$, $(0, 2)$ and $(2, \infty)$.

Interval	Sign of $f'(x)$	Nature of function
$(-\infty, -1)$	$(-)(-)(-) < 0$	Strictly decreasing
$(-1, 0)$	$(-)(+)(-) > 0$	Strictly increasing
$(0, 2)$	$(+)(+)(-) < 0$	Strictly decreasing
$(2, \infty)$	$(+)(+)(+) > 0$	Strictly increasing

(a) $F(x)$ is strictly increasing in $(-1, 0) \cup (2, \infty)$.

(b) $F(x)$ is strictly decreasing in $(-\infty, -1) \cup (0, 2)$.

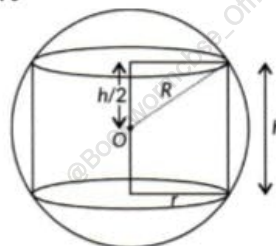
13.

Ans. Let r and h be the base radius and height of cylinder respectively.

$$\therefore \left(\frac{h}{2}\right)^2 + r^2 = R^2$$

.....(i)

Now, V = Volume of the cylinder inscribed in a sphere = $\pi r^2 h$



$$\Rightarrow V = \pi h \left(R^2 - \frac{h^2}{4} \right)$$

[Using (i)]

$$\Rightarrow V = \pi \left(R^2 h - \frac{h^3}{4} \right)$$

Now differentiating w.r.t. h , we get

$$\frac{dV}{dh} = \pi \left(R^2 - \frac{3h^2}{4} \right) \text{ and } \frac{d^2V}{dh^2} = \pi \left(0 - \frac{3}{4} \cdot 2h \right)$$

For maximum or minimum,

$$\frac{dV}{dh} = 0 \Rightarrow R^2 - \frac{3}{4} h^2 = 0$$

$$\Rightarrow h^2 = \frac{4}{3} R^2 \Rightarrow h = \frac{2R}{\sqrt{3}}$$

$$\text{For this value of } h, \frac{d^2V}{dh^2} = -\frac{3}{2} \pi \cdot \frac{2R}{\sqrt{3}} = -\sqrt{3} \pi R < 0$$

$\Rightarrow V$ is maximum

Also maximum value of V

$$= \pi \cdot \frac{2R}{\sqrt{3}} \left(R^2 - \frac{1}{4} \cdot \frac{4}{3} R^2 \right) = \pi \cdot \frac{2R}{\sqrt{3}} \cdot \frac{2}{3} R^2$$

$$= \frac{4\pi}{3\sqrt{3}} R^3 \text{ cu. units}$$

14.

Ans. Let ABC be a right angled triangle with $BC = x$, $AC = y$ such that $x + y = k$, where k is any constant.

Let θ be the angle between the base and the hypotenuse.

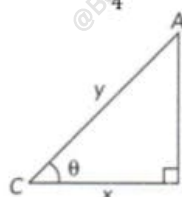
Let P be the area of the triangle.

$$P = \frac{1}{2} \times BC \times AB = \frac{1}{2} \times x \sqrt{y^2 - x^2}$$

$$\Rightarrow P^2 = \frac{x^2}{4} (y^2 - x^2)$$

$$\Rightarrow P^2 = \frac{x^2}{4} [(k-x)^2 - x^2]$$

$$\Rightarrow P^2 = \frac{k^2 x^2 - 2kx^3}{4}$$



Let $Q = P^2$ i.e., $Q = \frac{k^2 x^2 - 2kx^3}{4}$

∴ P is maximum when Q is maximum.

Differentiating Q w.r.t. x, we get

$$\frac{dQ}{dx} = \frac{2k^2 x - 6kx^2}{4} \dots\dots(i)$$

For maximum or minimum area,

$$\frac{dQ}{dx} = 0 \Rightarrow k^2 x - 3kx^2 = 0 \Rightarrow x = \frac{k}{3}$$

Differentiating (i) w.r.t. x, we get

$$\frac{d^2 Q}{dx^2} = \frac{2k^2 - 12kx}{4}$$

$$\therefore \left[\frac{d^2 Q}{dx^2} \right]_{x=\frac{k}{3}} = \frac{-k^2}{2} < 0$$

Thus, Q is maximum when $x = \frac{k}{3}$

⇒ P is maximum at $x = \frac{k}{3}$

Now, $x = \frac{k}{3} \Rightarrow y = k - \frac{k}{3} = \frac{2k}{3}$ [$\because x + y = k$]

$$\therefore \cos \theta = \frac{x}{y} = \frac{k/3}{2k/3} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

So, the area of $\triangle ABC$ is maximum when angle between the hypotenuse and base is $\frac{\pi}{3}$.

15. .

Soln. We have, $f(x) = -|x-1| + 5$ for all $x \in \mathbb{R}$

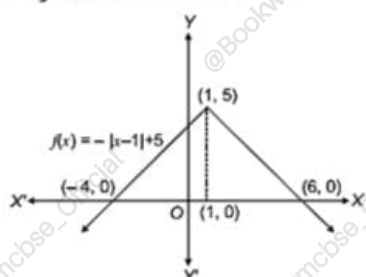
Clearly, $|x-1| \geq 0$ for all $x \in \mathbb{R}$

Taking minus sign both side, we get

$$-|x-1| \leq 0 \text{ for all } x \in \mathbb{R}$$

$$-|x-1| + 5 \leq 5 \text{ for all } x \in \mathbb{R}$$

$$\Rightarrow f(x) \leq 5 \text{ for all } x \in \mathbb{R}$$



So, 5 is the maximum value of $f(x)$.

Now, $f(x) = 5$

$$\Rightarrow -|x-1| + 5 = 5$$

$$\Rightarrow |x-1| = 0 \Rightarrow x = 1$$

Thus, $f(x)$ attains the maximum value 5 at $x = 1$.

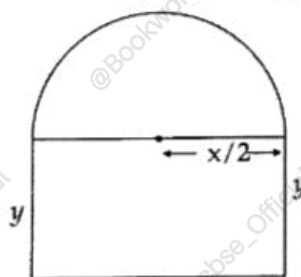
Since $f(x)$ can be made as small as possible. Therefore the minimum value of $f(x)$ does not exist.

16. .

Soln. Let length and breadth of rectangle be x and y.

$$\therefore P = 2y + x + \pi \frac{x}{2} \quad \text{(Given)}$$

$$\begin{aligned} \therefore A &= xy + \frac{1}{2} \pi \frac{x^2}{4} \\ &= \frac{x}{2} \left[P - x - \frac{\pi x}{2} \right] + \frac{\pi x^2}{8} \\ &= P \frac{x}{2} - \frac{x^2}{2} - \pi \frac{x^2}{4} + \pi \frac{x^2}{8} \end{aligned}$$



$$\begin{aligned} \frac{dA}{dx} &= \frac{P}{2} - x - \frac{\pi x}{2} + \frac{\pi x}{4} \\ &= \frac{P}{2} - x - \frac{\pi x}{4} \end{aligned}$$

$$\frac{dA}{dx} = 0$$

$$x = \frac{2P}{4 + \pi} \text{ and } y = \frac{P}{4 + \pi}$$

Or

$$\frac{d^2 A}{dx^2} = -1 - \pi/4 < 0$$

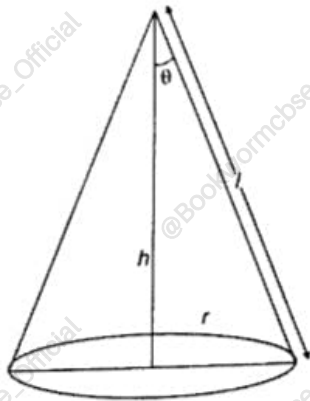
On Area is maximum, when length

$$= \frac{2P}{4 + \pi}$$

$$\text{Breadth} = \frac{P}{4 + \pi}$$

17. .

Soln.



$$r = l \sin \theta$$

$$h = l \cos \theta$$

$$V = \frac{1}{3} \pi r^2 h = \frac{\pi}{3} l^3 \sin^2 \theta \cos \theta$$

$$\text{Or } \frac{dV}{d\theta} = \frac{\pi}{3} l^3 [2 \sin \theta \cos^2 \theta - \sin^3 \theta]$$

$$\text{For maxima and minima, } \frac{dV}{d\theta} = 0$$

$$\sin \theta (2 \cos^2 \theta - \sin^2 \theta) = 0$$

$$\cos \theta = \frac{1}{\sqrt{3}} \text{ or } \theta = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

Or

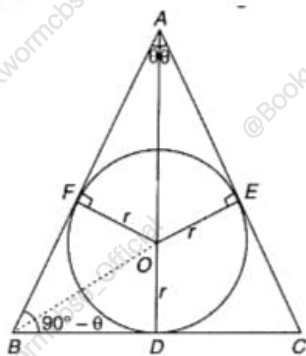
$$\therefore \frac{d^2V}{d\theta^2} \text{ is negative.}$$

Hence, volume of the cone is maximum when semi-

$$\text{vertical angle is } \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

18. .

Soln. Let $\triangle ABC$ be an isosceles triangle with $AB = AC$.



Let $\angle BAC = 2\theta$

AO bisects $\angle BAC$.

Join OE, OF, OD,

Where O is the centre of the circle and $OE = OF = OD = r$

Also, $OE \perp AC$, $OD \perp BC$ and $OF \perp AB$.

$$\text{In } \triangle AOE, \tan \theta = \frac{r}{AE}$$

$$\text{Or } AE = r \cot \theta$$

$$\text{Similarly } AF = r \cot \theta$$

$$\text{In } \triangle ABD, AD \perp BC \quad (\triangle ABC \text{ is isosceles})$$

$$\therefore \angle ABD = 90^\circ - \theta = \frac{\pi}{2} - \theta$$

$$OB \text{ bisects } \angle ABD, \therefore \angle OBF = \angle OBD = \frac{\pi}{4} - \frac{\theta}{2}$$

$$\text{In } \triangle OFB,$$

$$\text{Or } BF = r \cot \left(\frac{\pi}{4} - \frac{\theta}{2} \right) = r \cot \left(\frac{\pi - 2\theta}{4} \right)$$

$$\text{Similarly } BD = DC = CE = r \cot \left(\frac{\pi - 2\theta}{4} \right)$$

We have, perimeter of $\triangle ABC$

$$P = AB + BC + CA$$

$$= AE + EC + BD + DC + AF + BF$$

$$= 2r \cot \theta + 4r \cot \theta \left(\frac{\pi - 2\theta}{4} \right)$$

Differentiate with respect to θ ,

$$\therefore \frac{dP}{d\theta} = 2r(-\operatorname{cosec}^2 \theta)$$

$$+ 4r \left(-\operatorname{cosec}^2 \left(\frac{\pi - 2\theta}{4} \right) \times -\frac{1}{2} \right)$$

$$= -2r \operatorname{cosec}^2 \theta + 2r \operatorname{cosec}^2 \left(\frac{\pi - 2\theta}{4} \right)$$

$$\text{Or On equating, } \frac{dP}{d\theta} = 0 \Rightarrow 2r \operatorname{cosec}^2 \left(\frac{\pi - 2\theta}{4} \right)$$

$$= 2r \operatorname{cosec}^2 \theta$$

$$\text{Or } \sin^2 \left(\frac{\pi - 2\theta}{4} \right) = \sin^2 \theta$$

Or

$$\sin \left(\frac{\pi - 2\theta}{4} \right) = \sin \theta \text{ or } \sin \left(\frac{\pi - 2\theta}{4} \right) = -\sin \theta$$

$$\text{But } 0 \leq \theta \leq \frac{\pi}{4} \text{ or } \sin 2\theta \leq \sin \theta$$

$$\therefore \sin \left(\frac{\pi - 2\theta}{4} \right) \neq -\sin \theta$$

$$\& \sin \left(\frac{\pi - 2\theta}{4} \right) = \sin \theta$$

$$\text{or } \frac{\pi - 2\theta}{4} = \theta$$

$$\text{or } \pi - 2\theta = 4\theta \text{ or } \pi = 6\theta$$

$$\text{or } \theta = \frac{\pi}{6} \text{ or } 2\theta = \frac{\pi}{3}$$

Or $\triangle ABC$ is an equilateral triangle.

By second derivative test,

$$\frac{d^2P}{d\theta^2} = -2r\{2 \operatorname{cosec} \theta (-\operatorname{cosec} \theta \cot \theta)\} + 2r$$

$$\left[2 \operatorname{cosec} \left(\frac{\pi - 2\theta}{4} \right) \left\{ -\operatorname{cosec} \left(\frac{\pi - 2\theta}{4} \right) \cot \left(\frac{\pi - 2\theta}{4} \right) \right\} \right] \times \left(-\frac{1}{2} \right)$$

$$= 4r \operatorname{cosec}^2 \theta \cot \theta + \frac{4r}{2} \operatorname{cosec}^2 \left(\frac{\pi - 2\theta}{4} \right) \cot \left(\frac{\pi - 2\theta}{4} \right)$$

$$\therefore \frac{d^2P}{d\theta} \geq 0 \text{ at } \theta = \frac{\pi}{3}$$

$\therefore$ Perimeter is minimum when $\theta = \frac{\pi}{3}$

$$P = 2r \cot \frac{\pi}{6} + 4r \cot \left(\frac{\pi - 2(\pi/6)}{4} \right)$$

$$= 2r\sqrt{3} + 4r \cot \left(\frac{2\pi}{12} \right)$$

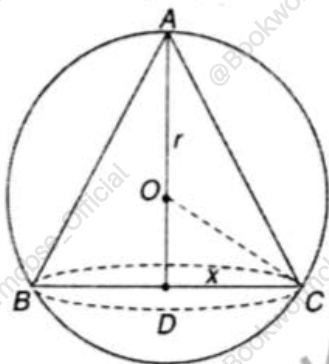
$$= 2r\sqrt{3} + 4r\sqrt{3}$$

$$\therefore P = 6\sqrt{3}r$$

19. .

Soln. Let radius of cone be x and its height be h .

$$\therefore OD = (h-r)$$



Volume of cone (V)

$$= \frac{1}{3} \pi x^2 h$$

.....(i)

$$\text{In } \triangle OCD, x^2 + (h-r)^2 = r^2 \text{ or } x^2 = r^2 - (h-r)^2$$

$$\therefore V = \frac{1}{3} \pi h \{ r^2 - (h-r)^2 \}$$

$$= \frac{1}{3} \pi (-h^3 + 2h^2r)$$

$$\text{Or } \frac{dV}{dh} = \frac{\pi}{3} (-3h^2 + 4hr)$$

$$\therefore \frac{dV}{dh} = 0 \text{ or } h = \frac{4r}{3}$$

$$\frac{d^2V}{dh^2} = \frac{\pi}{3} (-6h + 4r)$$

$$= \frac{\pi}{3} \left(-6 \left(\frac{4r}{3} \right) + 4r \right)$$

$$= -\frac{4\pi r}{3} < 0$$

$\therefore$ at $h = \frac{4r}{3}$, volume is maximum

Maximum volume

$$= \frac{1}{3} \pi \left\{ - \left(\frac{4r}{3} \right)^3 + 2 \left(\frac{4r}{3} \right)^2 r \right\}$$

$$= \frac{8}{27} \cdot \left(\frac{4}{3} \pi r^3 \right)$$

$$= \frac{8}{27} (\text{Volume of sphere})$$

20. .

Let the radius and height of cylinder be r and h respectively

$$\therefore V = \pi r^2 h \quad \dots\dots\dots(i)$$

$$r^2 = R^2 - \frac{h^2}{4}$$

But

$$\therefore \pi h \left(R^2 - \frac{h^2}{4} \right) = \pi \left(R^2 h - \frac{h^3}{4} \right)$$

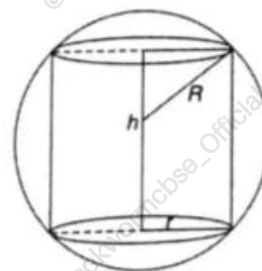
$$\text{or } \frac{dV}{dh} = \pi \left(R^2 - \frac{3h^2}{4} \right)$$

For maximum or minimum

$$\therefore \frac{dV}{dh} = 0 \text{ or } h^2 = \frac{4R^2}{3}$$

$$h = \frac{2R}{\sqrt{3}}$$

Or



$$\text{And } \frac{d^2V}{dh^2} = \pi \left(-\frac{6h}{4} \right) < 0$$

$$\text{Maximum volume} = \pi \cdot \left[R^2 \cdot \frac{2R}{\sqrt{3}} - \frac{1}{4} \left(\frac{2R}{\sqrt{3}} \right)^3 \right]$$

$$= \frac{4\pi R^3}{3\sqrt{3}} \text{ cubic units}$$

21. .

Soln. Let ABC be right-circular cone having radius ' r ' and height ' h '. If V and S are its volume and surface area (curved) respectively, then

$$S = \pi r l$$

$$\text{Or } S = \pi r \sqrt{h^2 + r^2} \quad \dots\dots\dots(i)$$

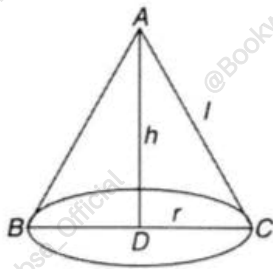
$$V = \frac{1}{3} \pi r^2 h \text{ or } h = \frac{3V}{\pi r^2}$$

Also, Putting the value of h in (i), we get

$$S = \pi r \sqrt{\frac{9V^2}{\pi^2 r^4} + r^2}$$

$$S^2 = \pi^2 r^2 \left(\frac{9V^2}{\pi^2 r^4} + r^2 \right)$$

Or [Maxima or Minima is same for S or S²]



$$S^2 = 9 \frac{V^2}{r^2} + \pi^2 r^4$$

Or Differentiating with respect to 'r'

$$(S^2)' = \frac{-18V^2}{r^3} + 4\pi^2 r^3 \quad \dots\dots\dots(ii)$$

$$\text{Now, } (S^2)' = 0$$

$$\frac{-18V^2}{r^3} + 4\pi^2 r^3 = 0$$

$$\text{Or } 4\pi^2 r^6 = 18V^2$$

Putting value of V,

$$4\pi^2 r^6 = 18 \times \frac{1}{9} \pi^2 r^4 h^2$$

$$\text{Or } 2r^2 = h^2$$

$$\text{Or } r = \frac{h}{\sqrt{2}}$$

Differentiating (ii) with respect to 'r', again

$$(S^2)'' = \frac{54V^2}{r^4} + 12\pi^2 r^2$$

$$\text{Or } (S^2)'' \Big|_{r=\frac{h}{\sqrt{2}}} > 0 \quad (\text{for any value of } r)$$

Hence, S² i.e., S is minimum for

$$r = \frac{h}{\sqrt{2}}$$

$$\text{Or } h = \sqrt{2}r.$$

i.e., for least curved surface, altitude is equal to $\sqrt{2}$ times the radius of the base.

Then,

If θ is the semi-vertical angle of cone, then

$$\cot \theta = \frac{h}{r} = \frac{\sqrt{2}r}{r}$$

$$\text{Or } \cot \theta = \sqrt{2}$$

$$\text{Or } \theta = \cot^{-1}(\sqrt{2})$$

22. .

Soln. Given x and y are the sides of two squares such that $y = x - x^2$.

Let area of the first square $(A_1) = x^2$

And area of the second square $(A_2) = y^2 = (x - x^2)^2$

$$\text{Now, } \frac{dA_1}{dt} = \frac{d}{dt} x^2 = 2x \frac{dx}{dt}$$

$$\text{Also, } \frac{dA_2}{dt} = \frac{d}{dt} (x - x^2)^2$$

$$= 2(x - x^2) \left(\frac{dx}{dt} - 2x \frac{dx}{dt} \right) = \frac{dx}{dt} (1 - 2x) 2(x - x^2)$$

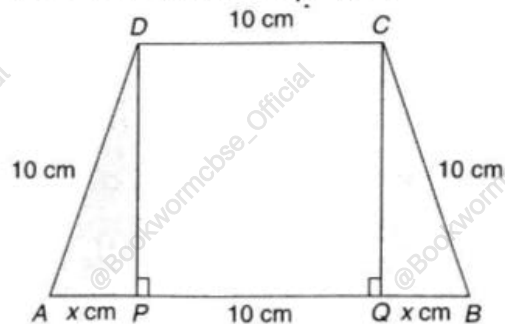
$$\therefore \frac{dA_2}{dA_1} = \frac{(dA_2/dt)}{(dA_1/dt)} = \frac{\frac{dx}{dt} (1 - 2x) 2(x - x^2)}{2x \frac{dx}{dt}}$$

$$= \frac{(1 - 2x) 2x(1 - x)}{2x} = (1 - 2x)(1 - x)$$

$$= 2x^2 - 3x + 1$$

23. .

Let A be the area of the trapezium.



$$\text{Then, } A \equiv A(x)$$

$$= \frac{1}{2} (\text{sum of parallel sides}) \times (\text{height})$$

$$= (x + 10) \sqrt{100 - x^2}$$

$$A'(x) = (x + 10) \frac{(-2x)}{2\sqrt{100 - x^2}} + \left(\sqrt{100 - x^2} \right)$$

Or

$$= \frac{-2x^2 - 10x + 100}{\sqrt{100 - x^2}}$$

$$\text{Now, } A'(x) = 0 \text{ gives } 2x^2 + 10x - 100 = 0,$$

i.e., $x=5$ and $x=-10$

So, $x=5$.

Now, $A''(x) =$

$$\frac{\sqrt{100-x^2}(-4x-10) - (2x^2-10x+100) \frac{(-2x)}{2\sqrt{100-x^2}}}{100-x^2}$$

$$= \frac{2x^3 - 300x - 1000}{(100-x^2)^{\frac{3}{2}}} \quad (\text{on simplification})$$

$$\text{Or } A''(5) = \frac{2(5)^3 - 300(5) - 1000}{(100-(5)^2)^{\frac{3}{2}}} = \frac{-2,250}{75\sqrt{75}} = \frac{-30}{\sqrt{75}} < 0$$

Thus, area of trapezium is maximum at $x=5$ and the maximum area is given by

$$A(5) = (5+10)\sqrt{100-(5)^2} = 15\sqrt{75} = 75\sqrt{3} \text{ cm}^2$$

24. .

Soln. Let foot of the ladder is at a distance x from the wall and height on the wall is y .

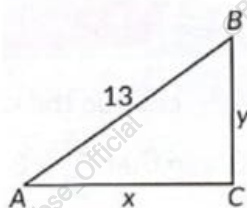
$$\text{Here, } x^2 + y^2 = (13)^2 \quad [\text{Using}]$$

Pythagoras theorem]

Differentiating with respect to t , we get

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\Rightarrow \frac{dy}{dt} = \frac{-x}{y} \frac{dx}{dt}$$



$$\text{When } x=5 \text{ m, } y^2 = (13)^2 - (5)^2 = 169 - 25 = 144$$

$$\therefore y=12 \text{ m}$$

$$\frac{dx}{dt} = 2 \text{ cm/sec}$$

Also, [Given]

$$\therefore \frac{dy}{dt} = \frac{-5}{12} \times 2 = \frac{-5}{6} \text{ cm/sec}$$

25. .

Soln. Let 'a' be the side of an equilateral triangle,

$$\text{Then } \frac{da}{dt} = 2 \text{ cm/sec}$$

Let 'A' be the area of an equilateral triangle,

$$A = \frac{\sqrt{3}}{4} a^2 \Rightarrow \frac{dA}{dt} = 2 \times \frac{\sqrt{3}}{4} a \frac{da}{dt} = \frac{\sqrt{3}}{2} a \frac{da}{dt}$$

$$\therefore \left(\frac{dA}{dt} \right)_{a=20} = \frac{\sqrt{3}}{2} \times 20 \times 2 = 20\sqrt{3} \text{ cm}^2 / \text{sec}$$

26. .

Soln. Let 'a' be the side of an equilateral triangle.

$$\frac{da}{dt} = 2 \text{ cm/sec}$$

Then

Let 'A' be the area of an equilateral triangle, then

$$A = \frac{\sqrt{3}}{4} a^2 \Rightarrow \frac{dA}{dt} = 2 \times \frac{\sqrt{3}}{4} a \frac{da}{dt} = \frac{\sqrt{3}}{2} a \frac{da}{dt}$$

$$\therefore \left(\frac{dA}{dt} \right)_{a=10} = \frac{\sqrt{3}}{2} \times 10 \times 2 = 10\sqrt{3} \text{ cm}^2 / \text{sec}$$

27. .

Soln. Let r , S and V respectively be the radius, surface area and volume of sphere at any time t .

$$\text{Then, } \frac{dV}{dt} = 3 \text{ cm}^3 / \text{sec}$$

$$\text{Now, } V = \frac{4}{3} \pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{3}{4\pi r^2}$$

$$\text{Also, } S = 4\pi r^2 \Rightarrow \frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$

$$\Rightarrow \frac{dS}{dt} = \frac{6}{r}$$

$$\therefore \left(\frac{dS}{dt} \right)_{r=2} = \frac{6}{2} = 3 \text{ cm}^2 / \text{sec}$$

$$28. \text{ Soln. We have, } f(x) = \frac{x^4}{4} - x^3 - 5x^2 + 24x + 12$$

.....(i)

$f(x)$ being polynomial function is continuous and derivable on R .

differentiating (i) w.r.t. x , we get

$$f'(x) = \frac{4x^3}{4} - 3x^2 - 10x + 24$$

$$= x^3 - 3x^2 - 10x + 24$$

$$= (x-2)(x^2-x-12) = (x-2)(x-4)(x+3)$$

(a) For increasing, $f'(x) > 0$

$$\Rightarrow (x-2)(x-4)(x+3) > 0$$

$$\Rightarrow x \in (4, \infty) \cup (-3, 2)$$

(b) For decreasing, $f'(x) < 0$

then

$$\Rightarrow (x-2)(x-4)(x+3) < 0$$

$$\Rightarrow x \in (2, 4) \cup (-\infty, -3)$$

29.

Soln. Here, $f(x) = x^2 - x + 1$; $x \in (-1, 1)$

$$\Rightarrow f'(x) = 2x - 1$$

$$f'(x) = 0 \Rightarrow x = \frac{1}{2}$$

$$\text{Now } f'(x) = 2\left(x - \frac{1}{2}\right) > 0 \text{ for } \frac{1}{2} < x < 1$$

$\Rightarrow f$ is strictly increasing in $\left(\frac{1}{2}, 1\right)$

$$\text{Also } f'(x) = 2\left(x - \frac{1}{2}\right) < 0 \text{ for } -1 < x < \frac{1}{2}$$

$\Rightarrow f$ is strictly decreasing in $\left(-1, \frac{1}{2}\right)$

Thus f is neither increasing nor decreasing in $(-1, 1)$.

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New

SURE SHOT QUESTIONS 2026

Chapter – 07 (Questions)

Integrals

Questions

1. Evaluate $\int \frac{(x-4)e^x}{(x-2)^3} dx$

2. Evaluate $\int \frac{dx}{\sqrt{5-4x-2x^2}} dx$

3. Find $\int \frac{1-x^2}{x(1-2x)} dx$

4. Evaluate: $\int \frac{\sin 2x}{(1+\sin x)(2+\sin x)} dx$

5. Find $\left(\int \frac{\log x}{(x+1)^2} dx \right)$

6. Evaluate: $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx$

7. Evaluate $\int \frac{dx}{x(x^5+3)}$

8. Find: $\int \frac{e^x-1}{e^x+1} dx$.

9. Find: $\int \frac{3x+5}{\sqrt{x^2+2x+4}} dx$.

10. Find: $\int \frac{\sin x}{\sin(x-2a)} dx$

11. Find $\int \frac{(3\sin\theta - 2)\cos\theta}{5 - \cos^2\theta - 4\sin\theta} d\theta$.

12. Evaluate: $\int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx$

13. Evaluate $\int_0^{\pi} e^{2x} \cdot \sin\left(\frac{\pi}{4} + x\right) dx$.

14. Find: $\int_0^{\pi/2} \frac{dx}{\cos^3 x \sqrt{2\sin 2x}}$.

15. Given $\int e^x (\tan x + 1) \sec x dx = e^x f(x) + C$ Write $f(x)$ satisfying the above.

16. Evaluate: $\int \frac{\tan \theta + \tan^3 \theta}{1 + \tan^3 \theta} d\theta$

17. Evaluate: $\int \frac{3x+5}{\sqrt{x^2-8x+7}} dx$

18. Find: $\int \frac{(5x-2)}{(3x^2+2x+1)} dx$

19. Evaluate: $\int e^x \frac{x^2+1}{(x+1)^2} dx$

20. Evaluate: $\int \frac{\sin(x-a)}{\sin(x+a)} dx$

21. Evaluate: $\int x \sin^{-1} x dx$

22. Evaluate: $\int e^x \left(\frac{\sin 4x-4}{1-\cos 4x} \right) dx$

23. Evaluate: $\int \frac{5x+3}{\sqrt{x^2+4x+10}} dx$

24. Evaluate: $\int \frac{x^2}{(x^2+4)(x^2+9)} dx$

25. Evaluate: $\int (\sqrt{\cot x} + \sqrt{\tan x}) dx$

26. Find: $\int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$

27. Find: $\int 5^{5^x} \cdot 5^{5^x} \cdot 5^x \cdot dx$

28. Find: $\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx$

29. Find : $\int \frac{\sin^{-1}\sqrt{x}-\cos^{-1}\sqrt{x}}{\sin^{-1}\sqrt{x}+\cos^{-1}\sqrt{x}} dx, x \in [0,1]$

30. Find: $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx$

31. Integrate the following w.r.t.x : $\frac{x^2-3x+1}{\sqrt{1-x^2}}$

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New

SURE SHOT QUESTIONS 2026

Chapter – 07 (Solutions)

Integrals

Questions

1. Soln. $I =$

$$\int \left[\frac{(x-2)-2}{(x-2)^3} \right] e^x dx = \int \frac{e^x}{(x-2)^2} dx - 2 \int \frac{e^x}{(x-2)^3} dx$$

$$= \frac{e^x}{(x-2)^2} + 2 \int \frac{e^x dx}{(x-2)^3} - 2 \int \frac{e^x dx}{(x-2)^3} = \frac{e^x}{(x-2)^2} + C$$

2.

Soln. We have, $\int \frac{dx}{\sqrt{5-4x-2x^2}} = \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\frac{5}{2}-2x-x^2}} =$

$$\frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\frac{5}{2}-(2x+x^2)}} = \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\frac{5}{2}+1-(1+2x+x^2)}}$$

[Add and subtract 1]

$$= \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\frac{7}{2}-(x+1)^2}} = \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\left(\sqrt{\frac{7}{2}}\right)^2 - (x+1)^2}}$$

$$= \frac{1}{\sqrt{2}} \sin^{-1} \frac{x+1}{\sqrt{\frac{7}{2}}} + c \left[\because \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a} + c \right]$$

$$= \frac{1}{\sqrt{2}} \sin^{-1} \left[\frac{\sqrt{2}(x+1)}{\sqrt{7}} \right] + c$$

3. Soln. Let

$$I = \int \frac{1-x^2}{x(1-2x)} dx$$

$$= \int \frac{1-x^2}{x-2x^2} dx$$

$\therefore \deg(\text{num}) = \deg(\text{den})$

$$\therefore \begin{array}{r} \frac{1/2}{-2x+x^2} \sqrt{-x^2+1} \\ -x^2+\frac{1}{2}x \\ + \\ -\frac{1}{2}x+1 \end{array}$$

$$I = \int \left(\frac{1}{2} + \frac{-\frac{1}{2}x+1}{x(1-2x)} \right) dx$$

$$I = \frac{1}{2}x + \int \frac{-\frac{1}{2}x+1}{x(1-2x)} dx \quad \dots\dots\dots(ii)$$

$$\text{Let } I_1 = \frac{-\frac{1}{2}x+1}{x(1-2x)} = \frac{A}{x} + \frac{B}{1-2x}$$

$$-\frac{1}{2}x + 1 = A(1-2x) + Bx \quad \dots\dots\dots(ii)$$

If $x = 0$

Then from equation (ii)

$$1 = A$$

$$x = \frac{1}{2}$$

If

Then from equation (ii)

$$-\frac{1}{4} + 1 = \frac{B}{2}$$

$$\frac{-1+4}{4} = \frac{B}{2}$$

$$\frac{3}{4} = \frac{B}{2}$$

$$B = \frac{3}{2}$$

$$I_1 = \int \frac{1}{x} dx + \frac{3}{2} \int \frac{dx}{1-2x}$$

$$= \log |x| + \frac{3}{2(-2)} \int \frac{-2}{1-2x} dx$$

$$= \log |x| - \frac{3}{4} \log |1-2x|$$

$$\left[\because \int \frac{f'(x)}{f(x)} dx = \log |f(x)| + c \right]$$

$$\therefore I = \frac{1}{2}x + \log |x| - \frac{3}{4} \log |1-2x| + c$$

4. Soln.

Put $\sin x = t$

Differentiate both sides w.r.t. x

$$\cos x dx = dt$$

$$\therefore I = \int \frac{2t}{(1+t)(2+t)} dt$$

$$\frac{2t}{(1+t)(2+t)} = \frac{A}{1+t} + \frac{B}{2+t}$$

Now,

$$2t = A(2+t) + B(1+t)$$

Put $t = -1$

Then $-2 = A$

Put $t = -2$

Then $-4 = B(1 - 2)$

$-4 = -B$

$B = 4$

$$\therefore I = \int \frac{-2}{1+t} dt + \int \frac{4}{2+t} dt$$

$$= -2 \log |1+t| + 4 \log |2+t| + c$$

$$I = -2 \log |1+\sin x| + 4 \log |2+\sin x| + c$$

5. Soln. Let $I = \int \frac{\log x}{(x+1)^2} dx$

$$= \int \log x \cdot \frac{1}{(x+1)^2} dx$$

II

Integrating by parts

$$I = \log x \int \frac{dx}{(x+1)^2} - \int \left(\frac{d}{dx} \log x \int \frac{dx}{(x+1)^2} \right) dx$$

$$\Rightarrow I = -\frac{\log x}{(x+1)} + \int \frac{1}{x(x+1)} dx$$

$$\left[\because \frac{d}{dx} \frac{1}{x^2} = -\frac{1}{x^2} \right]$$

$$= -\frac{\log x}{(x+1)} + \int \frac{(x+1) - x}{x(x+1)} dx$$

[Add and subtract x]

$$\Rightarrow I = -\frac{\log x}{(x+1)} + \int \left(\frac{1}{x} - \frac{1}{x+1} \right) dx$$

$$\Rightarrow I = -\frac{\log x}{(x+1)} + \log |x| - \log |x+1| + C$$

$$\therefore I = \log \left| \frac{x}{x+1} \right| - \frac{\log x}{(x+1)} + C$$

6. Soln. $\frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha}$

$$= \frac{-2 \sin \left(\frac{2x+2\alpha}{2} \right) \sin \left(\frac{2x-2\alpha}{2} \right)}{-2 \sin \left(\frac{x+\alpha}{2} \right) \sin \left(\frac{x-\alpha}{2} \right)}$$

$$= \frac{\sin(x+\alpha) \sin(x-\alpha)}{\sin \left(\frac{x+\alpha}{2} \right) \sin \left(\frac{x-\alpha}{2} \right)}$$

$$= \frac{\left[2 \sin \left(\frac{x+\alpha}{2} \right) \cos \left(\frac{x+\alpha}{2} \right) \right] \left[2 \sin \left(\frac{x-\alpha}{2} \right) \cos \left(\frac{x-\alpha}{2} \right) \right]}{\sin \left(\frac{x+\alpha}{2} \right) \sin \left(\frac{x-\alpha}{2} \right)}$$

$$[\because \sin 2\theta = 2 \sin \theta \cos \theta]$$

$$= 4 \cos \left(\frac{x+\alpha}{2} \right) \cos \left(\frac{x-\alpha}{2} \right)$$

$$= 2 \left[\cos \left(\frac{x+\alpha}{2} + \frac{x-\alpha}{2} \right) + \cos \left(\frac{x+\alpha}{2} - \frac{x-\alpha}{2} \right) \right]$$

$$[\because \cos(A+B) + \cos(A-B) = 2 \cos A \cos B]$$

Now,

$$\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} = \int (2 \cos x + 2 \cos \alpha) dx$$

$$= 2 \sin x + 2x \cos \alpha + C$$

$$\int \frac{dx}{x(x^5+3)} = \int \frac{x^4}{x^5(x^5+3)} dx$$

7. Soln.

Put, $(x^5+3) = t$

$$\Rightarrow 5x^4 dx = dt$$

$$\Rightarrow x^4 dx = \frac{dt}{5}$$

$$\therefore \int \frac{dx}{x(x^5+3)} = \frac{1}{5} \int \frac{dt}{t(t-3)}$$

Let $\frac{1}{t(t-3)} = \frac{A}{t} + \frac{B}{t-3}$

$$1 = A(t-3) + Bt$$

$$\Rightarrow 1 = At - 3A + Bt$$

$$\Rightarrow 1 = (A+B)t - 3A$$

On comparing the co-efficient, we get

$$1 = -3A$$

$$\Rightarrow A = -1/3$$

$$A+B=0$$

$$\Rightarrow B = 1/3$$

Now,

$$\frac{1}{5} \int \frac{dt}{t(t-3)} = \frac{1}{5} \int \left(\frac{-1}{3t} + \frac{1}{3(t-3)} \right) dt$$

$$= \frac{-1}{15} \log t + \frac{1}{15} \log(t-3) + C$$

$$= \frac{-1}{15} \left(\log \left(\frac{t}{t-3} \right) \right) + C$$

$$[\because \log(a/b) = \log a - \log b]$$

$$= \frac{-1}{15} \left(\log \frac{(x^5+3)}{(x^5+3-3)} \right) + C$$

$$= \frac{1}{15} \left(\log \frac{x^5}{x^5+3} \right) + C$$

8. .Ans.

$$\text{Let } I = \int \frac{e^{4x} - 1}{e^{4x} + 1} dx = \int \left(\frac{2e^{4x}}{e^{4x} + 1} - 1 \right) dx$$

$$= 2 \int \frac{e^{4x}}{e^{4x} + 1} dx - \int 1 dx = 2 \int \frac{e^{4x}}{e^{4x} + 1} dx - x + C$$

$$\text{Put } e^{4x} + 1 = t \Rightarrow 4e^{4x} dx = dt$$

$$\therefore I = \frac{2}{4} \int \frac{dt}{t} - x + C$$

$$\Rightarrow I = \frac{1}{2} \log |t| - x + C \Rightarrow I = \frac{1}{2} \log |e^{4x} + 1| - x + C$$

9. .Ans.

$$\text{Let } I = \int \frac{3x+5}{x^2+2x+4} dx$$

$$Nr = A \frac{d}{dx}(Dr) + B$$

$$\Rightarrow 3x+5 = A(2x+2) + B$$

$$\text{On comparing, } 2A = 3, 2A + B = 5$$

$$\Rightarrow A = \frac{3}{2} \text{ and } 2\left(\frac{3}{2}\right) + B = 5 \Rightarrow B = 2$$

$$\text{Now, } I = \int \frac{\frac{3}{2}(2x+2) + 2}{x^2+2x+4} dx$$

$$= \frac{3}{2} \int \frac{2x+2}{x^2+2x+4} dx + 2 \int \frac{dx}{x^2+2x+4}$$

$$[\text{Put } t = x^2 + 2x + 4 \Rightarrow dt = 2x + 2]$$

$$= \frac{3}{2} \int \frac{dt}{t} + 2 \int \frac{dx}{\sqrt{(x+1)^2 + (\sqrt{3})^2}}$$

$$= 3 \sqrt{x^2+2x+4} + 2 \log |x+1+\sqrt{x^2+2x+4}| + C$$

$$I = \int \frac{\sin x}{\sin(x-2a)} dx$$

10. .Soln. Let

$$\text{Put } x-2a = t$$

$$\Rightarrow x = 2a + t \Rightarrow dx = dt$$

$$\therefore I = \int \frac{\sin(t+2a)}{\sin t} dx$$

$$= \int \frac{(\sin t \cos 2a + \cos t \sin 2a)}{\sin t} dx$$

$$= \int (\cos 2a + \cot t \sin 2a) dx$$

$$= t \cos 2a + \sin 2a \log |\sin t| + c$$

$$= (x-2a) \cos 2a + \sin 2a \log |\sin(x-2a)| + c$$

11. .Soln. Let

$$I = \int \frac{(3 \sin \theta - 2) \cos \theta}{5 - \cos^2 \theta - 4 \sin \theta} d\theta$$

$$(\text{Put } \cos^2 \theta = 1 - \sin^2 \theta)$$

$$= 3 \int \frac{\sin \theta \cos \theta}{4 + \sin^2 \theta - 4 \sin \theta} d\theta - 2 \int \frac{\cos \theta}{4 + \sin^2 \theta - 4 \sin \theta} d\theta$$

$$= 3I_1 - 2I_2 \text{ (say)}$$

.....(i)

$$\text{Now, } I_1 = \int \frac{\sin \theta \cos \theta}{4 + \sin^2 \theta - 4 \sin \theta} d\theta$$

$$\text{Put } \sin^2 \theta = t \Rightarrow 2 \sin \theta \cos \theta d\theta = dt$$

$$\therefore I_1 = \frac{1}{2} \int \frac{dt}{4+t-4\sqrt{t}} = \frac{1}{2} \int \frac{dt}{(\sqrt{t}-2)^2}$$

$$\text{Put } \sqrt{t}-2 = u \Rightarrow \sqrt{t} = u+2$$

$$\Rightarrow \frac{1}{2\sqrt{t}} dt = du \Rightarrow dt = 2(u+2) du$$

$$\therefore I_1 = \int \frac{(u+2)}{u^2} du = \int \frac{du}{u} + 2 \int \frac{du}{u^2}$$

$$= \log u - \frac{2}{u} + C_1 = \log(\sqrt{t}-2) - \frac{2}{\sqrt{t}-2} + C_1$$

$$= \log(\sin \theta - 2) - \frac{2}{\sin \theta - 2} + C_1 \text{(ii)}$$

$$\text{Also, } I_2 = \int \frac{\cos \theta}{4 + \sin^2 \theta - 4 \sin \theta} d\theta$$

$$\text{Put } \sin \theta = m \Rightarrow \cos \theta d\theta = dm$$

$$\therefore I_2 = \int \frac{dm}{4+m^2-4m} = \int \frac{dm}{(m-2)^2} \text{(iii)}$$

From (i), (ii) and (iii), we get

$$I = 3 \log(\sin \theta - 2) - \frac{6}{\sin \theta - 2} + \frac{2}{\sin \theta - 2} + C$$

$$\text{Where } C = 3C_1 - 2C_2$$

$$\Rightarrow I = 3 \log(\sin \theta - 2) - \frac{4}{\sin \theta - 2} + C$$

12. .Soln. Let

$$I = \int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = \int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{1 - (1 - \sin 2x)}} dx$$

$$= \int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos x)^2}} dx$$

$$\text{Put } \sin x - \cos x = t \Rightarrow (\cos x + \sin x) dx = dt$$

$$\text{And when, } x = \frac{\pi}{6}, \text{ then } t = \frac{1-\sqrt{3}}{2} = -\alpha$$

$$\therefore l = \int_{-a}^a \frac{dt}{\sqrt{1-t^2}}$$

$$= [\sin^{-1}t]_{-a}^a = 2\sin^{-1}a = 2\sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$$

$$l = \int_0^{\pi} e^{2x} \cdot \sin\left(\frac{\pi}{4} + x\right) dx$$

13. Soln. Let

$$\text{Put } \frac{\pi}{4} + x = t \Rightarrow x = t - \frac{\pi}{4} \Rightarrow dx = dt$$

$$\text{When } x = 0, t = \frac{\pi}{4} \text{ and When } x = \pi, t = \frac{5\pi}{4}$$

$$\therefore l = \int_{\pi/4}^{5\pi/4} e^{2\left(t - \frac{\pi}{4}\right)} \sin t dt = e^{-\pi/2} \int_{\pi/4}^{5\pi/4} e^{2t} \sin t dt$$

$$= e^{-\pi/2} \left[\left(\sin t \frac{e^{2t}}{2} \right)_{\pi/4}^{5\pi/4} - \int_{\pi/4}^{5\pi/4} \cos \frac{e^{2t}}{2} dt \right]$$

$$= e^{-\pi/2} \left[\frac{1}{2} \left(e^{5\pi/2} \sin \frac{5\pi}{4} - e^{\pi/2} \sin \frac{\pi}{4} \right) - \left(\frac{e^{2t}}{4} \cos t \right)_{\pi/4}^{5\pi/4} - \int_{\pi/4}^{5\pi/4} \frac{e^{2t}}{4} \sin t dt \right]$$

$$= e^{-\pi/2} \left[\frac{1}{2} \left(\frac{-1}{\sqrt{2}} e^{5\pi/2} - \frac{1}{\sqrt{2}} e^{\pi/2} \right) - \frac{1}{4} \left(-\frac{1}{\sqrt{2}} e^{5\pi/2} - \frac{1}{\sqrt{2}} e^{\pi/2} \right) - \frac{l}{4} \right]$$

$$\Rightarrow l + \frac{1}{4}l = -\frac{1}{2\sqrt{2}}[e^{2\pi} + 1] + \frac{1}{4\sqrt{2}}[e^{2\pi} + 1]$$

$$\Rightarrow \frac{5}{4}l = \frac{(e^{2\pi} + 1)}{2\sqrt{2}} \left[\frac{1}{2} - 1 \right]$$

$$= -\frac{1}{4\sqrt{2}}[e^{2\pi} + 1] \Rightarrow l = -\frac{1}{5\sqrt{2}}(1 + e^{2\pi})$$

14. Soln.

Let

$$l = \int_0^{\pi/4} \frac{dx}{\cos^3 x \sqrt{2} \sin 2x} = \int_0^{\pi/4} \frac{dx}{\cos^3 x \sqrt{2} \cdot 2 \sin x \cos x}$$

$$= \frac{1}{2} \int \frac{dx}{\cos^{3\frac{1}{2}} x \sin^{\frac{1}{2}} x}$$

$$= \frac{1}{2} \int_0^{\pi/4} \frac{dx}{\cos^{\frac{7}{2}} x \tan^{\frac{1}{2}} x \cos^{\frac{1}{2}} x}$$

$$= \frac{1}{2} \int_0^{\pi/4} \frac{dx}{\cos^4 x \sqrt{\tan x}} = \frac{1}{2} \int_0^{\pi/4} \frac{\sec^4 x}{\sqrt{\tan x}} dx$$

$$\text{Put } \tan x = t \Rightarrow \sec^2 x dx = dt$$

$$\text{When } x = 0, \text{ then } t = 0 \text{ and when } x = \frac{\pi}{4} \text{ then } t = 1$$

$$\therefore l = \frac{1}{2} \int_0^1 \frac{(1+t^2) dt}{\sqrt{t}} = \frac{1}{2} \int_0^1 (t^{\frac{1}{2}} + t^{\frac{3}{2}}) dt$$

$$= \frac{1}{2} \left[\frac{t^{1/2}}{1/2} + \frac{t^{5/2}}{5/2} \right]_0^1 = \frac{1}{2} \left[2 + \frac{2}{5} \right]$$

$$= 1 + \frac{1}{5} = \frac{6}{5}$$

15. .

$$\text{Soln. Given, } \int e^x (\tan x + 1) \sec x dx = e^x f(x) + C$$

$$\Rightarrow \int e^x (\tan x \sec x + \sec x) dx = e^x f(x) + C$$

$$\Rightarrow \int e^x (\sec x + \tan x \sec x) dx = e^x f(x) + C$$

$$\Rightarrow e^x \sec x + C = e^x f(x) + C$$

$$\Rightarrow f(x) = \sec x$$

$$[\text{Note: } \int e^x [f(x) + f'(x)] dx = e^x \cdot f(x) + C, \text{ Here } f(x) = \sec x]$$

16. .

Soln. Let

$$I = \int \frac{\tan \theta + \tan^3 \theta}{1 + \tan^3 \theta} d\theta = \int \frac{\tan \theta (1 + \tan^2 \theta)}{1 + \tan^3 \theta} d\theta$$

$$= \int \frac{\tan \theta \sec^2 \theta}{1 + \tan^3 \theta} d\theta$$

$$\text{Let } \tan \theta = z \Rightarrow \sec^2 \theta d\theta = dz$$

$$\therefore I = \int \frac{z dz}{1 + z^3} = \int \frac{z dz}{(1+z)(z^2 - z + 1)}$$

$$\frac{z}{(1+z)(z^2 - z + 1)} = \frac{A}{1+z} + \frac{Bz+C}{z^2 - z + 1}$$

Now

$$\Rightarrow z = A(z^2 - z + 1) + (Bz + C)(1 + z)$$

Putting $z = -1$, we get, $A = -\frac{1}{3}$

Putting $z = 0$, we get, $C = \frac{1}{3}$

Putting $z = 1$, we get, $B = \frac{1}{3}$

$$\therefore \frac{z}{(1+z)(z^2-z+1)} = \frac{-1}{3(1+z)} + \frac{\frac{1}{3}z + \frac{1}{3}}{z^2-z+1}$$

$$\begin{aligned} \Rightarrow I &= -\frac{1}{3} \int \frac{dz}{1+z} + \frac{1}{3} \int \frac{z+1}{z^2-z+1} dz \\ &= -\frac{1}{3} \log |1+z| + \frac{1}{3 \times 2} \int \frac{2z-1+3}{z^2-z+1} dz \\ &= -\frac{1}{3} \log |1+z| + \frac{1}{6} \int \frac{2z-1}{z^2-z+1} dz + \frac{1}{2} \int \frac{dz}{z^2-z+1} \end{aligned}$$

$$I = -\frac{1}{3} \log |1+z| + \frac{1}{6} \log |z^2-z+1| + I_1 + C \quad \dots (i)$$

Where

$$I_1 = \frac{1}{2} \int \frac{dz}{z^2-z+1} = \frac{1}{2} \int \frac{dz}{\left(z - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= \frac{1}{2} \cdot \frac{2}{\sqrt{3}} + \tan^{-1} \left(\frac{z - \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right)$$

$$\therefore I = -\frac{1}{3} \log |1+z| + \frac{1}{6} \log |z^2-z+1| + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2z-1}{\sqrt{3}} \right) + C$$

Putting $z = \tan \theta$

$$\begin{aligned} \therefore I &= -\frac{1}{3} \log |1 + \tan \theta| + \frac{1}{6} \log |\tan^2 \theta - \tan \theta + 1| \\ &\quad + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2 \tan \theta - 1}{\sqrt{3}} \right) + C \end{aligned}$$

17. .

$$3x+5 = A \frac{d}{dx} (x^2-8x+7) + B$$

Soln. Let

$$\Rightarrow 3x+5 = A(2x-8) + B$$

$$\Rightarrow 3x+5 = 2Ax - 8A + B$$

Equating the coefficient of x and the constant, we get

$$2A = 3 \text{ and } -8A + B = 5$$

$$\Rightarrow A = \frac{3}{2} \text{ and } -8 \times \frac{3}{2} + B = 5 \Rightarrow B = 5 + 12 = 17$$

Hence,

$$\begin{aligned} \int \frac{3x+5}{\sqrt{x^2-8x+7}} dx &= \int \frac{3(2x-8)+17}{\sqrt{x^2-8x+7}} dx = \frac{3}{2} \int \frac{(2x-8)}{\sqrt{x^2-8x+7}} dx + 17 \int \frac{dx}{\sqrt{x^2-8x+7}} \\ &= \frac{3}{2} I_1 + 17 I_2 \quad \dots (i) \end{aligned}$$

$$I_1 = \int \frac{2x-8}{\sqrt{x^2-8x+7}} dx, I_2 = \int \frac{dx}{\sqrt{x^2-8x+7}}$$

Where

$$I_1 = \int \frac{2x-8}{\sqrt{x^2-8x+7}} dx$$

Now

$$\text{Put } x^2-8x+7 = z^2$$

$$\Rightarrow (2x-8) dx = 2z dz$$

$$\therefore I_1 = \int \frac{2z dz}{z} = 2 \int dz = 2z + C_1$$

$$I_1 = 2\sqrt{x^2-8x+7} + C_1 \quad \dots (ii)$$

$$I_2 = \int \frac{dx}{\sqrt{x^2-8x+7}} = \int \frac{dx}{\sqrt{x^2-2x+4+16-16+7}} = \int \frac{dx}{\sqrt{(x-4)^2-3^2}}$$

$$= \log \left| (x-4) + \sqrt{(x-4)^2-3^2} \right| + C_2$$

$$I_2 = \log \left| (x-4) + \sqrt{x^2-8x+7} \right| + C_2$$

Putting the value of I_1 and I_2 in (i), we get

$$\begin{aligned} \int \frac{3x+5}{\sqrt{x^2-8x+7}} dx &= \frac{3}{2} \sqrt{x^2-8x+7} + 17 \log \left| (x-4) + \sqrt{x^2-8x+7} \right| + (C_1 + C_2) \\ &= 3\sqrt{x^2-8x+7} + 17 \log \left| (x-4) + \sqrt{x^2-8x+7} \right| + C \end{aligned}$$

$$\left[\text{Note: } \int \frac{dx}{\sqrt{x^2-a^2}} = \log |x + \sqrt{x^2-a^2}| + C \right]$$

18. .

Soln. The given integral is in the form of

$$\int \frac{(px+q)}{ax^2+bx+c} dx.$$

Therefore, we express

$$(5x-2) = A \frac{d}{dx} (1+2x+3x^2) + B = A(2+6x) + B$$

Equating the coefficients of x and the constant term on both the sides, we get

$$6A = 5 \text{ and } 2A + B = -2 \quad \text{or} \quad A = \frac{5}{6} \text{ and } B = -\frac{11}{3}$$

$$\begin{aligned} \therefore \int \frac{(5x-2)}{(1+2x+3x^2)} dx &= \frac{5}{6} \int \frac{2+6x}{(1+2x+3x^2)} dx - \frac{11}{3} \int \frac{dx}{3x^2+2x+1} \\ &= \frac{5}{6} I_1 - \frac{11}{3} I_2, \quad (\text{say}) \end{aligned}$$

In I_1 putting $1+2x+3x^2 = t$, so that

$$(2+6x)dx = dt$$

$$\therefore I_1 = \int \frac{dt}{t} = \log |t| + C_1 = \log |3x^2 + 2x + 1| + C_1$$

And

$$I_2 = \int \frac{dx}{3x^2 + 2x + 1} = \int \frac{dx}{3\left(x^2 + \frac{2x}{3} + \frac{1}{3}\right)} = \frac{1}{3} \int \frac{dx}{\left(x + \frac{1}{3}\right)^2 + \left(\frac{\sqrt{2}}{3}\right)^2}$$

Putting $\left(x + \frac{1}{3}\right) = t$, so that $dx = dt$, we get

$$I_2 = \frac{1}{3} \int \frac{dt}{t^2 + \left(\frac{\sqrt{2}}{3}\right)^2} = \frac{1}{3 \cdot \frac{\sqrt{2}}{3}} \tan^{-1} \frac{t}{\frac{\sqrt{2}}{3}} + C_2$$

$$= \frac{1}{\sqrt{2}} \tan^{-1} \frac{3t}{\sqrt{2}} + C_2 = \frac{1}{\sqrt{2}} \tan^{-1} \frac{3\left(x + \frac{1}{3}\right)}{\sqrt{2}} + C_2$$

$$= \frac{1}{\sqrt{2}} \tan^{-1} \frac{(3x+1)}{\sqrt{2}} + C_2$$

Using (ii) and (iii) in (i), we get

$$\int \frac{(5x-2)}{(1+2x+3x^2)} dx = \frac{5}{6} \log |3x^2 + 2x + 1| - \frac{11}{3\sqrt{2}} \tan^{-1} \frac{(3x+1)}{\sqrt{2}} + C$$

Where, $C = C_1 + C_2$

19.

Soln. Let $I = \int e^x \frac{x^2+1}{(x+1)^2} dx = \int e^x \left(1 - \frac{2x}{(x+1)^2}\right) dx = \int e^x - 2 \left(\frac{e^x x dx}{(x+1)^2}\right)$

$$= e^x - 2 \int e^x \frac{x+1-1}{(x+1)^2} dx$$

$$= e^x - 2 \int e^x \left[\frac{1}{x+1} + \frac{-1}{(x+1)^2} \right] dx$$

$$= e^x - 2e^x \cdot \frac{1}{x+1} + C \quad [\text{Note:}]$$

$$\int e^x \{f(x) + f'(x)\} dx = e^x \cdot f(x) + C$$

20.

$$I = \int \frac{\sin(x-a)}{\sin(x+a)} dx$$

Soln. Let

$$\text{Let } X+a=t \Rightarrow X=t-a \Rightarrow dx=dt$$

$$\therefore I = \int \frac{\sin(t-2a)}{\sin t} dt$$

$$= \int \frac{\sin t \cdot \cos 2a - \cos t \cdot \sin 2a}{\sin t} dt$$

$$= \cos 2a \int dt - \int \sin 2a \cdot \cot t dt$$

$$= \cos 2a \cdot t - \sin 2a \cdot \log |\sin t| + C$$

$$= \cos 2a \cdot (x+a) - \sin 2a \cdot \log |\sin(x+a)| + C$$

$$= x \cos 2a + a \cos 2a - (\sin 2a) \log |\sin(x+a)| + C$$

21.

Soln. Let $I = \int x \sin^{-1} x dx$

$$= \sin^{-1} x \cdot \frac{x^2}{2} - \int \frac{x^2}{2\sqrt{1-x^2}} dx \quad [\text{By using integrating by part}]$$

$$= \frac{x^2}{2} \sin^{-1} x + \frac{1}{2} \int \frac{1-x^2-1}{\sqrt{1-x^2}} dx = \frac{x^2}{2} \sin^{-1} x + \frac{1}{2} \int \sqrt{1-x^2} dx - \frac{1}{2} \int \frac{dx}{\sqrt{1-x^2}}$$

$$= \frac{x^2}{2} \sin^{-1} x - \frac{1}{2} \sin^{-1} x + \frac{1}{2} \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x \right] + C$$

$$= \frac{x^2}{2} \sin^{-1} x - \frac{1}{2} \sin^{-1} x + \frac{x}{4} \sqrt{1-x^2} + \frac{1}{4} \sin^{-1} x + C$$

$$= \frac{x^2}{2} \sin^{-1} x - \frac{1}{4} \sin^{-1} x + \frac{x}{4} \sqrt{1-x^2} + C$$

22.

$$I = \int e^x \left(\frac{\sin 4x - 4}{1 - \cos 4x} \right) dx$$

Soln. Let

$$= \int e^x \left(\frac{2 \sin 2x \cos 2x - 4}{2 \sin^2 2x} \right) dx$$

$$[\because \sin 2x = 2 \sin x \cos x]$$

$$[\text{and } \cos 2x = 1 - 2 \sin^2 x]$$

$$= \int e^x (\cot 2x - 2 \operatorname{cosec}^2 2x) dx$$

Let

$$f(x) = \cot 2x \quad \therefore f'(x) = -2 \operatorname{cosec}^2 2x$$

$$\therefore I = \int e^x (f(x) + f'(x)) dx$$

$$\Rightarrow I = e^x \cdot f(x) + C = e^x \cdot \cot 2x + C$$

$$[\because \int e^x (f(x) + f'(x)) dx = e^x f(x) + C]$$

23.

Soln. We can express the N^r as

$$5x+3 = A \frac{d}{dx} (x^2+4x+10) + B$$

$$\Rightarrow 5x+3 = A(2x+4) + B \Rightarrow 5x+3 = 2Ax + (4A+B)$$

Equating the coefficients, we get

$$2A=5 \text{ and } 4A+B=3$$

$$A = \frac{5}{2} \Rightarrow 4 \times \frac{5}{2} + B = 3 \Rightarrow B = 3 - 10 = -7$$

$$\therefore 5x+3 = \frac{5}{2}(2x+4) + (-7)$$

$$\therefore I = \int \frac{5(2x+4)-7}{\sqrt{x^2+4x+10}} dx = \frac{5}{2} \int \frac{(2x+4)}{\sqrt{x^2+4x+10}} dx - 7 \int \frac{dx}{\sqrt{x^2+4x+10}}$$

$$I = \frac{5}{2} I_1 - 7 I_2$$

Where

$$I_1 = \int \frac{2x+4}{\sqrt{x^2+4x+10}} dx \text{ and } I_2 = \int \frac{dx}{\sqrt{x^2+4x+10}}$$

$$I_1 = \int \frac{(2x+4)}{\sqrt{x^2+4x+10}} dx$$

Now,

$$\text{Let } x^2+4x+10 = t \Rightarrow (2x+4) dx = dt$$

$$\therefore I_1 = \int \frac{dt}{\sqrt{t}} = \int t^{-1/2} dt = \frac{t^{-1/2+1}}{-1/2+1} + C_1 = 2\sqrt{t} + C_1$$

$$I_1 = 2\sqrt{x^2+4x+10} + C_1$$

Again,

$$I_2 = \int \frac{dx}{\sqrt{x^2+4x+10}} = \int \frac{dx}{\sqrt{(x+2)^2 + (\sqrt{6})^2}}$$

$$= \log |(x+2) + \sqrt{x^2+4x+10}| + C_2$$

Putting the value of I_1 and I_2 in (i), we get

$$I = \frac{5}{2} \times 2\sqrt{x^2+4x+10} - 7 \log |(x+2) + \sqrt{x^2+4x+10}| + \left(\frac{5}{2} C_1 - 7 C_2 \right)$$

$$= 5\sqrt{x^2+4x+10} - 7 \log |(x+2) + \sqrt{x^2+4x+10}| + C$$

24.

$$I = \int \frac{x^2}{(x^2+4)(x^2+9)} dx$$

Soln. Let

$$\text{Put } x^2 = t, \text{ we get}$$

$$\therefore \frac{x^2}{(x^2+4)(x^2+9)} = \frac{t}{(t+4)(t+9)}$$

$$\text{Now, } \frac{t}{(t+4)(t+9)} = \frac{A}{t+4} + \frac{B}{t+9} = \frac{A(t+9) + B(t+4)}{(t+4)(t+9)}$$

$$\Rightarrow t = (A+B)t + (9A+4B)$$

Equating the coefficients, we get

$$A+B=1, \quad 9A+4B=0$$

Solving above two equations, we get

$$A = -\frac{4}{5}, \quad B = \frac{9}{5}$$

$$\therefore \frac{x^2}{(x^2+4)(x^2+9)} = -\frac{4}{5(x^2+4)} + \frac{9}{5(x^2+9)}$$

$$I = -\frac{4}{5} \int \frac{dx}{x^2+2^2} + \frac{9}{5} \int \frac{dx}{x^2+3^2}$$

$$= -\frac{4}{5} \times \frac{1}{2} \tan^{-1} \frac{x}{2} + \frac{9}{5} \times \frac{1}{3} \tan^{-1} \frac{x}{3} + C$$

$$= -\frac{2}{5} \tan^{-1} \frac{x}{2} + \frac{3}{5} \tan^{-1} \frac{x}{3} + C$$

25.

$$\text{Soln. Let } I = \int (\sqrt{\cot x} + \sqrt{\tan x}) dx$$

$$I = \int \left(\frac{\sqrt{\cos x}}{\sqrt{\sin x}} + \frac{\sqrt{\sin x}}{\sqrt{\cos x}} \right) dx = \int \frac{(\cos x + \sin x)}{\sqrt{\sin x \cos x}} dx$$

$$\text{Let } \sin x - \cos x = t \Rightarrow (\cos x + \sin x) dx = dt$$

$$\text{Also } \therefore \sin x - \cos x = t$$

$$\Rightarrow (\sin x - \cos x)^2 = t^2$$

$$\Rightarrow \sin^2 x + \cos^2 x - 2 \sin x \cos x = t^2$$

$$\Rightarrow 1 - 2 \sin x \cos x = t^2$$

$$\Rightarrow \sin x \cos x = \frac{1-t^2}{2}$$

$$I = \int \frac{dt}{\sqrt{1-t^2}} = \sqrt{2} \int \frac{dt}{\sqrt{1-t^2}}$$

Therefore,

$$= \sqrt{2} \sin^{-1} t + C = \sqrt{2} \sin^{-1} (\sin x - \cos x) + C$$

26. .

$$I = \int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}} dx$$

Soln. Let
Putting $x+1=t^2 \Rightarrow dx=2t dt$

$$I = 2 \int \frac{t^2+1}{t^4+t^2+1} dt = 2 \int \frac{1+\left(\frac{1}{t}\right)^2}{t^2+\frac{1}{t^2}+1} dt \quad [\text{Dividing N}^r]$$

and D^r by t^2]

$$= 2 \int \frac{1+\left(\frac{1}{t}\right)^2}{\left(t-\frac{1}{t}\right)^2+3} dt$$

Now, Put $t-\frac{1}{t}=z \Rightarrow \left(1+\frac{1}{t^2}\right) dt = dz,$

we get

$$I = 2 \int \frac{dz}{z^2+3} = \frac{2}{\sqrt{3}} \tan^{-1} \frac{z}{\sqrt{3}} + C$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t-\frac{1}{t}}{\sqrt{3}} \right) + C = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t^2-1}{\sqrt{3}t} \right) + C$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3}(x+1)} \right) + C$$

27. .

Soln. Let $I = \int 5^{5^x} \cdot 5^x \cdot 5^x \cdot dx$

Putting $5^x = t \Rightarrow 5^x \cdot \log 5 dx = dt$ or $5^x \cdot dx = \frac{dt}{(\log 5)}$

Therefore,

$$I = \int 5^{5^x} \cdot 5^x \cdot 5^x \cdot dx = \int 5^{5^x} \cdot 5^t \cdot \frac{dt}{(\log 5)} = \frac{1}{(\log 5)} \int 5^{5^x} \cdot 5^t \cdot dt$$

Again, putting $5^t = u, 5^t dt = \frac{du}{(\log 5)}$

$$I = \frac{1}{(\log 5)} \int 5^{5^x} \cdot \frac{du}{(\log 5)}$$

Therefore,

$$= \frac{1}{(\log 5)^2} \int 5^u \cdot \frac{du}{(\log 5)^2 \cdot (\log 5)} + C$$

$$= \frac{5^u}{(\log 5)^3} + C = \frac{5^{5^x}}{(\log 5)^3} + C = \frac{5^{5^{5^x}}}{(\log 5)^3} + C$$

28. .

$$I = \int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx$$

Soln. Let

Putting $\sqrt{x} = \cos \theta$, i.e., $x = \cos^2 \theta \Rightarrow \theta = \cos^{-1} \sqrt{x}$
and $dx = -2 \cos \theta \sin \theta d\theta$, we get

$$\therefore I = \int \sqrt{\frac{1-\cos \theta}{1+\cos \theta}} (-2 \sin \theta \cos \theta) d\theta$$

$$= -2 \int \sqrt{\frac{2 \sin^2 \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}}} (\sin \theta \cos \theta) d\theta = -2 \int \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \left(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cos \theta \right) d\theta$$

$$= -2 \int 2 \sin^2 \frac{\theta}{2} \cos \theta d\theta = -2 \int (1 - \cos \theta) \cos \theta d\theta$$

$$= -2 \int (1 - \cos \theta) \cos \theta \cdot d\theta = -2 \sin \theta + \int (1 + \cos 2\theta) \cdot d\theta$$

$$= -2 \sin \theta + \int 1 \cdot d\theta + \int \cos 2\theta \cdot d\theta = -2 \sin \theta + \theta + \frac{\sin 2\theta}{2} + C$$

$$= -2 \sqrt{1 - \cos^2 \theta} + \theta + \frac{2 \sqrt{1 - \cos^2 \theta} \cdot \cos \theta}{2} + C$$

$$= -2 \sqrt{1 - x} + \cos^{-1} \sqrt{x} + \sqrt{x} \sqrt{1 - x} + C$$

29. .

$$I = \int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx, x \in [0, 1]$$

Soln. Let

$$\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x} = \frac{\pi}{2}$$

We know that

$$\Rightarrow \sin^{-1} \sqrt{x} = \frac{\pi}{2} - \cos^{-1} \sqrt{x}$$

$$\therefore I = \int \frac{\frac{\pi}{2} - 2 \cos^{-1} \sqrt{x}}{\frac{\pi}{2}} dx$$

$$= \int 1 \cdot dx - \frac{4}{\pi} \int 1 \cdot \cos^{-1} \sqrt{x} dx$$

$$= x - \frac{4}{\pi} \left[x \cos^{-1} \sqrt{x} - \int x \cdot \frac{-1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}} dx \right] + C$$

$$= x - \frac{4}{\pi} \left[x \cos^{-1} \sqrt{x} + \frac{1}{2} \int \frac{x}{\sqrt{1-x}} dx \right] + C$$

Put $x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta d\theta$

$$\begin{aligned}\therefore I &= x - \frac{4}{\pi} x \cos^{-1} \sqrt{x} - \frac{2}{\pi} \int \frac{\sin^2 \theta}{\sqrt{1-\sin^2 \theta}} \cdot 2 \sin \theta \cos \theta d\theta + C \\ &= x - \frac{4}{\pi} x \cos^{-1} \sqrt{x} - \frac{2}{\pi} \int \frac{\sin \theta}{\cos \theta} \cdot 2 \sin \theta \cos \theta d\theta + C \\ &= x - \frac{4}{\pi} x \cos^{-1} \sqrt{x} - \frac{2}{\pi} \int (1 - \cos 2\theta) d\theta + C \\ &= x - \frac{4}{\pi} x \cos^{-1} \sqrt{x} - \frac{2}{\pi} \left[\theta - \frac{\sin 2\theta}{2} \right] + C \\ &= x - \frac{4}{\pi} x \cos^{-1} \sqrt{x} - \frac{2}{\pi} [\theta - \sin \theta \cos \theta] + C \\ &= x - \frac{4}{\pi} x \cos^{-1} \sqrt{x} - \frac{2}{\pi} [\sin^{-1} \sqrt{x} - \sqrt{x} \sqrt{1-x}] + C\end{aligned}$$

30.

$$\begin{aligned}\text{Soln. Let } I &= \int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx \\ &= \sin^{-1} x \int \frac{x}{\sqrt{1-x^2}} dx - \int \left[\frac{d}{dx} (\sin^{-1} x) \int \frac{x}{\sqrt{1-x^2}} dx \right] dx \\ &\quad \text{(Applying integration by parts)}\end{aligned}$$

$$\text{Firstly, let us evaluate the integral } \int \frac{x}{\sqrt{1-x^2}} dx$$

$$\text{Put } t = 1 - x^2 \Rightarrow dt = -2x dx$$

$$\begin{aligned}\therefore \int \frac{x}{\sqrt{1-x^2}} dx &= -\frac{1}{2} \int \frac{dt}{\sqrt{t}} = -\sqrt{t} = -\sqrt{1-x^2} \\ \therefore I &= \sin^{-1} x (-\sqrt{1-x^2}) - \int \frac{1}{\sqrt{1-x^2}} (-\sqrt{1-x^2}) dx \\ &= -\sqrt{1-x^2} \sin^{-1} x + \int dx = -\sqrt{1-x^2} \sin^{-1} x + x + C\end{aligned}$$

31.

$$\begin{aligned}\text{Soln. Let } I &= \int \frac{x^2 - 3x + 1}{\sqrt{1-x^2}} dx \\ &= - \int \frac{(-x^2 + 3x - 1)}{\sqrt{1-x^2}} dx \\ &= - \int \frac{(1-x^2) + 3x - 2}{\sqrt{1-x^2}} dx \\ \text{i.e., } I &= - \int \sqrt{1-x^2} dx + \int \frac{-3x + 2}{\sqrt{1-x^2}} dx \\ &= - \int \sqrt{1-x^2} dx \\ &\quad + \frac{3}{2} \int \frac{-2x}{\sqrt{1-x^2}} dx + 2 \int \frac{1}{\sqrt{1-x^2}} dx \\ &= - \int \sqrt{1-x^2} dx + \frac{3 \times 2}{2} \sqrt{1-x^2} + 2 \sin^{-1} x + C \\ &= - \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x \right] + 3\sqrt{1-x^2} + 2 \sin^{-1} x + C \\ &= -\frac{x}{2} \sqrt{1-x^2} + \frac{3}{2} \sin^{-1} x + 3\sqrt{1-x^2} + C\end{aligned}$$

New

SURE SHOT QUESTIONS 2026

Chapter – 08 (Questions)

Application of Integrals

Questions

1. A farmer has a field of shape bounded by $x = y^2$ and $x = 3$, he wants to divide this into his two sons equally by a straight line $x = c$. Can you find c ?
2. Using integration, find the area of the region $\{(x, y): x^2 + y^2 \leq 2ax, y^2 > ax, x, y \geq 0\}$.
3. Using integration, find the area of the region bounded by the curve $x^2 = 4y$ and the line $x = 4y - 2$.
4. Using integration, find the area of the region bounded by the triangle whose vertices are $(-1, 2)$, $(1, 5)$ and $(3, 4)$.
5. Find the area of the smaller region bounded by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the line $\frac{x}{3} + \frac{y}{2} = 1$.
6. Find the area of the region $\{(x, y): x^2 + y^2 \leq 4, x + y \geq 2\}$.
7. Find the area of the region $\{(x, y): x^2 \leq y \leq |x|\}$.
8. Find the area bounded by the curve $y = x|x|$, x -axis and the lines $x = -3$ and $x = 3$.
9. If A_1 denotes the area of region bounded by $y^2 = 4x$, $x = 1$ and x -axis in the first quadrant and A_2 denotes the area of region bounded by $y^2 = 4x$, $x = 4$, find $A_1 \pm A_2$.
10. Find the area of the region bounded by the curve $4x^2 + y^2 = 36$ using integration.
11. Using integration, find the area of the region bounded by lines $x - y + 1 = 0$, $x = -2$, $x = 3$ and x -axis.
12. Using integration, find the area of the region bounded by the curves:
 $y = |x+1| + 1$, $x = -3$, $x = 3$, $y = 0$
13. Using method of integration, find the area of the region enclosed between two circles $x^2 + y^2$ and $(x - 2)^2 + y^2 = 4$.
14. Using integration, find the area of the region in the first quadrant enclosed by the X -axis, the line $y = x$ and the circle $x^2 + y^2 = 32$.
15. Find the area bounded by the circle $x^2 + y^2 = 16$ and the line $\sqrt{3}y = x$ in the first quadrant, using integration.
16. Using integration find the area of the triangular region whose sides have equations $y = 2x + 1$, $y = 3x + 1$ and $x = 4$.

New

SURE SHOT QUESTIONS 2026

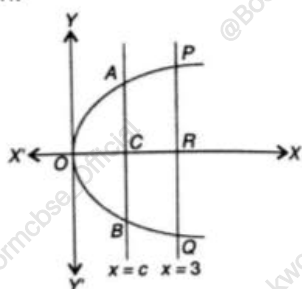
Chapter – 08 (Solutions)

Application of Integrals

Questions

1.

Soln.



Given,

Area OACBO = Area of APRQBCA

$$\text{Or Area OACBO} = 2 \int_0^c y \, dx = 2 \int_0^c \sqrt{x} \, dx$$

$$= 2 \left[\frac{x^{3/2}}{3/2} \right]_0^c = \frac{4}{3} c^{3/2}$$

$$\text{Area of APRQBCA} = 2 \int_c^3 y \, dx = 2 \int_c^3 \sqrt{x} \, dx$$

$$= 2 \left[\frac{x^{3/2}}{3/2} \right]_c^3 = \frac{4}{3} [3^{3/2} - c^{3/2}]$$

$$\therefore \frac{4}{3} c^{3/2} = \frac{4}{3} [3^{3/2} - c^{3/2}]$$

$$\text{or } c^{3/2} = 3^{3/2} - c^{3/2}$$

$$\text{or } 2c^{3/2} = 3^{3/2} \text{ or } 4c^3 = 3^3$$

$$\text{or } c = \sqrt[3]{27/4} = \frac{3}{2} \sqrt{2}$$

2. Soln.

$$\{(x, y) : x^2 + y^2 \leq 2ax, y^2 \geq ax, x, y \geq 0\}$$

Considering the inequations as equations

$$x^2 + y^2 = 2ax \quad \dots\dots\dots(i)$$

$$\text{Or } x^2 - 2ax + y^2 = 0$$

$$\text{Or } x^2 - 2ax + a^2 + y^2 = a^2$$

$$\text{Or } (x-a)^2 + y^2 = a^2 \quad \dots\dots\dots(ii)$$

It represents a circle whose

Centre is (a, 0) and radius r = a

$$y^2 = ax \quad \dots\dots\dots(iii)$$

Vertex (0, 0)

Axis along x-axis.

Point of intersection, from (i) and (iii)

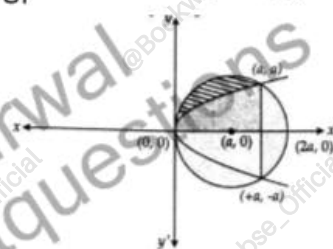
$$x^2 + ax = 2ax$$

$$\text{Or } x^2 - ax = 0$$

$$\text{Or } x(x-a) = 0$$

$$\text{Or } x = 0, x = a$$

$$\text{Or } y = 0, y = \pm a$$



Point of intersection are (0, 0) (a, a), (a, -a)

$$x^2 + y^2 \leq 2ax$$

(Area of shaded part)

$$\text{Required Area} = \int_0^a y \text{ of circle } dx - \int_0^a y \text{ of parabola } dx$$

$$= \int_0^a \sqrt{a^2 - (x-a)^2} \, dx - \int_0^a \sqrt{a} \sqrt{x} \, dx$$

$$A = \left[\frac{x-a}{2} \sqrt{a^2 - (x-a)^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x-a}{a} \right) \right]_0^a - \sqrt{a} \left[\frac{2x^{3/2}}{3} \right]_0^a$$

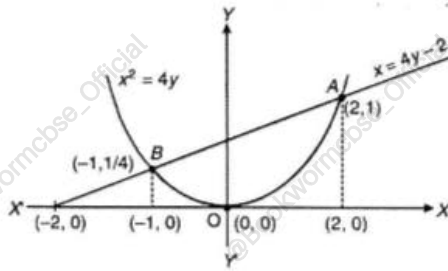
$$A = \left[0 - \left\{ \frac{-a}{2} \sqrt{0} + \frac{a^2}{2} \sin^{-1}(-1) \right\} \right] - \frac{2}{3} \sqrt{a} \left[a^{3/2} - 0 \right]$$

$$= \frac{a^2}{2} \left(\frac{\pi}{2} \right) - \frac{2}{3} \sqrt{a} \times a \sqrt{a}$$

$$= \frac{\pi a^2}{4} - \frac{2a^2}{3} \text{ sq. units}$$

3.

Soln. $\therefore$ Points of intersection are (2, 1) and (-1, 1/4).

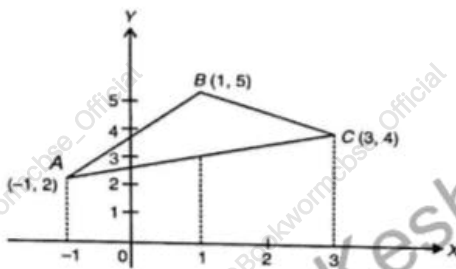


Area of region ($\Delta OABO$)

$$\begin{aligned}
 &= \int_{-1}^2 \frac{x+2}{4} dx - \int_{-1}^2 \frac{x^2}{4} dx \\
 &= \frac{1}{4} \left[\frac{(x+2)^2}{2} - \frac{x^3}{3} \right]_{-1}^2 \\
 &= \frac{1}{4} \left[\frac{16}{2} - \frac{5}{3} \right] = \frac{27}{24} \text{ sq. units} \\
 &= \frac{9}{8} \text{ sq. units}
 \end{aligned}$$

4. .

Soln.



Equation of:

$$y = \frac{1}{2}(3x+7)$$

AB is:

$$y = \frac{1}{2}(11-x)$$

BC is:

$$y = \frac{1}{2}(x+5)$$

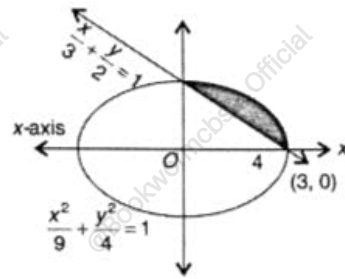
AC is:

Required Area

$$\begin{aligned}
 &= \frac{1}{2} \int_{-1}^1 (3x+7) dx + \frac{1}{2} \int_{1}^3 (11-x) dx - \frac{1}{2} \int_{-1}^3 (x+5) dx \\
 &= \left[\frac{1}{12} (3x+7)^2 \right]_{-1}^1 - \frac{1}{4} [(11-x)^2]_{1}^3 - \frac{1}{4} [(x+5)^2]_{-1}^3 \\
 &= 7 + 9 - 12 = 4 \text{ sq. units}
 \end{aligned}$$

5. .

Soln.

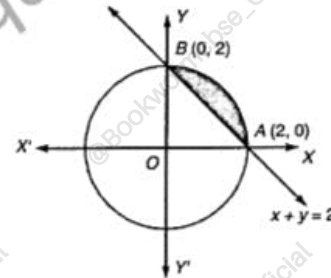


Area of shaded region

$$\begin{aligned}
 &= \int_0^3 \left\{ \frac{2}{3} \sqrt{9-x^2} - \frac{2}{3}(3-x) \right\} dx \\
 &= \frac{2}{3} \left[\frac{x}{2} \sqrt{9-x^2} + \frac{9}{2} \sin^{-1} \frac{x}{3} + \frac{(3-x)^2}{2} \right]_0^3 \\
 &= \frac{2}{3} \left[\left(0 + \frac{9}{2} \cdot \frac{\pi}{2} + 0 \right) - \left(0 + 0 + \frac{9}{2} \right) \right] \\
 &= \frac{2}{3} \left(9 \frac{\pi}{4} - \frac{9}{2} \right) \\
 &= 3 \left(\frac{\pi}{2} - 1 \right) \text{ sq. unit.}
 \end{aligned}$$

6. .

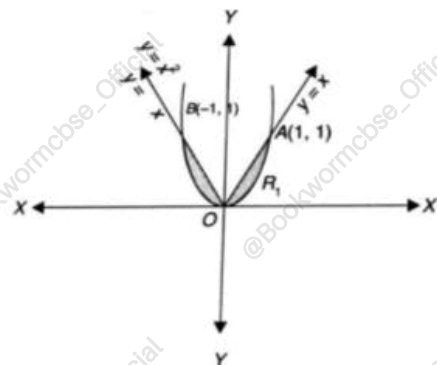
Soln. Finding points of intersection as $x = 0, 2$.



$$\begin{aligned}
 A &= \int_0^2 \sqrt{4-x^2} dx - \int_0^2 (2-x) dx \\
 &= \left[\frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1} \frac{x}{2} \right]_0^2 - \left[2x - \frac{x^2}{2} \right]_0^2 \\
 &= \left(0 + 2 \cdot \frac{\pi}{2} \right) - (4-2) \\
 &= (\pi - 2) \text{ sq. units}
 \end{aligned}$$

7. .

Soln. The required area is bounded between two curves $y = x^2$ and $y = |x|$. Both of these curves are symmetric about y-axis and shaded region in the fig. shows the region whose area is required. Therefore, the required area



$A = 2 \times$ Area of the region R_1

Now, to find the point of intersection of the curves

$y = |x|$ and $y = x^2$, we solve them simultaneously.

Clearly, the region R_1 is in the first quadrant, where $x > 0$

$$\therefore |x| = x \text{ or } y = x \quad \dots\dots\dots(i)$$

$$\text{and } y = x^2 \quad \dots\dots\dots(ii)$$

Solving these two equations, we get

$$x = x^2$$

Or either $x = 0$ or $x = 1$

The limits are, when $x = 0$, $y = 0$ and when $x = 1$, $y = 1$

So, the point of intersection of the curves are $O(0, 0)$ and $A(1, 1)$.

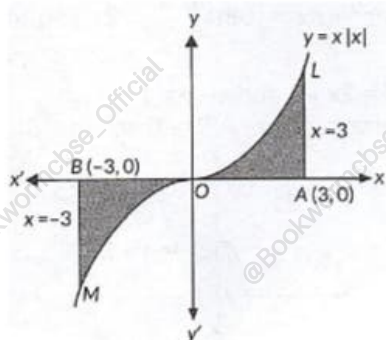
Now, required Area = $2 \times$ Area of line region $R_1 =$

$$\begin{aligned} & 2 \int_0^1 [(y \text{ of the line } y = x) - (y \text{ of the parabola } y = x^2)] dx \\ &= 2 \int_0^1 (x - x^2) dx = 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \\ &= 2 \left[\frac{1}{2} - \frac{1}{3} \right] = \frac{2}{3} \text{ Sq. units} \end{aligned}$$

8. .

Soln. The equation of the curve is

$$y = x|x| = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$$

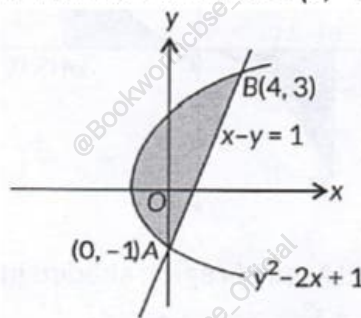


Required area = $2 \times$ (Area of region shaded in first quadrant)

$$= 2 \int_0^3 x^2 dx = 2 \times \left[\frac{x^3}{3} \right]_0^3 = 2 \times 9 = 18 \text{ sq. units}$$

Given $y^2 = 2x + 1$ and $x - y = 1$

Points of intersection are $A(0, -1)$ and $B(4, 3)$.



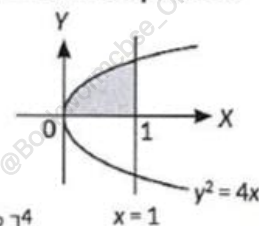
$$\begin{aligned} \therefore \text{ Required area} &= \int_{-1}^3 (1 + y) dy - \int_{-1}^3 \left(\frac{y^2 - 1}{2} \right) dy \\ &= \left[y + \frac{y^2}{2} \right]_{-1}^3 - \left[\frac{1}{2} \left(\frac{y^3}{3} - y \right) \right]_{-1}^3 \\ &= \left[3 + \frac{9}{2} - \left(-1 + \frac{1}{2} \right) \right] - \frac{1}{2} \left[9 - 3 - \left(-\frac{1}{3} + 1 \right) \right] = \frac{16}{3} \text{ sq. units} \end{aligned}$$

9. .

Ans.

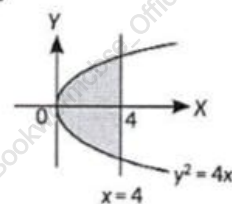
A_1 : Area bounded by $y^2 = 4x$, $x = 1$ in first quadrant

$$\begin{aligned} A_1 &= \int_0^1 \sqrt{4x} dx \\ &= 2 \left[\frac{2x^{3/2}}{3} \right]_0^1 = \frac{4}{3} \text{ sq. units} \end{aligned}$$



A_2 : Area bounded by $y^2 = 4x$, $x = 4$

$$\begin{aligned} \Rightarrow A_2 &= 2 \times \int_0^4 \sqrt{4x} dx = 2 \times 2 \left[\frac{2x^{3/2}}{3} \right]_0^4 \\ &= \frac{8}{3} [4^{3/2} - 0] = \frac{8}{3} \times 8 \\ &= \frac{64}{3} \text{ sq. units} \end{aligned}$$



$$\text{So, } A_1 : A_2 = \frac{4}{3} : \frac{64}{3} = 1 : 16$$

10. .

Ans. Given, equation of curve $4x^2 + y^2 = 36$

$$\Rightarrow \frac{x^2}{9} + \frac{y^2}{36} = 1$$

which is an ellipse with centre (0, 0).

$\therefore$ Area bounded by the ellipse

$$= 4 \int_0^3 |y| dx$$

$$= 4 \int_0^3 2\sqrt{9-x^2} dx = 8 \int_0^3 \sqrt{(3)^2 - x^2} dx$$

$$= 8 \left[\frac{x}{2} \sqrt{9-x^2} + \frac{3^2}{2} \sin^{-1}\left(\frac{x}{3}\right) \right]_0^3$$

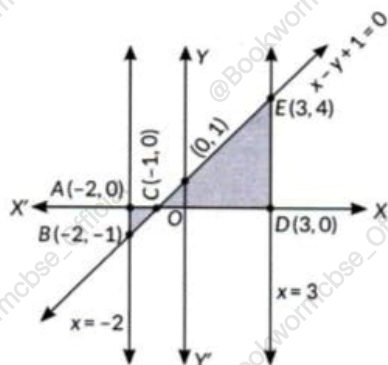
$$\left(\because \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) \right)$$

$$= 8 \left[0 + \frac{9}{2} \sin^{-1}\left(\frac{3}{3}\right) - 0 \right] = 8 \times \frac{9}{2} \sin^{-1}(1)$$

$$= 8 \times \frac{9}{2} \times \frac{\pi}{2} = 18\pi \text{ sq. units}$$

11. .

Soln.



Required area = ar($\triangle ABC$) + ar($\triangle CDE$)

$$= \left| \int_{-2}^{-1} (x+1) dx \right| + \left| \int_{-1}^3 (x+1) dx \right| = \left[\frac{x^2}{2} + x \right]_{-2}^{-1} + \left[\frac{x^2}{2} + x \right]_{-1}^3$$

$$= \left| \frac{1}{2} - 1 - (2 - 1) \right| + \left\{ \frac{9}{2} + 3 - \left(\frac{1}{2} - 1 \right) \right\} = \frac{1}{2} + 8$$

$$= \frac{17}{2} = 8.5 \text{ sq. units}$$

12. .

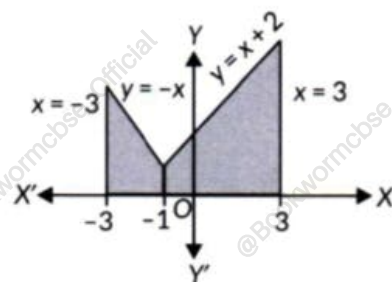
Soln. Here, $y = |x+1| + 1$

$$y = \begin{cases} x+2 & \text{if } x \geq -1 \\ -x & \text{if } x < -1 \end{cases}$$

We know draw the lines: $y = 0$, $x = 3$, $x = -3$ and $y = x + 2$ if $x \geq -1$ (i)

$y = -x$ if $x < -1$ (ii)

Lines (i) and (ii) intersect at $(-1, 1)$



$$\therefore \text{Required area} = \int_{-3}^{-1} (-x) dx + \int_{-1}^3 (x+2) dx$$

$$= - \left[\frac{x^2}{2} \right]_{-3}^{-1} + \left[\frac{x^2}{2} + 2x \right]_{-1}^3$$

$$= - \frac{1}{2} (1 - 9) + \frac{1}{2} (9 - 1) + 2(3 + 1)$$

$$= 4 + 4 + 8 = 16 \text{ sq. units.}$$

13. .

Soln. Equations of the given circles are,

$$x^2 + y^2 = 4 \quad \text{.....(i)}$$

$$(x-2)^2 + y^2 = 4 \quad \text{.....(ii)}$$

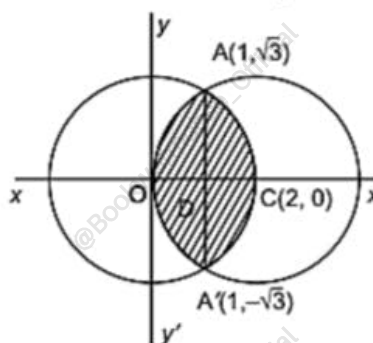
Equation (i) is a circle with centre O at the origin and radius 2. Equation (ii) is a circle with centre C(2, 0) and radius 2.

Solving equation (i) and (ii) we have

$$(x-2)^2 + y^2 = x^2 + y^2$$

$$\text{or } x^2 - 4x + 4 + y^2 = x^2 + y^2$$

Or $x = 1$ which gives $y = \pm\sqrt{3}$



Thus, the points of intersection of the given circles are $A(1, \sqrt{3})$ and $A'(1, -\sqrt{3})$

Required area of the enclosed region OACA'O between circles

$$= 2 [\text{area of region ODCAO}]$$

$$= 2 [\text{area of region ODAO} + \text{area of region OCAD}]$$

$$= 2 \left[\int_0^1 y dx + \int_1^2 y dx \right]$$

$$\begin{aligned}
 &= 2 \left[\int_0^1 \sqrt{4 - (x-2)^2} dx + \int_1^2 \sqrt{4 - x^2} dx \right] \text{ [from (i)]} \\
 &= 2 \left[\frac{1}{2} (x-2) \sqrt{4 - (x-2)^2} + \frac{1}{2} \times 4 \sin^{-1} \left(\frac{x-2}{2} \right) \right]_0^1 \\
 &\quad + 2 \left[\frac{1}{2} x \sqrt{4 - x^2} + \frac{1}{2} \times 4 \sin^{-1} \frac{x}{2} \right]_1^2 \\
 &= \left[(x-2) \sqrt{4 - (x-2)^2} + 4 \sin^{-1} \left(\frac{x-2}{2} \right) \right]_0^1 \\
 &\quad + \left[x \sqrt{4 - x^2} + 4 \sin^{-1} \frac{x}{2} \right]_1^2 \\
 &= \left[-\sqrt{3} + 4 \sin^{-1} \left(\frac{-1}{2} \right) - 4 \sin^{-1} (-1) \right] \\
 &\quad + \left[4 \sin^{-1} 1 - \sqrt{3} - 4 \sin^{-1} \frac{1}{2} \right] \\
 &= \left[\left(-\sqrt{3} - 4 \times \frac{\pi}{6} \right) + 4 \times \frac{\pi}{2} \right] + \left[4 \times \frac{\pi}{2} - \sqrt{3} - 4 \times \frac{\pi}{6} \right] \\
 &= \frac{8\pi}{3} - 2\sqrt{3}
 \end{aligned}$$

14. .

Soln. Given curves is

$$x^2 + y^2 = 32 \quad \dots\dots\dots (i)$$

$$x^2 + y^2 = (\sqrt{32})^2 = (4\sqrt{2})^2$$

It is a circle with centre (0, 0) and radius $4\sqrt{2}$.

Given line is $y = x$ (ii)

Solving equations (i) and (ii) for points of intersections,

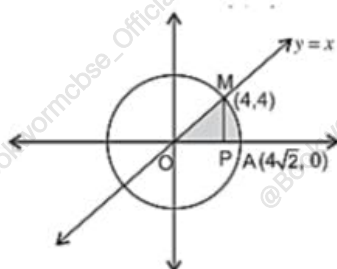
$$x^2 + x^2 = 32 \quad \text{[using (ii) in (i)]}$$

$$\Rightarrow x^2 = 16$$

$$\Rightarrow x = \pm 4$$

$$\Rightarrow x = 4 \quad \text{(first quadrant)}$$

$\therefore$ Point of intersection is (4, 4)



Required area = Area OMA

= Area OMP + Area MPA

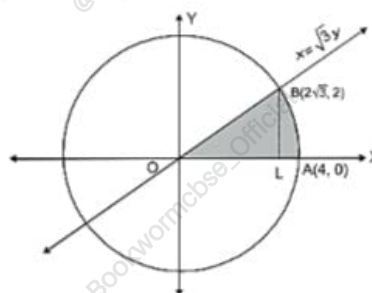
$$= \int_0^4 x dx + \int_4^{4\sqrt{2}} \sqrt{(4\sqrt{2})^2 - x^2} dx$$

$$\begin{aligned}
 &= \left[\frac{x^2}{2} \right]_0^4 + \left[\frac{x}{2} \sqrt{(4\sqrt{2})^2 - x^2} + \frac{(4\sqrt{2})^2}{2} \sin^{-1} \left(\frac{x}{4\sqrt{2}} \right) \right]_4^{4\sqrt{2}} \\
 &= \frac{16}{2} + \left[\left\{ \frac{4\sqrt{2}}{2} \sqrt{(4\sqrt{2})^2 - (4\sqrt{2})^2} + \frac{32}{2} \sin^{-1} 1 \right\} - \left\{ \frac{4}{2} \sqrt{(4\sqrt{2})^2 - (4)^2} + \frac{32}{2} \sin^{-1} \frac{1}{\sqrt{2}} \right\} \right] \\
 &= 8 + \left(2\sqrt{2}(0) + 16 \times \frac{\pi}{2} \right) - \left(2 \times 4 + 16 \times \frac{\pi}{4} \right) \\
 &= 8 + 8\pi - 8 - 4\pi \\
 &= 4\pi \text{ sq. units Ans.}
 \end{aligned}$$

15. .

Soln. Given, $x = \sqrt{3}y$

$$\text{And } x^2 + y^2 = 16,$$



$$\Rightarrow (\sqrt{3}y)^2 + y^2 = 16$$

$$\Rightarrow 4y^2 = 16$$

$$\Rightarrow y^2 = 4$$

$$\Rightarrow y = 2$$

$$\therefore x = \sqrt{3}y = 2\sqrt{3}$$

$\therefore B(2\sqrt{3}, 2)$ is the point of intersection in first quadrant.

Required area = Area under OBL + Area under LBA

$$\begin{aligned}
 &= \int_0^{2\sqrt{3}} \frac{x}{\sqrt{3}} dx + \int_{2\sqrt{3}}^4 \sqrt{16 - x^2} dx \\
 &= \frac{1}{\sqrt{3}} \left(\frac{x^2}{2} \right)_0^{2\sqrt{3}} + \left[\frac{x}{2} \sqrt{16 - x^2} + \frac{16}{2} \sin^{-1} \frac{x}{4} \right]_{2\sqrt{3}}^4 \\
 &= \frac{1}{2\sqrt{3}} (12 - 0) + \left(0 + 8 \sin^{-1} 1 \right) - \left(\frac{2\sqrt{3}}{2} \sqrt{16 - 12} + 8 \sin^{-1} \frac{2\sqrt{3}}{4} \right) \\
 &= \frac{6}{\sqrt{3}} + 8 \times \frac{\pi}{2} - 2\sqrt{3} - 8 \sin^{-1} \frac{\sqrt{3}}{2} \\
 &= \frac{6\sqrt{3}}{3} + 4\pi - 2\sqrt{3} - 8 \times \frac{\pi}{3} \\
 &= 2\sqrt{3} + \frac{4\pi}{3} - 2\sqrt{3} \\
 &= \frac{4\pi}{3} \text{ sq. units.} \quad \text{Ans.}
 \end{aligned}$$

16. .

Soln. Given equations are

$$y = 2x + 1 \quad \dots\dots(i)$$

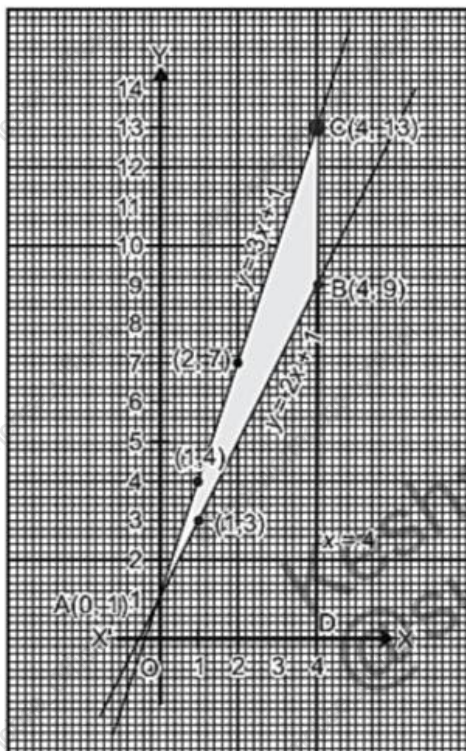
$$y = 3x + 1 \quad \dots\dots(ii)$$

Table for line (i)

X	0	1	2	4
Y	1	3	5	9

Table for line (ii),

X	0	1	2	4
Y	1	4	7	13



Area of triangular region ABC

= Area of the region OACDO – Area of the region OABDO

$$= \int_0^4 [y \text{ line(ii)} - y \text{ line(i)}] dx$$

$$= \int_0^4 [(3x+1) - (2x+1)] dx$$

$$= \int_0^4 (3x+1-2x-1) = \int_0^4 x dx$$

$$= \left[\frac{x^2}{2} \right]_0^4 = \left[\frac{(4)^2}{2} \right] = 8 \text{ sq. units}$$

New

SURE SHOT QUESTIONS 2026

Chapter – 09 (Questions)

Differential Equations

Questions

1. Solve the following differential equation :
 $(1 + x^2)dy + 2xy dx = \cot x dx; x \neq 0.$
2. Solve the differential equation : $(x + 1) \frac{dy}{dx} = 2e^{-1} - 1; y(0) = 0.$
3. Write the order & degree of the following differential equations
(i) $\frac{d^3y}{dx^3} + 2\left(\frac{d^2y}{dx^2}\right) - \frac{dy}{dx} + y = 0$
(ii) $\left(\frac{d^2y}{dx^2}\right)^2 + x^2\left(\frac{dy}{dx}\right)^3 = 0$
4. Find order & degree of differential equation:
 $\frac{d^4y}{dx^4} + \sin\left(\frac{d^3y}{dx^3}\right) = 0$
5. Solve the differential equation. $y \log y dx - x dy = 0.$
6. Find the general solution of the following differential equation: $e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$
7. Find the particular solution of the following differential equation :
 $\frac{dy}{dx} = 1 + x^2 + y^2 + x^2y^2$, given that $y=1$ when $x=0$
8. Solve : $(1 + e^{\frac{x}{y}})dx + e^{\frac{x}{y}}(1 - \frac{x}{y})dy$
9. Find the particular solution of the differential equation given by $2xy + y^2 - 2x^2 \frac{dy}{dx} = 0; y = 2$, when $x = 1.$
10. Find the general solution of the differential equation: $ydx = (x + 2y^2)dy$
11. Find the product of the order and the degree of the differential equation $\left[\frac{d}{dx}(xy^2)\right] \cdot \frac{dy}{dx} + y = 0.$
12. Find the integrating factor of
 $x \frac{dy}{dx} + (1 + x \cot x) y = x.$
13. Find the particular solution of the differential equation $(1 + x^2) \frac{dy}{dx} + 2xy = \tan x$, given $y(0) = 1.$
14. Solve the following differential equation:
 $x \cos\left(\frac{y}{x}\right) \frac{dy}{dx} = y \cos\left(\frac{y}{x}\right) + x; x \neq 0.$
15. Solve the following differential equation : $(1 + x)^2 \frac{dy}{dx} + y = \tan^{-1}x.$
16. Solve the differential equation: $(\tan^{-1} y - x)dy = (1 + y^2)dx$
17. Solve the differential equation: $(x^2 - 1) \frac{dy}{dx} + 2xy = \frac{2}{x^2 - 1}$, where
 $x \in (-\infty, -1) \cup (1, \infty).$

18. Solve the differential equation: $(1 + x^2) \frac{dy}{dx} + y = \tan^{-1} x$

19. Solve the differential equation: $x \frac{dy}{dx} \sin\left(\frac{y}{x}\right) + x - y \sin\left(\frac{y}{x}\right) = 0$ is homogeneous. Find the particular solution of this differential equation, given that $x = 1$, when $y = \pi/2$.

20. Solve the differential equation: $\cos^2 x \frac{dy}{dx} + y = \tan x$

21. Solve the differential equation: $\sqrt{1 + x^2 + y^2 + x^2 y^2} + xy \frac{dy}{dx} = 0$

22. Show that the following differential equation is homogeneous and then solve it. $y dx + x \log\left(\frac{y}{x}\right) dy - 2x dy = 0$

23. Show that the differential equation $(x - y) \frac{dy}{dx} = x + 2y$, is homogeneous and solve it.

24. Find the particular solution of the differential equation $(1 + x^3) \frac{dy}{dx} + 6x^2 y = (1 + x^2)$, given that $y = 1$ when $x = 1$.

25. Show that the differential equation $2ye^{x/y} dx + (y - 2xe^{x/y}) dy = 0$ is homogeneous. Find the particular solution of this differential equation, given that $x = 0$ when $y = 1$.

26. Find the particular solution of the differential equation $\frac{dy}{dx} = \frac{x(2\log x + 1)}{\sin y + y \cos y}$ given that $y = \pi/2$ when $x = 1$.

27. Solve the following differential equation: $x \cos\left(\frac{y}{x}\right) \frac{dy}{dx} = y \cos\left(\frac{y}{x}\right) + x; x \neq 0$.

28. Find the general solution of the following differential equation:

$$(1 + y^2) + (x - e^{\tan^{-1} y}) \frac{dy}{dx} = 0$$

29. Solve the differential equation: $(\tan^{-1} y - x) dy = (1 + y^2) dx$.

30. Find the particular solution of the differential equation $x \frac{dy}{dx} + y - x + xy \cot x = 0; x \neq 0$, given that when $x = \frac{\pi}{2}, y = 0$.

New

SURE SHOT QUESTIONS 2026

Chapter – 09 (Solutions)

Differential Equations

Questions

1. .

Soln. $(1+x^2)dy + 2xydx = \cot x dx$

Dividing with $(1+x^2)dx$ on both sides, we get

$$\frac{dy}{dx} + \left(\frac{2x}{1+x^2} \right) y = \frac{\cot x}{1+x^2}$$

On comparing this equation with

$$\frac{dy}{dx} + Py = Q,$$

Where $P = \frac{2x}{1+x^2}, Q = \frac{\cot x}{1+x^2}$

Now, $I.F. = e^{\int P dx} = e^{\int \frac{2x}{1+x^2} dx}$
 $= e^{\log(1+x^2)} = 1+x^2$

Solution is

$$y(I.F.) = \int Q \cdot (I.F.) dx + C$$

$$y(1+x^2) = \int \frac{\cot x}{(1+x^2) \cdot (1+x^2)} dx$$

$$\Rightarrow y(1+x^2) = \int \cot x dx + C$$

$$\Rightarrow y(1+x^2) = \log |\sin x| + C$$

2. .

Soln. Given, $(x+1)\frac{dy}{dx} = 2e^{-y} - 1$

$$\Rightarrow \frac{dy}{2e^{-y} - 1} = \frac{dx}{x+1}$$

$$\Rightarrow \int \frac{e^y dy}{2 - e^y} = \int \frac{dx}{x+1}$$

Put $2 - e^y = t$

$$-e^y dy = dt$$

$$e^y dy = -dt$$

$$\therefore \int -\frac{dt}{t} = \int \frac{dx}{x+1}$$

$$\Rightarrow -\log t = \log |x+1| + \log C$$

$$\Rightarrow -\log t^{-1} = \log [C(x+1)]$$

$$\Rightarrow \frac{1}{t} = C(x+1)$$

$$\Rightarrow \frac{1}{2 - e^y} = C(x+1) \dots \dots \dots (i)$$

Put $x = 0$ and $y = 0$ in (i), we get

$$\Rightarrow \frac{1}{2} = C$$

$$\therefore \frac{1}{2 - e^y} = \frac{1}{2}(x+1) \text{ Ans.}$$

3. .

Soln.

(i) Order = 3, degree = 1

(ii) Order = 2, degree = 2

4. .

Sol. Order is 4 but degree is not defined

5. .

Soln. Given $y \log y dx - x dy = 0$

$$\Rightarrow y \log y dx = x dy$$

$$\Rightarrow \frac{dx}{x} = \frac{dy}{y \log y}$$

$$\Rightarrow \int \frac{dy}{y \log y} = \int \frac{dx}{x}$$

Let $\log y = t \Rightarrow \frac{1}{y} dy = dt$

$$\Rightarrow \int \frac{dt}{t} = \log x + C_1$$

$$\Rightarrow \log t = \log x + \log C \text{ [Where } C_1 = \log C]$$

$$\Rightarrow t = Cx$$

$$\Rightarrow \log y = Cx$$

$$\Rightarrow y = e^{Cx}$$

6. .

Soln. Given differential equation,

$$e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0$$

$$\Rightarrow e^x \tan y dx = -(1 - e^x) \sec^2 y dy$$

$$\therefore dy = \frac{e^x}{e^x - 1} \cdot \frac{\tan y}{\sec^2 y} dx \Rightarrow \frac{\sec^2 y dy}{\tan y} = \frac{e^x}{e^x - 1} dx$$

Integrating both sides, we get

$$\int \frac{\sec^2 y dy}{\tan y} = \int \frac{e^x}{e^x - 1} dx$$

$$\Rightarrow \log |\tan y| = \log |e^x - 1| + \log C$$

$$\therefore \log |\tan y| = \log |(e^x - 1)C|$$

$$\therefore \tan y = (e^x - 1)C$$

7.

Soln. Given $\frac{dy}{dx} = 1 + x^2 + y^2 + x^2y^2$

$$\Rightarrow \frac{dy}{dx} = (1 + x^2) + y^2(1 + x^2)$$

$$\Rightarrow \frac{dy}{dx} = (1 + x^2)(1 + y^2)$$

$$\Rightarrow (1 + x^2) dx = \frac{dy}{(1 + y^2)}$$

Integrating both sides, we get

$$\int (1 + x^2) dx = \int \frac{dy}{(1 + y^2)}$$

$$\Rightarrow \int dx + \int x^2 dx = \int \frac{dy}{(1 + y^2)}$$

$$\Rightarrow x + \frac{x^3}{3} + C = \tan^{-1} y$$

Putting $y = 1$ and $x = 0$, we get

$$\Rightarrow \tan^{-1}(1) = 0 + 0 + C$$

$$\Rightarrow C = \tan^{-1}(1) = \frac{\pi}{4}$$

Therefore, required particular solution is

$$\tan^{-1} y = x + \frac{x^3}{3} + \frac{\pi}{4}$$

8.

Soln. We have, $(1 + e^{x/y}) dx + e^{x/y} \left(1 - \frac{x}{y}\right) dy = 0$

$$\Rightarrow (1 + e^{x/y}) dx = -e^{x/y} \left(1 - \frac{x}{y}\right) dy$$

$$\therefore \frac{dx}{dy} = \frac{-e^{x/y} \left(1 - \frac{x}{y}\right)}{1 + e^{x/y}} = g\left(\frac{x}{y}\right) \dots\dots(i)$$

Here, RHS of differential equation is of the form

$g\left(\frac{x}{y}\right)$, so it is a homogeneous function of degree zero.

Now, we put $x = vy$ and, $\frac{dx}{dy} = v + y \frac{dv}{dy}$

From (i), we get $v + y \frac{dv}{dy} = \frac{-e^v(1-v)}{1+e^v}$

$$y \frac{dv}{dy} = \frac{-e^v(1-v)}{1+e^v} - v = \frac{-(v+e^v)}{1+e^v}$$

$$\Rightarrow \frac{1+e^v}{-(v+e^v)} dv = \frac{dy}{y}$$

$$-\log |v+e^v| + \log C = \log |y| \Rightarrow \log C = \log |y| + \log |v+e^v|$$

$$\log C = \log |y(v+e^v)| \Rightarrow \log \left| y \left(\frac{x}{y} + e^{x/y} \right) \right|$$

$$\Rightarrow C = y \left(\frac{x}{y} + e^{x/y} \right) \text{ or } C = x + ye^{x/y}$$

Hence, $x + ye^{x/y} = C$ is the required solution.

9.

Ans. The given differential equation is

$$2xy + y^2 - 2x^2 \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{2xy + y^2}{2x^2}$$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx} \therefore v + x \frac{dv}{dx} = \frac{2x \cdot vx + v^2 x^2}{2x^2}$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{2v + v^2}{2} \Rightarrow x \frac{dv}{dx} = \frac{2v + v^2}{2} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2}{2} \Rightarrow \int \frac{dv}{v^2} = \frac{1}{2} \int \frac{dx}{x}$$

$$\Rightarrow -\frac{1}{v} = \frac{1}{2} \log |x| + C \Rightarrow -\frac{x}{y} = \frac{1}{2} \log |x| + C$$

Put $x = 1$ and $y = 2$ in above equation, we get

$$-\frac{1}{2} = \frac{1}{2} \log |1| + C \Rightarrow C = -\frac{1}{2}$$

$\therefore$ Particular solution of given differential equation is

$$-\frac{x}{y} = \frac{1}{2} \log |x| - \frac{1}{2}$$

$$\Rightarrow y = \frac{-2x}{\log |x| - 1} \Rightarrow y = \frac{2x}{1 - \log |x|}$$

10.

Ans. The given differential equation is $y dx = (x + 2y^2) dy$

$$\Rightarrow y dx = x dy + 2y^2 dy \Rightarrow y dx - x dy = 2y^2 dy$$

$$\Rightarrow \frac{y dx - x dy}{y^2} = 2 dy = d\left(\frac{x}{y}\right) = 2 dy$$

Integrating both sides, we get

$$\int d\left(\frac{x}{y}\right) = 2 \int dy \Rightarrow \frac{x}{y} = 2y + C, \text{ which is the required general solution.}$$

11.

Soln.

The given differential equation is

$$\left[\frac{d}{dx}(xy^2) \right] \cdot \frac{dy}{dx} + y = 0$$

$$\Rightarrow \left[x \cdot 2y \frac{dy}{dx} + y^2 \cdot 1 \right] \frac{dy}{dx} + y = 0$$

$$\Rightarrow 2x \left(\frac{dy}{dx} \right)^2 + y^2 \left(\frac{dy}{dx} \right) + y = 0$$

∴ Its order is 1 and degree is 2.

∴ Required product = $1 \times 2 = 2$.

12. .

$$\text{Soln. We have, } x \frac{dy}{dx} + (1 + x \cot x) y = x$$

$$\Rightarrow \frac{dy}{dx} + \frac{(1 + x \cot x)}{x} y = 1$$

Clearly it is a linear differential equation of the form,

$$\frac{dy}{dx} + Py = Q, \text{ where } P = \frac{1 + x \cot x}{x} \text{ and } Q = 1$$

$$\therefore \text{I.F.} = e^{\int \left(\frac{1}{x} + \cot x \right) dx}$$

$$= e^{\log x + \log \sin x} = e^{\log(x \sin x)} = x \sin x.$$

13. .

$$\text{Soln. We have, } (1 + x^2) \frac{dy}{dx} + 2xy = \tan x$$

$$\Rightarrow \frac{dy}{dx} + \frac{2xy}{1+x^2} = \frac{\tan x}{1+x^2} \quad \dots\dots\dots(i)$$

Clearly, it is a linear differential equation of the form

$$\frac{dy}{dx} + Py = Q, \text{ Where } P = \frac{2x}{1+x^2} \text{ and } Q = \frac{\tan x}{1+x^2}$$

$$\therefore \text{I.F.} = \int \frac{2x}{1+x^2} dx = e^{\log(1+x^2)} = 1+x^2$$

∴ Solution of (i) is

$$y(1+x^2) = \int (1+x^2) \frac{\tan x}{(1+x^2)} dx + C$$

$$\Rightarrow y(1+x^2) = \int \tan x dx + C$$

$$\Rightarrow y(1+x^2) = \log |\sec x| + C$$

Also given, $y(0) = 1$

$$\therefore 1 = \log \sec 0 + C \Rightarrow C = 1$$

Particular solution is

$$y(1+x^2) = \log |\sec x| + 1$$

14. .

Soln. We have,

$$x \cos \left(\frac{y}{x} \right) \frac{dy}{dx} = \left(\frac{y}{x} \right) \cos \left(\frac{y}{x} \right) + x, \quad x \neq 0$$

$$\Rightarrow \cos \left(\frac{y}{x} \right) \frac{dy}{dx} = \left(\frac{y}{x} \right) \cos \left(\frac{y}{x} \right) + 1$$

.....(i)

This is a linear homogeneous differential equation

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Now (i) becomes

$$\cos v \cdot \left[v + x \frac{dv}{dx} \right] = v \cos v + 1$$

$$\Rightarrow x \cos v \frac{dv}{dx} = 1 \Rightarrow \cos v dv = \frac{dx}{x}$$

Integrating both sides, we get

$$\sin v = \log x + C \Rightarrow \sin \left(\frac{y}{x} \right) = \log x + C$$

Is the required solution.

15. .

Soln. Given differential equation is,

$$(1+x^2) \frac{dy}{dx} + y = \tan^{-1} x \quad \dots\dots(i)$$

Divide by $(1+x^2)$ in above equation, we get

$$\frac{dy}{dx} + \frac{1}{1+x^2} y = \frac{\tan^{-1} x}{1+x^2}$$

The above equation is of the form $\frac{dy}{dx} + Py = Q$

$$P = \frac{1}{1+x^2} \text{ and } Q = \frac{\tan^{-1} x}{1+x^2}$$

Where,

$$\therefore \text{I.F.} = e^{\int P dx}$$

$$= e^{\int \frac{1}{1+x^2} dx}$$

$$= e^{\tan^{-1} x}$$

Solution of the equation is given by,

$$y \cdot \text{I.F.} = \int Q \cdot \text{I.F.} dx + c$$

$$y e^{\tan^{-1} x} = \int \frac{e^{\tan^{-1} x} \tan^{-1} x}{1+x^2} dx + c$$

.....(ii)

$$\text{Put } \tan^{-1} x = t$$

Differentiate both sides w.r.t. x

$$\frac{1}{1+x^2} = \frac{dt}{dx}$$

$$\Rightarrow \frac{dx}{1+x^2} = dt$$

∴ Equation (ii) becomes

$$ye^{\tan^{-1}x} = \int e^t \cdot t dt + c$$

$$ye^{\tan^{-1}x} = te^t - \int 1 \cdot e^t \cdot dt + c$$

$$= te^t - e^t + c$$

$$= \tan^{-1}x e^{\tan^{-1}x} - e^{\tan^{-1}x} + c$$

$$\Rightarrow ye^{\tan^{-1}x} = e^{\tan^{-1}x} (\tan^{-1}x - 1) + c$$

$$\Rightarrow y = \tan^{-1}x - 1 + ce^{-\tan^{-1}x}$$

16. .

Soln. The given differential equation can be written as

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1}y}{1+y^2} \quad \dots\dots\dots(i)$$

Now, (i) is a linear differential equation of the form

$$\frac{dy}{dx} + PX = Q,$$

$$P = \frac{1}{1+y^2} \text{ and } Q = \frac{\tan^{-1}y}{1+y^2}$$

Where,

Therefore,

Thus, the solution of the given differential equation is

$$xe^{\tan^{-1}y} = \int \left(\frac{\tan^{-1}y}{1+y^2} \right) e^{\tan^{-1}y} dy + C \quad \dots\dots\dots(ii)$$

$$I = \int \left(\frac{\tan^{-1}y}{1+y^2} \right) e^{\tan^{-1}y} dy$$

Let

Substituting

$$\tan^{-1}y = t \text{ so that } \left(\frac{1}{1+y^2} \right) dy = dt, \text{ we get}$$

$$I = \int t e^t dt = t e^t - \int 1 \cdot e^t dt = t e^t - e^t = e^t (t-1)$$

$$\text{Or } I = e^{\tan^{-1}y} (\tan^{-1}y - 1)$$

Substituting the value of I in equation (ii), we get

$$xe^{\tan^{-1}y} = e^{\tan^{-1}y} (\tan^{-1}y - 1) + C$$

Or

$$x = (\tan^{-1}y - 1) + C e^{-\tan^{-1}y}$$

Which is the general solution of the given differential equation.

17. .

Soln. The given differential equation is

$$(x^2-1) \frac{dy}{dx} + 2xy = \frac{2}{x^2-1}$$

$$\Rightarrow \frac{dy}{dx} + \frac{2x}{x^2-1} y = \frac{2}{(x^2-1)^2} \quad \dots\dots\dots(i)$$

This is a linear differential equation of the form

$$\frac{dy}{dx} + Py = Q, \text{ where } P = \frac{2x}{x^2-1} \text{ and } Q = \frac{2}{(x^2-1)^2}$$

$$\therefore IF = e^{\int P dx} = e^{\int \frac{2x}{x^2-1} dx} = e^{\log(x^2-1)} = (x^2-1)$$

Multiplying both sides of (i) by $IF = (x^2-1)$, we get

$$(x^2-1) \frac{dy}{dx} + 2xy = \frac{2}{x^2-1}$$

Integrating both sides, we get

$$y(x^2-1) = \int \frac{2}{x^2-1} dx +$$

$$C \left[U \sin g : y(IF) = \int Q \cdot (IF) dx + C \right]$$

$$\Rightarrow y(x^2-1) = \frac{2}{2} \log \left| \frac{x-1}{x+1} \right| + C$$

$$\Rightarrow y(x^2-1) = \log \left| \frac{x-1}{x+1} \right| + C$$

This is the required solution.

18. .

$$\text{Soln. We have, } (1+x^2) \frac{dy}{dx} + y = \tan^{-1}x$$

Dividing each term by $(1+x^2)$, we get

$$\frac{dy}{dx} + \frac{1}{1+x^2} \cdot y = \frac{\tan^{-1}x}{1+x^2}$$

Clearly, it is linear equation of the form

$$\frac{dy}{dx} + P \cdot y = Q, \text{ where } P = \frac{1}{1+x^2} \text{ and } Q = \frac{\tan^{-1}x}{1+x^2}$$

∴ Integrating factor,

$$IF = e^{\int P dx} = e^{\int \frac{1}{1+x^2} dx} = e^{\tan^{-1}x}$$

Therefore, solution of given differential equation is

$$y \times I.F. = \int Q \times I.F. dx$$

$$\Rightarrow y \cdot e^{\tan^{-1}x} = \int \frac{\tan^{-1}x}{1+x^2} \cdot e^{\tan^{-1}x} dx$$

$$\text{Put } I = \int \frac{\tan^{-1}x e^{\tan^{-1}x}}{1+x^2} dx$$

$$\text{Put } e^{\tan^{-1}x} = t \Rightarrow \frac{e^{\tan^{-1}x}}{1+x^2} dx = dt$$

Also

$$\tan^{-1}x = \log t$$

$$\Rightarrow I = \int \log t dt = t \log t - t + C$$

[Integrating by parts]

$$\Rightarrow I = e^{\tan^{-1} x} \cdot \tan^{-1} x - e^{\tan^{-1} x} + C$$

Hence, required solution is

$$y e^{\tan^{-1} x} = e^{\tan^{-1} x} (\tan^{-1} x - 1) + C$$

$$\Rightarrow y = (\tan^{-1} x - 1) + C e^{-\tan^{-1} x}$$

19.

Soln. Given differential equation is

$$x \frac{dy}{dx} \sin \frac{y}{x} + x - y \sin \frac{y}{x} = 0$$

Dividing both sides by $x \sin \frac{y}{x}$, we get

$$\frac{dy}{dx} + \operatorname{cosec} \frac{y}{x} - \frac{y}{x} = 0 \Rightarrow \frac{dy}{dx} = \frac{y}{x} - \operatorname{cosec} \frac{y}{x}$$

$$\text{Let } F(x, y) = \frac{y}{x} - \operatorname{cosec} \frac{y}{x}$$

$$\therefore F(\lambda x, \lambda y) = \frac{\lambda y}{\lambda x} - \operatorname{cosec} \frac{\lambda y}{\lambda x} = \lambda^0 \left[\frac{y}{x} - \operatorname{cosec} \frac{y}{x} \right] = \lambda^0 F(x, y)$$

Hence, differential equation (i) is homogenous.

$$\text{Let } y = vx \Rightarrow \frac{y}{x} = v \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Now, equation (i) becomes

$$v + x \frac{dv}{dx} = \frac{vx}{x} - \operatorname{cosec} \frac{vx}{x}$$

$$v + x \frac{dv}{dx} = v - \operatorname{cosec} v \Rightarrow x \frac{dv}{dx} = -\operatorname{cosec} v$$

$$\Rightarrow -\sin v dv = \frac{dx}{x} \Rightarrow -\int \sin v dv = \int \frac{dx}{x}$$

$$\Rightarrow \cos v = \log |x| + C \Rightarrow \cos \frac{y}{x} = \log |x| + C$$

Putting $y = \frac{\pi}{2}$, $x = 1$ in (ii), we get

$$\therefore \cos \frac{\pi}{2} = \log 1 + C \Rightarrow 0 = 0 + C \Rightarrow C = 0$$

Hence, particular solution is

$$\cos \frac{y}{x} = \log |x| + 0 \quad \text{i.e.,} \quad \cos \frac{y}{x} = \log |x|$$

20.

Soln. Given differential equation is,

$$\cos^2 x \frac{dy}{dx} + y = \tan x \Rightarrow \frac{dy}{dx} + y \sec^2 x = \tan x \sec^2 x$$

Given differential equation is a linear differential

equation of the type $\frac{dy}{dx} + Py = Q$ where

$$P = \sec^2 x, Q = \tan x \sec^2 x$$

$$IF = e^{\int P dx} = e^{\int \sec^2 x dx} = e^{\tan x}$$

$\therefore$ Solution is given by

$$e^{\tan x} y = \int \tan x \sec^2 x e^{\tan x} dx$$

$$I = \int \tan x \sec^2 x e^{\tan x} dx \quad \text{Put}$$

$$\tan x = t, \sec^2 x dx = dt, \text{ we get}$$

$$I = \int t e^t dt$$

$$\therefore = t e^t - \int e^t dt = t e^t - e^t + C \quad [\text{Integrating}$$

by parts]

$$= \tan x e^{\tan x} - e^{\tan x} + C \quad \text{Hence,}$$

$$e^{\tan x} y = e^{\tan x} (\tan x - 1) + C \Rightarrow y = \tan x - 1 + C e^{-\tan x}$$

21.

$$\text{Soln. Given } \sqrt{1+x^2+y^2+x^2y^2} + xy \frac{dy}{dx} = 0$$

By simplifying the equation, we get

$$xy \frac{dy}{dx} = -\sqrt{1+x^2+y^2+x^2y^2}$$

$$\Rightarrow xy \frac{dy}{dx} = -\sqrt{(1+x^2)(1+y^2)} = -\sqrt{1+x^2} \sqrt{1+y^2}$$

$$\Rightarrow \frac{y}{\sqrt{1+y^2}} dy = -\frac{\sqrt{1+x^2}}{x} dx$$

Integrating both sides, we get

$$\int \frac{y}{\sqrt{1+y^2}} dy = -\int \frac{\sqrt{1+x^2}}{x} dx \quad \dots\dots\dots(i)$$

$$\text{Let } 1+y^2 = t \Rightarrow 2y dy = dt, \text{ (For LHS) And}$$

$$1+x^2 = m^2 \Rightarrow 2x dx = 2m dm \Rightarrow x dx = m dm \text{ (For RHS)}$$

$$\therefore (i) \Rightarrow \frac{1}{2} \int \frac{1}{\sqrt{t}} dt = -\int \frac{m}{m^2-1} m dm$$

$$\Rightarrow \frac{1}{2} t^{1/2} + \int \frac{m^2}{m^2-1} dm = 0$$

$$\Rightarrow \sqrt{t} + \int \frac{m^2+1-1}{m^2-1} dm = 0$$

$$\Rightarrow \sqrt{t} + \int \left(1 + \frac{1}{m^2-1}\right) dm = 0$$

$$\Rightarrow \sqrt{t} + m + \frac{1}{2} \log \left| \frac{m-1}{m+1} \right| = 0$$

Now, substituting these value of t and m, we get

$$\sqrt{1+y^2} + \sqrt{1+x^2} + \frac{1}{2} \log \left| \frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \right| + C = 0$$

22. .

$$y dx + x \log \left(\frac{y}{x} \right) dy - 2x dy = 0$$

Soln. We have

Simplifying the above equation, we get

$$\left[x \log \left(\frac{y}{x} \right) - 2x \right] dy = -y dx \Rightarrow \frac{dy}{dx} = \frac{y}{2x - x \log \left(\frac{y}{x} \right)} \dots \dots \dots (i)$$

$$\text{Let } F(x, y) = \frac{y}{2x - x \log \left(\frac{y}{x} \right)}$$

$$F(\mu x, \mu y) = \frac{\mu y}{2\mu x + \mu x \log \left(\frac{\mu y}{\mu x} \right)} = \frac{y}{2x - x \log \left(\frac{y}{x} \right)} = \mu^0 F(x, y)$$

∴ Function F(x, y) is homogeneous and hence the equation is homogeneous.

$$\text{Let } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Substituting in equation (i), we get

$$v + x \frac{dv}{dx} = \frac{vx}{2x - x \log v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v}{2 - \log v} - v \Rightarrow x \frac{dv}{dx} = \frac{v \log v - v}{2 - \log v}$$

$$\Rightarrow \frac{2 - \log v}{v \log v - v} dv = \frac{dx}{x}$$

Integrating both sides, we get

$$\int \frac{2 - \log v}{v \log v - v} dv = \int \frac{dx}{x} \Rightarrow \int \frac{1 + (1 + \log v)}{v(\log v - 1)} dv = \int \frac{dx}{x}$$

$$\Rightarrow \int \frac{dv}{v(\log v - 1)} - \int \frac{dv}{v} = \int \frac{dx}{x}$$

$$\log v - 1 = m \Rightarrow \frac{1}{v} dv = dm$$

Let

$$\Rightarrow \int \frac{1}{m} dm - \int \frac{1}{v} dv = \int \frac{dx}{x}$$

$$\Rightarrow \log |m| - \log |v| = \log |x| + \log |C|$$

$$\Rightarrow \log \left| \frac{m}{v} \right| = \log |Cx| \Rightarrow \frac{m}{v} = Cx \Rightarrow (\log v - 1) = v Cx$$

$$\Rightarrow \left[\log \left(\frac{y}{x} \right) - 1 \right] = Cy$$

Which is the required solution.

23. .

$$(x-y) \frac{dy}{dx} = x+2y$$

Soln. Given,

By simplifying the above equation, we get

$$\frac{dy}{dx} = \frac{x+2y}{x-y} \dots \dots \dots (i)$$

$$\text{Let } F(x, y) = \frac{x+2y}{x-y}$$

$$\text{Then } F(\lambda x, \lambda y) = \frac{\lambda x + 2\lambda y}{\lambda x - \lambda y} = \frac{\lambda(x+2y)}{\lambda(x-y)} = \lambda^0 F(x, y)$$

F(x, y) is homogeneous function and hence given differential equation is homogeneous.

$$y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Now, Let

Substituting these values in equation (i), we get

$$v + x \frac{dv}{dx} = \frac{x+2vx}{x-vx}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+2v}{1-v} - v = \frac{1+2v-v+v^2}{1-v} = \frac{1+v+v^2}{1-v}$$

$$\Rightarrow \frac{1-v}{1+v+v^2} dv = \frac{dx}{x}$$

By integrating both sides, we get

$$\int \frac{1-v}{v^2+v+1} dv = \int \frac{dx}{x} \dots \dots \dots (ii)$$

$$\text{LHS } \int \frac{1-v}{v^2+v+1} dv$$

$$\text{Let } 1-v = A(2v+1) + B = 2Av + (A+B)$$

Comparing coefficients of both sides, we get

$$2A = -1, \quad A+B=1$$

$$\text{Or } A = -\frac{1}{2}, \quad B = \frac{3}{2}$$

$$\begin{aligned} \therefore \int \frac{1-v}{v^2+v+1} dv &= \int \frac{-\frac{1}{2}(2v+1) + \frac{3}{2}}{v^2+v+1} dv \\ &= -\frac{1}{2} \int \frac{2v+1}{v^2+v+1} dv + \frac{3}{2} \int \frac{dv}{v^2+v+1} \\ &= -\frac{1}{2} \int \frac{(2v+1)}{v^2+v+1} dv + \frac{3}{2} \int \frac{dv}{\left(v+\frac{1}{2}\right)^2 + \frac{3}{4}} \end{aligned}$$

$$= -\frac{1}{2} \log |v^2 + v + 1| + \frac{3}{2} \times \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{v+1}{\sqrt{3}/2} \right) \text{ Now,}$$

substituting it in equation (ii), we get

$$-\frac{1}{2} \log |v^2 + v + 1| + \sqrt{3} \tan^{-1} \left(\frac{2v+1}{\sqrt{3}} \right) = \log x + C$$

$$\Rightarrow -\frac{1}{2} \log \left| \frac{y^2}{x^2} + \frac{y}{x} + 1 \right| + \sqrt{3} \tan^{-1} \left(\frac{\frac{2y}{x} + 1}{\sqrt{3}} \right) = \log x + C$$

$$\Rightarrow -\frac{1}{2} \log |x^2 + xy + y^2| + \frac{1}{2} \log x^2 + \sqrt{3} \tan^{-1} \left(\frac{2y+x}{\sqrt{3}x} \right) = \log x + C$$

$$\Rightarrow -\frac{1}{2} \log |x^2 + xy + y^2| + \sqrt{3} \tan^{-1} \left(\frac{2y+x}{\sqrt{3}x} \right) = C$$

24. .

Soln. The given differential equation is

$$(1+x^3) \frac{dy}{dx} + 6x^2y = (1+x^2) \Rightarrow \frac{dy}{dx} + \frac{6x^2}{(1+x^3)}y = \frac{1+x^2}{1+x^3}$$

It is in the form of $\frac{dy}{dx} + Py = Q$, where

$$P = \frac{6x^2}{(1+x^3)}, Q = \frac{1+x^2}{1+x^3}$$

$$\begin{aligned} \therefore IF &= e^{\int P dx} = e^{\int \frac{6x^2}{1+x^3} dx} \\ &= e^{2 \int \frac{3x^2}{1+x^3} dx} = e^{2 \int \frac{dt}{t}} \quad [\text{Let } 1+x^3 = t \Rightarrow 3x^2 dx = dt] \\ &= e^{2 \log t} = e^{\log t^2} = t^2 \\ &= (1+x^3)^2 \end{aligned}$$

Therefore, general solution is

$$\begin{aligned} y(1+x^3)^2 &= \int \frac{1+x^2}{1+x^3} \times (1+x^3)^2 dx + C \\ &= \int (1+x^2)(1+x^3) dx + C = \int (x^5 + x^3 + x^2 + 1) dx + C \end{aligned}$$

$$y(1+x^3)^2 = \frac{x^6}{6} + \frac{x^4}{4} + \frac{x^3}{3} + x + C$$

Putting $y = 1, x = 1$, we get

$$\therefore 4 = \frac{1}{6} + \frac{1}{4} + \frac{1}{3} + 1 + C \Rightarrow C = 4 - \frac{1}{6} - \frac{1}{4} - \frac{1}{3} - 1 = \frac{9}{4}$$

$\therefore$ Required particular solution is

$$y(1+x^3)^2 = \frac{x^6}{6} + \frac{x^4}{4} + \frac{x^3}{3} + x + \frac{9}{4}$$

25. .

Soln. Given: $2ye^{xy} dx + (y - 2xe^{xy}) dy = 0$

$$\Rightarrow \frac{dx}{dy} = -\frac{y - 2xe^{xy}}{2ye^{xy}} \Rightarrow \frac{dx}{dy} = \frac{2xe^{xy} - y}{2ye^{xy}}$$

$$\text{Let } F(x, y) = \frac{2xe^{xy} - y}{2ye^{xy}}$$

$$\therefore F(\lambda x, \lambda y) = \frac{2\lambda x e^{\lambda x \lambda y} - \lambda y}{2\lambda y e^{\lambda x \lambda y}} = \lambda^0 \frac{2xe^{xy} - y}{2ye^{xy}} = \lambda^0 F(x, y)$$

Hence, given differential equation is homogeneous.

$$\text{Now, } \frac{dx}{dy} = \frac{2xe^{xy} - y}{2ye^{xy}} \quad \dots\dots\dots(i)$$

$$\text{Let } x = vy \Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$\therefore (i) \Rightarrow v + y \frac{dv}{dy} = \frac{2vy.e^y - y}{2y.e^y}$$

$$\Rightarrow y \frac{dv}{dy} = \frac{y(2v.e^y - 1)}{2y.e^y} - v \Rightarrow y \frac{dv}{dy} = \frac{2v.e^y - 1}{2e^y} - v$$

$$\Rightarrow y \frac{dv}{dy} = -\frac{1}{2e^y} \Rightarrow 2ye^y dv = -dy$$

$$\Rightarrow 2 \int e^y dv = - \int \frac{dy}{y} \Rightarrow 2e^y = -\log y + C$$

$$\Rightarrow 2e^1 + \log y = C$$

When $x = 0, y = 1$

$$\therefore 2e^0 + \log 1 = C \text{ or } C = 2$$

Hence, the required solution is

$$2e^{xy} + \log y = 2 \Rightarrow \log C = 2$$

26. .

Soln. Given differential equation is

$$\frac{dy}{dx} = \frac{x(2 \log x + 1)}{\sin y + y \cos y}$$

$$\Rightarrow (\sin y + y \cos y) dy = x(2 \log x + 1) dx$$

$$\Rightarrow \int \sin y dy + \int y \cos y dy = 2 \int x \log x dx + \int x dx$$

$$\Rightarrow \int \sin y dy + [y \sin y - \int \sin y dy] = 2 \left[\log x \frac{x^2}{2} - \int \frac{1}{x} \frac{x^2}{2} dx \right] + \int x dx$$

$$\Rightarrow \int \sin y dy + y \sin y - \int \sin y dy = x^2 \log x - \int x dx + \int x dx + C$$

$$\Rightarrow y \sin y = x^2 \log x + C \quad \dots\dots\dots(i)$$

It is general solution.

For particular solution, we put $y = \frac{\pi}{2}$ when $x = 1$

$$(i) \text{ Becomes } \frac{\pi}{2} \sin \frac{\pi}{2} = 1 \cdot \log 1 + C$$

$$\frac{\pi}{2} = C \quad [\because \log 1 = 0]$$

Putting the value of C in (i), we get the required particular solution

$$y \sin y = x^2 \log x + \frac{\pi}{2}$$

27.

Soln. We have,

$$x \cos\left(\frac{y}{x}\right) \frac{dy}{dx} = y \cos\left(\frac{y}{x}\right) + x; x \neq 0$$

$$\Rightarrow \cos\left(\frac{y}{x}\right) \cdot \frac{dy}{dx} = \left(\frac{y}{x}\right) \cos\left(\frac{y}{x}\right) + 1 \dots\dots\dots(i)$$

This is a linear homogeneous differential equation

$$\text{Put } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Now (i) becomes

$$\cos v \left[v + x \frac{dv}{dx} \right] = v \cos v + 1$$

$$\Rightarrow x \cos v \frac{dv}{dx} = 1 \Rightarrow \cos v dv = \frac{dx}{x}$$

Integrating both sides, we get

$$\sin v = \log x + C \Rightarrow \sin\left(\frac{y}{x}\right) = \log x + C$$

Is the required solution.

28.

$$(1 + y^2) + (x - e^{\tan^{-1} y}) \frac{dy}{dx} = 0$$

$$\text{Soln. We have, } (1 + y^2) + (x - e^{\tan^{-1} y}) \frac{dy}{dx} = 0$$

$$\Rightarrow (x - e^{\tan^{-1} y}) \frac{dy}{dx} = -(1 + y^2)$$

$$\Rightarrow \frac{dx}{dy} = \frac{x - e^{\tan^{-1} y}}{-(1 + y^2)} \Rightarrow \frac{dx}{dy} + \frac{1}{1 + y^2} \cdot x = \frac{e^{\tan^{-1} y}}{1 + y^2}$$

This is a linear differential equation of the form

$$\frac{dx}{dy} + Px = Q, \text{ where } P = \frac{1}{1 + y^2} \text{ and } Q = \frac{e^{\tan^{-1} y}}{1 + y^2}$$

$$\therefore I.F. = e^{\int \frac{dy}{1 + y^2}} = e^{\tan^{-1} y}$$

$\therefore$ Solution is

$$x \cdot e^{\tan^{-1} y} = \int \frac{(e^{\tan^{-1} y})^2}{1 + y^2} dy + C$$

$$= \int \frac{e^{2 \tan^{-1} y}}{1 + y^2} dy + C$$

$$\Rightarrow x \cdot e^{\tan^{-1} y} = \frac{e^{2 \tan^{-1} y}}{2} + C_1$$

$$\Rightarrow x = \frac{e^{\tan^{-1} y}}{2} + C_1 e^{-\tan^{-1} y}$$

29.

$$\text{Soln. We have, } (\tan^{-1} y - x) dy = (1 + y^2) dx$$

$$\Rightarrow \frac{dx}{dy} = \frac{\tan^{-1} y - x}{1 + y^2} \Rightarrow \frac{dx}{dy} + \frac{1}{1 + y^2} \cdot x = \frac{\tan^{-1} y}{1 + y^2}$$

This is a linear differential equation of the form

$$\frac{dx}{dy} + Px = Q, \text{ where } P = \frac{1}{1 + y^2} \text{ and } Q = \frac{\tan^{-1} y}{1 + y^2}$$

$$I.F. = e^{\int \frac{1}{1 + y^2} dy} = e^{\tan^{-1} y}$$

$\therefore$ Required solution is,

$$x \cdot e^{\tan^{-1} y} = \int \frac{e^{\tan^{-1} y} \cdot \tan^{-1} y}{1 + y^2} dy + C$$

$$\text{Put } \tan^{-1} y = t \Rightarrow \left(\frac{1}{1 + y^2} \right) dy = dt$$

$\therefore$ (i) becomes,

$$x \cdot e^{\tan^{-1} y} = \int e^t \cdot t dt + C$$

$$\Rightarrow x \cdot e^{\tan^{-1} y} = t \cdot e^t - \int 1 \cdot e^t dt + C$$

$$\Rightarrow x \cdot e^{\tan^{-1} y} = t \cdot e^t - e^t + C$$

$$\Rightarrow x \cdot e^{\tan^{-1} y} = \tan^{-1} y \cdot e^{\tan^{-1} y} - e^{\tan^{-1} y} + C$$

$$\Rightarrow x = \tan^{-1} y - 1 + C e^{-\tan^{-1} y}$$

30.

Soln. We have,

$$x \frac{dy}{dx} + y - x + xy \cot x = 0, (x \neq 0)$$

$$\Rightarrow x \frac{dy}{dx} + (1 + x \cot x) \cdot y = x$$

$$\Rightarrow \frac{dy}{dx} + \frac{1 + x \cot x}{x} \cdot y = 1 \dots\dots(i)$$

This is linear D.E. of the form $\frac{dy}{dx} + Py = Q$

$$P = \frac{1 + x \cot x}{x} = \frac{1}{x} + \cot x \text{ and } Q = 1$$

Where

$$\therefore \text{Now } I.F. = e^{\int P dx} = e^{\log(x \sin x)} = x \sin x$$

$$\therefore \text{The solution of (i) is } y \cdot x \sin x = \int 1 \cdot x \sin x dx + C$$

$$= x(-\cos x) + \int 1 \cdot \cos x dx + C$$

$$\Rightarrow xy \sin x = -x \cos x + \sin x + C$$

The required solution

$$y \cdot x \sin x = x(-\cos x) + \sin x + C$$

$\dots\dots\dots(i)$

$$\text{Putting } x = \frac{\pi}{2}, y = 0 \text{ in (ii), we get}$$

$$\therefore 0 = -\frac{\pi}{2} \cos \frac{\pi}{2} + \sin \frac{\pi}{2} + C \Rightarrow C = -1$$

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New

SURE SHOT QUESTIONS 2026

Chapter – 10 (Questions)

Vector Algebra

Questions

1. The two vectors $\hat{j} + \hat{k}$ and $3\hat{i} - \hat{j} + 4\hat{k}$ represent the two sides AB and AC, respectively of a $\triangle ABC$. Find the length of the median through A.

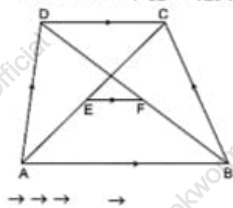
2. If $\vec{a} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$, Find the unit vector in the direction of $\vec{a} + \vec{b}$.

3. Using vectors find the area of triangle ABC with vertices $A(1, 2, 3)$, $B(2, -1, 4)$ and $C(4, 5, -1)$.

4. Write the position vector of the point which divides the join of points with position vector $3\vec{a} - 2\vec{b}$ and $2\vec{a} + 3\vec{b}$ in the ratio 2:1.

5. If ABCD is a quadrilateral and E and F are the mid-points of AC and BD, prove that

$$\vec{AB} + \vec{AD} + \vec{CB} + \vec{CD} = 4\vec{EF}.$$



6. The position vectors of vertices of $\triangle ABC$ are $A(2\hat{i} - \hat{j} + \hat{k})$, $B(\hat{i} - 3\hat{j} + 5\hat{k})$ and $C(3\hat{i} - 4\hat{j} - 4\hat{k})$. Find all the angles of $\triangle ABC$.

7. Find the projection of vector $(\vec{b} + \vec{c})$ on vector $\vec{a}$. Where $\vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + \hat{k}$.

8. If $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$, then find a unit vector along the vector $\vec{a} \times \vec{b}$.

9. Find the area of a parallelogram whose adjacent sides are determined by the vectors $\vec{a} = \hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} - 7\hat{j} + \hat{k}$.

10. The two adjacent sides of a parallelogram are $2\hat{i} - 4\hat{j} - 5\hat{k}$ and $2\hat{i} + 2\hat{j} + 3\hat{k}$. Find the two unit vectors parallel to its diagonals. Using the diagonal vectors, find the area of the parallelogram.

11. Find a unit vector in the direction of $\vec{a} = 3\hat{i} - 2\hat{j} + 6\hat{k}$.

12. Find the projection of the vector $\hat{i} - \hat{j}$ on the vector $\hat{i} + \hat{j}$.

13. Find the projection of $\vec{a}$ on $\vec{b}$, if $\vec{a} \cdot \vec{b} = 8$ and $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$.

14. Find 'λ' when the projection of $\vec{a} = \lambda\hat{i} + \hat{j} + 4\hat{k}$ on $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$ is 4 units.

15. If $\vec{a}, \vec{b}$ are two vectors such that $|\vec{a} + \vec{b}| = |\vec{a}|$, then prove that $2\vec{a} + \vec{b}$ is perpendicular to $\vec{b}$.

16. Write the direction ratio's of the vector $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$ and hence calculate its direction cosines.

17. Find the vector $\vec{p}$ which is perpendicular to both $\vec{a} = 4\hat{i} + 5\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{p} \cdot \vec{q} = 21$, where $\vec{q} = 3\hat{i} + \hat{j} - \hat{k}$.

18. If $\vec{a} = \hat{i} - \hat{j} + 7\hat{k}$ and $\vec{b} = 5\hat{i} - \hat{j} + \lambda\hat{k}$, then find the value of λ so that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular vectors.

19. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $|\vec{a}| = 3, |\vec{b}| = 4$ and $|\vec{c}| = 5$ and each one of them is perpendicular to the sum of the other two, then find $|\vec{a} + \vec{b} + \vec{c}|$.

20. If vectors $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j}$ are such that $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{c}$, then find the value of λ .

21. Find the direction cosines of the vector joining the points A(1, 2, -3) and B(-1, -2, 1) directed from B to A.

22. Find the area of parallelogram whose adjacent sides are determined by the vector $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} - \hat{k}$.

23. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$, find a vector $\vec{c}$ such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$.

24. Find a unit vector perpendicular to both of the vector

$3\vec{a} + 2\vec{b}$ and $3\vec{a} - 2\vec{b}$, where $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.

25. If the vector $-\hat{i} + \hat{j} - \hat{k}$ bisects the angle between the vector $\vec{c}$ and the vector $3\hat{i} + 4\hat{j}$, then find the unit vector in the direction of $\vec{c}$.

26. If $\vec{a}, \vec{b}$ are unit vectors such that the vector $\vec{a} + 3\vec{b}$ is perpendicular to $7\vec{a} - 5\vec{b}$ and $\vec{a} - 4\vec{b}$ is perpendicular to $7\vec{a} - 2\vec{b}$, then find the angle between $\vec{a}$ and $\vec{b}$.

27. Using vectors, find the area of the triangle ABC with vertices A(1, 2, 3), B(2, -1, 4) and C(4, 5, -1).

28. If $\hat{i} - \hat{j} + \hat{k}, 2\hat{i} - 5\hat{j}, 3\hat{i} - 2\hat{j} - 3\hat{k}$ and $\hat{i} - 6\hat{j} - \hat{k}$ respectively are the position vectors of points A, B, C and D, then find the angle between the straight lines AB and CD. Find whether $\overline{AB}$ and $\overline{CD}$ are collinear or not.

29. Let $\vec{a} = 4\hat{i} + 5\hat{j} - \hat{k}, \vec{b} = \hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{c} = 3\hat{i} - \hat{j} - \hat{k}$. Find a vector $\vec{d}$ which is perpendicular to both $\vec{c}$ and $\vec{b}$ and $\vec{d} \cdot \vec{a} = 21$.

30. Vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are such that $\vec{a} + \vec{b} - \vec{c} = 0$ and $|\vec{a}| = 3, |\vec{b}| = 5$ and $|\vec{c}| = 7$. Find the angle between $\vec{a}$ and $\vec{b}$.

New

SURE SHOT QUESTIONS 2026

Chapter – 10 (Solutions)

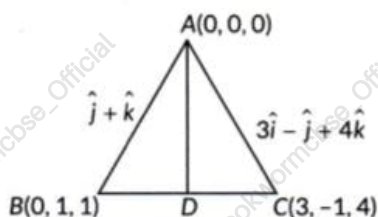
Vector Algebra

Questions

1.

Soln. Take A to be as origin (0, 0, 0).

∴ Coordinates of B are (0, 1, 1) and coordinates of C are (3, -1, 4).



Let D be the mid point of BC and AD is a median of $\triangle ABC$.

∴ Coordinates of D are $\left(\frac{3}{2}, 0, \frac{5}{2}\right)$

$$\begin{aligned} \text{So, length of AD} &= \sqrt{\left(\frac{3}{2}-0\right)^2 + (0)^2 + \left(\frac{5}{2}-0\right)^2} \\ &= \sqrt{\frac{9}{4} + \frac{25}{4}} = \frac{\sqrt{34}}{2} \text{ units} \end{aligned}$$

2.

Soln. We have, $\vec{a} + \vec{b} = (2\hat{i} + 4\hat{j} - 5\hat{k}) + (\hat{i} + 2\hat{j} + 3\hat{k})$

$$\begin{aligned} &= 3\hat{i} + 6\hat{j} - 2\hat{k} \\ \Rightarrow |\vec{a} + \vec{b}| &= \sqrt{(3)^2 + (6)^2 + (-2)^2} \\ &= \sqrt{9 + 36 + 4} \\ &= \sqrt{49} = 7 \end{aligned}$$

Hence, the unit vector in the direction of $\vec{a} + \vec{b}$

$$\begin{aligned} &= \frac{\vec{a} + \vec{b}}{|\vec{a} + \vec{b}|} \\ &= \frac{1}{7}(3\hat{i} + 6\hat{j} - 2\hat{k}) \\ &= \frac{3}{7}\hat{i} + \frac{6}{7}\hat{j} - \frac{2}{7}\hat{k} \end{aligned}$$

3.

Soln. Vertices of the given $\triangle ABC$ are A(1,2,3)

B(2, -1, 4) and C(4, 5, -1)

$$\begin{aligned} \vec{AB} &= \vec{OB} - \vec{OA} = (2\hat{i} - \hat{j} + 4\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) \\ &= \hat{i} - 3\hat{j} + \hat{k} \end{aligned}$$

$$\begin{aligned} \vec{AC} &= \vec{OC} - \vec{OA} = (4\hat{i} + 5\hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) \\ &= 3\hat{i} + 3\hat{j} - 4\hat{k} \end{aligned}$$

$$\text{Required area} = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 1 \\ 3 & 3 & -4 \end{vmatrix} = 9\hat{i} + 7\hat{j} + 12\hat{k}$$

$$\begin{aligned} |\vec{AB} \times \vec{AC}| &= \sqrt{9^2 + 7^2 + 12^2} \\ &= \sqrt{81 + 49 + 144} = \sqrt{274} \end{aligned}$$

$$\therefore \text{Area of } \triangle ABC = \frac{1}{2} \sqrt{274} \text{ sq. units}$$

4.

Soln. Let A and B be the given points with position vectors $3\vec{a} - 2\vec{b}$ and $2\vec{a} + 3\vec{b}$ respectively.

Let P and Q be the points dividing AB in the ratio 2 : 1 internally and externally respectively. Then,

$$\begin{aligned} \text{Position vector of P} &= \frac{1(3\vec{a} - 2\vec{b}) + 2(2\vec{a} + 3\vec{b})}{1+2} \\ &= \frac{7\vec{a}}{3} + \frac{4\vec{b}}{3} \end{aligned}$$

$$\begin{aligned} \text{Position vector of Q} &= \frac{1(3\vec{a} - 2\vec{b}) - 2(2\vec{a} + 3\vec{b})}{1-2} \\ &= \vec{a} + 8\vec{b} \end{aligned}$$

5.

Soln. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ be the position vectors of the vertices of the quadrilateral ABCD relative to some chosen point O as the origin.

Since, E and F are the mid - points of the sides AC and BD respectively, so the position vectors of E and F are

$$\frac{\vec{a} + \vec{c}}{2} \text{ and } \frac{\vec{b} + \vec{d}}{2}$$

We have, $\vec{AB} + \vec{AD} + \vec{CB} + \vec{CD}$
 $= (\vec{b} - \vec{a}) + (\vec{d} - \vec{a}) + (\vec{b} - \vec{c}) + (\vec{d} - \vec{c})$
 $= 2(\vec{b} + \vec{d} - \vec{c} - \vec{a})$

$$= 4 \left(\frac{\vec{b} + \vec{d}}{2} \right) - 4 \left(\frac{\vec{c} + \vec{a}}{2} \right)$$

$$= 4 [\text{Position vector of F} - \text{Position vector of E}]$$

$$= 4\vec{EF}$$

6. .

Ans. We have.

$$\text{Position vector of A} = 2\hat{i} - \hat{j} + \hat{k}$$

$$\text{Position vector of B} = \hat{i} - 3\hat{j} - 5\hat{k}$$

$$\text{Position vector of C} = 3\hat{i} - 4\hat{j} - 4\hat{k}$$

$$\text{Now, } \vec{AB} = \hat{i} - 3\hat{j} - 5\hat{k} - 2\hat{i} + \hat{j} - \hat{k} \Rightarrow \vec{AB} = -\hat{i} - 2\hat{j} - 6\hat{k}$$

$$\text{Similarly, } \vec{BC} = 3\hat{i} - 4\hat{j} - 4\hat{k} - \hat{i} + 3\hat{j} + 5\hat{k} = 2\hat{i} - \hat{j} + \hat{k}$$

$$\text{and } \vec{AC} = 3\hat{i} - 4\hat{j} - 4\hat{k} - 2\hat{i} + \hat{j} - \hat{k} = \hat{i} - 3\hat{j} - 5\hat{k}$$

$$\text{Now, } \vec{AB} \cdot \vec{AC} = |\vec{AB}| |\vec{AC}| \cos \angle BAC$$

$$\Rightarrow \cos \angle BAC = \frac{(-\hat{i} - 2\hat{j} - 6\hat{k}) \cdot (\hat{i} - 3\hat{j} - 5\hat{k})}{|-\hat{i} - 2\hat{j} - 6\hat{k}| \cdot |\hat{i} - 3\hat{j} - 5\hat{k}|}$$

$$= \frac{-1 + 6 + 30}{\sqrt{1+4+36} \sqrt{1+9+25}} = \frac{35}{\sqrt{41} \sqrt{35}}$$

$$\Rightarrow \angle BAC = \cos^{-1} \left(\sqrt{\frac{35}{41}} \right)$$

$$\text{Similarly, } \cos \angle ABC = \frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| |\vec{BC}|}$$

$$\Rightarrow \cos \angle ABC = \frac{(\hat{i} + 2\hat{j} + 6\hat{k}) \cdot (2\hat{i} - \hat{j} + \hat{k})}{\sqrt{1+4+36} \sqrt{4+1+1}}$$

$$= \frac{2 - 2 + 6}{\sqrt{41} \sqrt{6}} = \frac{6}{\sqrt{41} \sqrt{6}}$$

$$\Rightarrow \angle ABC = \cos^{-1} \left(\sqrt{\frac{6}{41}} \right)$$

$$\text{And, } \cos \angle ACB = \frac{\vec{CA} \cdot \vec{CB}}{|\vec{CA}| |\vec{CB}|} = \frac{(-\hat{i} + 3\hat{j} + 5\hat{k}) \cdot (-2\hat{i} + \hat{j} - \hat{k})}{\sqrt{1+9+25} \sqrt{4+1+1}} = 0$$

$$\Rightarrow \angle ACB = \cos^{-1}(0) = \frac{\pi}{2}$$

7. .

Ans.

$$\text{Given } \vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}, \vec{b} = \hat{i} + 3\hat{j} + \hat{k} \text{ and } \vec{c} = \hat{i} + \hat{k}$$

$$\vec{b} + \vec{c} = (\hat{i} + 3\hat{j} + \hat{k}) + (\hat{i} + \hat{k}) = 2\hat{i} + 3\hat{j} + 2\hat{k}$$

$$\text{Projection of vector } (\vec{b} + \vec{c}) \text{ on vector } \vec{a} = \frac{(\vec{b} + \vec{c}) \cdot \vec{a}}{|\vec{a}|}$$

$$= \frac{(2\hat{i} + 3\hat{j} + 2\hat{k}) \cdot (2\hat{i} + 2\hat{j} + \hat{k})}{\sqrt{4+4+1}} = \frac{4+6+2}{\sqrt{9}} = \frac{12}{3} = 4$$

8. .

$$\text{Soln. We have, } \vec{a} = 4\hat{i} - \hat{j} + \hat{k} \text{ and } \vec{b} = 2\hat{i} - 2\hat{j} +$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 1 \\ 2 & -2 & 1 \end{vmatrix}$$

$$= \hat{i}(-1+2) - \hat{j}(4-2) + \hat{k}(-8+2) = \hat{i} - 2\hat{j} - 6\hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{1^2 + (-2)^2 + (-6)^2} = \sqrt{41}$$

$$\text{Unit vector along } \vec{a} \times \vec{b} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

$$= \frac{\hat{i} - 2\hat{j} - 6\hat{k}}{\sqrt{41}} = \frac{1}{\sqrt{41}}\hat{i} - \frac{2}{\sqrt{41}}\hat{j} - \frac{6}{\sqrt{41}}\hat{k}$$

9. .

$$\text{Soln. Area of parallelogram} = |\vec{a} \times \vec{b}| = \left| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 3 \\ 2 & -7 & 1 \end{vmatrix} \right|$$

$$= [(-1+21)\hat{i} - (1-6)\hat{j} + (-7+2)\hat{k}]$$

$$= |20\hat{i} + 5\hat{j} - 5\hat{k}|$$

$$= \sqrt{(20)^2 + (5)^2 + (-5)^2}$$

$$= \sqrt{400 + 25 + 25} = \sqrt{450} = 15\sqrt{2} \text{ sq. units}$$

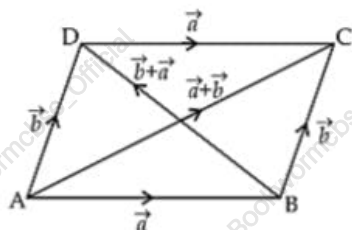
10. .

Soln. Let ABCD be a parallelogram such that

$$\vec{AB} = \vec{a} = 2\hat{i} - 4\hat{j} - 5\hat{k}$$

And

$$\vec{BC} = \vec{b} = 2\hat{i} + 2\hat{j} + 3\hat{k}$$



Then,

$$\vec{AB} + \vec{BC} = \vec{AC}$$

$$\Rightarrow \vec{AC} = \vec{a} + \vec{b} = 4\hat{i} - 2\hat{j} - 2\hat{k}$$

and $\vec{AB} + \vec{BD} = \vec{AD}$

$$\Rightarrow \vec{BD} = \vec{AD} - \vec{AB}$$

$$\Rightarrow \vec{BD} = \vec{b} - \vec{a} = 0\hat{i} + 6\hat{j} + 8\hat{k}$$

Now, $\vec{AC} = 4\hat{i} - 2\hat{j} - 2\hat{k}$

$$|\vec{AC}| = \sqrt{(4)^2 + (-2)^2 + (-2)^2} = \sqrt{24} = 2\sqrt{6}$$

and $|\vec{BD}| = 0\hat{i} + 6\hat{j} + 8\hat{k}$

$$|\vec{BD}| = \sqrt{(0)^2 + (6)^2 + (8)^2} = \sqrt{100} = 10$$

Unit vector along $\vec{AC}$

$$= \frac{\vec{AC}}{|\vec{AC}|} = \frac{1}{2\sqrt{6}} (4\hat{i} - 2\hat{j} - 2\hat{k})$$

$$= \frac{1}{\sqrt{6}} (2\hat{i} - \hat{j} - \hat{k})$$

Unit vector along $\vec{BD}$

$$= \frac{\vec{BD}}{|\vec{BD}|} = \frac{1}{10} (6\hat{j} + 8\hat{k}) = \frac{1}{5} (3\hat{j} + 4\hat{k})$$

Now, area of parallelogram

$$= \frac{1}{2} |\vec{AC} \times \vec{BD}|$$

$$\Rightarrow \vec{AC} \times \vec{BD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -2 & -2 \\ 0 & 6 & 8 \end{vmatrix}$$

$$= 4\hat{i} + 32\hat{j} + 24\hat{k}$$

Area of parallelogram

$$= \frac{1}{2} |\vec{AC} \times \vec{BD}|$$

$$= 2\sqrt{101} \text{ sq. units Ans.}$$

11. .

Soln. $\vec{a} = 3\hat{i} - 2\hat{j} + 6\hat{k}$

Unit vector in the direction of

$$\vec{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{3\hat{i} - 2\hat{j} + 6\hat{k}}{\sqrt{3^2 + (-2)^2 + 6^2}} = \frac{1}{7} (3\hat{i} - 2\hat{j} + 6\hat{k})$$

12. .

Soln. Let $\vec{a} = \hat{i} - \hat{j}$ and $\vec{b} = \hat{i} + \hat{j}$

Projection of

$$\vec{a} \text{ on } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{(\hat{i} - \hat{j}) \cdot (\hat{i} + \hat{j})}{\sqrt{1+1}} = \frac{1-1}{\sqrt{2}} = 0$$

13. .

Soln. $\vec{a} \cdot \vec{b} = 8$

$$\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$$

We know projection of $\vec{a}$ on $\vec{b} =$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{8}{\sqrt{4+36+9}} = \frac{8}{7}$$

14. .

Soln. Projection of $\vec{a}$ on $\vec{b} = 4$, (given)

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = 4$$

$$\therefore \text{or } \frac{(\lambda\hat{i} + \hat{j} + 4\hat{k}) \cdot (2\hat{i} + 6\hat{j} + 3\hat{k})}{\sqrt{4+36+9}} = 4$$

$$\text{or } 2\lambda + 6 + 12 = 7 \times 4$$

$$\text{or } 2\lambda = 28 - 18 = 10$$

$$\text{or } \lambda = \frac{10}{2} = 5.$$

15. .

Soln. Given, $|\vec{a} + \vec{b}| = |\vec{a}|$

$$|\vec{a} + \vec{b}|^2 = |\vec{a}|^2$$

Or $|\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = |\vec{a}|^2$

$$|\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = 0 \dots (i)$$

Now, $(2\vec{a} + \vec{b}) \cdot \vec{b} = 2\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b}$

$$\vec{b} \cdot (2\vec{a} + \vec{b}) = 0$$

From eqn. (i), $2\vec{a} + \vec{b} \perp \vec{c}$.

16. .

Soln. We know that, the direction ratio's a, b, c of a vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ are just the respective components x, y and z of the vector. So, for the given vector, we have a = 1, b = 1 and c = -2 Further, if l, m and n are the direction cosines of the given vector, then

$$l = \frac{a}{|\vec{r}|} = \frac{1}{\sqrt{6}}, m = \frac{b}{|\vec{r}|} = \frac{1}{\sqrt{6}}, n = \frac{c}{|\vec{r}|} = \frac{-2}{\sqrt{6}} \text{ as } |\vec{r}| = \sqrt{6}$$

Thus, the direction cosines are $\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right)$.

17. .

Soln. Any vector perpendicular to both $\vec{\alpha}$ and $\vec{\beta}$ =

Parallel to $(\vec{\alpha} \times \vec{\beta})$

$$\therefore \vec{p} = \lambda (\vec{\alpha} \times \vec{\beta}) = \lambda \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 5 & -1 \\ 1 & -4 & 5 \end{vmatrix}$$

$$= \lambda [i(25 - 4) - j(20 + 1) + k(-16 - 5)]$$

$$= \lambda [21\hat{i} - 21\hat{j} - 21\hat{k}]$$

$$p \cdot q = 21 \text{ (Given)}$$

$$\lambda(21\hat{i} - 21\hat{j} - 21\hat{k}) \cdot (3\hat{i} + \hat{j} - \hat{k}) = 21$$

$$\lambda(63 - 21 + 21) = 21$$

$$\lambda = \frac{1}{3}$$

$$\therefore \vec{p} = \lambda(21\hat{i} - 21\hat{j} - 21\hat{k})$$

$$\vec{p} = \frac{1}{3}(21\hat{i} - 21\hat{j} - 21\hat{k})$$

$$\vec{p} = 7\hat{i} - 7\hat{j} - 7\hat{k}$$

18. .

Soln. Given, $\vec{a} = \hat{i} - \hat{j} + 7\hat{k}$

And $\vec{b} = 5\hat{i} - \hat{j} + \lambda\hat{k}$

$$\begin{aligned} \vec{a} + \vec{b} &= (\hat{i} - \hat{j} + 7\hat{k}) + (5\hat{i} - \hat{j} + \lambda\hat{k}) \\ &= 6\hat{i} - 2\hat{j} + (7 + \lambda)\hat{k} \end{aligned}$$

$$\begin{aligned} \text{And } \vec{a} - \vec{b} &= (\hat{i} - \hat{j} + 7\hat{k}) - (5\hat{i} - \hat{j} + \lambda\hat{k}) \\ &= -4\hat{i} + (7 - \lambda)\hat{k} \end{aligned}$$

Since $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ are perpendicular vectors,

$$\therefore (\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0$$

$$\{6\hat{i} - 2\hat{j} + (7 + \lambda)\hat{k}\} \cdot \{-4\hat{i} + (7 - \lambda)\hat{k}\} = 0$$

$$\text{or } -24 + (7 + \lambda)(7 - \lambda) = 0$$

$$\text{or } 49 - \lambda^2 - 24 = 0$$

$$\text{or } \lambda^2 = 49 - 24 = 25$$

$$\text{or } \lambda = \pm 5 \text{ units}$$

19. .

Soln. Since $\vec{a} \perp (\vec{b} + \vec{c})$, therefore

$$\vec{a} \cdot (\vec{b} + \vec{c}) = 0$$

$$\text{Or } \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} = 0$$

$$\text{Or } \vec{a} \cdot \vec{a} + (\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}) = \vec{a} \cdot \vec{a} + 0$$

$$\text{Or } \vec{a} \cdot (\vec{a} + \vec{b} + \vec{c}) = |\vec{a}|^2$$

$$\text{Or } \vec{a} \cdot (\vec{a} + \vec{b} + \vec{c}) = 3^2 = 9 \quad \dots\dots\dots(i)$$

$$\text{Similarly, } \vec{b} \cdot (\vec{a} + \vec{b} + \vec{c}) = |\vec{b}|^2 = 16 \quad \dots\dots\dots(ii)$$

$$\text{And } \vec{c} \cdot (\vec{a} + \vec{b} + \vec{c}) = |\vec{c}|^2 = 25 \quad \dots\dots\dots(iii)$$

Adding eqn. (i), (ii) and (iii), we get

$$\vec{a} \cdot (\vec{a} + \vec{b} + \vec{c}) + \vec{b} \cdot (\vec{a} + \vec{b} + \vec{c}) + \vec{c} \cdot (\vec{a} + \vec{b} + \vec{c}) = 50$$

$$\text{or } (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) = 50$$

$$\text{or } |\vec{a} + \vec{b} + \vec{c}|^2 = 50$$

$$\text{or } |\vec{a} + \vec{b} + \vec{c}| = \sqrt{50} = 5\sqrt{2}$$

20. .

Soln. Given, $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and

$$\vec{c} = 3\hat{i} + \hat{j}$$

Also, $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{c}$.

$$\therefore (\vec{a} + \lambda\vec{b}) \cdot \vec{c} = 0 \quad \dots\dots\dots(i)$$

[$\therefore$ when $\vec{a} \perp \vec{b}$, then $\vec{a} \cdot \vec{b} = 0$]

$$\text{Now, } \vec{a} + \lambda\vec{b} = (2\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(-\hat{i} + 2\hat{j} + \hat{k})$$

$$\text{Or } \vec{a} + \lambda\vec{b} = \hat{i}(2 - \lambda) + \hat{j}(2 + 2\lambda) + \hat{k}(3 + \lambda)$$

Then from Eq. (i), we get

$$[\hat{i}(2 - \lambda) + \hat{j}(2 + 2\lambda) + \hat{k}(3 + \lambda)] \cdot [3\hat{i} + \hat{j}] = 0$$

$$\text{or } 3(2 - \lambda) + 1(2 + 2\lambda) = 0$$

$$\text{or } 8 - \lambda = 0$$

$$\therefore \lambda = 8$$

21.

Soln. $\vec{BA} = 2\hat{i} + 4\hat{j} - 4\hat{k}$

Or d-ratios of $\vec{BA}$ are 2, 4, -4

$\therefore$ Direction cosines are: $\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}$

22.

Soln. $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 2 & -1 & -1 \end{vmatrix}$
 $= \hat{i}(1+2) - \hat{j}(-1-4) + \hat{k}(-1+2)$
 $= 3\hat{i} + 5\hat{j} + \hat{k}$

Area of $|\vec{a} \times \vec{b}| = \sqrt{9+25+1}$
 $= \sqrt{35}$ sq. units

23.

Soln. Let $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$

Given $\vec{a} = \hat{i} + \hat{j} + \hat{k}$

$\vec{b} = \hat{j} - \hat{k}$

According to the question,

$\vec{a} \cdot \vec{c} = 3$

Or $(\hat{i} + \hat{j} + \hat{k}) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) = 3$

Or $x + y + z = 3$ (i)

And $\vec{a} \times \vec{c} = \vec{b}$

Or $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ x & y & z \end{vmatrix} = \vec{b} = \hat{j} - \hat{k}$

Or $(z-y)\hat{i} + (x-z)\hat{j} + (y-x)\hat{k} = \hat{j} - \hat{k}$

On equating the coefficients of like terms, we get

$z-y=0$, or $y=z$ (ii)

$x-z=1$ (iii)

And $y-x=-1$ (iv)

Solving eqns. (i), (ii), (iii) and (iv), we get

$x = 5/3$

And $y = 2/3 = z$

Hence, $\vec{c} = \frac{5}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$

24.

Soln. Given $\vec{a} = \hat{i} + \hat{j} + \hat{k}$

$\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$

$3\vec{b} = 3\hat{i} + 3\hat{j} + 3\hat{k}$

And $2\vec{b} = 2\hat{i} + 4\hat{j} + 6\hat{k}$

Let $\vec{c}$ be the vector perpendicular to both $(3\vec{a} + 2\vec{b})$

and $(3\vec{a} - 2\vec{b})$.

$\therefore 3\vec{a} + 2\vec{b} = 5\hat{i} + 7\hat{j} + 9\hat{k}$ and $3\vec{a} - 2\vec{b} = \hat{i} - \hat{j} - 3\hat{k}$

$\therefore \vec{c} = (3\vec{a} + 2\vec{b}) \times (3\vec{a} - 2\vec{b})$

$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 7 & 9 \\ 1 & -1 & -3 \end{vmatrix}$

$= \hat{i}(-21+9) - \hat{j}(-15-9) + \hat{k}(-5-7)$

$= -12\hat{i} + 24\hat{j} - 12\hat{k}$

25.

Soln. Let $x\hat{i} + y\hat{j} + z\hat{k}$ be the unit vector along $\vec{c}$.

Since $-\hat{i} + \hat{j} - \hat{k}$ bisects the angle between $\vec{c}$ and

$3\hat{i} + 4\hat{j}$.

Therefore, $\lambda(-\hat{i} + \hat{j} - \hat{k}) = (x\hat{i} + y\hat{j} + z\hat{k}) + \frac{3\hat{i} + 4\hat{j}}{5}$

$x + \frac{3}{5} = -\lambda$

$y + \frac{4}{5} = -\lambda$

And $z = -\lambda$

Now, $x^2 + y^2 + z^2 = 1$

$\left[\because x\hat{i} + y\hat{j} + z\hat{k} \text{ is a unit vector} \right]$

Or $\left(-\lambda - \frac{3}{5}\right)^2 + \left(\lambda - \frac{4}{5}\right)^2 + \lambda^2 = 1$

Or $3\lambda^2 - \frac{2}{5}\lambda = 0$

Or $\lambda = 0$ or $\lambda = \frac{2}{15}$

But $\lambda \neq 0$, because $\lambda = 0$ implied that the given vectors are parallel.

Or $\lambda = -\frac{2}{15}$

$y = \frac{-10}{15}$

And $z = \frac{-2}{15}$

Hence, $x\hat{i} + y\hat{j} + z\hat{k} = -\frac{1}{15}(11\hat{i} + 10\hat{j} + 2\hat{k})$

26.

Soln. Let angle between $\vec{a}$ and $\vec{b}$ be θ .

Given, $(\vec{a} + 3\vec{b}) \perp (7\vec{a} - 5\vec{b})$

Or $(\vec{a} + 3\vec{b}) \cdot (7\vec{a} - 5\vec{b}) = 0$

Or $7|\vec{a}|^2 + 16(\vec{a} \cdot \vec{b}) - 15|\vec{b}|^2 = 0$

Or $7 + 16 \cos \theta - 15 = 0$

$\left[\because |\vec{a}|^2 = |\vec{b}|^2 = 1 \right]$

Or $\cos \theta = \frac{8}{16} = \frac{1}{2}$

Or $\theta = \frac{\pi}{3}$

Also, given that $(\vec{a} - 4\vec{b}) \perp (7\vec{a} - 2\vec{b})$

Or $(\vec{a} - 4\vec{b}) \cdot (7\vec{a} - 2\vec{b}) = 0$

Or $7|\vec{a}|^2 + 8|\vec{b}|^2 - 30(\vec{a} \cdot \vec{b}) = 0$

Or $15 - 30 \cos \theta = 0$

Or $\cos \theta = \frac{1}{2}$

Or $\theta = \frac{\pi}{3}$

27.

Soln. Given, $\triangle ABC$ with vertices

$A(1,2,3) = \hat{i} + 2\hat{j} + 3\hat{k}, B(2,-1,4) = 2\hat{i} - \hat{j} + 4\hat{k},$

$C(4,5,-1) = 4\hat{i} + 5\hat{j} - \hat{k}$

Now $\vec{AB} = \vec{OB} - \vec{OA} = (2\hat{i} - \hat{j} + 4\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$

$= \hat{i} - 3\hat{j} + \hat{k}.$

$\vec{AC} = \vec{OC} - \vec{OA} = (4\hat{i} + 5\hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$
 $= 3\hat{i} + 3\hat{j} - 4\hat{k}$

$\therefore (\vec{AB} \times \vec{AC}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 1 \\ 3 & 3 & -4 \end{vmatrix} = 9\hat{i} + 7\hat{j} + 12\hat{k}$

Hence, area of $\triangle ABC = \frac{1}{2} |\vec{AB} \times \vec{AC}|$

$= \frac{1}{2} |9\hat{i} + 7\hat{j} + 12\hat{k}|$

$= \frac{1}{2} \sqrt{9^2 + 7^2 + 12^2} = \frac{1}{2} \sqrt{81 + 49 + 144}$

$= \frac{1}{2} \sqrt{274} \text{sq. units}$

28.

Soln. Given, position vector of $A = \hat{i} + \hat{j} + \hat{k}$

Position vector of $B = 2\hat{i} + 5\hat{j}$

Position vector of $C = 3\hat{i} + 2\hat{j} - 3\hat{k}$

Position vector of $D = \hat{i} - 6\hat{j} - \hat{k}$

$\therefore \vec{AB} = (2\hat{i} + 5\hat{j}) - (\hat{i} + \hat{j} + \hat{k}) = \hat{i} + 4\hat{j} - \hat{k}$

And

$\vec{CD} = (\hat{i} - 6\hat{j} - \hat{k}) - (3\hat{i} + 2\hat{j} - 3\hat{k}) = -2\hat{i} - 8\hat{j} + 2\hat{k}$

Now, $|\vec{AB}| = \sqrt{(1)^2 + (4)^2 + (-1)^2} = \sqrt{18}$

$|\vec{CD}| = \sqrt{(-2)^2 + (-8)^2 + (2)^2} = \sqrt{4 + 64 + 4}$

$= \sqrt{72} = 2\sqrt{18}$

Let θ be the angle between $\vec{AB}$ and $\vec{CD}$.

$\therefore \cos \theta = \frac{\vec{AB} \cdot \vec{CD}}{|\vec{AB}| |\vec{CD}|} = \frac{(\hat{i} + 4\hat{j} - \hat{k}) \cdot (-2\hat{i} - 8\hat{j} + 2\hat{k})}{(\sqrt{18})(2\sqrt{18})}$

$= \frac{-2 - 32 - 2}{36} = \frac{-36}{36} = -1$

$\Rightarrow \cos \theta = -1 \Rightarrow \theta = \pi$

Since, angle between $\vec{AB}$ and $\vec{CD}$ is 180° .

$\therefore \vec{AB}$ and $\vec{CD}$ are collinear.

29.

Soln. Let $\vec{d} = x\hat{i} + y\hat{j} + z\hat{k}$

Now, it is given that, $\vec{d}$ is perpendicular to

$\vec{b} = \hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j} - \hat{k}$

$\therefore \vec{d} \cdot \vec{b} = 0$ and $\vec{d} \cdot \vec{c} = 0$

$\Rightarrow x - 4y + 5z = 0$

.....(i)

And $3x + y - z = 0$

.....(ii)

Also, $\vec{d} \cdot \vec{a} = 21$, where $\vec{a} = 4\hat{i} + 5\hat{j} - \hat{k}$

$\Rightarrow 4x + 5y - z = 21$

.....(iii)

Eliminating z from (i) and (ii), we get

$16x + y = 0$

.....(iv)

Eliminating z from (ii) and (iii), we get

$x + 4y = 21$

.....(v)

Solving (iv) and (v), we get

$x = \frac{-1}{3}, y = \frac{16}{3}$

Putting the values of x and y in (i), we get $z = \frac{13}{3}$

$\therefore \vec{d} = \frac{-1}{3}\hat{i} + \frac{16}{3}\hat{j} + \frac{13}{3}\hat{k}$ is the required vector.

30. .

Soln. Given $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $|\vec{a}| = 3, |\vec{b}| = 5, |\vec{c}| = 7$

We have $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

$$\begin{aligned}\Rightarrow \vec{a} + \vec{b} &= -\vec{c} \Rightarrow |\vec{a} + \vec{b}| = |-\vec{c}| \\ \Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + 2(\vec{a} \cdot \vec{b}) &= |\vec{c}|^2 \\ \Rightarrow 9 + 25 + 2|\vec{a}||\vec{b}|\cos\theta &= 49 \\ \Rightarrow 2 \times 3 \times 5 \times \cos\theta &= 49 - 34 = 15 \\ \Rightarrow \cos\theta &= \frac{15}{30} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3} = 60^\circ\end{aligned}$$

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New

SURE SHOT QUESTIONS 2026

Chapter – 11 (Questions)

Three-Dimensional Geometry

Questions

1. Write the direction ratios of the vector $3\vec{a} + 2\vec{b}$ where

$$\vec{a} = \hat{i} + \hat{j} - 2\hat{k} \text{ and } \vec{b} = 2\hat{i} - 4\hat{j} + 5\hat{k}.$$

2. A line passes through the point with position vector $2\hat{i} - 3\hat{j} + 4\hat{k}$ and makes angles 60° , 120° , and 45° with x , y and z - axis respectively. Find the equation of the line in the Cartesian form.

3. Write the Cartesian equation of the following line given in vector form.

$$\vec{r} = (2\hat{i} + \hat{j} - 4\hat{k}) + \lambda(\hat{i} - \hat{j} - \hat{k})$$

4. Find the equation of a plane that cuts the coordinates axes at $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$.

5. Write the vector equation of the line passing through the point $(1, -1, 2)$ and parallel to the line whose equations are $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z+1}{-2}$

6. If the equation of the line AB is

$$\frac{3-x}{3} = \frac{y+2}{-2} = \frac{z+2}{6}$$

7. Find the shortest distance between the following pair of skew lines:

$$\frac{x-1}{2} = \frac{2-y}{3} = \frac{z+1}{4}, \quad \frac{x+2}{-1} = \frac{y-3}{2} = \frac{z}{3}$$

8. Show that the lines

$$\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(3\hat{i} - \hat{j}) \text{ and } \vec{r} = (4\hat{i} - \hat{k}) + \mu(2\hat{i} + 3\hat{k})$$

intersect. Also find their point of intersection.

9. Find the shortest distance between the following lines :

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} \text{ and } \frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$$

10. Find the vector equation of the line passing through the point $(2, 3, -5)$ and making equal angles with the co-ordinate axes.

11. Find the equation of the line which bisects the line segment joining points $A(2, 3, 4)$ and $B(4, 5, 8)$ and is perpendicular to the lines

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7} \text{ and } \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$$

12. The equations of a line are $5x - 3 = 15y - r = 7 = 3 - 10z$. Write the direction cosines of the line and find the coordinates of a point through which it passes.

13. The cartesian equation of a line AB is:

$$\frac{2x-1}{12} = \frac{y+2}{2} = \frac{z-3}{3}$$

Find the direction cosines of a line parallel to line AB.

14. Show that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and

$$\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$$

intersect. Also find their point of intersection. [Delhi 2014]

15. Find the equation of a line passing through the point $(1, 2, -4)$ and perpendicular to two lines

$$\vec{r} = (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$$

and $\vec{r} = (15\hat{i} - 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$.

16. Find the equation of the plane passing through the points $(-1, 2, 0)$, $(2, 2, -1)$ and parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$.

17. Find the value of p , so that the lines $l_1 : \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2}$ and $l_2 : \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$ are perpendicular to each other. Also find the equations of a line passing through a point $(3, 2, -4)$ and parallel to line l_1 .

18. Find the vector and cartesian equations of the line passing through the point $(1, 2, -4)$ and perpendicular to the two lines

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7} \text{ and } \frac{x-15}{3} = \frac{y-29}{8}$$

$$= \frac{z-5}{-5}$$

19. Find the coordinates of the foot of perpendicular and the length of the perpendicular drawn from the point $P(5, 4, 2)$ to the line $\vec{r} = (-\hat{i} + 3\hat{j} + \hat{k}) + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$. Also find the image of P in this line.

20. Prove that the line through $A(0, -1, -1)$ and $B(4, 5, 1)$ intersects the line through $C(3, 9, 4)$ and $D(-4, 4, 4)$.

21. Show that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ are in intersection.

22. Show that the lines $\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$; $\vec{r} = 5\hat{i} - 2\hat{j} + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$ are intersecting. Hence find their point of intersection.

23. If a line makes 60° and 45° angles with the positive directions of x -axis and z -axis respectively, then find the angle that it makes with the positive direction of y -axis. Hence, write the direction cosines of the line.

24. Find the value of λ , So that the lines $\frac{1-x}{3} = \frac{7y-14}{\lambda} = \frac{z-3}{2}$ and $\frac{7-7x}{3\lambda} = \frac{y-5}{1} = \frac{6-z}{5}$ are at right angles. Also, find whether the lines are intersecting or not.

25. Find the vector and cartesian equations of the line through the point $(1, 2, -4)$ and perpendicular to the two lines

$$\vec{r} = (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$$

and

$$\vec{r} = (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k}).$$

26. Find the value of p , so that the lines $l_1 : \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2}$ and $l_2 : \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$ are perpendicular to each other. Also find the equation of a line passing through a point $(3, 2, -4)$ and parallel to line l_1 .

27. Find the direction cosines of the line $\frac{x+2}{2} = \frac{2y-7}{6} = \frac{5-z}{6}$. Also, find the vector equation of the line through the point $A(-1, 2, 3)$ and parallel to the given line.

New

SURE SHOT QUESTIONS 2026

Chapter – 11 (Solutions)

Three-Dimensional Geometry

Questions

1. .

Soln. Getting

$$3\vec{a} + 2\vec{a} = 7\hat{i} - 5\hat{j} + 4\hat{k}$$

∴ D.R's are 7, -5, 4.

2. .

Soln. D-Cosines of line are $\frac{1}{2}, -\frac{1}{2}, \frac{1}{\sqrt{2}}$

Equation of line is:

$$\frac{x-2}{1/2} = \frac{y+3}{-1/2} = \frac{z-4}{1/\sqrt{2}}$$

Or $\sqrt{2}(z-4)$

3. .

Soln. Vector form of a line is given as

$$\vec{r} = 2\hat{i} + \hat{j} - 4\hat{k} + \lambda(\hat{i} - \hat{j} - \hat{k})$$

Direction ratios of above equation are (1, -1, -1) and point through which the line passes is (2, 1, -4).

∴ Cartesian equation is

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

$$= \frac{x-2}{1} = \frac{y-1}{-1} = \frac{z+4}{-1} \text{ or } x-2 = 1-y = -z-4$$

4. .

Soln. The equation of such plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

5. .

Soln. Here, D.R's or required line are (1, 2, -2)

Passing through (1, -1, 2)

Therefore, vector equation of line will be

$$\vec{r}(\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} + 2\hat{j} - 2\hat{k})$$

Or $\frac{x-1}{1} = \frac{y+1}{2} = \frac{z-2}{-2}$

6. .

Soln. Given,

$$\frac{3-x}{-3} = \frac{y+2}{-2} = \frac{z+2}{6}$$

$$\frac{x-3}{3} = \frac{y+2}{-2} = \frac{z+2}{6}$$

Or Direction ratios are 3, -2, 6.

DRF Parallel line 3, -2, 6

$$\text{D.C's } l, m, n : l = \frac{a}{\sqrt{a^2 + b^2 + c^2}},$$

$$h = \frac{b}{\sqrt{a^2 + b^2 + c^2}},$$

$$n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

$$\text{D.C's} = \frac{-2}{\sqrt{9+4+36}}, \frac{6}{\sqrt{9+4+36}}, \frac{6}{\sqrt{9+4+36}}$$

$$\text{D.C's are } \frac{-2}{7}, \frac{6}{7}, \frac{6}{7}$$

7. .

Soln. $\vec{r}_1 = \hat{i} + 2\hat{j} - \hat{k} + \lambda(2\hat{i} - 3\hat{j} + 4\hat{k})$

And $\vec{r}_2 = -2\hat{i} + 3\hat{j} + \mu(-\hat{i} + 2\hat{j} + 3\hat{k})$

Let $\vec{a}_1 = \hat{i} + 2\hat{j} - \hat{k}, \vec{b}_1 = 2\hat{i} - 3\hat{j} + 4\hat{k}$

And $\vec{a}_2 = -2\hat{i} + 3\hat{j}, \vec{b}_2 = -\hat{i} + 2\hat{j} + 3\hat{k}$

$$\vec{a}_2 - \vec{a}_1 = -3\hat{i} + \hat{j} + \hat{k}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 4 \\ -1 & 2 & 3 \end{vmatrix}$$

$$= -17\hat{i} - 10\hat{j} + \hat{k}$$

The required shortest distance

$$= \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

$$= \frac{(-3\hat{i} + \hat{j} + \hat{k}) \cdot (-17\hat{i} - 10\hat{j} + \hat{k})}{\sqrt{(-17)^2 + (-10)^2 + (1)^2}}$$

$$= \frac{42}{\sqrt{390}} \text{ units}$$

8.

Soln. General points on the lines are

$$(1+3\lambda)\hat{i} + (1-\lambda)\hat{j} - \hat{k}$$

And $(4+2\mu)\hat{i} + 3(3\mu-1)\hat{k}$

Lines intersect if

$$1+3\lambda = 4+2\mu \dots\dots (i)$$

$$1-\lambda = 0 \dots\dots (ii)$$

$$\text{and } 3\mu-1 = -1 \dots\dots (iii)$$

For some λ & μ

From (ii) & (iii), $\lambda=1, \mu=0$

Substituting in equation (i)

Since, $1+3(1)=4+2(0)$ is true

$\therefore$ lines intersect

Point of intersection is:

$$4\hat{i} - \hat{k} \text{ or } (4, 0, -1)$$

9. Soln.

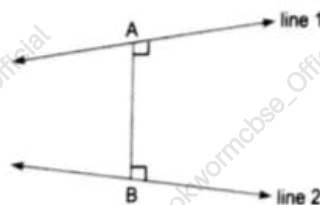
Let $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} = \lambda$ and

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} = k$$

Now, let's take a point on first line as

$$A(\lambda+3, -2\lambda+5, \lambda+7) \text{ and let}$$

$$B(7k-1, -6k-1, k-1) \text{ be point on the second line}$$



The direction ratio of the line AB

$$7k-\lambda-4, -6k+2\lambda-6, k-\lambda-8$$

Now, as AB is the shortest distance between line 1 and

line 2 so, $(7k-\lambda-4) \times 1 + (-6k+2\lambda-6) \times (-2) + (k-\lambda-8) \times 1 = 0 \dots\dots (i)$

And $(7k-\lambda-4) \times 7 + (-6k+2\lambda-6) \times (-6) + (k-\lambda-8) \times 1 = 0 \dots\dots (ii)$

Solving equation (i) and (ii), we get

$$\lambda = 0 \text{ and } k = 0$$

$$\therefore A = (3, 5, 7) \text{ and } B = (-1, -1, -1)$$

$$\therefore AB = \sqrt{(3+1)^2 + (5+1)^2 + (7+1)^2} = \sqrt{16+36+64}$$

$$= \sqrt{116} \text{ units} = 2\sqrt{29} \text{ units}$$

10.

Ans. Let the line makes angle α, β and γ with the co-ordinate axes.

$$\text{Given, } \alpha = \beta = \gamma$$

$$\text{Now, } \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\Rightarrow 3\cos^2\alpha = 1$$

$$\Rightarrow \cos\alpha = \pm \frac{1}{\sqrt{3}} \therefore (l, m, n) = \pm \frac{1}{\sqrt{3}}(1, 1, 1)$$

Dr's of the line is $(1, 1, 1)$

$\therefore$ Required vector equation of line passing through $(2, 3, -5)$ and having dr's $(1, 1, 1)$ is

$$\vec{r} = (2\hat{i} + 3\hat{j} - 5\hat{k}) + \lambda(\hat{i} + \hat{j} + \hat{k}), \lambda \in \mathbb{R}$$

11.

Let the required equation of the line be

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} \quad \dots(i)$$

Now, line (i) passes through the mid-point of A(2, 3, 4) and B(4, 5, 8).

$$\therefore x_1 = \frac{2+4}{2} = 3, y_1 = \frac{3+5}{2} = 4, z_1 = \frac{4+8}{2} = 6$$

$$\Rightarrow (x_1, y_1, z_1) = (3, 4, 6) \quad \dots(ii)$$

Also, line (i) is perpendicular to the lines

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7} \quad \text{and} \quad \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$$

$$\therefore 3a - 16b + 7c = 0 \quad \dots(iii)$$

$$3a + 8b - 5c = 0 \quad \dots(iv)$$

Solving (iii) and (iv), we get

$$\frac{a}{16 \times 5 - 7 \times 8} = \frac{b}{7 \times 3 - 3 \times (-5)} = \frac{c}{3 \times 8 - 3 \times (-16)}$$

$$\Rightarrow \frac{a}{80-56} = \frac{b}{21+15} = \frac{c}{24+48} \Rightarrow \frac{a}{24} = \frac{b}{36} = \frac{c}{72}$$

$$\Rightarrow \frac{a}{2} = \frac{b}{3} = \frac{c}{6}$$

$$\text{So, } a = 2, b = 3, c = 6 \quad \dots(v)$$

Using (i), (ii) and (v), we get the required equation of line is

$$\frac{x-3}{2} = \frac{y-4}{3} = \frac{z-6}{6}$$

12. .

Soln.

The given line is $5x - 3 = 15y + 7 = 3 - 10z$

$$\Rightarrow \frac{x-\frac{3}{5}}{\frac{1}{5}} = \frac{y+\frac{7}{15}}{\frac{1}{15}} = \frac{z-\frac{3}{10}}{-\frac{1}{10}}$$

$$\text{Its direction ratios are } \frac{1}{5}, \frac{1}{15}, -\frac{1}{10}$$

i.e., Its direction ratios are proportional to 6, 2, -3.

$$\text{Now, } \sqrt{6^2 + 2^2 + (-3)^2} = 7$$

$$\therefore \text{Its direction cosines are } \frac{6}{7}, \frac{2}{7}, -\frac{3}{7}$$

13. .

Soln.

The cartesian equation of line AB is

$$\frac{2x-1}{12} = \frac{y+2}{2} = \frac{z-3}{3}$$

$$\text{Can be rearranged as } \frac{x-\frac{1}{2}}{6} = \frac{y+2}{2} = \frac{z-3}{3}$$

$$\text{So, } a = 6, b = 2, c = 3$$

$$\Rightarrow \sqrt{a^2 + b^2 + c^2} = \sqrt{6^2 + 2^2 + 3^2} = 7$$

$$\therefore \text{Required direction cosines are } \frac{6}{7}, \frac{2}{7}, \frac{3}{7}$$

14. .

Soln. Any point on the line

$$\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7} = r \quad (\text{say}) \quad \dots(i)$$

$$\text{Is } (3r-1, 5r-3, 7r-5).$$

Any point on the line

$$\frac{x-2}{11} = \frac{y-4}{3} = \frac{z-6}{5} = k \quad (\text{say}) \quad \dots(ii)$$

$$\text{Is } (k+2, 3k+4, 5k+6)$$

For lines (i) and (ii) to intersect, we must have

$$3r-1 = k+2, 5r-3 = 3k+4, 7r-5 = 5k+6$$

$$\text{On solving these, we get } r = \frac{1}{2}, k = -\frac{3}{2}$$

$\therefore$ Lines (i) and (ii) intersect and their point of

$$\text{intersection is } \left(\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2} \right)$$

15. .

Soln. Let the direction ratios of required line be a, b, c, since, the line is perpendicular to

$$\vec{r} = (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$$

$$\text{And } \vec{r} = (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$$

$$\therefore 3a - 16b + 7c = 0$$

$$\text{and } 3a + 8b - 5c = 0.$$

Solving by cross multiplication, we get

$$\frac{a}{80-56} = \frac{b}{21+15} = \frac{c}{24+48}$$

$$\Rightarrow \frac{a}{24} = \frac{b}{36} = \frac{c}{72}$$

$$\therefore \text{the direction ratios of line : } 2, 3, 6.$$

Hence, required line through the point (1, 2, -4) is

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$$

16. .

Soln. The equation of a plane passing through (-1, 2, 0) is

$$a(x+1) + b(y-2) + c(z-0) = 0 \quad \dots(i)$$

It passes through (2, 2, -1)

$$\therefore a(2+1)+b(2-2)+c(-1-0)=0$$

$$3a+0b-c=0$$

.....(ii)

The given lines is

$$\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$$

$$\frac{x-1}{1} = \frac{y+\frac{1}{2}}{1} = \frac{z+1}{-1}$$

i.e.,

$\therefore$ d. r. s of line are 1, 1, -1

The plane (i) is parallel to the given line

$$a+b-c=0$$

.....(iii)

Solving (ii) and (iii) by cross multiplication, we get

$$\frac{a}{1} = \frac{b}{2} = \frac{c}{3}$$

i.e., direction ratios of normal to the plane are 1, 2, 3.

$$\therefore 1(x+1)+2(y-2)+3(z-0)=0$$

$$\text{i.e., } x+2y+3z=3$$

17. .

Soln. The given lines are

$$l_1: \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2}$$

$$\Rightarrow l_1: \frac{x-1}{-3} = \frac{y-2}{p/7} = \frac{z-3}{2}$$

.....(i)

$$l_2: \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$$

And

$$\Rightarrow l_2: \frac{x-1}{-3p/7} = \frac{y-5}{1} = \frac{z-6}{-5}$$

.....(ii)

Since l_1 and l_2 are perpendicular

$$\therefore (-3) \cdot \left(-\frac{3p}{7}\right) + \left(\frac{p}{7}\right) \cdot 1 + 2 \cdot (-5) = 0$$

$$\Rightarrow \frac{10p}{7} - 10 = 0$$

$$\Rightarrow p = 7$$

$\therefore$ Equation of the line passes through (3, 2, -4)

and parallel to l_1 is

$$\frac{x-3}{-3} = \frac{y-2}{1} = \frac{z+4}{2}$$

18. .

Soln. Let the cartesian equation of line passing through (1, 2, -4) be

$$\frac{x-1}{a} = \frac{y-2}{b} = \frac{z+4}{c}$$

.....(i)

Given lines are

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$$

.....(ii)

$$\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$$

.....(iii)

Obviously parallel vectors $\vec{b}_1, \vec{b}_2$ and $\vec{b}_3$ of (i), (ii) and (iii) respectively are given as

$$\vec{b}_1 = a\hat{i} + b\hat{j} + c\hat{k}; \vec{b}_2 = 3\hat{i} - 16\hat{j} + 7\hat{k}; \vec{b}_3 = 3\hat{i} + 8\hat{j} - 5\hat{k}$$

According to question

$$(i) \perp (ii) \Rightarrow \vec{b}_1 \perp \vec{b}_2 \Rightarrow \vec{b}_1 \cdot \vec{b}_2 = 0$$

$$(i) \perp (iii) \Rightarrow \vec{b}_1 \perp \vec{b}_3 \Rightarrow \vec{b}_1 \cdot \vec{b}_3 = 0$$

$$\text{Hence, } 3a - 16b + 7c = 0$$

.....(iv)

$$\text{And } 3a + 8b - 5c = 0$$

.....(v)

From equation (iv) and (v), we get

$$\frac{a}{80-56} = \frac{b}{21+15} = \frac{c}{24+48}$$

$$\Rightarrow \frac{a}{24} = \frac{b}{36} = \frac{c}{72} \Rightarrow \frac{a}{2} = \frac{b}{3} = \frac{c}{6} = \lambda$$

(say)

$$\Rightarrow a = 2\lambda, b = 3\lambda, c = 6\lambda$$

Putting the value of a, b, c in (i), we get the required cartesian equation of line as

$$\frac{x-1}{2\lambda} = \frac{y-2}{3\lambda} = \frac{z+4}{6\lambda} \Rightarrow \frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$$

Hence, vector equation is

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$$

19. .

Soln. Given line is

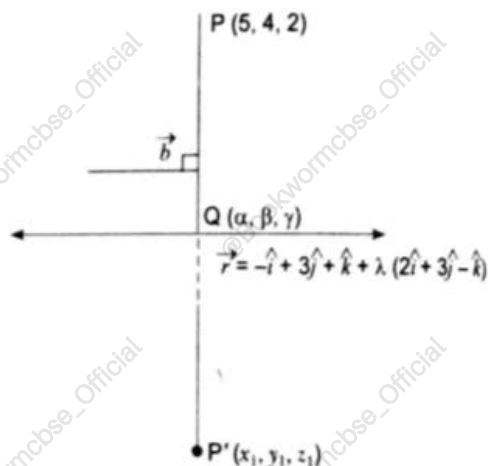
$$\vec{r} = -\hat{i} + 3\hat{j} + \hat{k} + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$$

It can be written in cartesian form as

$$\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$$

.....(i)

Let Q (α, β, γ) be the foot of perpendicular drawn from



P(5, 4, 2) to the line (i) and P' (x₁, y₁, z₁) be the image of P on the line (i)

∴ Q(α, β, γ) lie on line (i)

$$\therefore \frac{\alpha+1}{2} = \frac{\beta-3}{3} = \frac{\gamma-1}{-1} = \lambda \text{ (say)}$$

$$\Rightarrow \alpha = 2\lambda - 1; \beta = 3\lambda + 3 \text{ and } \gamma = -\lambda + 1$$

.....(ii)

$$\text{Now, } \vec{PQ} = (\alpha - 5)\hat{i} + (\beta - 4)\hat{j} + (\gamma - 2)\hat{k}$$

Parallel vector of line (i) $\vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$.

$$\text{Obviously } \vec{PQ} \perp \vec{b} \Rightarrow \vec{PQ} \cdot \vec{b} = 0$$

$$2(\alpha + 5) + 3(\beta - 4) + (-1)(\gamma - 2) = 0$$

$$\Rightarrow 2\alpha - 10 + 3\beta - 12 - \gamma + 2 = 0$$

$$\Rightarrow 2\alpha + 3\beta - \gamma - 20 = 0$$

$$\Rightarrow 2(2\lambda - 1) + 3(3\lambda + 3) - (-\lambda + 1) - 20 = 0$$

$$= 0 \text{ [Putting value of } \alpha, \beta, \gamma \text{ from (ii)]}$$

$$\Rightarrow 4\lambda - 2 + 9\lambda + 9 + \lambda - 1 - 20 = 0 \Rightarrow 14\lambda - 14 = 0 \Rightarrow \lambda = 1$$

Hence the coordinates of foot of perpendicular Q are (2 × 1 - 1, 3 × 1 + 3, -1 + 1) i.e., (1, 6, 0)

∴ Length of perpendicular =

$$\sqrt{(5-1)^2 + (4-6)^2 + (2-0)^2}$$

$$= \sqrt{16+4+4} = \sqrt{24} = 2\sqrt{6} \text{ units.}$$

Also, since Q is mid - point of PP'

$$\therefore 1 = \frac{x_1 + 5}{2} \Rightarrow x_1 = -3$$

$$6 = \frac{y_1 + 4}{2} \Rightarrow y_1 = 8$$

$$0 = \frac{z_1 + 2}{2} \Rightarrow z_1 = -2$$

Therefore required image is (-3, 8, -2).

20. .

Soln. The equation of line AB is given by

$$\frac{x-0}{4-0} = \frac{y+1}{5+1} = \frac{z+1}{1+1} = \lambda \text{ (say)}$$

$$\Rightarrow x = 4\lambda, y = 6\lambda - 1, z = 2\lambda - 1$$

The coordinates of a general point on AB are (4λ, 6λ, -1, 2λ - 1)

The equation of line CD is given by

$$\frac{x-3}{3+4} = \frac{y-9}{9-4} = \frac{z-4}{4-4} = \mu \text{ (say)}$$

$$\Rightarrow x = 7\mu + 3, y = 5\mu + 9, z = 4$$

The coordinates of a general point on CD are (7μ + 3, 5μ + 9, 4)

If the line AB and CD intersect then they have a common point. So, for some values of λ and μ, we must have

$$4\lambda = 7\mu + 3, 6\lambda - 1 = 5\mu + 9, 2\lambda - 1 = 4$$

$$\Rightarrow 4\lambda - 7\mu = 3 \text{(i), } 6\lambda - 5\mu = 10 \text{(ii) and } \lambda = \frac{5}{2} \text{(iii)}$$

Substituting $\lambda = \frac{5}{2}$ in (ii), we get $\mu = 1$

Since $\lambda = \frac{5}{2}$ and $\mu = 1$ satisfy (i), so the given lines AB and CD intersect.

21. .

Sol. Any point on the line

$$\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7} = r \text{ (say)}$$

.....(i)

Is (3r - 1, 5r - 3, 7r - 5).

Any point on the line

$$\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5} = k \text{ (say)}$$

.....(ii)

Is (k + 2, 3k + 4, 5k + 6)

For lines (i) and (ii) to intersect, we must have

$$3r - 1 = k + 2, 5r - 3 = 3k + 4, 7r - 5 = 5k + 6$$

$$\text{On solving these, we get } r = \frac{1}{2}, k = -\frac{3}{2}$$

∴ Lines (i) and (ii) intersect and their point of

$$\text{intersection is } \left(\frac{1}{2}, \frac{1}{2}, -\frac{3}{2} \right)$$

22. .

Soln. The given lines are

$$\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k}) \text{ and}$$

$$\vec{r} = 5\hat{i} - 2\hat{j} + \mu(3\hat{i} + 2\hat{j} + 6\hat{k})$$

$$\Rightarrow \vec{r} = (3 + \lambda)\hat{i} + (2 + 2\lambda)\hat{j} + (2\lambda - 4)\hat{k} \quad \dots(i)$$

$$\text{and } \vec{r} = (5 + 3\mu)\hat{i} + (2\mu - 2)\hat{j} + 6\mu\hat{k} \quad \dots(ii)$$

If these lines intersect, they must have a common point.

So, we must have

$$(3 + \lambda)\hat{i} + (2 + 2\lambda)\hat{j} + (2\lambda - 4)\hat{k} = (5 + 3\mu)\hat{i} + (2\mu - 2)\hat{j} + 6\mu\hat{k}$$

$$\Rightarrow 3 + \lambda = 5 + 3\mu \Rightarrow \lambda - 3\mu = 2,$$

$$2 + 2\lambda = 2\mu - 2 \Rightarrow \lambda - \mu = -2$$

$$\text{and } 2\lambda - 4 = 6\mu \Rightarrow \lambda - 3\mu = 2$$

$$\Rightarrow \lambda = -4, \mu = -2.$$

$\therefore$ The given lines intersect and their point of intersection is $(-1, -6, -12)$.

23. .

Soln. Let θ be the angle made by the line with y-axis.

$$\text{Then, } \cos^2 60^\circ + \cos^2 45^\circ + \cos^2 \theta = 1$$

$$\Rightarrow \frac{1}{4} + \frac{1}{2} + \cos^2 \theta = 1$$

$$\Rightarrow \cos^2 \theta = 1 - \frac{1}{2} - \frac{1}{4} \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

$\therefore$ Direction cosines of the line are

$$\langle \cos 60^\circ, \cos 45^\circ, \cos 60^\circ \rangle \text{ i.e., } \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2}$$

24. .

Soln. The given lines are

$$l_1: \frac{x-1}{-3} = \frac{y-2}{\lambda/7} = \frac{z-3}{2}$$

$$\text{And } l_2: \frac{x-1}{-3\lambda/7} = \frac{y-5}{1} = \frac{z-6}{-5}$$

$$\text{Now, } l_1 \perp l_2$$

[Given]

$$\therefore (-3)\left(-\frac{3\lambda}{7}\right) + \frac{\lambda}{7} - 10 = 0$$

$$\Rightarrow \frac{9\lambda}{7} + \frac{\lambda}{7} - 10 = 0 \Rightarrow \frac{10\lambda}{7} = 10 \Rightarrow \lambda = 7$$

Since for $\lambda = 7$, given lines are at right angle.

$\therefore$ Lines are intersecting.

25. .

Soln. The given lines are

$$\vec{r} = (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$$

$$\text{and } \vec{r} = (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$$

Equation of any line through $(1, 2, -4)$ with d.r's l, m, n is

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + p(l\hat{i} + m\hat{j} + n\hat{k}) \quad \dots(i)$$

Since, the required line is perpendicular to both the given lines.

$$\therefore 3l - 16m + 7n = 0 \text{ and } 3l + 8m - 5n = 0$$

$$\Rightarrow \frac{l}{80-56} = \frac{m}{21+15} = \frac{n}{24+48} \Rightarrow \frac{l}{2} = \frac{m}{3} = \frac{n}{6}$$

$\therefore$ From (i), the required line is

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + p(2\hat{i} + 3\hat{j} + 6\hat{k})$$

Here, the position vector of passing point is

$$\vec{a} = \hat{i} + 2\hat{j} - 4\hat{k} \text{ and parallel vector is } \vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

$\therefore$ Cartesian equation of line is given by

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$$

26. .

Soln. The given lines are

$$l_1: \frac{x-1}{-3} = \frac{y-2}{p/7} = \frac{z-3}{2}$$

$$l_2: \frac{x-1}{-3p/7} = \frac{y-5}{1} = \frac{z-6}{-5}$$

$\therefore l_1$ is perpendicular to l_2 .

$$\therefore (-3)\left(-\frac{3p}{7}\right) + \frac{p}{7} \cdot 1 + 2(-5) = 0$$

$$\Rightarrow \frac{9p}{7} + \frac{p}{7} = 10 \Rightarrow \frac{10p}{7} = 10 \Rightarrow p = 7$$

Now, eq. of the line passing through $(3, 2, -4)$ and parallel to l_1 is

$$\frac{x-3}{-3} = \frac{y-2}{1} = \frac{z+4}{2}$$

27. .

$$\text{Soln. The given line is } \frac{x+2}{2} = \frac{2y-7}{6} = \frac{5-z}{6}$$

$$\Rightarrow \frac{x+2}{2} = \frac{y-\frac{7}{2}}{3} = \frac{z-5}{-6} \quad \dots(i)$$

Its d.r's are 2, 3, -6

$$\therefore \sqrt{2^2 + 3^2 + (-6)^2} = 7$$

$$\therefore \text{Its d.c's are } \frac{2}{7}, \frac{3}{7}, -\frac{6}{7}$$

Eq. of a line through $(-1, 2, 3)$ and parallel to (i) is

$$\frac{x+1}{2} = \frac{y-2}{3} = \frac{z-3}{-6} = \lambda \text{ (say)}$$

$\therefore$ Vector equation of a line passing through $(-1, 2, 3)$ and parallel to (i) is given by

$$\vec{r} = (-\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} - 6\hat{k}).$$

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New

SURE SHOT QUESTIONS 2026

Chapter – 12 (Questions)

Linear Programming

Questions

- Find graphically, the maximum value of $z = 2x + 5y$, subject to constraints given below:
 $2x + 4y \leq 8$
 $3x + y \leq 6$
 $x + y \leq 4$
 $x \geq 0, y \geq 0$
- Solve the following linear programming problem graphically :
Maximize $Z = 7x + 10y$
Subject to the constraints
 $4x + 6y \leq 240$
 $6x + 3y \leq 240$
 $x \geq 10$
 $x \geq 0, y \geq 0$
- Solve the following LPP graphically: Minimize $z = 5x + 7y$ Subject to the constraints $2x + y \geq 8$
 $x + 2y \geq 10$
 $x, y \geq 0$
- Solve the following LPP graphically: Minimise $Z = 5x + 10y$ Subject to constraints $x + 2y \leq 120$, $x + y \geq 60$, $x - 2y \geq 0$ and $x, y \geq 0$
- Solve the following linear programming problem graphically :
Maximize $Z = 2x + 3y$ subject to the constraints:
 $x - y \leq 6$
 $x \geq 2$
 $y \leq 3$
 $x, y \geq 0$
- Solve the following L.P.P. graphically:
Minimize: $Z = 6x + 3y$
Subject to constraints : $4x + y \geq 80$; $x + 5y \geq 115$;
 $3x + 2y \leq 150$; $x, y \geq 0$.
- Solve the following linear programming problem graphically:
Maximise $z = -3x - 5y$
Subject to the constraints
[2023]
 $-2x + y \leq 4$
 $x + y \geq 3$
 $x - 2y \leq 2$
 $x \geq 0, y \geq 0$
- Solve the following linear programming problem graphically.
Maximize: $P = 70x + 40y$
Subject to: $3x + 2y \leq 9$, $3x + y \leq 9$, $x \geq 0, y \geq 0$.
- Maximise $Z = x + 2y$ Subject to the constraints:
 $x + 2y \geq 100$, $2x - y < 0$, $2x + y \leq 200$, $x, y \geq 0$
Solve the above LPP graphically.
- Let $z = X + y$, then the maximum of z subject to constraints $y \geq |x| - 1$, $y \leq 1 - |x|$, $x \geq 0$, $y \geq 0$ is
- Find graphically, the maximum value of $z = 2x + 5y$, subject to constraints given below:
 $2x + 4y \leq 8$, $3x + y \leq 6$, $x + y \leq 4$; $x \geq 0, y \geq 0$
- Maximise $z = 8x + 9y$ subject to the constraints given below: $2x + 3y \leq 6$, $3x - 2y \leq 6$, $y \leq 1$; $x, y \geq 0$
- Minimise $Z = 30x + 20y$ subject to $x + y \leq 8$, $x + 4y \geq 12$, $5x + 8y \geq 20$, $x, y \geq 0$
 $O(0, 0)$, $A\left(\frac{20}{3}, \frac{4}{3}\right)$ and $C(0, 8)$.

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New

SURE SHOT QUESTIONS 2026

Chapter – 12 (Solutions)

Linear Programming

Questions

1.

Soln. We first convert the inequalities into equations to obtain lines

$$2x + 4y = 8 \quad \text{.....(i)}$$

$$3x + y = 6 \quad \text{.....(ii)}$$

$$x + y = 4 \quad \text{.....(iii)}$$

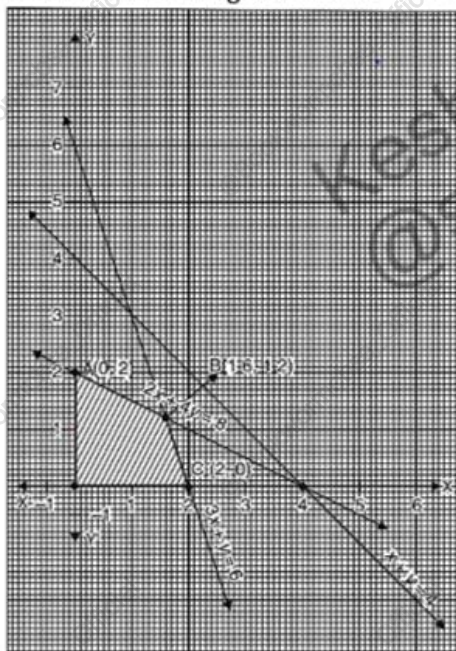
$$x = 0$$

$$\text{And } y = 0$$

We need to maximize the objective function

$$z = 2x + 5y$$

These lines are drawn and the feasible region of the L.P.P. is the shaded region:



The point of intersection of (i) and (ii) is B (1.6, 1.2)

The coordinates of the corner points of the feasible region are O(0, 0), A(0, 2), B(1.6, 1.2) and C(2, 0).

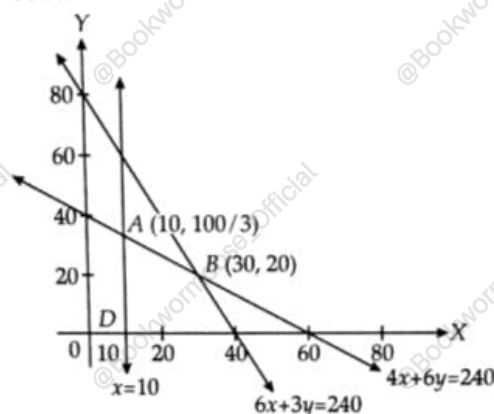
The value of the objective function at these points are given in the following table:

Corner Points	Value of the objective function $z = 2x + 5y$
O(0, 0)	$2 \times 0 + 5 \times 0 = 0$
A (0, 2)	$2 \times 0 + 5 \times 2 = 10$ maximum
B(1.6, 1.2)	$2 \times 1.6 + 5 \times 1.2 = 9.2$
C(2, 0)	$2 \times 2 + 5 \times 0 = 4$

Out of these values of z, the maximum value of z is 10 which is attained at the point (0, 2). Thus the maximum value of z is 10.

2.

Soln.



Maximise $z = 7x + 10y$, subject to $4x + 6y \leq 240$;
 $6x + 3y \leq 240$; $x \geq 10$, $x \geq 0$, $y \geq 0$

Correct graph of three lines

For correct shading

$$Z(A) = Z\left(10, \frac{200}{6}\right) = 70 + 10 \times \frac{100}{3} = 403\frac{1}{3}$$

$$Z(B) = Z(30, 20) = 210 + 200 = 410$$

$$Z(C) = Z(40, 0) = 280 + 0 = 280$$

$$Z(D) = Z(10, 0) = 70 + 0 = 70$$

or $\text{Max}(= 410)$ at $x = 30, y = 20$

3. .

Soln. We have, minimize $z = 5x + 7y$,

Subject to constraints,

$$2x + y \geq 8, x + 2y \geq 10, x, y \geq 0$$

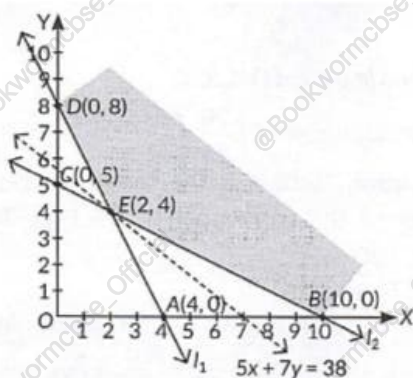
To solve LPP graphically, we convert inequations into equations.

Now,

$$l_1 : 2x + y = 8, l_2 : x + 2y = 10 \text{ and } x = 0, y = 0$$

l_1 and l_2 intersect at $E(2, 4)$

Let us draw the graph of these equations as shown below.



The corner points of the feasible region are $D(0, 8)$, $B(10, 0)$ and $E(2, 4)$.

Corner points	Value of $z = 5x + 7y$
$D(0, 8)$	56
$B(10, 0)$	50
$E(2, 4)$	38 (Minimum)

From the table, we find that 38 is the minimum value of z at $E(2, 4)$. Since the region is unbounded, so we draw the graph of inequality $5x + 7y < 38$ to check whether the resulting open half plane has any point common with the feasible region. Since it has no point in common. So, the minimum value of z is obtained at $E(2, 4)$ and the minimum value of $z = 38$.

4. .

Soln. We have, Minimise $Z = 5x + 10y$,

Subject to constraints:

$$x + 2y \leq 120$$

$$x + y \geq 60$$

$$x - 2y \geq 0$$

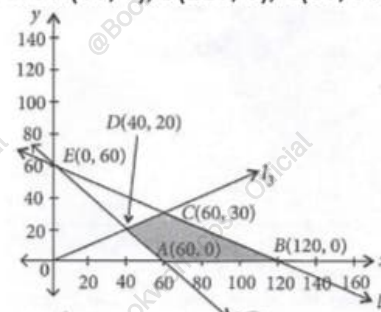
and $x, y \geq 0$

To solve L.P.P graphically, we convert inequations into equations.

$$l_1 : x + 2y = 120, l_2 : x + y = 60, l_3 : x - 2y = 0 \text{ and } x = 0, y = 0$$

l_1 and l_2 intersect at $E(0, 60)$, l_1 and l_3 intersect at $C(60, 30)$, l_2 and l_3 intersect at $D(40, 20)$.

The shaded region ABCD is the feasible region and is bounded. The corner points of the feasible region are $A(60, 0)$, $B(120, 0)$, $C(60, 30)$ and $D(40, 20)$.

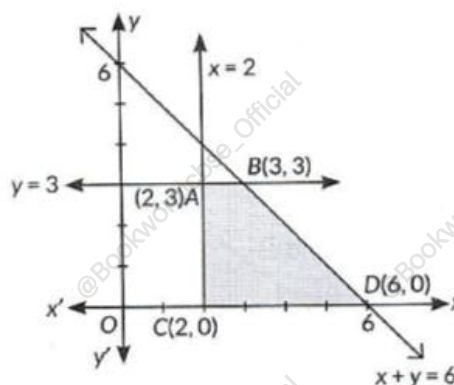


Corner points	Value of $Z = 5x + 10y$
$A(60, 0)$	300 ← (Minimum)
$A(120, 0)$	600
$C(60, 30)$	600
$D(40, 20)$	400

Hence, Z is minimum at $A(60, 0)$ i.e., 300.

5. .

Ans. Consider the graph given below:



Corner points are $A(2, 3)$, $B(3, 3)$, $C(2, 0)$ and $D(6, 0)$

$$Z_A = 4 + 9 = 13$$

$$Z_B = 6 + 9 = 15$$

$$Z_C = 4$$

$$Z_D = 12$$

Maximum value of z is 15 at $(3, 3)$.

6.

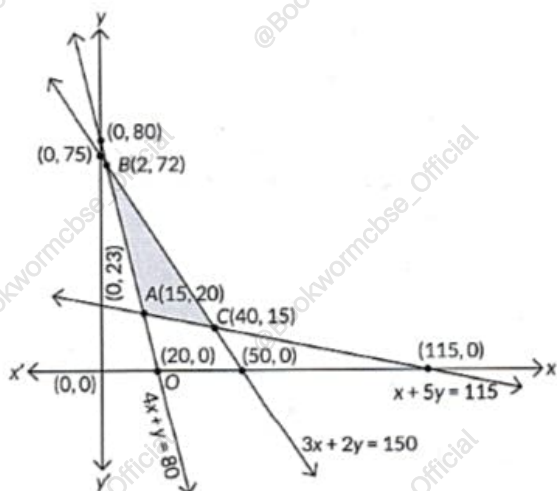
Ans. Given, L.P.P.

Minimise, $Z = 6x - 3y$

Subject to constraints:

$$4x + y \geq 80; x + 5y \geq 115; 3x + 2y \leq 150; x, y \geq 0$$

Graph of the lines $4x + y = 80$, $x + 5y = 115$ and $3x + 2y = 150$ is given below.



Intersecting point of lines $x + 5y = 115$ and $4x + y = 80$ is $A(15, 20)$.

Intersecting point of lines $3x + 2y = 150$ and $4x + y = 80$ is $B(2, 72)$.

Intersecting point of lines $3x + 2y = 150$ and $x + 5y = 115$ is $C(40, 15)$.

$\therefore$ The corner points of the feasible region ABC are $A(15, 20)$, $B(2, 72)$ and $C(40, 15)$.

The value of Z at these points are as follows:

Corner point	$Z = 6x + 3y$
$A(15, 20)$	$6 \times 15 + 3 \times 20 = 150 \leftarrow \text{Minimum}$
$B(2, 72)$	$6 \times 2 + 3 \times 72 = 228$
$C(40, 15)$	$6 \times 40 + 3 \times 15 = 285$

7.

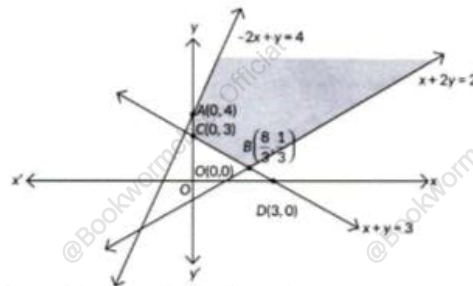
Soln. We have, maximise $z = -3x - 5y$

Converting the given inequations into equations, we get

$$-2x + y = 4 \quad \text{.....(i)}$$

$$x + y = 3 \quad \text{.....(ii)}$$

$$x - 2y = 2 \quad \text{.....(iii)}$$



We draw the graph of these lines.

As, $x \geq 0$, $y \geq 0$ so the solution lies in first quadrant.

From graph, corner point of feasible region are $A(0, 4)$, $B(8/3, 1/3)$ and $C(0, 3)$

The value of z at these corner points are shown as:

Corner points	$Z = -3x - 5y$
$A(0, 4)$	-20
$B(8/3, 1/3)$	-29/3
$C(0, 3)$	-15

Hence maximum value of $Z = \frac{-29}{3}$.

8.

Soln. We have, maximize $P = 70x + 40y$

Subject to: $3x + 2y \leq 9$

$$3x + y \leq 9$$

$$x \geq 0, y \geq 0$$

Convert all inequations into equation, we get

$$3x + 2y = 9 \quad \text{.....(i)}$$

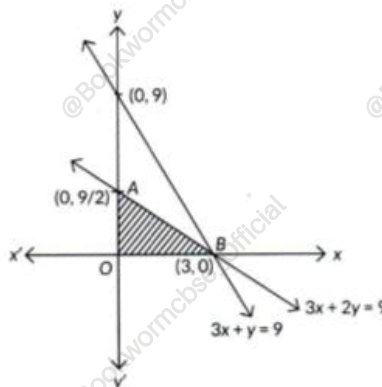
$$3x + y = 9 \quad \text{.....(ii)}$$

$$x = 0 \text{ and } y = 0$$

Solving (i) and (ii), we get

$$x = 3, y = 0$$

So, point of intersection of equation (i) and (ii) are $(3, 0)$.



The given shaded region is the feasible region.

The corner points of the feasible region are $O(0, 0)$, $A(0, 9/2)$ and $B(3, 0)$.

Corner points	Value of $p = 70x + 40y$
$O(0, 0)$	$70 \times 0 + 40 \times 0 = 0$
$A(0, 9/2)$	$70 \times 0 + 40 \times \frac{9}{2} = 180$
$B(3, 0)$	$70 \times 3 + 40 \times 0 = 210$ (maximum)

So, P is maximum at point B(3, 0).

9.

Soln. Maximise $Z = x + 2y$,

Subject to constraints: $x + 2y \geq 100$, $2x - y < 0$,

$2x + y \leq 200$ and $x, y \geq 0$.

Converting the inequations into equations, we obtain the lines

$l_1: x + 2y = 100$ (i)

$l_2: 2x - y = 0$ (ii)

$l_3: 2x + y = 200$ (iii)

$l_4: x = 0$ (iv)

and $l_5: y = 0$ (v)

By intercept form, we get

$l_1: \frac{x}{100} + \frac{y}{50} = 1$

$\Rightarrow$ The line l_1 meets the coordinate axes at (100, 0) and (0, 50).

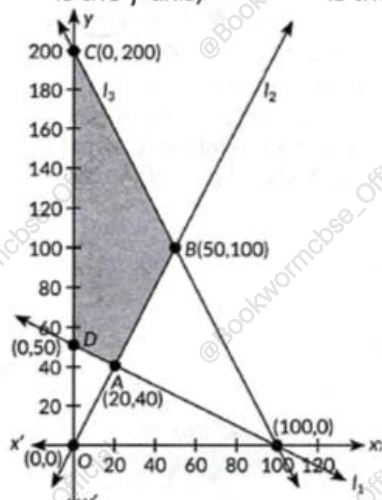
$l_2: 2x = y$

$\Rightarrow$ The line l_2 passes through origin and (50, 100).

$l_3: \frac{x}{100} + \frac{y}{200} = 1$

$\Rightarrow$ The line l_3 meets the coordinates axes at (100, 0) and (0, 200).

$l_4: x = 0$ is the y-axis, $l_5: y = 0$ is the x-axis.



Now, plotting the above points on the graph, we get the feasible region of the LPP as shaded region ABCD. The coordinates of the corner points of the feasible region ABCD are A(20, 40), B(50, 100), C(0, 200), D(0, 50),

Now, $Z_A = 20 + 2 \times 40 = 100$

$Z_B = 50 + 2 \times 100 = 250$, $Z_C = 0 + 2 \times 200 = 400$

$Z_D = 0 + 2 \times 50 = 100$

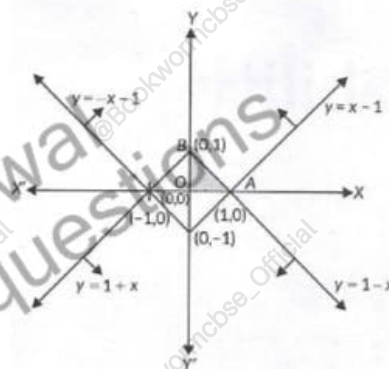
$\therefore$ Z is maximum at C(0, 200) and having value 400.

10.

Soln. Max $z = x + y$

Subject to

$y \geq \begin{cases} x-1, & x \geq 0 \\ -x-1, & x < 0 \end{cases}$, $y \leq \begin{cases} 1-x, & x \geq 0 \\ 1+x, & x < 0 \end{cases}$ and $x, y \geq 0$



The common region OAB is showing with shades.

Corner Points	Value of $z = x + y$
$O(0, 0)$	$z = 0 + 0 = 0$
$A(1, 0)$	$z = 1 + 0 = 1$ (Maximum)
$B(0, 1)$	$z = 0 + 1 = 1$ (Maximum)

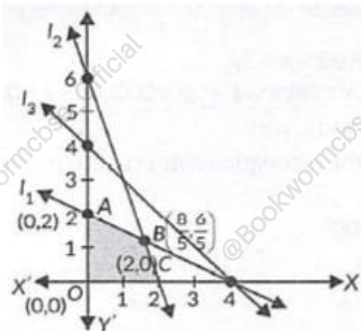
From table, maximum value of $z = 1$.

11.

Soln. Let

$l_1: 2x + 4y = 8$, $l_2: 3x + y = 6$, $l_3: x + y = 4$; $x = 0$, $y = 0$

Solving l_1 and l_2 we get $B\left(\frac{8}{5}, \frac{6}{5}\right)$



Shaded portion OADC is the feasible region, where coordinates of the corner points are $O(0, 0)$, $A(0, 2)$, $D(1.5, 1)$, $C(2, 0)$.

$$B\left(\frac{8}{5}, \frac{6}{5}\right), C(2, 0)$$

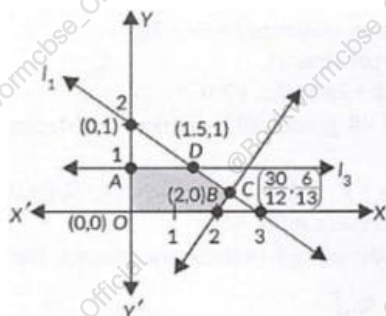
The value of objective function at these points are:

Corner Points	Value of the objective function $z = 2x + 5y$
$O(0, 0)$	$2 \times 0 + 5 \times 0 = 0$
$A(0, 2)$	$2 \times 0 + 5 \times 2 = 10$ (Maximum)
$B\left(\frac{8}{5}, \frac{6}{5}\right)$	$2 \times \frac{8}{5} + 5 \times \frac{6}{5} = 9.2$
$C(2, 0)$	$2 \times 2 + 5 \times 0 = 4$

$\therefore$ The maximum value of z is 10, which is at $A(0, 2)$.

12.

Soln. Let $l_1: 2x + 3y = 6$, $l_2: 3x - 2y = 6$, $l_3: y = 1$; $x = 0$, $y = 0$



Solving l_1 and l_3 , we get $D(1.5, 1)$

Solving l_1 and l_2 , we get $C\left(\frac{30}{13}, \frac{6}{13}\right)$

Shaded portion OADC is the feasible region, where coordinates of the corner points are $O(0, 0)$, $A(0, 1)$, $D(1.5, 1)$, $C\left(\frac{30}{13}, \frac{6}{13}\right)$, $B(2, 0)$.

The value of the objective function at these points are:

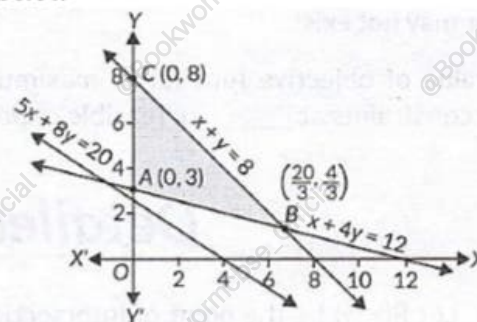
Corner Points	Value of the objective function $z = 8x + 9y$
$O(0, 0)$	$8 \times 0 + 9 \times 0 = 0$
$A(0, 1)$	$8 \times 0 + 9 \times 1 = 9$
$D(1.5, 1)$	$8 \times 1.5 + 9 \times 1 = 21$
$C\left(\frac{30}{13}, \frac{6}{13}\right)$	$8 \times \frac{30}{13} + 9 \times \frac{6}{13} = 22.6$ (Maximum)
$B(2, 0)$	$8 \times 2 + 9 \times 0 = 16$

The maximum value of z is 22.6, which is at

$$C\left(\frac{30}{13}, \frac{6}{13}\right).$$

13. Soln. Converting the given inequation into equation, we get $x + y = 8$, $x + 4y = 12$, $5x + 8y = 20$

Let us draw the graph of these equations as shown below



The point of intersection of the lines $x + 4y = 12$ and

$$x + y = 8 \text{ is } B = \left(\frac{20}{3}, \frac{4}{3}\right)$$

We have, corner points $A(0, 3)$, $B\left(\frac{20}{3}, \frac{4}{3}\right)$ and $C(0, 8)$.

Now, $Z = 30x + 20y$

$$\therefore Z(0, 3) = 30(0) + 20(3) = 60$$

$$Z\left(\frac{20}{3}, \frac{4}{3}\right) = 30\left(\frac{20}{3}\right) + 20\left(\frac{4}{3}\right) = 226.6$$

$$Z(0, 8) = 30(0) + 20(8) = 160$$

$\therefore$ Minimum value of Z is 60 which is attained at point $A(0, 3)$.

New

SURE SHOT QUESTIONS 2026

Chapter – 13 (Questions)

Probability

Questions

1. A purse contains 3 silver and 6 copper coins and a second purse contains 4 silver and 3 copper coins. If a coin is drawn at random from one of the two purses, find the probability that it is a silver coin.
2. A die, whose faces marked 1, 2, 3 in red and 4, 5, 6 in green, is tossed. Let A be the event "number obtained is even" and B be the event "number obtained is red". Find if A and B are independent events.
3. Three persons A, B and C apply for a job of Manager in a private Company. Chances of their selection (A, B and C) are in the ratio 1:2:4, the probabilities that A, B and C can introduce changes to improve to improve profits of the company are 0.8, 0.5 and 0.3 respectively. If the change does not take place, find the probability that it is due to the appointment of C.
4. A and B throw a pair of dice alternately. A wins the game if he gets a total of 7 and B wins the game if he gets a total of 10. If A starts the game, then find the probability that B wins.
5. If $P(E) = \frac{7}{13}$, $P(F) = \frac{9}{13}$ and $P(\bar{E}/\bar{F}) = \frac{4}{13}$, then evaluate :
 - (i) $P(\bar{E}/F)$
 - (ii) $P(E/F)$
6. A DIE IS ROLLED. If $E = \{1, 3, 5\}$, $F = \{2, 3\}$ and $G = \{2, 3, 4, 5\}$, find (a) $P(E \cup F)/G$
(b) $P(E \cap F)/G$
7. If $P(F) = 1/5$ and $P(E) = 1/2$, E and F are independent events. Find $P(\overline{E} \cap \overline{F})$.
8. A card from a well shuffled deck of 52 playing cards is lost. From the remaining cards of the pack, a card is drawn at random and is found to be a king. Find the probability of the lost card being a King.
9. A biased die is twice as likely to show an even number as an odd number. If such a die is thrown twice, find the probability distribution of the number of sixes. Also, find the mean of the distribution.
10. The probability that A hits the target is $\frac{1}{3}$ and the probability that B hits it, is $\frac{2}{5}$. If both try to hit the target independently, find the probability that the target is hit.
11. There are two bags. Bag contains 1 red and 3 white balls, and Bag II contains 3 red and 5 white balls. A bag is selected at random and a ball is drawn from it. Find the probability that the ball so drawn is red in colour.
12. A problem in mathematics is given to 4 students A, B, C, D. Their chances of solving the problem, respectively are, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ and $\frac{2}{3}$. What is the probability that (i) the problem will be solved? (ii) at most one of them solve the problem?
13. A speaks truth in 75% of the cases, while B in 90% of the cases. In what percent of cases are they likely to contradict each other in stating the same fact? Do you think that statement of B is always true?

14. A bag contains $(2n + 1)$ coins. It is known that $(n - 1)$ of these coins have a head on both sides, whereas the rest of the coins are fair. A coin is picked up at random from the bag and is tossed. If the probability that the toss results in a head is $\frac{31}{42}$, determine the value of n .
15. A and B throw a pair of dice alternatively, till one of them gets a total of 10 and wins the game. Find their respective probabilities of winning, if A starts first.
16. A bag A contains 4 black and 6 red balls and bag B contains 7 black and 3 red balls. A die is thrown. If 1 or 2 appears on it, then bag is chosen, otherwise bag. If two balls are drawn at random (without replacement) from the selected bag, find the probability of one of them being red and other black.
17. Of the students in a school, it is known that 30% have 100% attendance and 70% students are irregular. Previous year results report that 70% of all students who have 100% attendance attain A grade and 10% irregular students attain A grade in their annual examination. At the end of the year, one student is chosen at random from the school and he was found to have an A grade. What is the probability that the student has 100% attendance?
18. A bag contains 3 red and 7 black balls. Two balls are selected at random one-by-one without replacement. If the second selected ball happens to be red, what is the probability that the first selected ball is also red?
19. Bag A contains 5 black and 3 red balls while bag B contains 4 black and 4 red balls. Two balls are transferred at random from bag A to bag B and then a ball is drawn from bag B at random. If the ball drawn from bag B is found to be red, find the probability that two red balls were transferred from A to B.
20. In a factory which manufactures bolts, machines A, B and C manufacture respectively 30%, 50% and 20% of the bolts. Of their outputs, 3%, 4% and 1% respectively are defective bolts. A bolt is drawn at random from the product and is found to be defective. Find the probability that this is not manufactured by machine B.
21. Three machines E_1, E_2, E_3 in a certain factory produces 50%, 25% and 25% respectively, of the total daily output of electric tubes. It is known that 4% of the tube produced on each of machines E_1 and E_2 are defective and that 5% of those produced on E_3 are defective. If one tube is picked up at random from a day's production, calculate the probability that it is defective.
22. A card from a pack of 52 playing cards is lost. From the remaining cards of the pack, three cards are drawn at random (without replacement) and are found to be all spades. Find the probability of the lost card being spade.
23. Of the students in a college, it is known that 60% reside in hostel and 40% are day scholars (not residing in hostel). Previous year results report that 30% of all students who reside in hostel attain A grade and 20% of day scholars attain A grade in their annual examination. At the end of the year, one student is chosen at random from the college and he has an 'A' grade. What is the probability that the student is a hosteler.
24. Assume that each born child is equally likely to be a boy or a girl. If a family has two children, what is the conditional probability that both are girls? Given that
- the youngest is a girl,
 - atleast one is a girl.

New

SURE SHOT QUESTIONS 2026

Chapter – 13 (Solutions)

Probability

Questions

1. .

Soln.

Let E_1 , E_2 and A denote the events defined as follow:

E_1 = selecting a purse 1

E_2 = selecting a purse 2

A = drawing a silver coin

Since one of two purses is selected randomly

$$\therefore P(E_1) = \frac{1}{2} \text{ and } P(E_2) = \frac{1}{2}$$

$$P(A/E_1) = \frac{3}{9} = \frac{1}{3} \text{ and } P(A/E_2) = \frac{4}{7}$$

Now,

Using the total law of probability, we have

Required probability,

$$P(A) = P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)$$

$$\Rightarrow P(A) = \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{4}{7} = \frac{1}{6} + \frac{2}{7} = \frac{19}{42}$$

2. .

Soln. Since, A be the event of number obtained is even

Then, $A = \{2, 4, 6\}$

And B be the event of number obtained is red then,

$$B = \{1, 2, 3\}$$

$$\therefore A \cap B = \{2\}$$

So,

$$P(A) = \frac{3}{6} = \frac{1}{2}; P(B) = \frac{3}{6} = \frac{1}{2}; P(A \cap B) = \frac{1}{6}$$

$$P(A \cap B) \neq P(A) \cdot P(B)$$

$$\frac{1}{6} \neq \frac{1}{4}$$

Hence, the events A and B are not independent events.

3. .

Soln. Let E_1 , E_2 and E_3 be the events denoting the selection of A , B and C as managers respectively.

$$P(E_1) = \text{Probability of selection of } A = \frac{1}{7}$$

$$P(E_2) = \text{Probability of selection of } B = \frac{2}{7}$$

$$P(E_3) = \text{Probability of selection of } C = \frac{4}{7}$$

Let A be the event denoting the change not taking place.

$P(A/E_1)$ = Probability that A does not introduce change = 0.2

$P(A/E_2)$ = Probability that B does not introduce change = 0.5

$P(A/E_3)$ = Probability that C does not introduce change = 0.7

$\therefore$ Required probability = $P(E_3/A)$

By Bayes' theorem, we have

$$P(E_3/A) = \frac{P(E_3)P(A/E_3)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2) + P(E_3)P(A/E_3)}$$

$$= \frac{\frac{4}{7} \times 0.7}{\frac{1}{7} \times 0.2 + \frac{2}{7} \times 0.5 + \frac{4}{7} \times 0.7}$$

$$= \frac{2.8}{0.2 + 1 + 2.8}$$

$$= \frac{2.8}{4} = 0.7$$

4. .

Soln. Total of 7 on the dice can be obtained in the following ways:

$$(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)$$

$$\text{Probability of getting a total of } 7 = \frac{6}{36} = \frac{1}{6}$$

$$\text{Probability of not getting a total of } 7 = 1 - \frac{1}{6} = \frac{5}{6}$$

Total of 10 on the dice can be obtained in the following ways:

$$(4, 6), (6, 4), (5, 5)$$

Probability of getting a total of 10

$$= \frac{3}{36} = \frac{1}{12}$$

Probability of not getting a total of 10

$$= 1 - \frac{1}{12} = \frac{11}{12}$$

Let E and F be the two events, defined as follows:

E = Getting a total of 7 in a single throw of a dice

F = Getting a total of 10 in a single throw of a dice

$$P(E) = \frac{1}{6}, P(\bar{E}) = \frac{5}{6}$$

$$P(F) = \frac{1}{12}, P(\bar{F}) = \frac{11}{12}$$

A wins if he gets a total of 7 in 1st, 3rd or 5th throws.

Probability of A getting a total of 7 in the 1st throw

$$= \frac{1}{6}$$

A will get the 3rd throw if he fails in the 1st throw and B fails in the 2nd throw.

Probability of A getting a total of 7 in the 3rd throw

$$= P(\bar{E}) P(\bar{F}) P(E) = \frac{5}{6} \times \frac{11}{12} \times \frac{1}{6}$$

Similarly, probability of getting a total of 7 in the 5th

grow = $P(\bar{E}) P(\bar{F}) P(\bar{E}) P(\bar{E}) P(E)$

$$= \frac{5}{6} \times \frac{11}{12} \times \frac{5}{6} \times \frac{11}{12} \times \frac{1}{6} \text{ and so on}$$

Probability of winning of A

$$= \frac{1}{6} + \left(\frac{5}{6} \times \frac{11}{12} \times \frac{1}{6} \right) + \left(\frac{5}{6} \times \frac{11}{12} \times \frac{5}{6} \times \frac{11}{12} \times \frac{1}{6} \right) + \dots$$

$$= \frac{\frac{1}{6}}{1 - \frac{5}{6} \times \frac{11}{12}} = \frac{12}{17}$$

∴ Probability of winning of B = 1 - Probability of winning of A

$$= 1 - \frac{12}{17} = \frac{5}{17}$$

5. .
(i)

Soln. $P(E) = \frac{7}{13}$

$$P(F) = \frac{9}{13}$$

And $P(E \cap F) = \frac{4}{13}$ (Given)

$$(i) \quad P(\bar{E}/F) = \frac{P(\bar{E} \cap F)}{P(F)} \\ = \frac{P(F) - P(E \cap F)}{P(F)}$$

$$= 1 - \frac{P(E \cap F)}{P(F)} = 1 - \frac{\frac{4}{13}}{\frac{9}{13}}$$

$$\text{Or } P(\bar{E}/F) = 1 - \frac{4}{9} = \frac{5}{9}$$

$$(ii) \quad P(\bar{E}/\bar{F}) = \frac{P(\bar{E} \cap \bar{F})}{P(\bar{F})}$$

$$= \frac{P(\overline{E \cup F})}{P(\bar{F})}$$

$$= \frac{1 - P(E \cup F)}{1 - P(F)}$$

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

$$= \frac{7}{13} + \frac{9}{13} - \frac{4}{13} \\ = \frac{12}{13}$$

$$P(E/F) = \frac{1 - \frac{12}{13}}{1 - \frac{9}{13}}$$

$$= \frac{\frac{1}{13}}{\frac{4}{13}} = \frac{1}{4}$$

6. .

Soln. Given, E = (1, 3, 5), F = (2, 3), G = (2, 3, 4, 5)

$$P(E) = \frac{3}{6}$$

$$P(F) = \frac{2}{6}$$

$$P(G) = \frac{4}{6} = \frac{2}{3}$$

$$P(E \cup F) = \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{2}{3}$$

$$P(E \cap F) = \frac{1}{6}$$

$$(i) \quad (E \cup F) \cap G = (2, 3, 5)$$

$$P[(E \cup F) \cap G] = \frac{3}{6}$$

$$P[(E \cup F)/G] = \frac{P[(E \cup F) \cap G]}{P(G)}$$

$$= \frac{\frac{3}{6}}{\frac{2}{3}} = \frac{3}{4}$$

$$(ii) \quad (E \cap F) \cap G = 3$$

$$P[(E \cap F) \cap G] = \frac{1}{6}$$

$$P[(E \cap F)/G] = \frac{P[(E \cap F) \cap G]}{P(G)}$$

$$= \frac{\frac{1}{6}}{\frac{2}{3}} = \frac{1}{4}$$

7. .

Soln. $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

If E and F are independent, then

$$= P(E \cap F) = P(E) \times P(F)$$

$$P(E \cup F) = P(E) + P(F) - P(E) \times P(F)$$

$$P(E \cup F) = \frac{1}{2} + \frac{1}{5} - \frac{1}{2} \times \frac{1}{5}$$

$$= \frac{1}{2} + \frac{1}{5} - \frac{1}{10} = \frac{5+2-1}{10}$$

$$= \frac{6}{10} = \frac{3}{5}$$

$$P(E \cup F) = 1 - P(\overline{E \cup F})$$

$$= 1 - \frac{3}{5} = \frac{2}{5}$$

8.

Ans. Let E_1 = Lost card is king

E_2 = Lost card is non king

A = Card drawn is king

$$P(E_1) = \frac{4}{52} = \frac{1}{13}, P(E_2) = \frac{12}{13}, P(A|E_1) = \frac{3}{51}, P(A|E_2) = \frac{4}{51}$$

$$P(E_1|A) = \frac{P(A|E_1)P(E_1)}{P(A|E_1)P(E_1) + P(A|E_2)P(E_2)}$$

$$\Rightarrow P(E_1|A) = \frac{\frac{3}{51} \times \frac{1}{13}}{\frac{3}{51} \times \frac{1}{13} + \frac{4}{51} \times \frac{12}{13}} = \frac{3}{3+48} = \frac{3}{51} = \frac{1}{17}$$

9.

Ans.

Let $P(1) = P(3) = P(5) = p$, so $P(2) = P(4) = P(6) = 2p$

As $P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1 \Rightarrow 9p = 1 \Rightarrow p = \frac{1}{9}$

$P(\text{Getting } 6) = \frac{2}{9}, P(\text{Not getting six}) = \frac{7}{9}$

Let X represents the number of sixes.

Possible values of X are 0, 1 or 2

$$\text{Now, } P(X=0) = \frac{7}{9} \times \frac{7}{9} = \frac{49}{81}, P(X=1) = 2 \times \frac{7}{9} \times \frac{2}{9} = \frac{28}{81}$$

$$P(X=2) = \frac{2}{9} \times \frac{2}{9} = \frac{4}{81}$$

Required probability distribution of number of sixes is

X	0	1	2
P(X)	$\frac{49}{81}$	$\frac{28}{81}$	$\frac{4}{81}$

$$\therefore \text{Mean of } X = \sum X P(X) = 0 + \frac{28}{81} + \frac{8}{81} = \frac{36}{81} = \frac{4}{9}$$

10.

Ans.

11.

Soln. Let E_1 be the event that bag I is chosen, E_2 be the event that bag II is chosen and A be the event that red ball is drawn.

Clearly, E_1 and E_2 are mutually exclusive and exhaustive events.

Since, one of the bag is chosen at random

$$\therefore P(E_1) = \frac{1}{2} \text{ and } P(E_2) = \frac{1}{2}$$

$$P(A|E_1) = \frac{1}{4} \text{ and } P(A|E_2) = \frac{3}{8}$$

By using law of total probability, we get

$$P(A) = P(E_1)P(A|E_1) + P(E_2)P(A|E_2)$$

$$= \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{3}{8} = \frac{1}{8} + \frac{3}{16} = \frac{5}{16}$$

12.

Soln. Let

E be the event = A solves the problem

F be the event = B solves the problem

G be the event = C solves the problem

H be the event = D solves the problem

$$P(E) = \frac{1}{3}, P(\bar{E}) = \frac{2}{3}$$

$$P(F) = \frac{1}{2}, P(\bar{F}) = \frac{1}{2}$$

$$P(G) = \frac{1}{5} P(\bar{G}) = \frac{4}{5}$$

$$P(H) = \frac{2}{3} P(\bar{H}) = \frac{1}{3}$$

(i) The probability = $P(E \cup F \cup G \cup H)$

$$= 1 - P(\bar{E} \cap \bar{F} \cap \bar{G} \cap \bar{H})$$

$$= 1 - P(\bar{E}) \times P(\bar{F}) \times P(\bar{G}) \times P(\bar{H})$$

$$= 1 - \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{1}{3}$$

$$= \frac{13}{15}$$

(ii) The required probability = $P(\bar{E}) \times P(\bar{F}) \times P(\bar{G})$

$$\times P(\bar{H}) + P(E) \times P(\bar{F}) \times P(\bar{G}) \times P(\bar{H}) + P(\bar{E}) \times P(F) \times P(\bar{G})$$

$$\times P(\bar{H}) + P(\bar{E}) \times P(\bar{F}) \times P(G) \times P(\bar{H}) + P(\bar{E}) \times P(F) \times P(G) \times P(H)$$

$$= \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{1}{3} + \frac{1}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{1}{3} + \frac{2}{3} \times \frac{1}{4} \times \frac{4}{5} \times \frac{1}{3}$$

$$+ \frac{2}{3} \times \frac{3}{4} \times \frac{1}{5} \times \frac{1}{3} + \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{2}{3}$$

$$= \frac{2}{15} + \frac{1}{15} + \frac{2}{45} + \frac{1}{30} + \frac{4}{15}$$

$$= \frac{7}{15} + \frac{1}{30} + \frac{2}{45}$$

$$= \frac{42+3+4}{90} = \frac{49}{90}$$

13.

Soln. Let A and B denote 'A speaks the truth' and 'B speaks the truth' respectively.

$$P(A) = \frac{75}{100}$$

Now

$$P(\bar{A}) = 1 - P(A)$$

$$= 1 - \frac{75}{100} = \frac{25}{100}$$

$$P(B) = \frac{90}{100}$$

$$P(\bar{B}) = 1 - P(B)$$

$$= 1 - \frac{90}{100} = \frac{10}{100}$$

Required probability = $P(A) P(\bar{B}) + P(\bar{A}) P(B)$

$$= \frac{75}{100} \times \frac{10}{100} + \frac{25}{100} \times \frac{90}{100}$$

$$= \frac{30}{100} = 30\%$$

Hence, they are likely to contradict each other in 30% of the cases in stating the same fact. Though B speaks truth in 90% of the cases but he also lies in 10% of the cases. So, his statement is not always true.

14.

Soln. No. of coins with head on both sides = $(n - 1)$

No. of fair coins = $(n + 2)$

Let event

E_1 = Picking a coin with head on both sides

E_2 = Picking a fair coin

A : getting a head on tossing the coin

$$P(E_1) = \frac{n-1}{2n+1},$$

$$P(E_2) = \frac{n+2}{2n+1}$$

$$P(E_1) = 1, P(A/E_2) = 1/2$$

$$P(A) = P(E_1) P(A/E_1) + P(E_2) P(A/E_2)$$

$$= \frac{n-1}{2n+1} \cdot 1 + \frac{n+2}{2n+1} \cdot \frac{1}{2}$$

$$= \frac{3n}{2(2n+1)}$$

$$\text{or } \frac{3n}{2(2n+1)} = \frac{31}{42} \text{ or } n = 31.$$

15.

Soln. Let A_i and B_i be the events of throwing 10 by A and B in the respectively i^{th} turn, then

$$P(A_i) = P(B_i) = \frac{1}{12}$$

$$\text{And } P(\bar{A}_i) = P(\bar{B}_i) = \frac{11}{12}$$

Probability of winning A, when A starts first

$$= \frac{1}{12} + \left(\frac{11}{12}\right)^2 \frac{1}{12} + \left(\frac{11}{12}\right)^4 \frac{1}{12} + \dots$$

$$= \frac{1}{12} \left[1 + \left(\frac{11}{12}\right)^2 + \left(\frac{11}{12}\right)^4 + \dots \right]$$

Probability of winning of

$$B = 1 - P(A) = 1 - \frac{12}{23} = \frac{11}{23}$$

16.

Soln. Let E_1 = Selecting bag A

And E_2 = Selecting bag B

$$\therefore P(E_1) = \frac{1}{3}$$

$$\text{and } P(E_2) = \frac{2}{3}$$

Let A = Getting one red and one black ball

$$\therefore P(A/E_1) = \frac{{}^4C_1 \cdot {}^6C_1}{{}^{10}C_2} = \frac{8}{15}$$

$$(A/E_2) = \frac{{}^7C_1 \cdot {}^3C_1}{{}^{10}C_2} = \frac{7}{15}$$

And

$$P(A) = P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)$$

$$= \frac{1}{3} \cdot \frac{8}{15} + \frac{2}{3} \cdot \frac{7}{15}$$

$$= \frac{22}{45}$$

17. .

Soln. Let E_1 : Selecting a student with 100% attendance

E_2 : Selecting a student who is not regular

A : selected student attains A grade.

$$P(E_1) = \frac{30}{100} \text{ and } P(E_2) = \frac{70}{100}$$

$$P(A/E_1) = \frac{70}{100} \text{ and } P(A/E_2) = \frac{10}{100}$$

$$P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)}$$

$$= \frac{\frac{30}{100} \times \frac{70}{100}}{\frac{30}{100} \times \frac{70}{100} + \frac{70}{100} \times \frac{10}{100}} = \frac{3}{4}$$

18. .

Soln. Let A be the event of drawing a red ball in first draw and B be the event of drawing a red ball in second draw.

$$\therefore P(A) = \frac{{}^3C_1}{{}^{10}C_1} = \frac{3}{10}$$

Now, $P(B/A)$ = Probability of drawing a red ball in the second draw, when a red ball already has been drawn

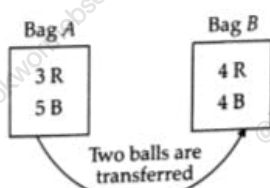
$$\text{in the first draw} = \frac{{}^2C_1}{{}^9C_1} = \frac{2}{9}$$

$\therefore$ The required probability = $P(A \cap B)$

$$= P(A) \cdot P(B/A) = \frac{3}{10} \times \frac{2}{9} = \frac{1}{15}$$

19. .

Soln.



Let us define the following events:

E_1 = one red and one black ball is transferred

E_2 = two red balls are transferred

E_3 = two black balls are transferred

E = drawn ball is red.

Then,

$$P(E_1) = \frac{{}^3C_1 \times {}^5C_1}{{}^8C_2} = \frac{3 \times 5}{28} = \frac{15}{28}$$

$$P(E_2) = \frac{{}^3C_2}{{}^8C_2} = \frac{3}{28}$$

$$P(E_3) = \frac{{}^5C_2}{{}^8C_2} = \frac{10}{28}$$

$$P\left(\frac{E}{E_1}\right) = \frac{5}{10}, P\left(\frac{E}{E_2}\right) = \frac{6}{10}, P\left(\frac{E}{E_3}\right) = \frac{4}{10}$$

Now, required probability = $P\left(\frac{E_2}{E}\right)$

$$= \frac{P(E_2) \cdot P\left(\frac{E}{E_2}\right)}{P(E_1) \cdot P\left(\frac{E}{E_1}\right) + P(E_2) \cdot P\left(\frac{E}{E_2}\right) + P(E_3) \cdot P\left(\frac{E}{E_3}\right)}$$

$$= \frac{\frac{3}{28} \cdot \frac{6}{10}}{\frac{15}{28} \cdot \frac{5}{10} + \frac{3}{28} \cdot \frac{6}{10} + \frac{10}{28} \cdot \frac{4}{10}}$$

$$= \frac{18}{75 + 18 + 40} = \frac{18}{133}$$

20. .

Soln. Let E_1 : Event that the selected bolt is manufactured by machine A,

E_2 : Event that the selected bolt is manufactured by machine B,

E_3 : Event that the selected bolt is manufactured by machine C,

And E : Event that the selected bolt is defective.

Then, we have

$$P(E_1) = 30\% = \frac{30}{100}$$

$$P(E_2) = 50\% = \frac{50}{100}$$

$$\text{and } P(E_3) = 20\% = \frac{20}{100}$$

Also, given that 3%, 4% and 1% bolts manufactured by machines A, B and C respectively are defective.

So,

$$P\left(\frac{E}{E_1}\right) = 3\% = \frac{3}{100}$$

$$P\left(\frac{E}{E_2}\right) = 4\% = \frac{4}{100}$$

$$P\left(\frac{E}{E_3}\right) = 1\% = \frac{1}{100}$$

Now, the probability that selected bolt which is defective, is manufactured by machine B

$$P\left(\frac{E_2}{E}\right) = \frac{P(E_2) \cdot P\left(\frac{E}{E_2}\right)}{P(E_1) \cdot P\left(\frac{E}{E_1}\right) + P(E_2) \cdot P\left(\frac{E}{E_2}\right) + P(E_3) \cdot P\left(\frac{E}{E_3}\right)}$$

$$= \frac{\frac{50}{100} \times \frac{4}{100}}{\frac{30}{100} \times \frac{3}{100} + \frac{50}{100} \times \frac{4}{100} + \frac{20}{100} \times \frac{1}{100}}$$

$$= \frac{200}{90 + 200 + 20} = \frac{200}{310}$$

∴ The probability that selected bolt which is defective, is not manufactured by machine B

$$= 1 - P\left(\frac{E_2}{E}\right) = 1 - \frac{200}{310} = \frac{110}{310} = \frac{11}{31}$$

21. .

Soln. Let A be the event that the picked up tube is defective.

Let A_1, A_2, A_3 be events such that

A_1 = event of producing tube by machine E_1

A_2 = event of producing tube by machine E_2

A_3 = event of producing tube by machine E_3

$$P(A_1) = \frac{50}{100} = \frac{1}{2}, \quad P(A_2) = \frac{25}{100} = \frac{1}{4}, \quad P(A_3) = \frac{25}{100} = \frac{1}{4}$$

$$P\left(\frac{A}{A_1}\right) = \frac{4}{100} = \frac{1}{25}$$

Also,

$$P\left(\frac{A}{A_2}\right) = \frac{4}{100} = \frac{1}{25} \quad \text{and} \quad P\left(\frac{A}{A_3}\right) = \frac{5}{100} = \frac{1}{20}$$

Now, $P(A)$ is required.

From concept of total probability,

$$P(A) = P(A_1) \cdot P\left(\frac{A}{A_1}\right) + P(A_2) \cdot P\left(\frac{A}{A_2}\right) + P(A_3) \cdot P\left(\frac{A}{A_3}\right)$$

$$= \frac{1}{2} \times \frac{1}{25} + \frac{1}{4} \times \frac{1}{25} + \frac{1}{4} \times \frac{1}{20} = \frac{1}{50} + \frac{1}{100} + \frac{1}{80}$$

$$= \frac{8+4+5}{400} = \frac{17}{400} = 0.0425$$

22. .

Soln. Let, E_1 : Event that lost card is a spade

E_2 : Event that lost card is a not spade

A : Event that three spades are drawn without replacement from 51 cards

$$P(E_1) = \frac{13}{52} = \frac{1}{4}, \quad P(E_2) = 1 - \frac{1}{4} = \frac{3}{4}$$

$$P(A/E_1) = \frac{{}^{12}C_3}{{}^{51}C_3}, \quad P(A/E_2) = \frac{{}^{13}C_3}{{}^{51}C_3}$$

$$P(E_1/A) = \frac{\frac{1}{4} \cdot \frac{{}^{12}C_3}{{}^{51}C_3}}{\frac{1}{4} \cdot \frac{{}^{12}C_3}{{}^{51}C_3} + \frac{3}{4} \cdot \frac{{}^{13}C_3}{{}^{51}C_3}}$$

$$= \frac{10}{49}$$

23. .

Soln. Let E_1 : selected student is a hosteler

E_2 : selected student is a day scholar

A : selected student attain 'A' grade in exam.

$$P(E_1) = \frac{60}{100}, \quad P(E_2) = \frac{40}{100}$$

$$P(A/E_1) = \frac{30}{100}, \quad P(A/E_2) = \frac{20}{100}$$

$$P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)}$$

$$= \frac{\frac{60}{100} \cdot \frac{30}{100}}{\frac{60}{100} \cdot \frac{30}{100} + \frac{40}{100} \cdot \frac{20}{100}} = \frac{9}{13}$$

24. .

(i)

Soln.

Let G_i ($i = 1, 2$) and B_i ($i = 1, 2$) denote the i^{th} child is a girl or a boy respectively

Then sample space is,

$$S = \{G_1G_2, G_1B_2, B_1G_2, B_1B_2\}$$

Let A be the event that both children are girls, B be the event that the youngest child is a girl and C be the event that at least one of the children is a girl.

$$\text{Then } A = \{G_1G_2\}, B = \{G_1G_2, B_1G_2\}$$

$$\text{And } C = \{B_1G_2, G_1G_2, G_1B_2\}$$

$$\Rightarrow A \cap B = \{G_1G_2\} \text{ and } A \cap C = \{G_1G_2\}$$

(i) Required probability = $P(A/B) =$

$$\frac{P(A \cap B)}{P(B)} = \frac{1/4}{2/4} = \frac{1}{2}$$

$$\text{Required probability} = P(A/C) = \frac{P(A \cap C)}{P(C)} = \frac{1/4}{3/4} = \frac{1}{3}$$

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